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INTRODUCTION

• Processing of Seismic Data • Inversion of Seismic Data • Interpretation of Seismic Data
• From Seismic Exploration to Seismic Monitoring

The Classical Greeks had a love for wisdom —
It came down to us as philo·sophia.
And I have a passion for the seismic method —
Let this be an ode to philo·seismos.
O how sweet it is —
Listening to the echos from the earth.

The seismic method has three principal applications:

(a) Delineation of near-surface geology for engineering studies, and coal and mineral exploration within a depth of up to 1 km: The seismic method applied to the near-surface studies is known as *engineering seismology.*
(b) Hydrocarbon exploration and development within a depth of up to 10 km: The seismic method applied to the exploration and development of oil and gas fields is known as *exploration seismology.*
(c) Investigation of the earth's crustal structure within a depth of up to 100 km: The seismic method applied to the crustal and earthquake studies is known as *earthquake seismology.*

This book is devoted to application of the reflection seismic method to the exploration and development of oil and gas fields.

Conventional processing of reflection seismic data yields an earth image represented by a seismic section which usually is displayed in time. Figure I-1 shows a seismic section from the Gulf of Mexico, nearly 40 km in length. Approximate depth scale indicates a sedimentary section of interbedded sands and shales down to 8 km. Note from this earth image a salt sill embedded in the sedimentary sequence. This allocthonous salt sill has a rugose top and a relatively smooth base. Note the folding and faulting of the sedimentary section above the salt.

The reflection seismic method has been used to delineate near-surface geology for the purpose of coal and mineral exploration and engineering studies, especially in recent years with increasing acceptance. Figure I-2a shows a seismic section along a 500-m traverse across a bedrock valley with steep flanks. The lithologic column based on borehole data indicates a sedimentary sequence of clay, sand, and gravel deposited within the valley. The bedrock is approximately 15 m below the surface at the fringes of the valley and 65 m below the surface at the bottom of the valley. The strong reflection at the sediment-bedrock boundary is a result of the contrast between the low-velocity sediments above and the high-velocity Precambrian quartz pegmatite below.

The reflection seismic method also has been used to delineate the crustal structure down to the Moho

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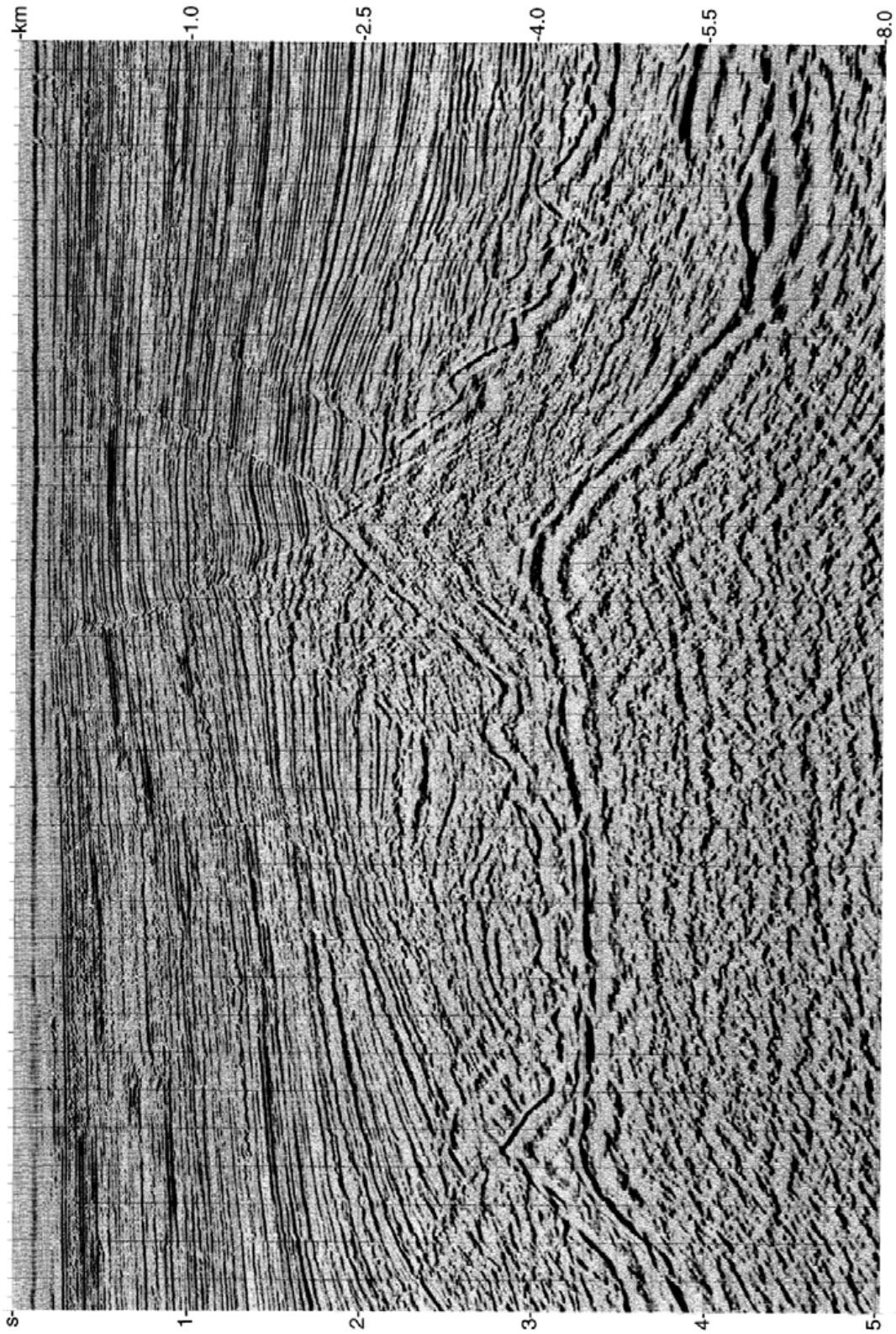

FIG. I-1. A seismic section from the Gulf of Mexico used for hydrocarbon exploration. (Data courtesy Schlumberger Geco-Prakla and TGS.)

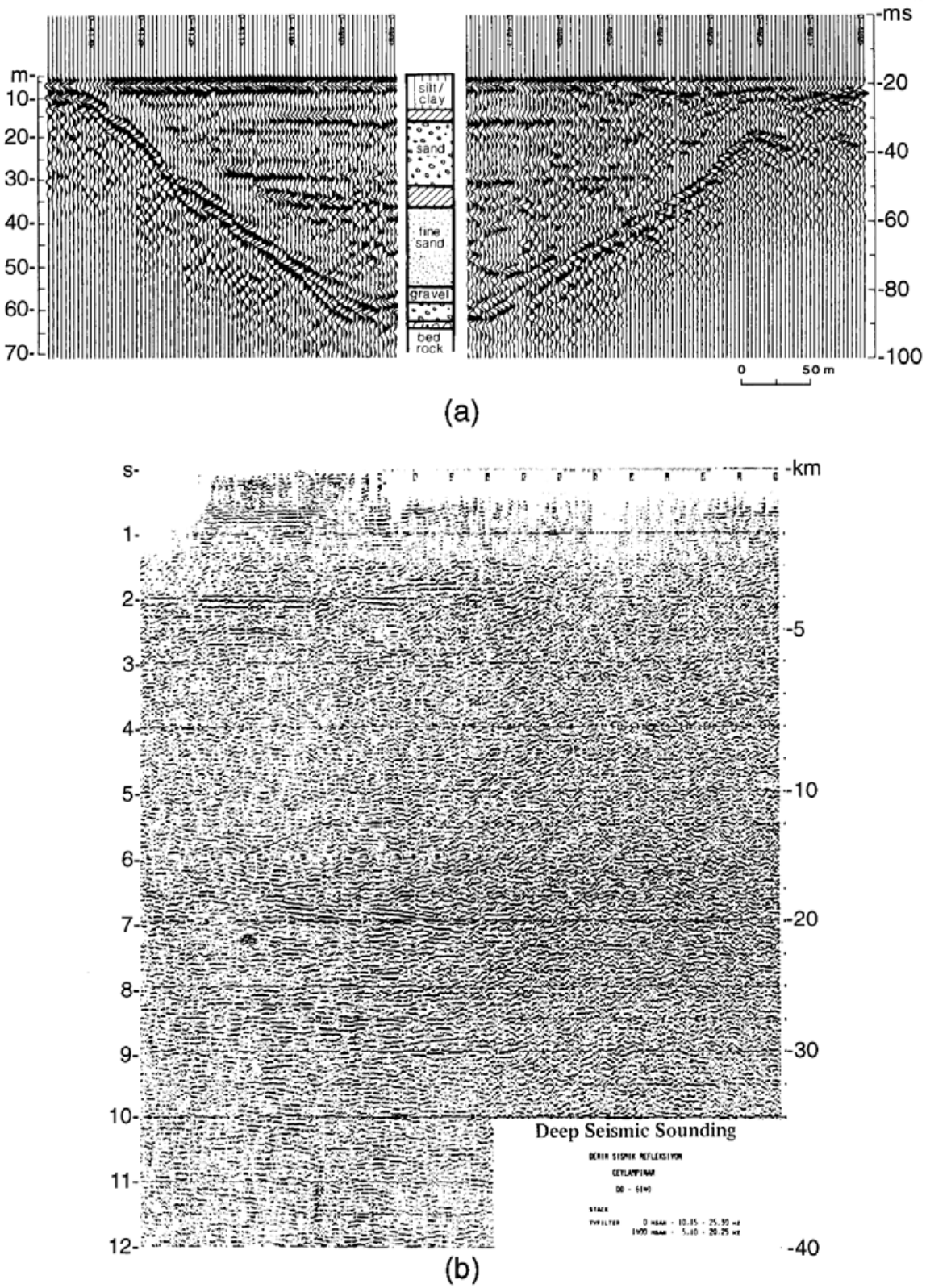

FIG. I-2. (a) A shallow reflection seismic section from Ontario (Pullan and Hunter, 1990), and (b) a deep reflection seismic section from southeast Turkey (Yilmaz, 1976).

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discontinuity and below. Figure I-2b shows a seismic section recorded on land along a 15-km traverse. Based on regional control, it is known that the section consists of sediments down to about 4 km. The reflection event at 6.5-7 s, which corresponds to a depth range of 15-20 km, can be postulated as the crystalline basement. The group of reflections between 8-10 s, which corresponds to a depth range of 25-35 km, represents a transition zone in the lower crust — most likely, the Moho discontinuity, itself.

Common-midpoint (CMP) recording is the most widely used seismic data acquisition technique. By providing redundancy, measured as the fold of coverage in the seismic experiment, it improves signal quality. Figure I-3 shows seismic data collected along the same traverse in 1965 with single-fold coverage and in 1995 with twelve-fold coverage. These two different vintages of data have been subjected to different treatments in processing; nevertheless, the fold of coverage has caused the most difference in the signal level of the final sections.

Seismic data processing strategies and results are strongly affected by field acquisition parameters. Additionally, surface conditions have a significant impact on the quality of data collected in the field. Part of the seismic section shown in Figure I-4 between midpoints A and B is over an area covered with karstic limestone. Note the continuous reflections between 2 and 3 s outside the limestone-covered zone. These reflections abruptly disappear under the problem zone in the middle. The lack of events is not the result of a subsurface void of reflectors. Rather, it is caused by a low signal-to-noise (S/N) ratio resulting from energy scattering and absorption in the highly porous surface limestone.

Surface conditions also have an influence on how much energy from a given source type can penetrate into the subsurface. Figure I-5 shows a seismic section along a traverse over a karstic topography with a highly weathered near-surface. In data acquisition, surface charges have been used to the right of midpoint A, and charges have been placed in holes to the left of midpoint A. In the absence of source coupling using surface charges, there is very little energy that can penetrate into the subsurface through the weathered near-surface layer. As a result, note the lack of coherent reflections to the right of midpoint A. On the other hand, improved source coupling using downhole charges has resulted in better penetration of the energy into the subsurface in the remainder of the section.

Besides surface conditions, environmental and demographic restrictions can have a significant impact on field data quality. The part of the seismic section shown in Figure I-6 between midpoints A and B is through a village. In the village, the vibroseis source was not operated with full power. Hence, not enough energy penetrated into the earth. Although surface conditions were similar along the entire line, the risk of property damage resulted in poor signal quality in the middle portion of the line.

Other factors, such as weather conditions, care taken during recording, and the condition of the recording equipment, also influence data quality. Almost always, seismic data are collected often in less-than-ideal conditions. Hence, we can only hope to attenuate the noise and enhance the signal in processing to the extent allowed by the quality of the data acquisition.

In addition to field acquisition parameters, seismic data processing results also depend on the techniques used in processing. A conventional processing sequence almost always includes the three principal processes — *deconvolution, CMP stacking*, and *migration.*

Processing of Seismic data

We begin with a review of the fundamentals of digital signal processing in Chapter 1. Seismic data recorded in digital form by each channel of the recording instrument are represented by a time series. Processing algorithms are designed for and applied to either single-channel time series, individually, or multichannel time series. The Fourier transform constitutes the foundation of much of the digital signal processing applied to seismic data. Aside from sections on the one- and two-dimensional Fourier transforms and their applications, Chapter 1 also includes a section on a worldwide assortment of recorded seismic data. By referring to the field data examples, we examine characteristics of the seismic signal — primary reflections from layer boundaries and random and coherent noise such as multiple reflections, reverberations, linear noise associated with guided waves and point scatterers. Chapter 1 concludes with a section on the basic processing sequence and guidelines for quality control in processing.

The next three chapters are devoted to the three principal processes — deconvolution, CMP stacking, and migration. We study deconvolution in Chapter 2. Deconvolution often improves temporal resolution by collapsing the seismic wavelet to approximately a spike and suppressing reverberations on some field data (Figure I-7). The problem with deconvolution is that the accuracy of its output may not always be self-evident unless it can be compared with well data. The main reason for this is that our model for deconvolution is nondeterministic in character.

We study the second principal process, CMP stacking, in Chapter 3 with the accompanying subjects on velocity analysis, normal-moveout (NMO), and statics

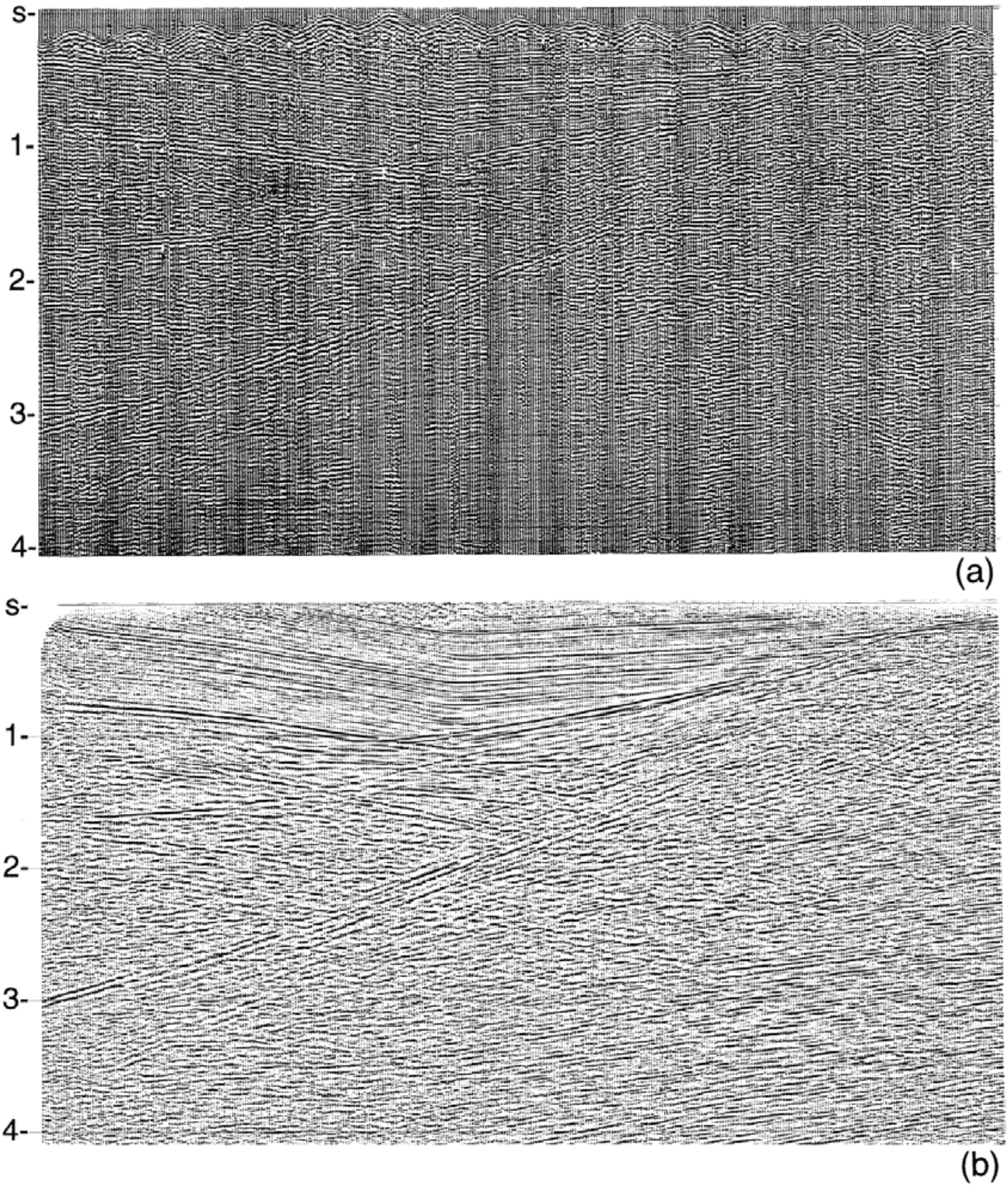

FIG. I-3. (a) A single-fold section obtained in 1965, and (b) a twelve-fold section obtained in 1995 along the same line traverse. (Data courtesy Turkish Petroleum Corp.)

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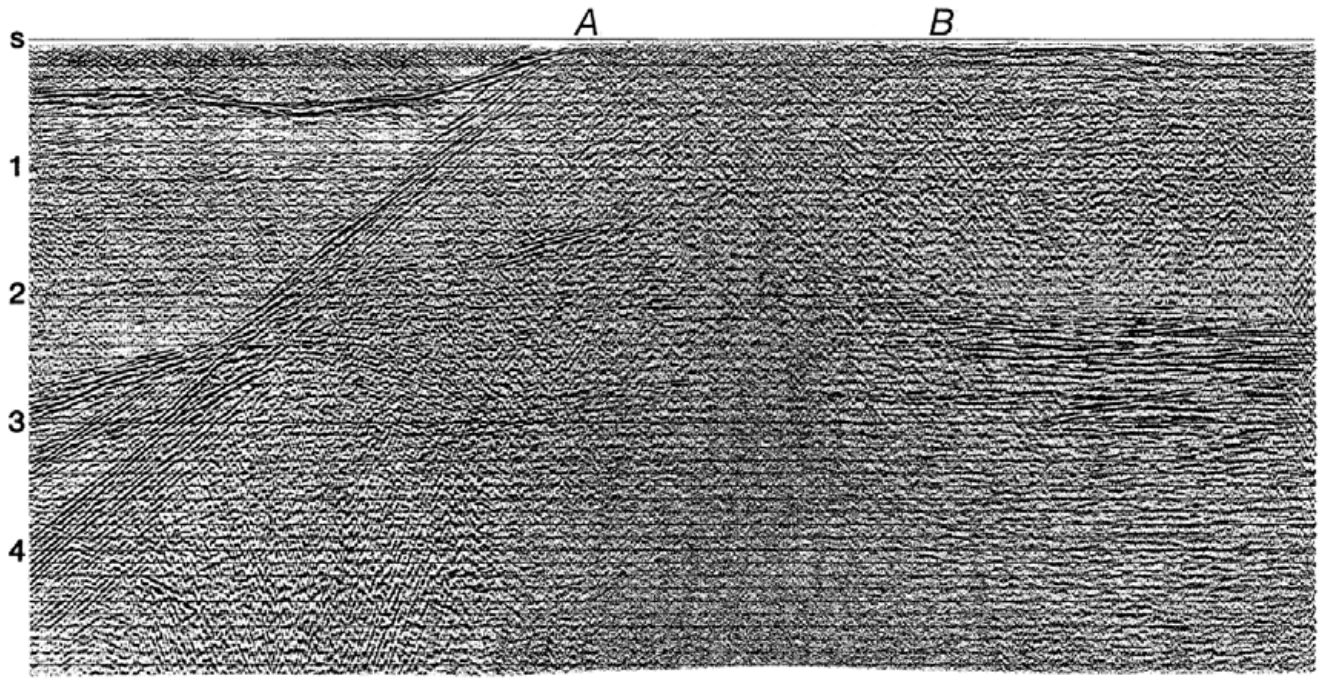

FIG. I-4. The poor signal between midpoints A and B on this seismic section is caused by a karstic limestone on the surface.

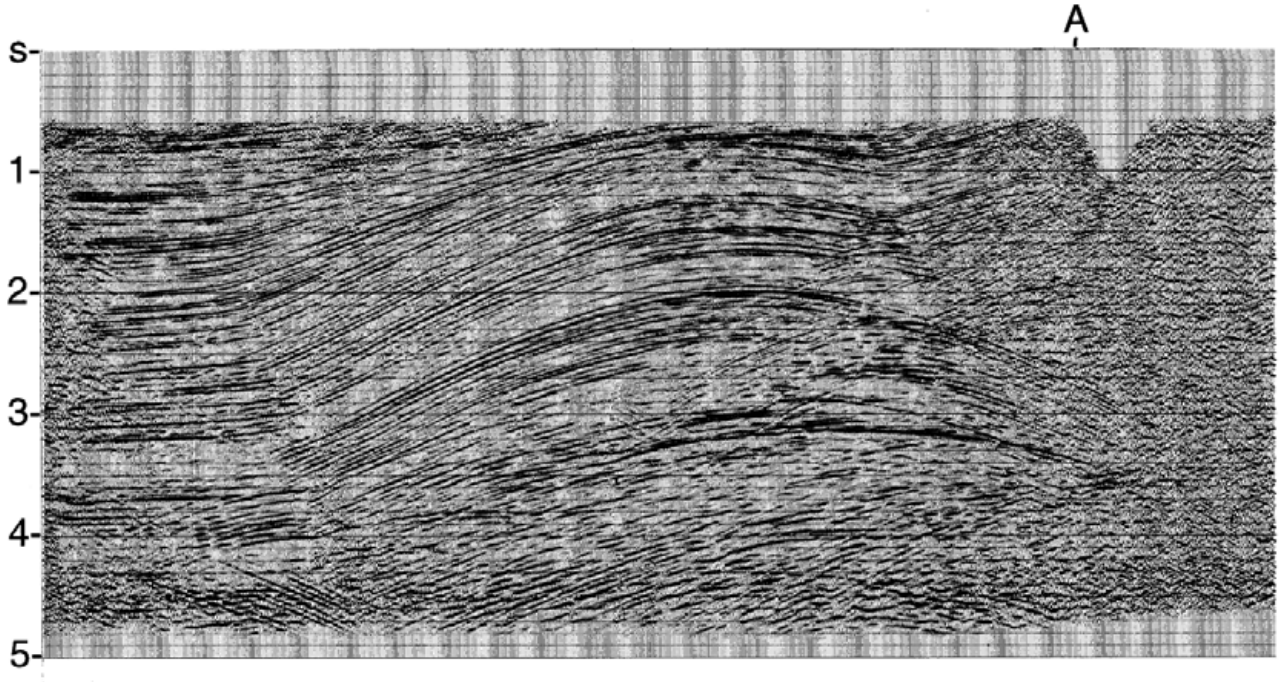

FIG. I-5. The lack of coherent reflections to the right of midpoint A on this seismic section results from the surface charges used during recording. By using charges placed in holes below the karstic limestone in the near surface, signal penetration has been improved to the left of midpoint A.)

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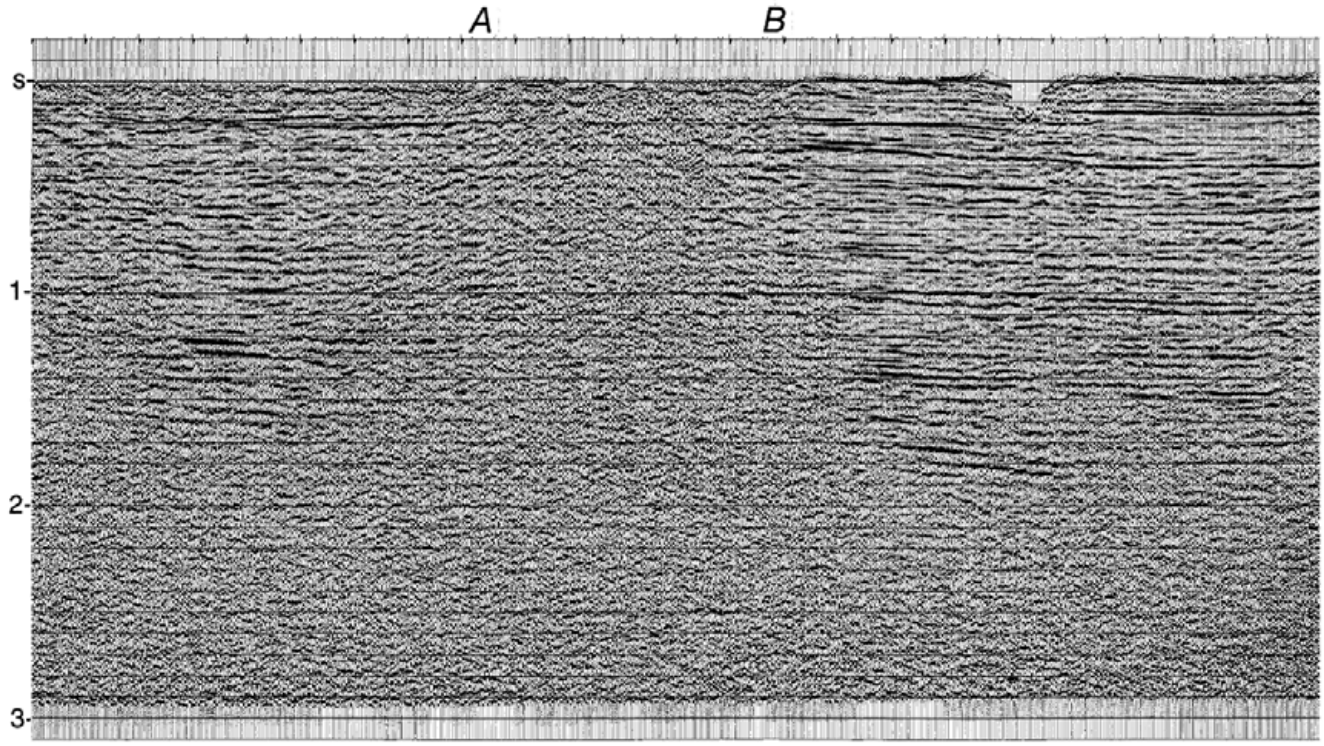

FIG. I-6. A village is situated between midpoints A and B. The poor signal in that zone of the seismic section is caused by operating the vibroseis source at low power.

corrections. Common-midpoint stacking is the most robust of the three principal processes. By using redundancy in CMP recording, stacking can attenuate uncorrelated noise significantly, thereby increasing the S/N ratio (Figure I-3). It also can attenuate a large part of the coherent noise in the data, such as guided waves and multiples.

The normal moveout (NMO) correction before stacking is done using the primary velocity function. Because multiples have larger moveout than primaries, they are undercorrected and, hence, attenuated during stacking (Figure I-8).

The main problem with CMP stacking is that it is based on the hyperbolic moveout assumption. Although it may be violated in areas with severe structural complexities, seismic data acquired in many parts of the world seem to satisfy this assumption reasonably well.

Data acquired on land must be corrected for elevation differences at shot and receiver locations and traveltime distortions caused by a near-surface weathering layer. The corrections usually are in the form of vertical traveltime shifts to a flat datum level (statics corrections). Because of uncertainties in near-surface model estimation, there always remains some residual statics which need to be removed from data before stacking (Figure I-9).

Finally, we study the third principal process, migration, in Chapter 4. Migration collapses diffractions and moves dipping events to their supposedly true subsurface locations (Figure I-10). In other words, migration is an imaging process. Because it is based on the wave equation, migration also is a deterministic process. The migration output often is self-evident — you can tell whether the output is migrated properly. When the output is not self-evident, this uncertainty often can be traced to the imprecision of the velocity information available for input to the migration program. Other factors that influence migration results include type of input data — two-dimensional (2-D) or three-dimensional (3-D), migration strategies — time or depth, post- or prestack, and algorithms and associated parameters. Two-dimensional migration does not correctly position events with 3-D orientation in the subsurface. Note the accurate imaging of the erosional unconformity (event A) in Figure I-10. However, this event is intercepted by event B which is most likely associated with the same unconformity, only that it is out-of-the-plane of recording along the line traverse.

Events with conflicting dips require an additional step — dip-moveout (DMO) correction, prior to CMP stacking (Figure I-11). Dip-moveout correction is the

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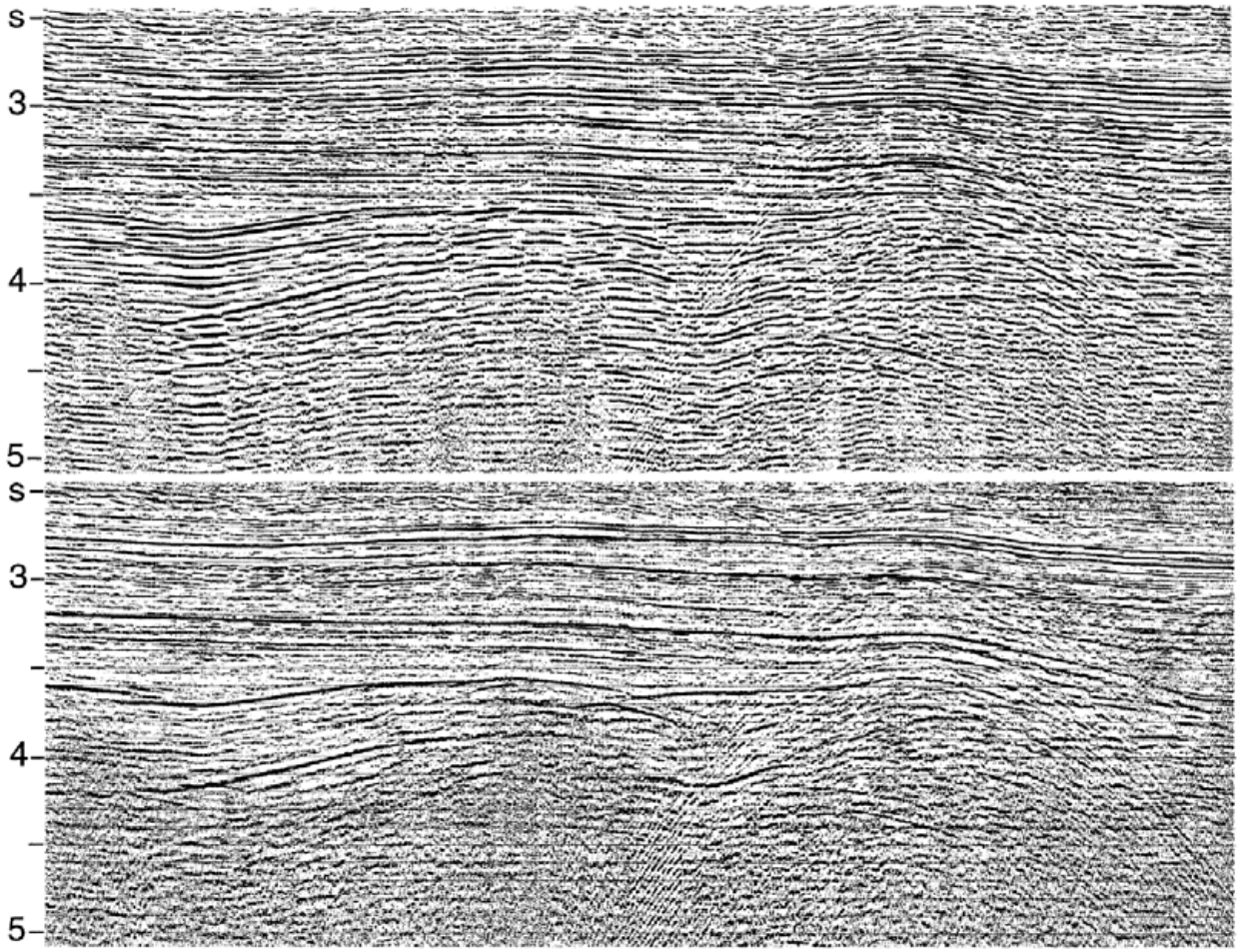

FIG. I-7. A seismic section without (top) and with (bottom) deconvolution. Note the improved vertical resolution on the deconvolved section as a result of wavelet compression and removal of reverberations. (Data courtesy Enterprise Oil.)

subject for Chapter 5. Conflicting dips with different stacking velocities often are associated with fault blocks and salt flanks. Specifically, the moveout associated with steeply dipping fault-plane reflections or reflections off a salt flank is in conflict with the moveout associated with reflections from gently dipping strata. Following NMO correction, DMO correction is applied to data so as to preserve events with conflicting dips during stacking. Migration of a DMO stack then yields an improved image of fault blocks (Figure I-11) and salt flanks (Figure I-1).

The rigorous solution to the problem of conflicting dips with different stacking velocities is migration before stack. Because this topic is closely related to DMO correction, it also is covered in Chapter 5.

We study in Chapter 6 various techniques for attenuating random noise, coherent noise, and multiple reflections. Techniques to attenuate random noise exploit part of the signal uncorrelated from trace to trace. Techniques to attenuate coherent linear noise exploit the linearity in the frequency-wavenumber and slant-stack domains. Finally, techniques to attenuate multiples exploit their periodicity in the common-midpoint, slant-stack and velocity-stack domains. Multiples also can be attenuated by using techniques that exploit the velocity discrimination between primaries and multiples in the same domains.

After reviewing the fundamentals of signal processing (Chapter 1), studying the three principal processes — deconvolution (Chapter 2), CMP stacking (Chapter 3) and migration (Chapter 4), and reviewing dip-moveout correction (Chapter 5) and the noise and multiple attenuation techniques (Chapter 6), we then move on to processing of 3-D seismic data in Chapter 7. The principal objective for 3-D seismic exploration is to obtain an earth image in three dimensions. Clearly, all

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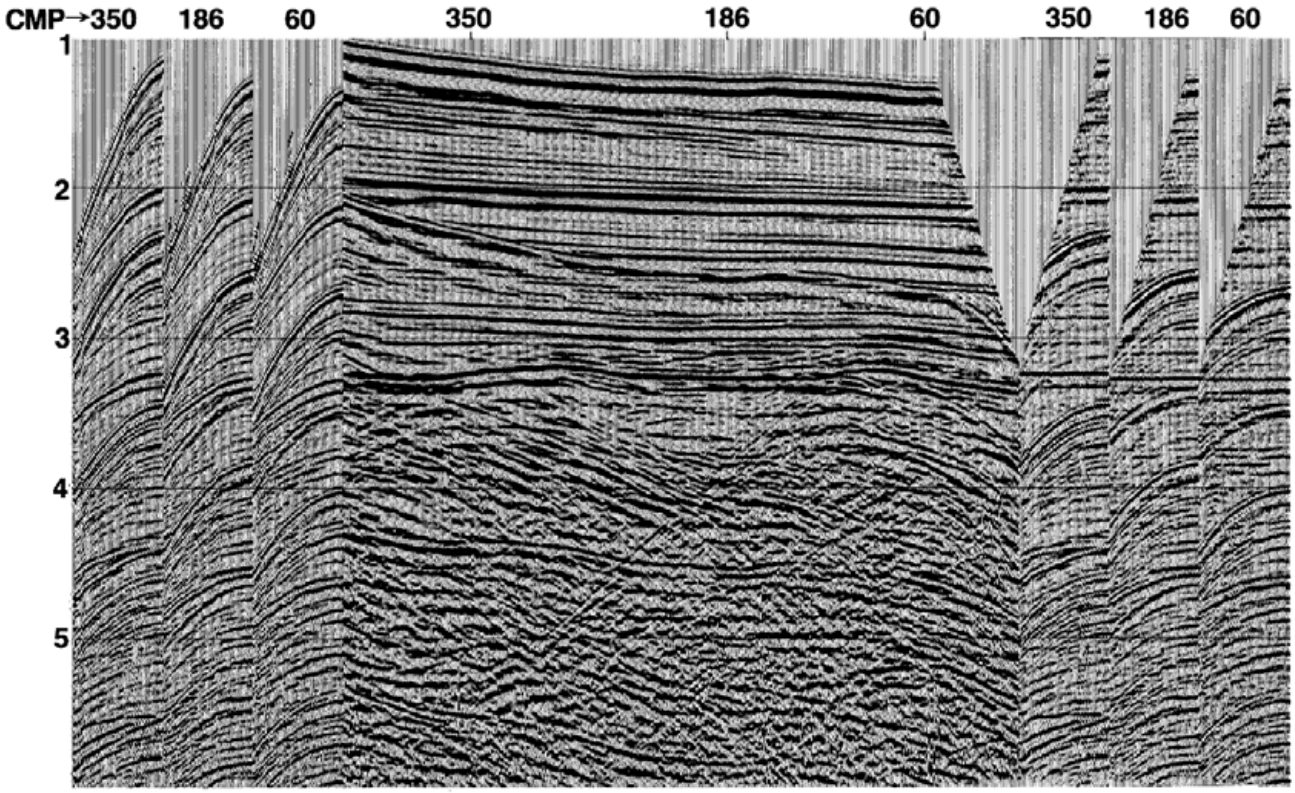

FIG. I-8. Three CMP gathers before (left) and after (right) NMO correction. Note that the primaries have been flattened and the multiples have been undercorrected after NMO correction. As a result, multiple energy has been attenuated on the stacked section (center) relative to primary energy. (Data courtesy Petro-Canada Resources.)

of the 2-D processing techniques covered in Chapters 1 through 6 are either directly applicable to 3-D seismic data or need to be extended to the third dimension, such as migration and dip-moveout correction.

There is a fundamental problem with seismic data processing. Even when starting with the same raw data, the result of processing by one organization seems to be different from that of another organization. The example shown in Figure I-12 demonstrates this problem. The same data have been processed by six different contractors. Note the significant differences in frequency content, S/N ratio, and degree of structural continuity from one section to another. These differences often stem from differences in the choice of parameters and the detailed aspects of implementation of processing algorithms. For example, all the contractors have applied residual statics corrections in generating the sections in Figure I-12. However, the programs each contractor has used to estimate residual statics most likely differ in the handling of the correlation window, selecting the traces used for crosscorrelation with the pilot trace, and statistically treating the correlation peaks.

One other aspect of seismic data processing is the generation of artifacts while trying to enhance signal. A good seismic data analysis program not only performs the task for which it is written, but also generates minimum numerical artifacts. One of the features that makes a production program different from a research program, which is aimed at testing whether the idea works or not, is refinement of the algorithm in the production program to minimize artifacts. Processing can be hazardous if artifacts overpower the intended action of the program.

The ability of the seismic data analyst invariably is as important as the effectiveness of the algorithms in determining the quality of the final product from data processing. There are many examples of good processing using mediocre software. There are also examples of poor processing using good software. The example shown in Figure I-12 rigorously demonstrates how implementational differences in processing algorithms and differences in the analyst's skills can influence results of processing.

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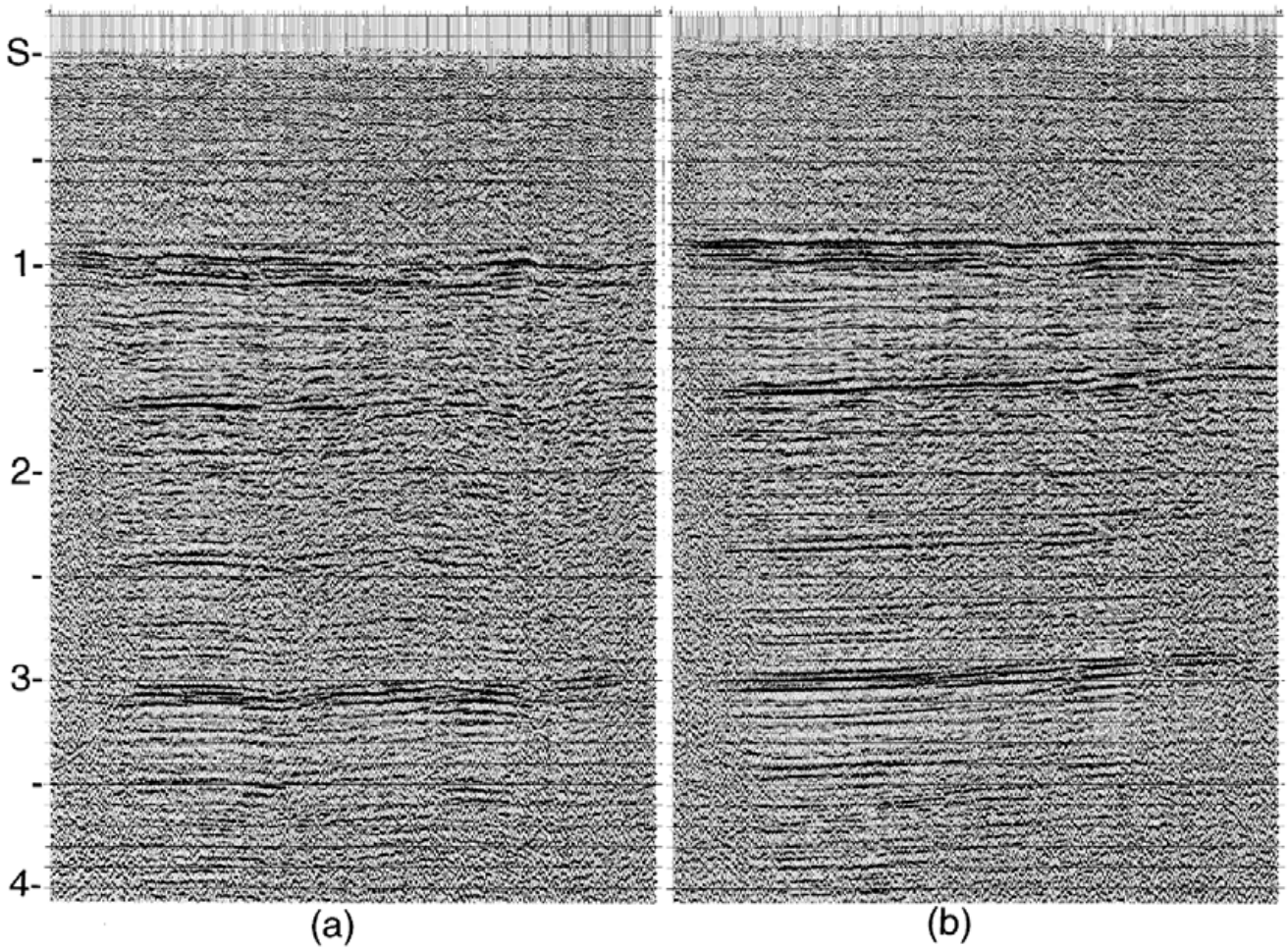

FIG. I-9. A portion of a CMP-stacked section (a) before, and (b) after residual statics corrections. Note the removal of traveltime distortions caused by the near-surface layer and improvement in the continuity of events after residual statics corrections.

Inversion of Seismic data

A narrow meaning of seismic inversion — commonly referred to as trace inversion, is acoustic impedance estimation from a broad-band time-migrated CMP-stacked data. A broad meaning of seismic inversion — commonly referred to as elastic inversion, is the grand scheme of estimating elastic parameters directly from observed data. Nevertheless, in practice, applications of inversion methods can be grouped in two categories — *data modeling* and *earth modeling*. Much of what we do in seismic data processing described in Chapters 1 through 7 is based on data modeling.

Applications of seismic inversion for data modeling include deconvolution (Chapter 2), refraction and residual statics corrections (Chapter 3) and the discrete Radon transform (Chapter 6). The discrete Radon transform is an excellent example to demonstrate the benefits of data modeling in seismic data processing. Consider a 2-D operator $\mathbf{L^T}$ that corresponds to moveout correction to a CMP gather using a range of constant velocities and summing the trace amplitudes along the offset axis. (T stands for matrix transpose.) As a result, the data represented by the CMP gather is transformed from the offset space (offset versus two-way traveltime) to velocity space (velocity versus two-way zero-offset time). The gather in the output domain is called the velocity stack. The stack amplitudes on the velocity-stack gather exhibit smearing along the velocity axis. This is caused by discrete sampling along the offset axis and finite cable length. The operator $\mathbf{L^T}$ alone does not account for these effects. Instead, we must use its generalized linear inverse $(\mathbf{L^T L})^{-1}\mathbf{L^T}$. Application of the operator $\mathbf{L^T}$ is within the framework of conventional processing, whereas application of the operator $(\mathbf{L^T L})^{-1}\mathbf{L^T}$ is within the framework of seismic

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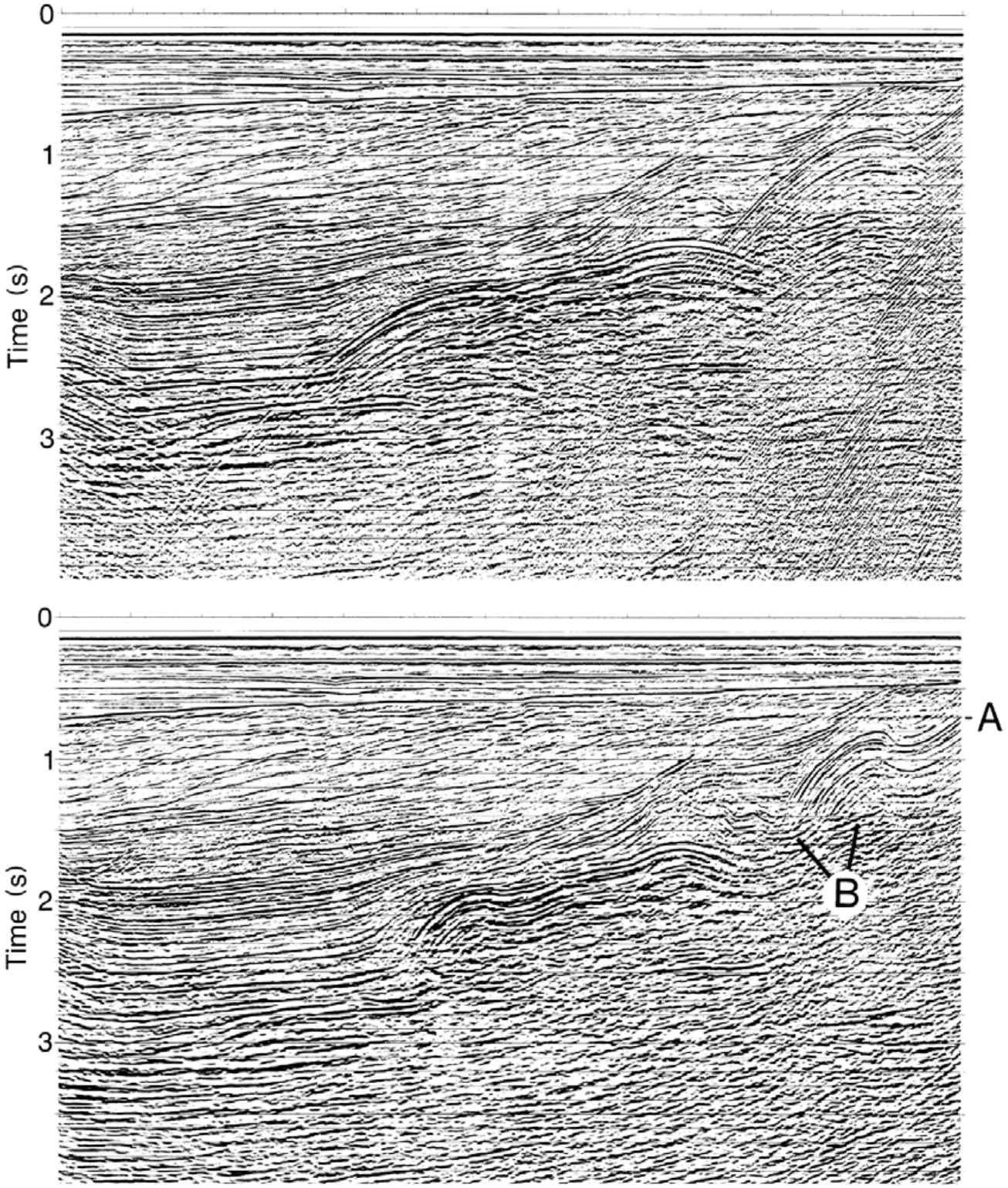

FIG. I-10. A portion of a CMP-stacked section before (top) and after (bottom) migration. Note the accurate imaging of the erosional unconformity (A). Nevertheless, the out-of-the-plane event (B) associated with this unconformity can only be imaged accurately by 3-D migration.

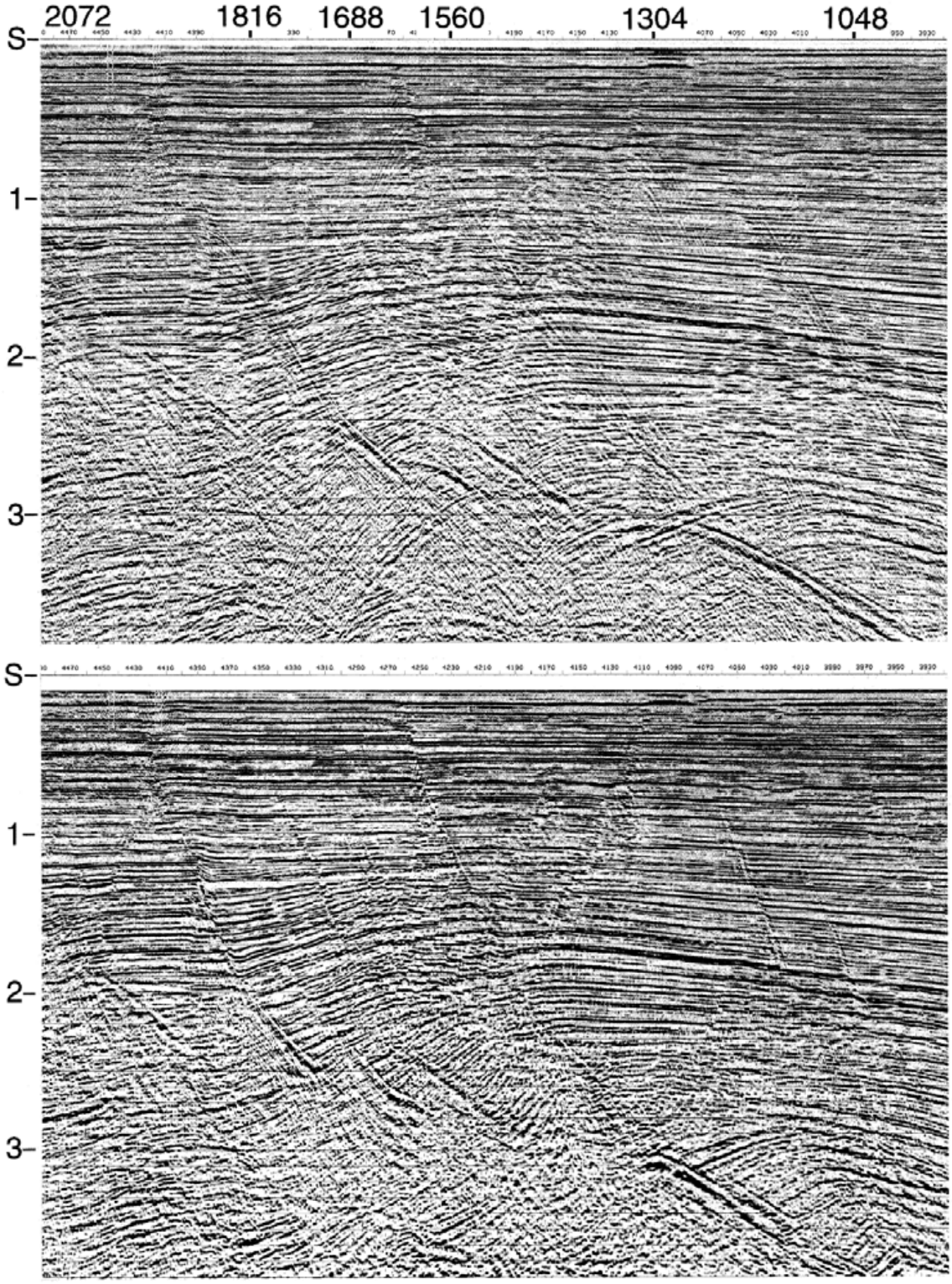

FIG. I-11. A portion of a CMP-stacked section, which has been corrected for dip moveout, before (top) and after (bottom) migration. Dip-moveout correction preserves diffractions and fault-plane reflections which conflict with gently-dipping reflections. These conflicting events are otherwise attenuated by conventional stacking. (Data courtesy Schlumberger Geco-Prakla and TGS.)

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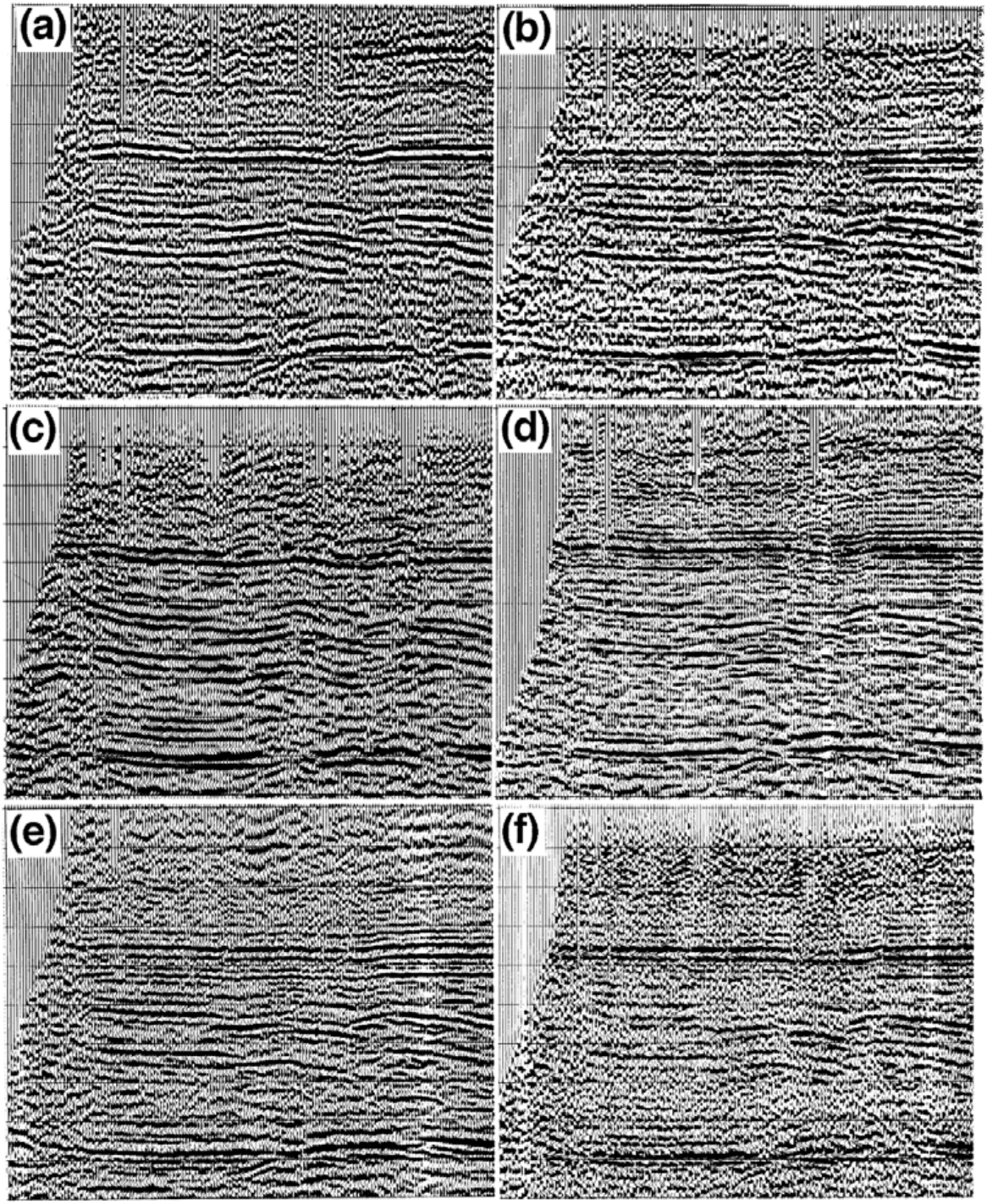

FIG. I-12. A seismic line processed by six different contractors. (Data courtesy British Petroleum Development, Ltd.; Carless Exploration Ltd.; Clyde Petroleum Plc.; Goal Petroleum Plc.; Premier Consolidated Oilfields Plc.; and Tricentrol Oil Corporation Ltd.)

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inversion. The CMP gather can be reconstructed by applying inverse moveout correction and summing over the velocity axis. This inverse transformation is represented by the operator $\mathbf{L}$. Reconstruction of the CMP gather from the velocity-stack gather is one example of data modeling. Data modeling using the velocity-stack gather computed by the processing operator $\mathbf{L^T}$ does not faithfully restore the amplitudes of the original CMP gather, whereas data modeling using the inversion operator $(\mathbf{L^T L})^{-1}\mathbf{L^T}$ does.

Just as there is a difference between processing and inversion in data modeling, also, there exists a difference between processing and inversion in earth modeling. The primary objective in processing is to obtain an earth model in time with an accompanying earth image in time — a time-migrated section or volume of data (Figure I-13). Representation of an earth model in time usually is in the form of a *velocity field*, which has to be smoothly varying both in time and space. Whereas the primary objective in inversion is to obtain an earth model in depth with an accompanying earth image in depth — a depth-migrated section or volume of data (Figure I-14). Representation of an earth model in depth usually is in the form of a detailed *velocity-depth model*, which can include layer boundaries with velocity contrast (Figure I-14).

Chapters 8 and 9 are devoted to earth imaging and modeling in depth, respectively. Results of conventional processing of seismic data often are displayed in the form of an unmigrated (Figure I-15a) and migrated CMP-stacked section (Figure I-15b), with the vertical axis as time, which is different from the recording time of seismic wavefields. For unmigrated data, the vertical axis of the CMP-stacked section represents times of reflection events in the unmigrated position in the subsurface. These event times are associated with normal-incidence raypaths from coincident source-receiver locations at the surface to reflectors in the subsurface and back. For migrated data, the vertical axis represents times of reflection events in the migrated position. These event times are associated with vertical-incidence raypaths from coincident source-receiver locations at the surface to reflectors in the subsurface and back. As long as there are no lateral velocity variations, seismic imaging of the subsurface can be achieved using time migration techniques and the result can be displayed in time. This time-migrated section can then be converted to depth along vertical raypaths.

When there are mild to moderate lateral velocity variations, time migration can still yield a reasonably accurate image of the subsurface. Nevertheless, depth conversion must be done along image rays to accommodate for the lateral mispositioning of the events as a result of time migration.

In the presence of strong to severe lateral velocity variations, however, time migration no longer is valid. Instead, seismic imaging of the subsurface must be done using depth migration techniques so as to properly account for lateral velocity variations and the result must be displayed in depth.

The depth-migrated section (Figure I-15c) can be considered a close representation of the structural cross-section of the subsurface only if the velocity-depth model is sufficiently accurate. In the example shown in Figure I-15, the picked horizons correspond to layer boundaries with significant velocity contrast. The zone of interest is base Zechstein (the red horizon) and the underlying Carboniferous sequence. The green horizon just below 2 km is the top Zechstein. This formation consists of two units of anhydrite-dolomite with a thickness of approximately 100 m — the shallow unit very close to top-Zechstein and concordant with it, and the deeper unit which manifests itself with a very complex geometry as seen in the migrated sections.

An earth model in depth usually is described by two sets of parameters — *layer velocities* and *reflector geometries* (Figure I-16). Practical methods to delineate reflector geometries described in Chapter 8, and to estimate layer velocities described in Chapter 9 can be appropriately combined to construct earth models in depth from seismic data.

In practice, smoothness of earth models derived from processing means that we can make a straight-ray assumption and usually do not have to honor ray bending at layer boundaries. In contrast, detailed definition of earth models derived from inversion with a more stringent requirement in accuracy means that we do have to honor ray bending at layer boundaries and account for vertical and lateral velocity gradients within the layers themselves. Hence, to a large extent, processing can be automated, while inversion requires interpretive pause at each layer boundary.

There is a fundamental problem with inversion applied to earth modeling in depth — *velocity-depth ambiguity*. This means that an error in depth is indistinguishable from an error velocity. To resolve velocity-depth ambiguity as much as possible, one needs to do an independent estimate of layer velocities and reflector geometries using prestack data. As a result of velocity-depth ambiguity, an output from inversion is an estimated velocity-depth model with a measure of uncertainty in layer velocities and reflector geometries. It is now widely accepted in the industry that results of inversion are geologically plausable only when there is a sound interpretation effort put into the data analysis.

It is the limited accuracy in velocity estimation that has led to the acceptance of time sections to be the standard mode of display in seismic exploration. Facing the challenge of improving the accuracy in velocity estimation should make the depth sections increasingly more acceptable. Specifically, improving the accuracy means the ability to resolve detailed velocity

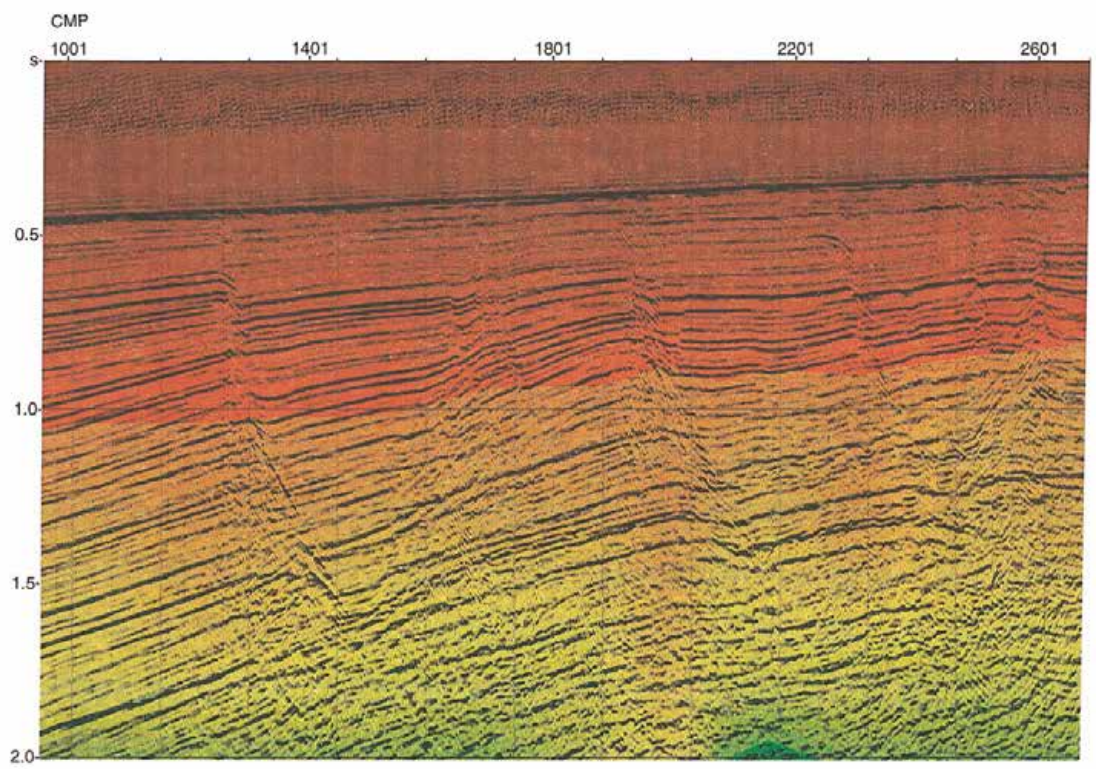

FIG. I-13. An earth image in time obtained by poststack time migration of a CMP-stacked section with the color-coded earth model in time represented by a velocity field.

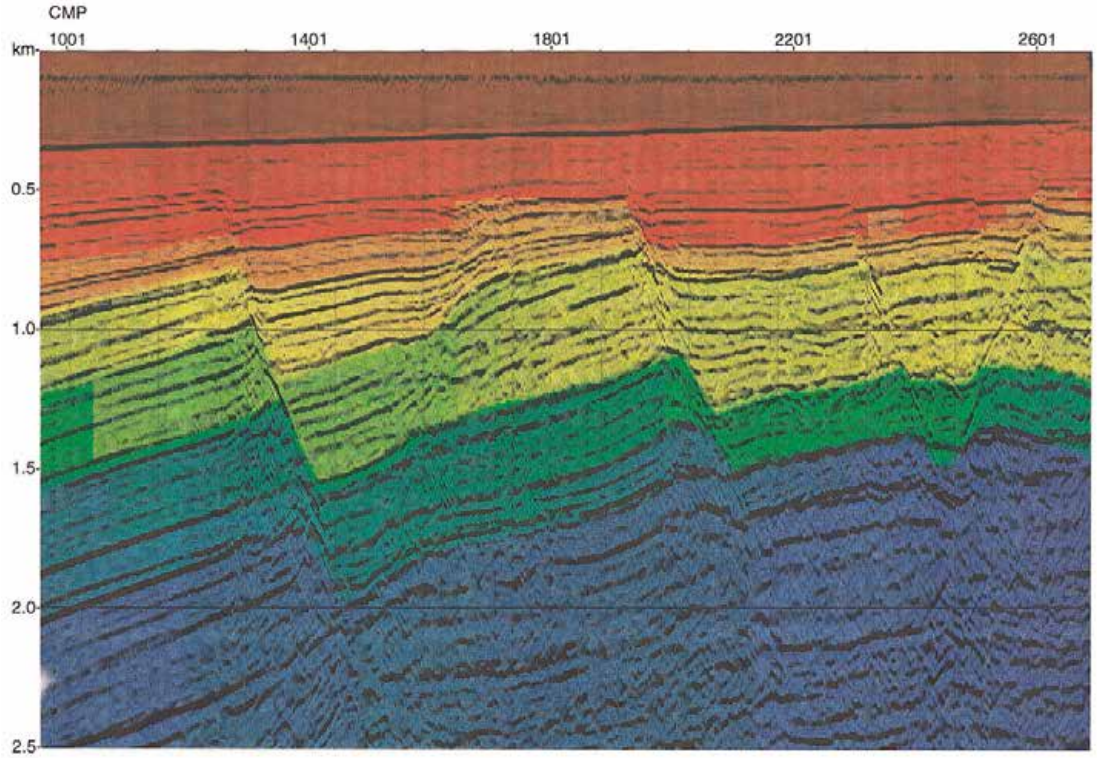

FIG. I-14. An earth image in depth obtained by prestack depth migration with the color-coded earth model in depth represented by a velocity-depth model.

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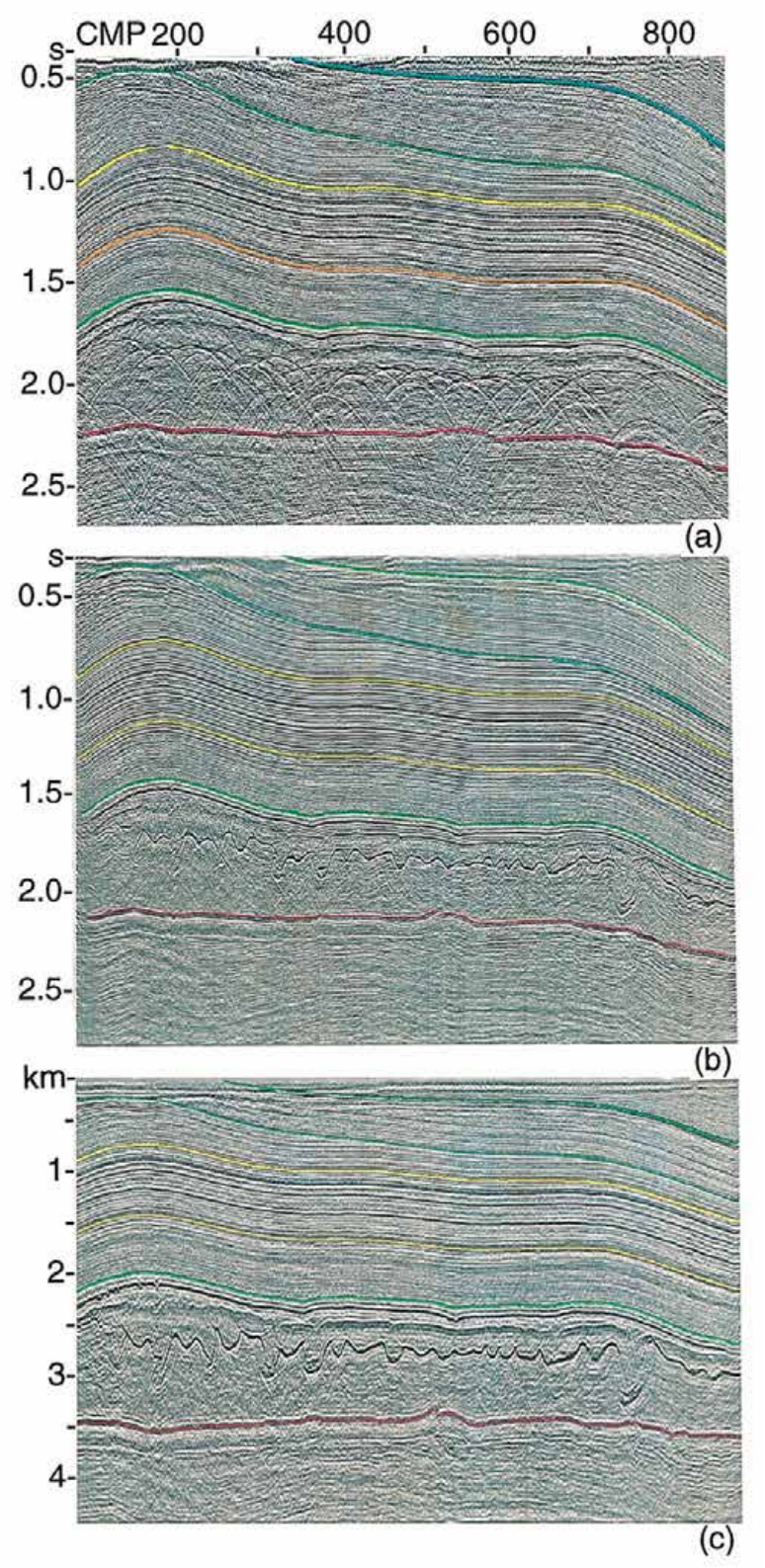

FIG. I-15. (a) A cross-section from an unmigrated volume of CMP-stacked data; (b) the same cross-section after 3-D poststack time migration; and (c) after 3-D poststack depth migration. See text for details. (Data courtesy Amoco Production (UK) Ltd.)

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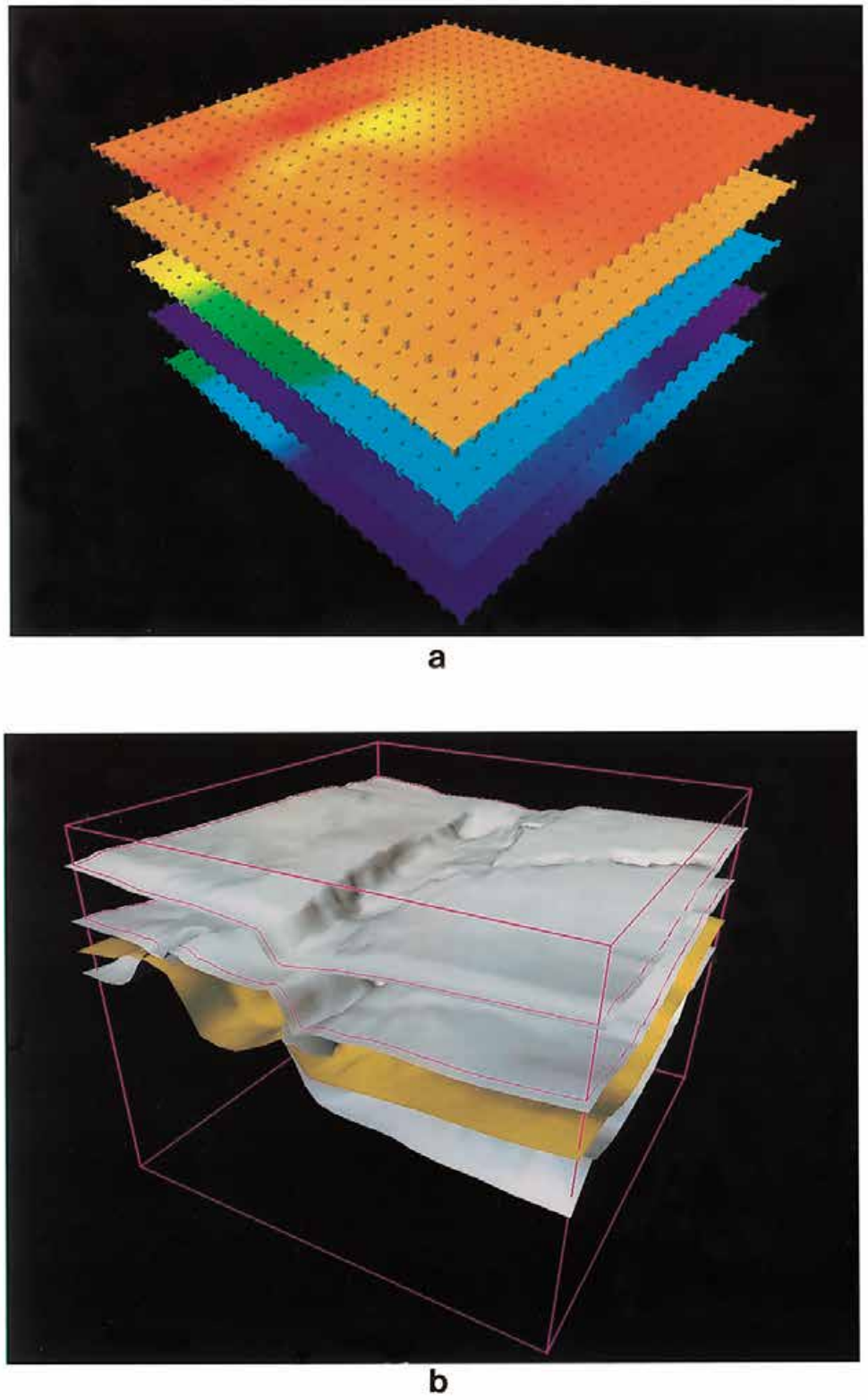

FIG. I-16. An earth model in depth is described by two sets of parameters — (a) layer velocities, and (b) reflector geometries.

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variations in the vertical and lateral directions, associated with both structural and stratigraphic targets.

Earth modeling in depth usually involves implementation of an inversion procedure layer by layer starting from the top (Figure I-17). First, estimate a velocity field (the color-coded surface and the vertical cross-section) for the first layer, for instance, using 3-D coherency inversion. Then delineate the reflector geometry (the silver surface) associated with the base of the layer, for instance, using 3-D poststack depth migration (Figure I-17a). Next, estimate a velocity field for the second layer and delineate the reflector geometry associated with the base of the layer (Figure I-17b). Alternate between layer velocity estimation and reflector geometry delineation, one layer at a time, to complete the construction of the earth model in depth (Figure I-17c). This layer-by-layer, structure-dependent estimation of earth models in depth is needed when there are distinct layer boundaries with significant velocity contrast (as in many parts of the North Sea). In practice, an iterative, structure-independent estimation of earth models in depth also is used in the case of a background velocity field with not-so-distinct layer boundaries (as in the Gulf of Mexico).

Practical methods of layer velocity estimation include Dix conversion and inversion of stacking velocities, coherency inversion, and analysis of image gathers from prestack depth migration (Chapter 9). Velocity nodes at analysis locations for the layer under consideration (Figure I-18a) are assigned to the normal-incidence reflection points over the surface associated with the base of the layer (Figure I-18b). A velocity field for the layer is then created by spatial interpolation of the velocity nodes. This layer velocity field is assigned to the layer together with a similar field for a vertical velocity gradient whenever it is available from well data.

Practical methods of reflector geometry delineation include vertical-ray and image-ray depth conversion of time horizons interpreted from time-migrated data, commonly known as vertical stretch and map migration, respectively. Additionally, reflector geometries in depth can be delineated by interpreting post- and prestack depth-migrated data. By interpreting cross-sections from the volume of depth-migrated data at appropriate intervals, horizon strands are created (Figure I-19a). These strands then are interpolated spatially to create the surface that represents the reflector geometry associated with the layer boundary included in the earth model in depth (Figure I-19b).

In Chapter 10, we present case studies for 2- and 3-D earth modeling and imaging in depth applicable to structural plays. These cases involve exploration and development objectives that require solving specific problems such as imaging beneath diapiric structures associated with salt tectonics, imaging beneath imbricate structures associated with overthrust tectonics, target reflectors below an irregular water-bottom topography, fault shadows, and shallow velocity anomalies.

A concise, but sufficiently rigorous, review of seismic wave propagation is given in Chapter 11. This also is intended to remind the reader of the two components of observed seismic data that can be used in inversion — traveltimes and amplitudes, to estimate the earth parameters. It is generally favorable to do the inversion of reflection traveltimes and amplitudes separately. The former is more robust and stable in the presence of noise. The latter is more sensitive to ambient noise and is prone to producing unstable solutions, and therefore, it may require more stringent constraints.

In Chapter 11, we review inversion of amplitudes of acoustic wavefields, specifically, prestack amplitude inversion to derive the attributes associated with amplitude variation with offset (AVO) and poststack amplitude inversion to estimate an acoustic impedance (AI) model of the earth. We broadly associate traveltime inversion with the estimation of a structural model of a reservoir that describes the geometry of the layer boundaries and faults. Whereas, we broadly associate amplitude inversion with the estimation of a stratigraphic model of the reservoir that describes the lateral and vertical variations of the AVO and AI attributes within the layers themselves. The latter can then be transcribed into petrophysical parameters — pore pressure, porosity, permeability, and fluid saturation, and it combined with the structural model to create a model of the reservoir. Therefore, seismic inversion is a true pronouncement of integration between petroleum geology, petroleum engineering, and exploration seismology. Only the exploration seismologists *timespeak*, while the peroleum geologists and engineers *depthspeak*. To achieve integration, they all must be fluent in the same language — *depthspeak*.

Interpretation of Seismic data

When you pick semblance peaks from a velocity spectrum (Section 3.2) to determine the moveout velocity function, you implicitly make a judgment as to what is primary and what is multiple. When you pick a coherency semblance spectrum (Section 9.2) to determine the interval velocity profile, you make a judgment as to what degree of lateral velocity variations needs to be honored. These are but two examples of interpretive work involved in processing and inversion of seismic data, respectively.

What is known as traditional seismic interpreta-

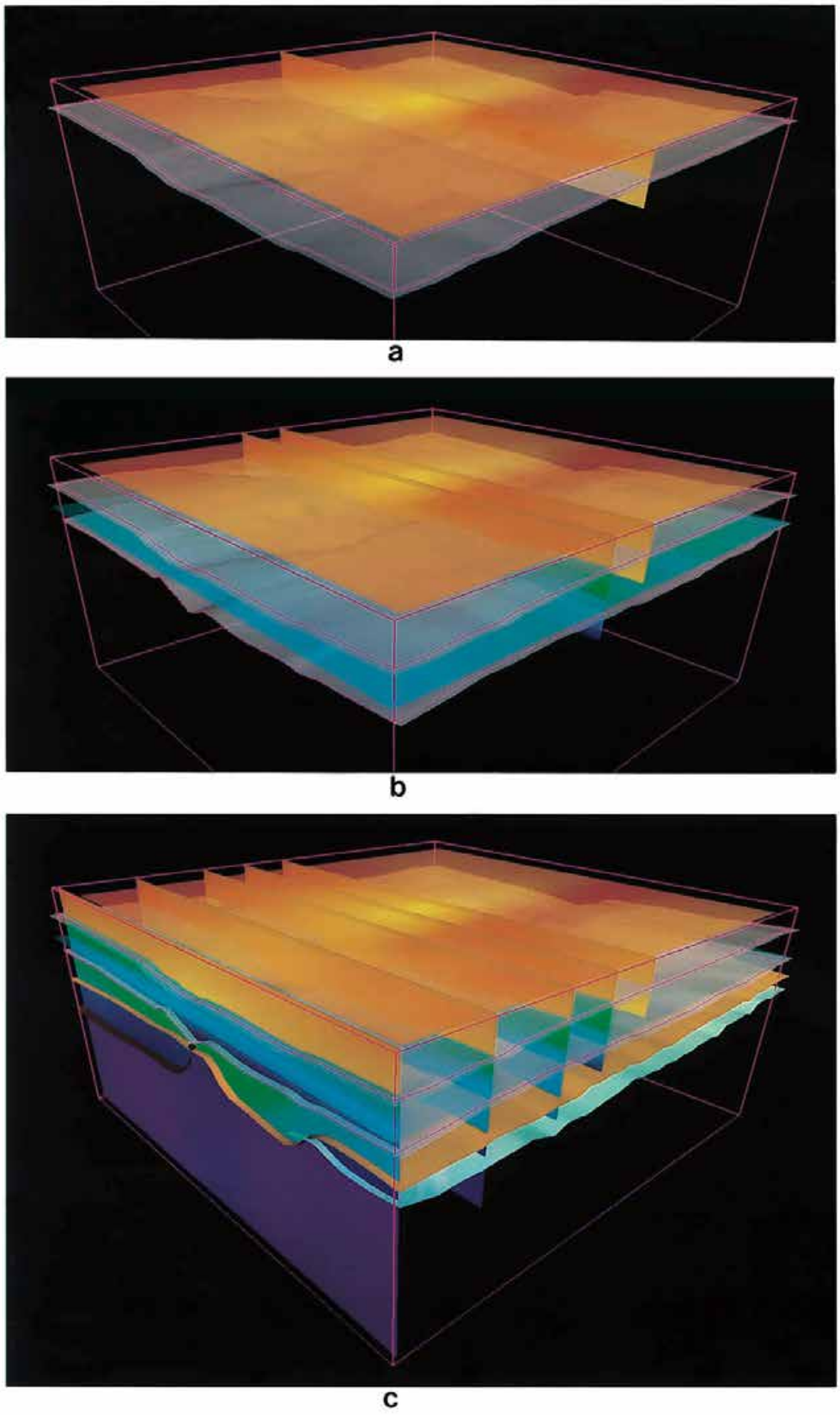

FIG. I-17. Layer-by-layer estimation of an earth model in depth. See text for details.

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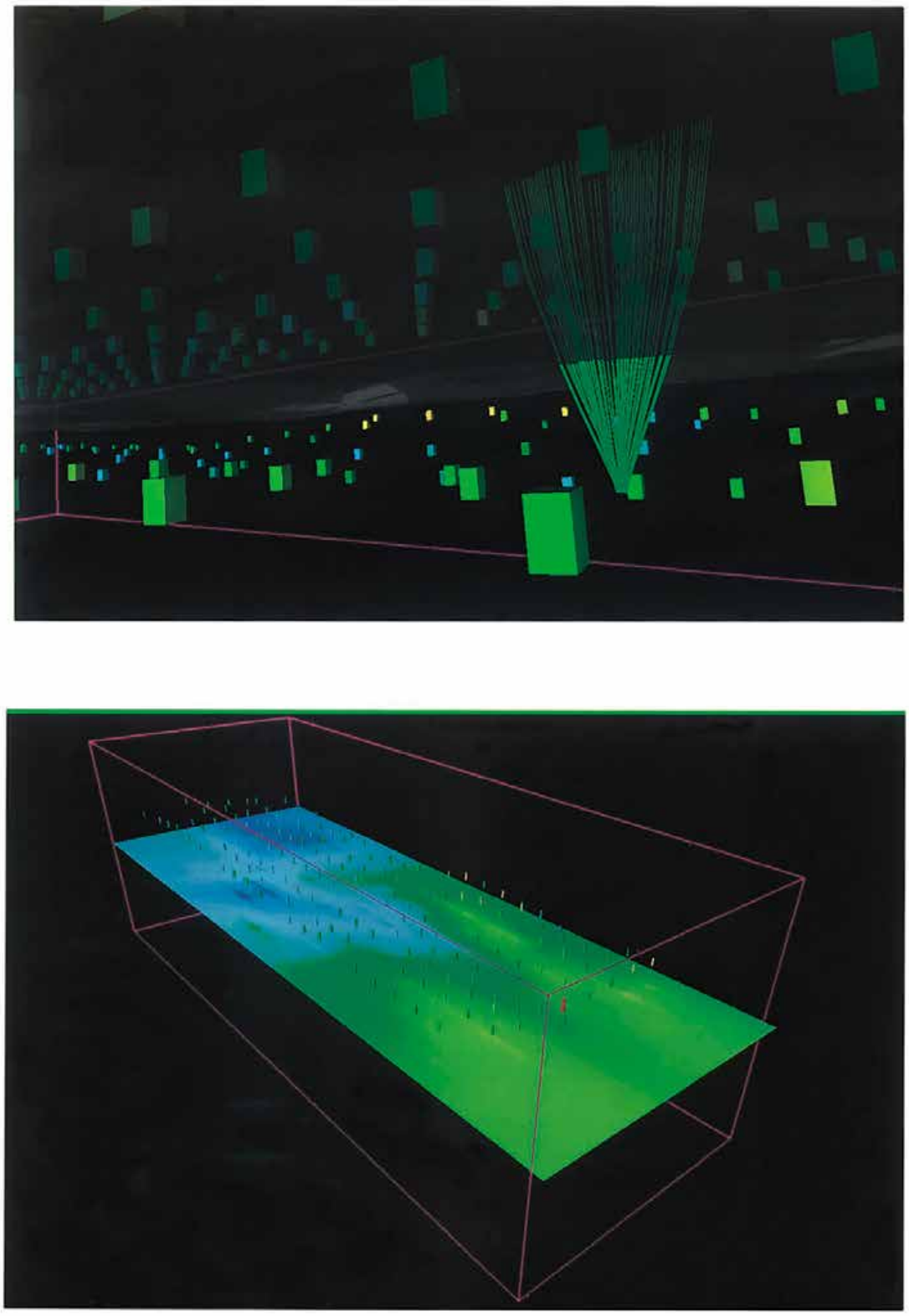

FIG. I-18. Estimated velocities for a layer represented by the color-coded velocity nodes (top) and the velocity field derived from the nodes (bottom).

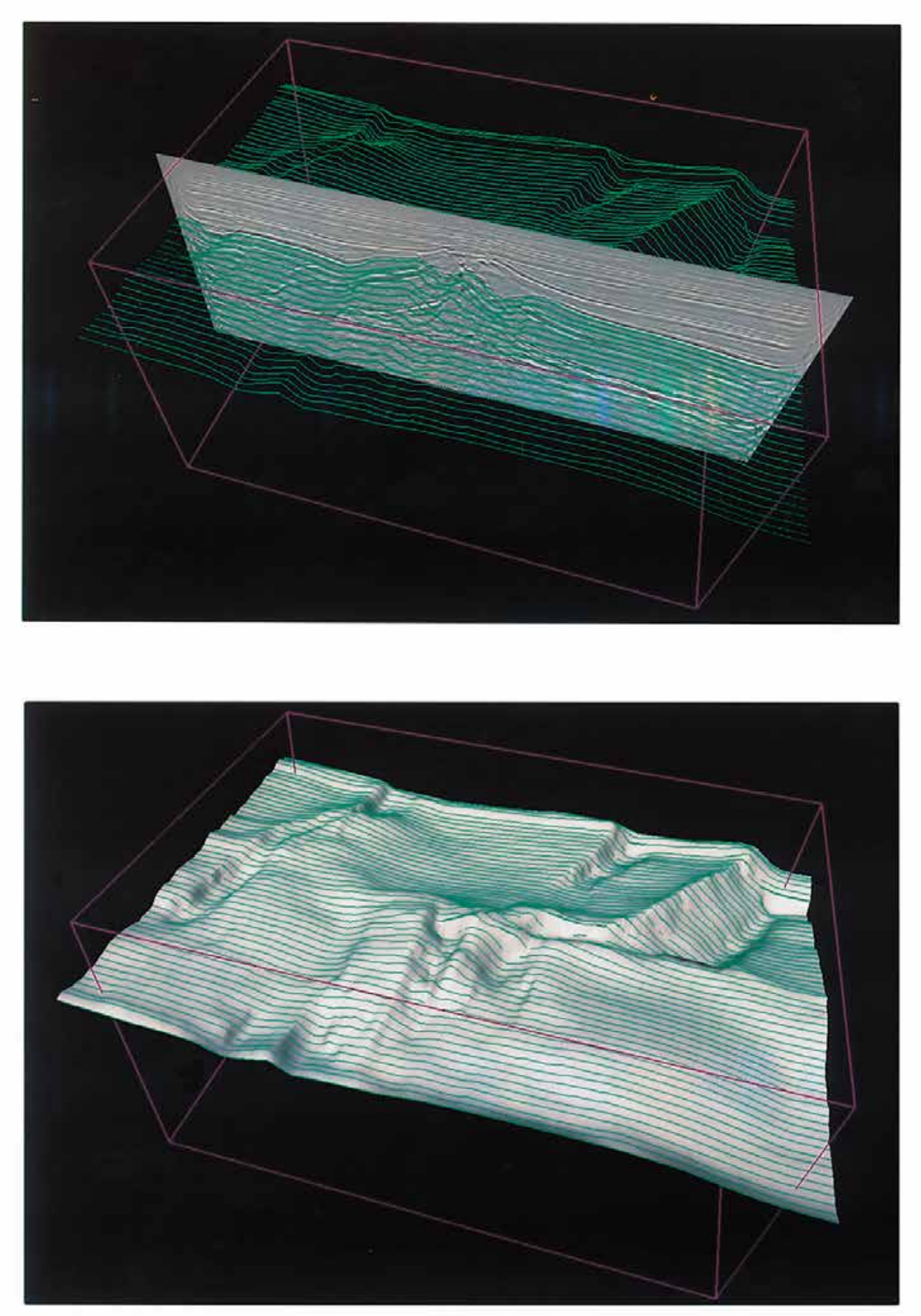

FIG. I-19. Reflector geometry delineation: (top) depth horizon strands created by interpreting selected cross-sections (displayed is one such section) from the depth-migrated volume of data, and (bottom) the surface that represents the reflector boundary created by spatial interpolation of the strands.

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tion, however, involves picking a reflection time surface associated with a layer boundary from a time-migrated volume of data or a reflector from a depth-migrated volume of data to determine the structure map for that layer boundary (Figure I-19). The power of 3-D visualization of image volumes, velocity volumes, and attribute volumes, such as those associated with AVO analysis and acoustic impedance estimation, have dramatically changed the way seismic interpretation is done now. Interperetation no longer is picking traveltimes to determine the structural geology of the area of interest, but also involves manipulation of amplitudes contained in the data volumes to derive information about the depositional environment, depositional sequence boundaries, and the internal constitution of the sequence units themselves. Interpretation of 3-D seismic data is covered in Section 7.5, while further examples are provided with the case studies in Sections 10.8 and 10.9.

From Seismic Exploration to Seismic Monitoring

The seismic industry has been impressively dynamic and creative during its 60-year history. Although it is relatively a small sector within the oil and gas industry at large, it has made the most significant impact on increasing proven reserves and reserve-production ratios worldwide.

We shall now sketch a brief historiography of the seismic industry before we look ahead. The evolution of the seismic industry can be described briefly in decades of development and forward leaps from one theme to another as outlined in Table I-1.

In the 1960s, the digital revolution profoundly changed seismic acquisition. We were then able to record more data by increasing the number of channels and fold of coverage. The digital revolution brought about the need to use digital computers to analyze the recorded data. That came about in the 1970s when we switched from calculators to computers. Many of the data processing algorithms, including deconvolution, velocity analysis, refraction, and residual statics corrections, normal-moveout corection and stacking, and even migration, were implemented in those years. The computer before the seventies was a person using the calculator; now the computer is a machine and the person became the seismic analyst.

Table I-1. The milestones in the seismic industry.

1960s	From *analog to digital*
1970s	From *calculators to computers*
1980s	From *2-D to 3-D*
1990s	From *time to depth*
2000s	From *3-D to 4-D*
	From *4-D to 4-C*
	From *isotropy to anisotropy*

In the 1980s, the seismic industry took another big step forward; it was now beginning to provide the oil and gas industry with 3-D images of the subsurface.We need only to examine the global reserve-production curves over the past decades to see that the 3-D revolution gave a big jump from 35 to 45 years for oil and from 50 to 65 years for gas. The seismic industry was already pushing the computer industry to the limit with its need for power to handle large-scale data volumes acquired by 3-D surveys.

Finally, in the 1990s, the seismic industry was capable of providing the oil and gas industry with images of the subsurface, not just in 3-D, but also in depth. It took years of exhaustive experimental research to test and field-prove numerous methods to accurately estimate an earth model in depth and use it to efficiently create an earth image in depth. Once again, the seismic industry has challenged the computer industry to provide cost-effective solutions for numerically intensive applications with large input-output operations, such as 3-D prestack depth migration.

As the seismic industry made one breakthrough after another during its history, it also created new challenges for itself. Now we record not just P-waves but also converted S-waves for a wide range of objectives. Using the multicomponent seismic method, commonly known as the 4-C seismic method, we are now able to see through gas plumes caused by the reservoir below. We are able to sometimes better image the subsalt and subbasalt targets with the 4-C seismic method. Using the converted S-waves, we are able to detect the oil-water contact, and the top or base of the reservoir unit that we sometimes could not delineate using only P-waves. We even go further now and attempt to identify fluid types in reservoir rocks, discriminate sand from shale, and map hydrocarbon saturation, again using the 4-C seismic method. Our ultimate objective is to use the seismic method, in addition to the production and geologic data, to characterize oil and gas reservoirs accurately.

Just as we may characterize oil and gas reservoirs seismically, we may also seismically monitor them. Given a set of time-lapse 3-D seismic survey data, which constitutes the basis of the 4-D seismic method, we can track flow paths and fluid distribution in the reservoirs throughout their lifetime. And finally, we have to acknowledge that the earth is anisotropic. By accounting

for anisotropy, we can map fractures and increase the accuracy of velocity esitmation and imaging techniques.

Accompanying all of these new frontiers for the seismic industry is the availability of a dazzling 3-D visualization technology that now enables us to perform volume-based processing (Section 5.4) and inversion and interpretation (Sections 10.8 and 10.9). Keep the following principle in mind when analyzing large volumes of data: *Before you get more data, get the most out of your data.*

The topics on the 4-D and 4-C seismic methods, and anisotropy discussed in Chapter 11 are for the road immediately ahead in the seismic industry with the aim of a rigorous, seismically driven reservoir characterization and monitoring.

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1 Fundamentals of Signal Processing

• **Introduction** • **The 1-D Fourier Transform** • Analog versus Digital Signal • Frequency Aliasing • Phase Considerations • Time-Domain Operations • Convolution • Crosscorrelation and Autocorrelation • Vibroseis Correlation • Frequency Filtering • Practical Aspects of Frequency Filtering • Bandwidth and Vertical Resolution • Time-Variant Filtering • **The 2-D Fourier Transform** • Spatial Aliasing • **Worldwide Assortment of Shot Records** • Wave Types • **Gain Applications** • Geometric Spreading Correction • Programmed Gain Control • RMS Amplitude AGC • Instantaneous AGC • Relative Trace Balancing • **Basic Data Processing Sequence** • Preprocessing • Deconvolution • CMP Sorting • Velocity Analysis • Normal-Moveout Correction • Multiple Attenuation • Dip-Moveout Correction • CMP Stacking • Poststack Processing • Migration • Residual Statics Corrections • Quality Control in Processing • Parsimony in Processing • **Exercises** • **Appendix A: A Mathematical Review of the Fourier Transform** • The 1-D Fourier Transform • The z-Transform • The 2-D Fourier Transform • **References**

1.0 INTRODUCTION

The Fourier transform is fundamental to seismic data analysis. It applies to almost all stages of processing. A seismic trace represents a seismic wavefield recorded at a receiver location. The digital form of a seismic trace is a time series which can be completely described as a discrete sum of a number of sinusoids — each with a unique peak amplitude, frequency, and a phase-lag (relative alignment). The *analysis* of a seismic trace into its sinusoidal components is achieved by the *forward* Fourier transform. Conversely, the *synthesis* of a seismic trace from the individual sinusoidal components is achieved by the *inverse* Fourier transform. A brief mathematical review of the Fourier transform is given in Appendix A.

Seismic data processing algorithms often can be described or implemented more simply in the frequency domain than in the time domain. In Section 1.1, the one-dimensional (1-D) Fourier transform is introduced and some basic properties of time series in both time and frequency domains are described. Many of the processing techniques — single- and multichannel, involve an operand (seismic trace) and an operator (filter). A simple application of Fourier analysis is in the design of zero-phase frequency filters, typically in the form of band-pass filtering.

The two-dimensional (2-D) Fourier transform (Section 1.2) is a way to decompose a seismic wavefield, such as a common-shot gather, into its plane-wave components, each with a certain frequency propagating at a certain angle from the vertical. Therefore, the 2-D Fourier transform can describe processes like migration and frequency-wavenumber ($f-k$) filtering. A common application of $f-k$ filtering is the rejection of coherent linear noise by dip filtering, and attenuation of multiples based on velocity discrimination between primaries and multiples in the $f-k$ domain (Section 6.2).

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In Section 1.3, 40 common-shot gathers recorded in different parts of the world with different types of sources and recording instruments are introduced. Various types of seismic energy are described on these shot records — reflections, refractions, coherent noise such as multiples, guided waves, side-scattered energy and ground roll, and ambient random noise.

Seismic data often require application of a gain function — time-variant scaling of amplitudes, for various reasons. The scaling function commonly is derived from the data. Gain types are discussed in Section 1.4. At an early stage in processing, gain is applied to data to correct for wavefront divergence — decay in amplitudes caused by geometric spreading of seismic waves. Seismic data often are gained for display purposes; for instance, by applying automatic gain control (AGC), which brings up weak reflection zones in seismic data. However, an AGC-type gain can destroy signal character and must, therefore, be considered with caution.

Finally, in Section 1.5, a summary of the basic data processing sequence is presented with field data examples. There are three primary stages in seismic data processing; each is aimed at improving seismic resolution — the ability to separate two events that are very close together, either spatially or temporally:

(a) *Deconvolution* is performed along the time axis to increase temporal resolution by compressing the basic seismic wavelet to approximately a spike and suppressing reverberating wavetrains.
(b) *Stacking* compresses the offset dimension, thus reducing seismic data volume to the plane of the zero-offset seismic section and increasing the signal-to-noise ratio.
(c) *Migration* commonly is performed on the stacked section (which is assumed to be equivalent to a zero-offset section) to increase lateral resolution by collapsing diffractions and moving dipping events to their supposedly true subsurface positions.

Secondary processes are implemented at certain stages to condition the data and improve the performance of deconvolution, stacking, and migration. When coherent noise is dip filtered, for example, deconvolution and velocity analysis may be improved. Residual statics corrections also improve velocity analysis and, hence, the quality of the stacked section.

1.1 THE 1-D FOURIER TRANSFORM

Consider the following experiment. Hold a spring at one end and attach a weight to the other end. Pull the weight down a certain amount, say 0.8 units of distance. Release the weight. Assume that the spring is elastic — it bounces up and down *ad infinitum*. Set the time to zero at the onset of motion. Displacement of the weight as a function of time should vary between the peak amplitudes $(+0.8, -0.8)$. If you had a device that could trace the amplitude of the displacement as a function of time, it would produce a sinusoidal curve as shown in Figure 1.1-1 (frame 1). Measure the time interval between two consecutive peaks; you will find that it is 0.080 s (80 ms). This time interval is called the *period* of the spring and it depends on the spring constant — a measure of spring stiffness. We say that the spring has completed one cycle of motion in a single period of time. Count the number of cycles within one second. This should be 12.5 cycles, which is called the *frequency* associated with the spring motion. One cycle per second (cps) is one hertz (Hz). Note that $1/0.080$ s $= 12.5$ Hz; that is, frequency is the inverse of the period.

To continue, repeat the above experiment using a spring with a higher stiffness. Give the second spring a peak displacement amplitude of 0.4 units. The motion of the spring is traced as another sinusoid in Figure 1.1-1 (frame 2). The period and frequency of the spring are 0.040 s and 25 Hz, respectively. To keep track of these measurements, plot the peak amplitude of each spring as a function of frequency. These are the *amplitude spectra* shown in Figure 1.1-1.

Working with two identical springs, release spring 1 from a peak amplitude displacement of 0.8 units and set the time to zero at the onset of the motion. When spring 1 passes through the zero amplitude position, set spring 2 in motion from the same peak amplitude displacement (0.8). The motion of spring 1 is plotted in frame 1, while the motion of spring 2 is plotted in frame 3 of Figure 1.1-1. Because the springs were set to motion with the same peak amplitude displacement, the amplitude spectra of the two sinusoidal time functions should be identical. However, a difference is noted between the time functions in frames 1 and 3. In particular, when the sinusoid in frame 1 takes the peak amplitude value, the sinusoid in frame 3 takes the zero amplitude value. There was a time delay (20 ms) equivalent to one-quarter of a full cycle in setting spring 2 in motion relative to spring 1. This time delay is the difference between the two sinusoids shown in frames 1 and 3. A full cycle is equivalent to 360 degrees or 2π radians. Therefore, a time delay of one-quarter of a cycle is equivalent to a +90-degree *phase-lag*.

Phase is defined as the negative of phase-lag (Robinson and Treitel, 1980). Thus, *a negative time shift corresponds to a positive phase value.* Note that in Figure 1.1-1, if we apply a time shift of one-quarter of a full cycle (20 ms) to the sinusoid in frame 3 in the negative time direction, we obtain the sinusoid in frame 1.

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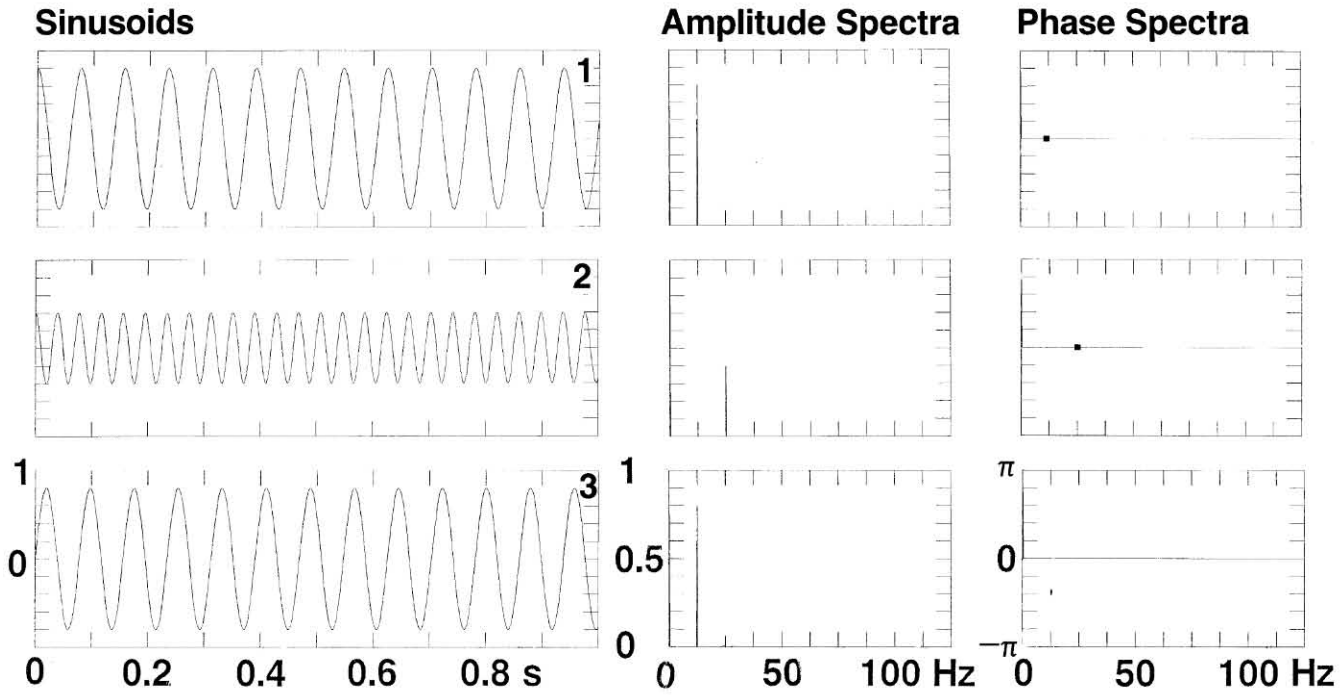

FIG. 1.1-1. Tracing the motion of a spring in time yields a sinusoidal curve where positive amplitude corresponds to spring motion in the upward direction. The peak amplitude represents the maximum displacement of the weight at the end of the spring from the unstretched position. The time between the two consecutive peaks is the period of the sinusoid, the inverse of which is called frequency. Amplitude spectra distinguish sinusoids 1 and 2, which have different peak amplitudes and frequencies. The time delay of the onset of one spring relative to another is defined as phase-lag. Phase spectra (the negative of phase-lag spectra) distinguish sinusoids 1 and 3.

Although their amplitude spectra are identical, these two sinusoids can be distinguished based on their *phase spectra* as seen in Figure 1.1-1.

The experiment is completed. What is learned? First, the motion of an elastic spring can be described by a sinusoidal time function. Second, and more important, a complete description of a sinusoidal motion is given by its frequency, peak amplitude, and phase. This experiment teaches us how to describe spring motion as a function of time and frequency.

Now imagine an ensemble of many springs, each with a sinusoidal motion with a specific frequency, peak amplitude, and phase. The sinusoidal responses of all the members are shown in Figure 1.1-2. Suppose the motions of the individual springs are superimposed by adding all the traces. The result is a *time-dependent* signal that is represented by the first trace in Figure 1.1-2 (as indicated by the asterisk). The superposition (*synthesis*) allows us to transform the motion from frequency to time domain. This transformation is reversible; that is, the time-domain signal can be broken down (*analyzed*) into its sinusoidal components in the frequency domain.

Mathematically, this two-way process is achieved by the Fourier transform. In practice, the standard algorithm used on digital computers is the fast Fourier transform (FFT). Analysis of a time-dependent signal into its frequency components is done by forward Fourier transform, while synthesis of all the frequency components to the time-domain signal is done by inverse Fourier transform.

Figure 1.1-3 is a display of the Fourier transform of the time-dependent signal from Figure 1.1-2. The amplitude and phase spectra constitute a more condensed frequency-domain representation of the sinusoids in Figure 1.1-2. We can clearly see the parallelism between the two types of displays. In particular, the amplitude spectrum in Figure 1.1-3 has a large and a relatively small peak at about 20 and 40 Hz, respectively. Darker bands corresponding to larger peak amplitudes occur in Figure 1.1-2 at about the same frequencies. On the other hand, zones of weak amplitudes at about 30 Hz and at the low- and high-frequency ends of the spectrum also

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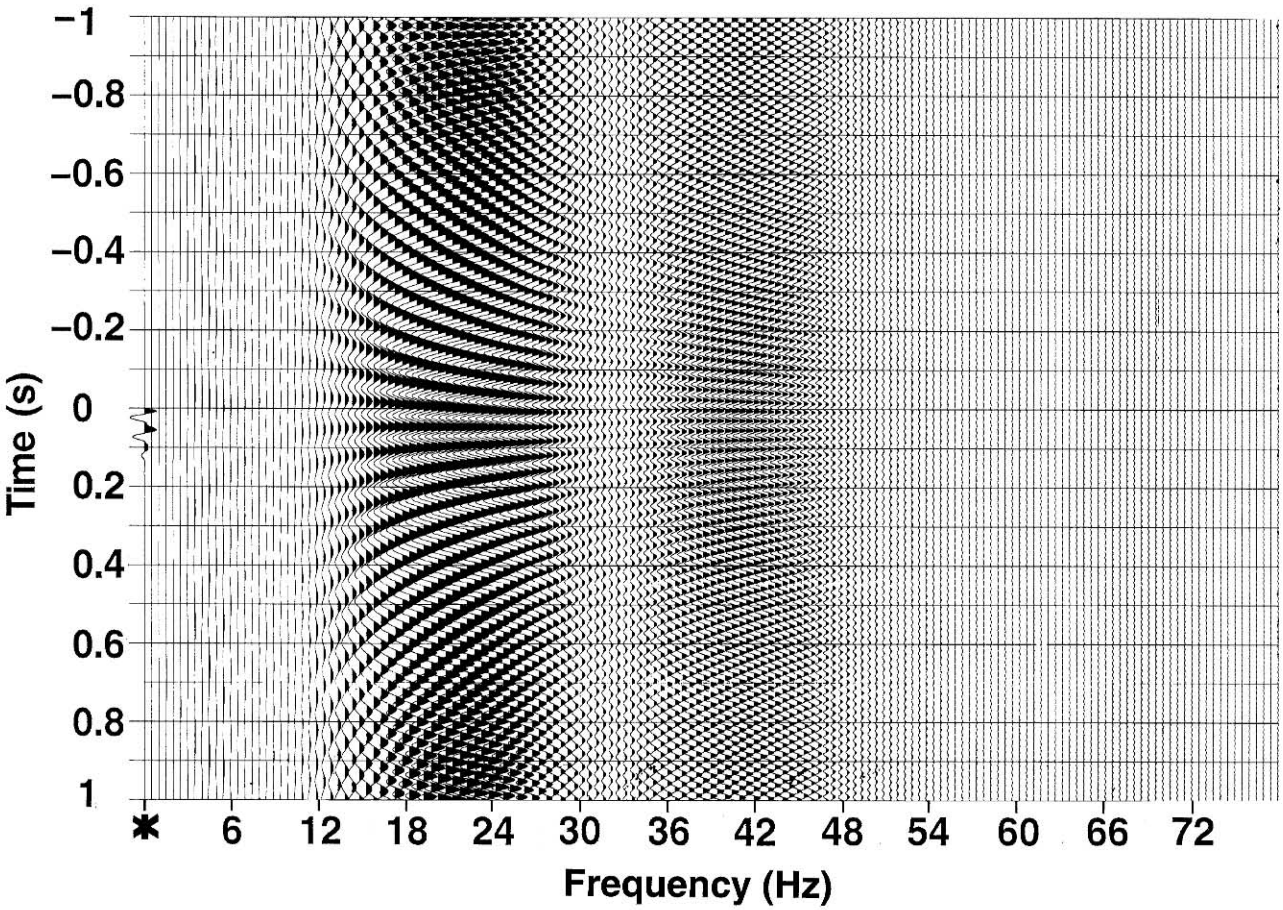

FIG. 1.1-2. An ensemble of sinusoidal motions with different frequency, peak amplitude, and phase-lag can be superimposed to synthesize a time-dependent waveform on the trace as indicated by the asterisk.

are apparent in both types of representations. Remember that the amplitude spectrum curve represents the peak amplitudes of the individual sinusoidal components as a function of frequency.

Now examine the phase spectrum. From the spring experiment, recall that the time delay of a particular frequency component also was expressed as a phase-lag. To better trace phase-lag as a function of frequency, a part of Figure 1.1-2 is magnified in Figure 1.1-4. Follow the positive-peak trend denoted by P. Note that the peaks fall above the zero timing line on the negative side of time axis at the low-frequency end of the spectrum. They then cross over to the positive side of the time axis at about 20 Hz and stay on that side over the rest of the frequency axis. The path that the peaks follow in Figure 1.1-4 can be plotted as the phase spectrum of Figure 1.1-3. If all the peaks were aligned along the zero timing line in Figure 1.1-4, then the corresponding time-domain signal would have a zero-phase spectrum. In this case, all the sinusoids would reinforce each other, causing a maximum peak value at zero time (Figure 1.1-11).

The physical significance of the amplitude spectrum is easier to understand than that of the phase spectrum. These two spectra are discussed further in this chapter. Basic mathematical details of the Fourier transform are given in Appendix A at the end of this chapter.

Analog versus Digital Signal

A seismic signal is a continuous time function. In digital recording, the continuous (analog) seismic signal is sampled at a fixed rate in time, called the *sampling interval* (or sampling rate). Typical values of sampling intervals range between 1 and 4 ms for most reflection seismic work. High-resolution studies require sampling intervals as small as 0.25 ms.

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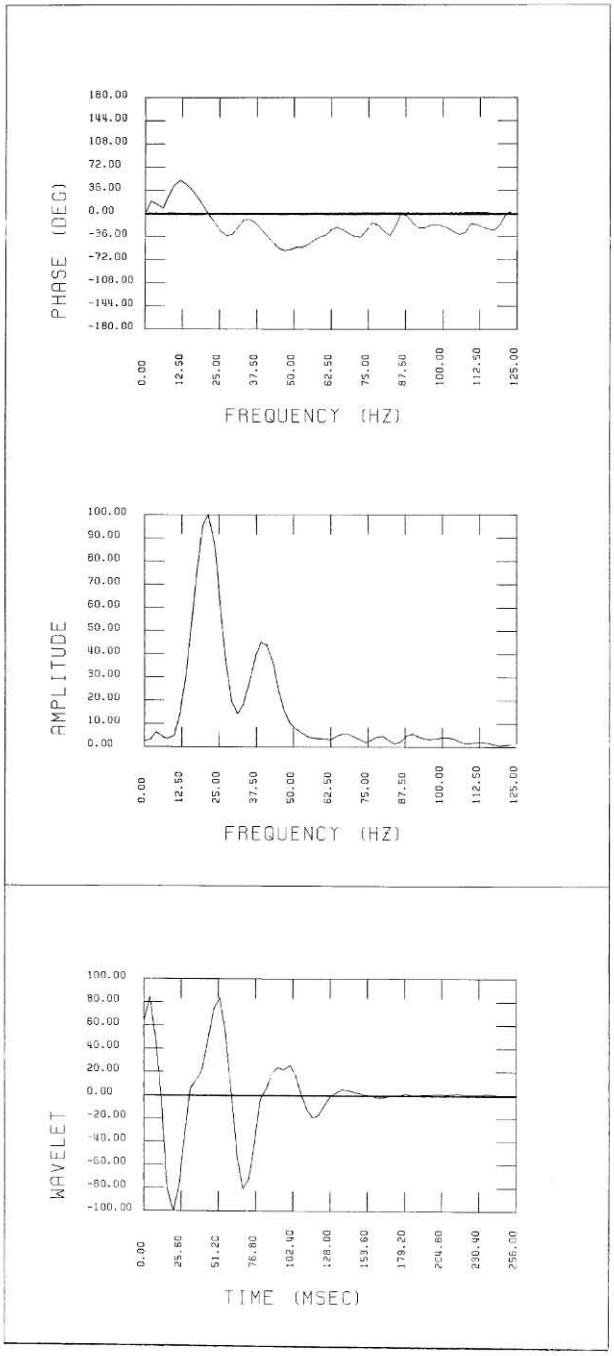

FIG. 1.1-3. The information from Figure 1.1-2 can be condensed into amplitude and phase spectra. Each point along the amplitude spectrum curve corresponds to the peak amplitude of the sinusoid at that frequency plotted as a trace in Figure 1.1-2. Note the equivalence of the two peaks in the amplitude spectrum with the two high-amplitude zones in Figure 1.1-2. Each point along the phase spectrum corresponds to the time delay of a peak or trough along the sinusoid at that frequency with respect to the timing line at $t = 0$. Note the equivalence of the phase curve with the trend of a positive peak from trace to trace in Figure 1.1-4.

Figure 1.1-5 shows a continuous signal in time. The discrete samples that might actually be recorded are shown by dots. A discrete time function is called a *time series*. The bottom curve in Figure 1.1-5 is an attempted reconstruction of the original analog signal, which is shown as the curve on top. Note that the reconstructed signal lacks the details present in the original analog signal. These details correspond to high-frequency components that were lost by sampling. If a smaller sampling interval were chosen, then the reconstructed signal would more accurately represent the original signal. For the extreme case of a zero sampling interval, the continuous signal can be represented exactly.

Is there a measure of the restorable frequency bandwidth of the digitized data? Figure 1.1-6 shows a time series, such as a seismic trace, with a 2-ms sampling interval and the corresponding amplitude spectrum. In general, given the sampling interval Δt, the highest frequency that can be restored accurately is called the Nyquist frequency and is given by

$$f_{Nyq} = \frac{1}{2\Delta t}. \qquad (1-1)$$

For the time series in Figure 1.1-6, $\Delta t = 2$ ms; therefore, the Nyquist frequency is 250 Hz. The original time series was resampled to obtain a series with 4- and 8-ms sampling intervals. The corresponding Nyquist frequencies are 125 and 62.5 Hz, respectively. Figure 1.1-6 also shows the series (as reconstructed back to 2 ms for plotting purposes) sampled at 4 and 8 ms with their amplitude spectra. Note that the coarser the sampling interval, the smoother the series. Smoothness results from a loss of high frequencies as seen in the amplitude spectra. Frequency components between 125 and 250 Hz, which are present in the time series with the 2-ms sampling interval, seem to be absent in the series resampled to 4 ms. Likewise, frequency components between 62.5 and 250 Hz seem to be absent from the series resampled to 8 ms. Can these frequencies be recovered? No. *Once a continuous signal is digitized, the highest frequency that can be restored accurately is the Nyquist frequency.*

We may think that when the time series sampled at 4 or 8 ms is interpolated back to a 2-ms sampling interval, those high frequencies should return. As stated earlier, the time series in Figure 1.1-6 with 4- and 8-ms sampling intervals actually were reconstructed by interpolation back to 2 ms to get the same number of samples as the original series for plotting with the same scale. *Interpolation does not recover the frequencies lost by sampling; it only generates extra samples.*

The implication for sampling the continuous signal in the field is an important one. If the earth signal had

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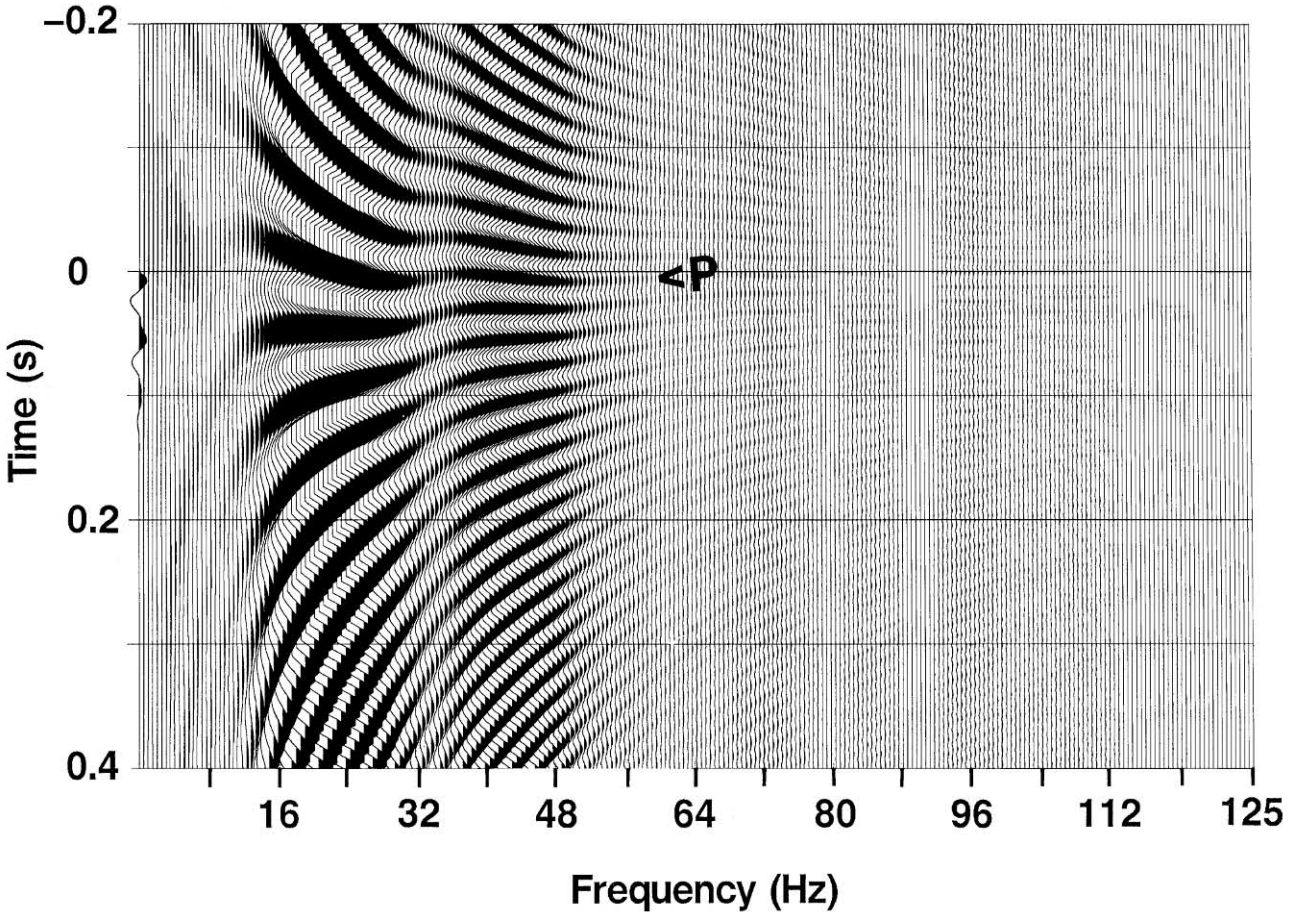

FIG. 1.1-4. A portion of Figure 1.1-2 is enlarged to better delineate the trend of the phase curve from trace to trace — from one frequency component to another. Compare the trend indicated by the positive peak P with the phase spectrum in Figure 1.1-3.

frequencies, say up to 150 Hz, then the 4-ms sampling interval would cause a loss of the band between 125 and 150 Hz.

While maximum recoverable signal frequency is the Nyquist frequency for a 1-D digitized data set, such as a single seismic trace, the situation can be different for two- or more dimensional data. Consider the process of moveout correction and stacking (Section 3.1). Given the sampling rate for the stacked trace, say 4 ms, data samples are searched on each input trace in the common-midpoint (CMP) gather based on the hyperbolic moveout equation. Since each input trace also is sampled at regular intervals, say 4 ms, the computed input sample location would normally fall in between two samples on a given input trace. By interpolation, the required sample value can be computed and placed on the output sample location (Section 3.1). Such an output-driven process would faithfully preserve frequencies below the Nyquist, only. Consider the alternative process in which a sample in the input trace is placed at the exact time location on the output trace (Wisecup, 1998). The resulting output stacked trace from this input-driven process would contain all the samples from all the input traces at exact time locations with irregular intervals. Such random sampling then yields a potential Nyquist frequency greater than the Nyquist frequency associated with the input traces.

Frequency Aliasing

Consider the sinusoid in Figure 1.1-7. This signal is resampled as before to 4 and 8 ms. The amplitude spectra indicate that all three have the same frequency, 25 Hz. Nothing happened to the signal after resampling it to a coarser sampling interval. Now examine the higher frequency sinusoid (75 Hz) in Figure 1.1-8. It appears the

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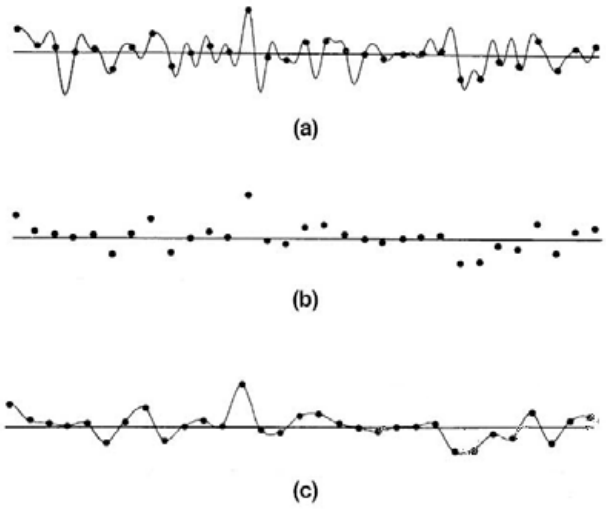

FIG. 1.1-5. (a) A continuous analog signal, (b) a digitized form of this signal, (c) reconstructed analog signal. When digitized, a continuous analog signal loses frequencies above the Nyquist frequency. (Adapted from Rothman, 1981)

same at both 2- and 4-ms sampling. However, resampling to 8 ms changed the signal and made it appear to be a lower frequency sinusoid. The resampled signal has a frequency of 50 Hz as seen in the amplitude spectrum. The Nyquist frequency for an 8-ms sampling interval is 62.5 Hz. The true signal frequency is 75 Hz. As a result of resampling, the signal with 75-Hz frequency *folded back* onto the spectrum and appeared at its *alias frequency* of 50 Hz. Finally, a 150-Hz sinusoid resampled to 4 and 8 ms is shown in Figure 1.1-9. This time, the 4-ms sampling made the signal appear as a 100-Hz signal, while the 8-ms sampling made it appear as a 25-Hz signal. By using a single-frequency sinusoid, we see that frequencies above the Nyquist really are not lost after sampling, but appear at frequencies below the Nyquist.

Now consider the superposition of two sinusoids with frequencies of 12.5- and 75-Hz as shown in Figure 1.1-10. Digitization of this signal at 2- and 4-ms sampling intervals does not alter the original signal, since its frequency components are below the Nyquist frequencies associated with 2- and 4-ms sampling intervals — 250 and 125 Hz, respectively. However, when the

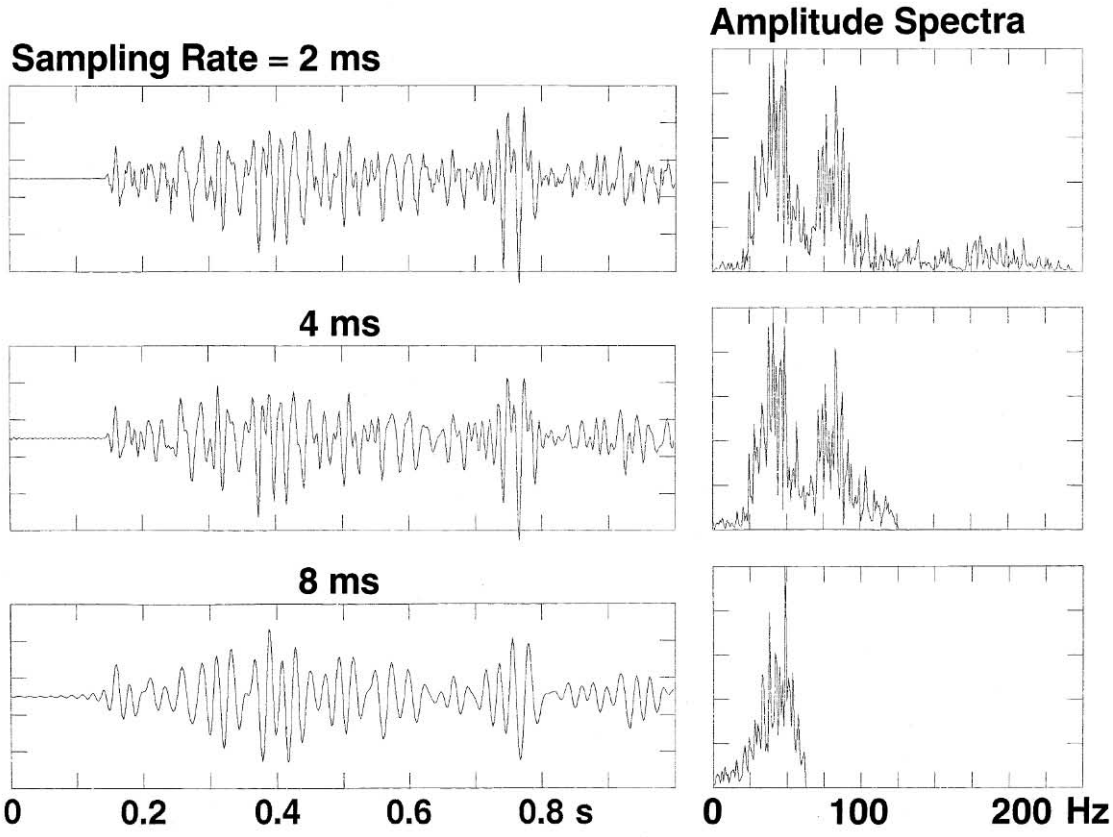

FIG. 1.1-6. A time series sampled at 2 ms has a Nyquist frequency of 250 Hz. Resampling to 4 and 8 ms confines the frequency band to 125 and 62.5 Hz, respectively. Note the loss of high frequencies at larger sampling intervals.

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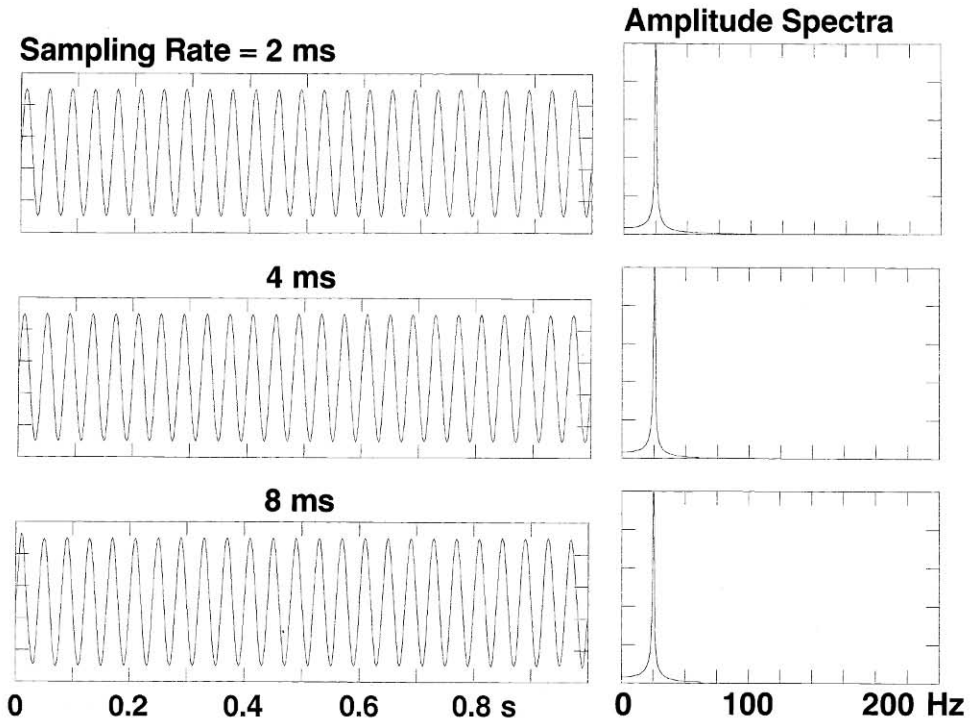

FIG. 1.1-7. A 25-Hz sinusoid sampled at 2 ms remains unchanged when resampled at 4 or 8 ms.

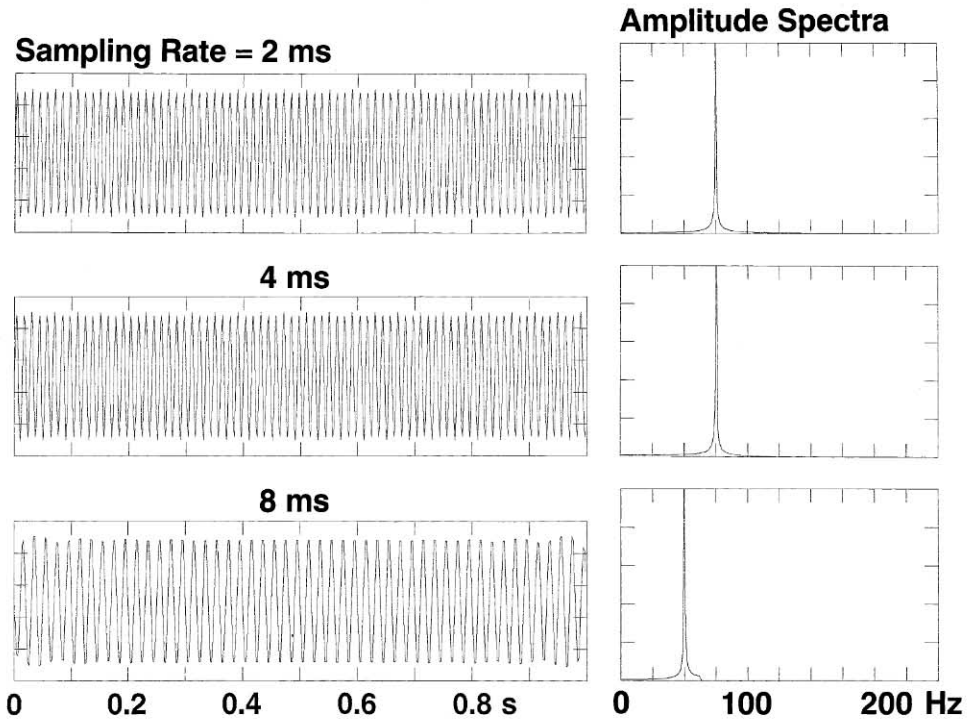

FIG. 1.1-8. A 75-Hz sinusoid sampled at 2 ms remains unchanged when resampled at 4 ms, but appears as a 50-Hz sinusoid when resampled at 8 ms. Hence, the latter is the alias of the original sinusoid.

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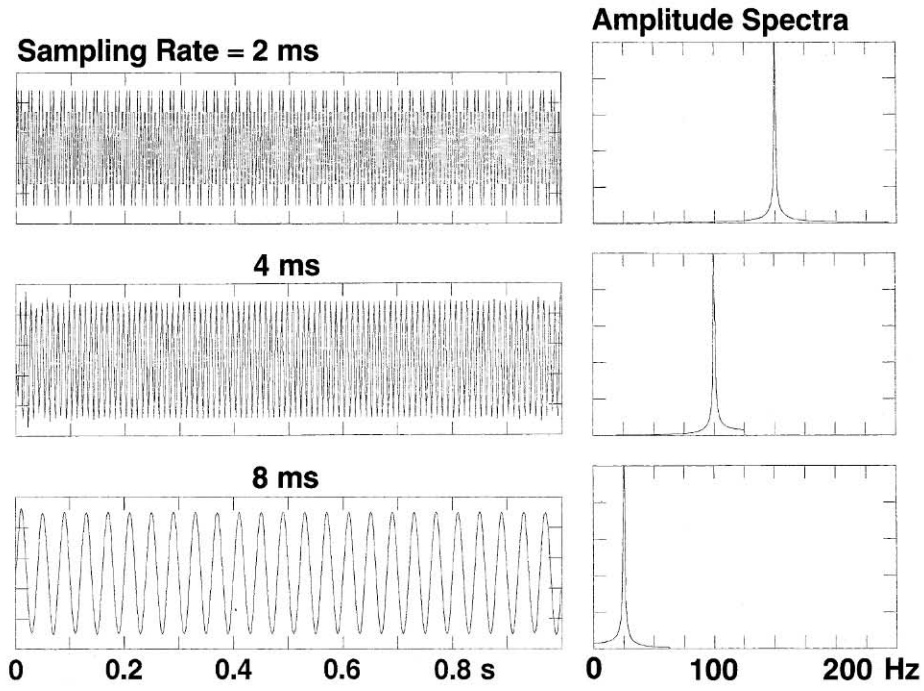

FIG. 1.1-9. A 150-Hz sinusoid sampled at 2 ms appears as a 100-Hz sinusoid when resampled at 4 ms and as a 25-Hz sinusoid when resampled at 8 ms. (Amplitude modulation in the 2-ms sinusoid is a result of limitations in plotting very high-frequency signals.)

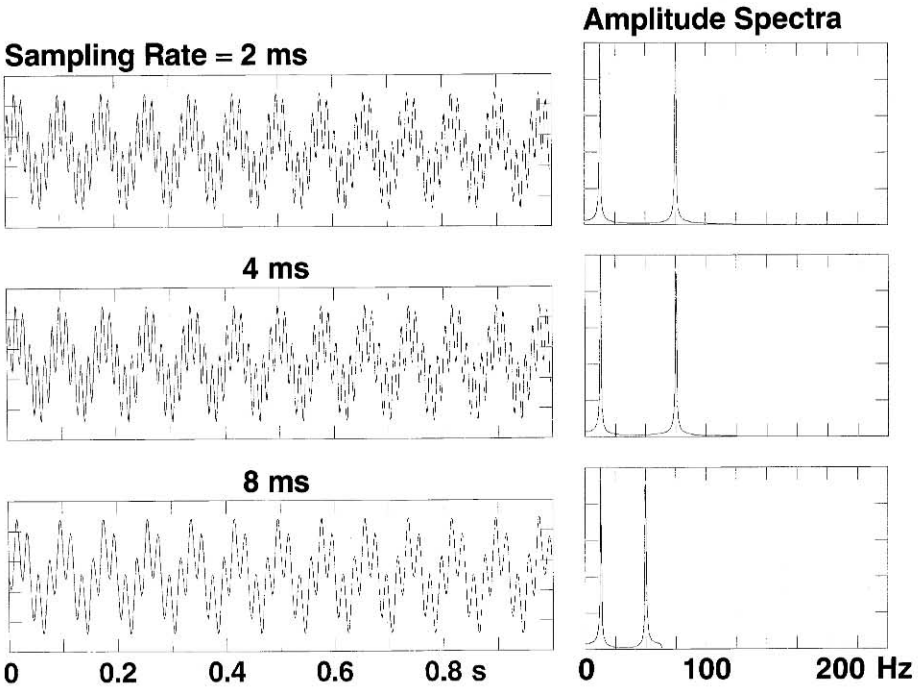

FIG. 1.1-10. A time series synthesized from two sinusoids with 12.5 and 75 Hz at 2-ms sampling rate remains unchanged when resampled at 4 ms. However, at 8 ms, its high-frequency component shifts from 75 to 50 Hz, while its low-frequency component remains the same.

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signal is digitized at a coarser sampling interval, such as 8 ms, the amplitude spectrum changes. The 12.5-Hz component is not affected, because 8-ms sampling still is sufficient to sample this low-frequency component. On the other hand, the 75-Hz component is seen as a lower-frequency component (50 Hz). Once again, note that those frequencies in the original signal above the Nyquist frequency corresponding to the chosen sampling interval are folded back onto the amplitude spectrum of the digitized version of the signal.

This analysis can be extended to many sinusoids of different frequencies. In particular, the discrete time series derived from a too coarse sampling (undersampling) of a continuous signal actually contains contributions from high-frequency components of that continuous signal. Those high frequencies fold back onto the spectrum of the discrete time series and appear as lower frequencies. The phenomenon that is caused by undersampling the continuous signal is termed *frequency aliasing*.

To compute the alias frequency f_a, use the following relation

$$f_a = |2mf_N - f_s|, \qquad (1-2)$$

where f_N is the folding frequency, f_s is the signal frequency, and m is an integer such that $f_a < f_N$. For example, suppose that $f_s = 65$ Hz, $f_N = 62.5$ Hz, which corresponds to 8-ms sampling rate. The alias frequency then is $f_a = |2 \times 62.5 - 65| = 60$ Hz.

In conclusion, undersampling has two effects:

(a) band limiting the spectrum of the continuous signal, with the maximum frequency being the Nyquist, and
(b) contamination of the digital signal spectrum by high frequencies beyond the Nyquist, which may have been present in the continuous signal.

Nothing can be done about the first problem. The second problem is of practical importance. To keep the recoverable frequency band between zero and the Nyquist frequency free from aliased frequencies, a high-cut antialiasing filter is applied in the field before analog-to-digital conversion of seismic signals. This filter eliminates those frequency components that would have been aliased during sampling. Typically, the high-cut antialiasing filter has a cutoff frequency that is either three-quarters or half of the Nyquist frequency. This filter rolls off steeply so that aliases of frequencies above the Nyquist are highly attenuated.

Phase Considerations

In the beginning of this section, a time-dependent signal was synthesized from its frequency components (Figure 1.1-2). Consider a signal with a zero-phase spectrum. Figure 1.1-11 shows sinusoids with frequencies ranging from approximately 1 to 32 Hz. All of these sinusoids have zero-phase lag; thus, the peak amplitudes align at $t = 0$. The time-domain signal on the trace identified by an asterisk in Figure 1.1-11 is synthesized by summing all these sinusoids — a process described by inverse Fourier transform. Such a time-domain signal is called a *wavelet*. A wavelet usually is considered a transient signal, that is, a signal with a finite duration. It has a start time and an end time, and its energy is confined between these two time positions. The wavelet that was just constructed is symmetric around $t = 0$ and has a (positive) peak amplitude at $t = 0$. Such a wavelet is called *zero phase*. In fact, the wavelet was synthesized using the zero-phase sinusoids of equal peak amplitude.

A zero-phase wavelet is symmetric with respect to zero time and peaks at zero time. Figure 1.1-12 shows the result of applying a linear phase shift to the sinusoids in Figure 1.1-11. Linear phase shift is described by a function that represents a line in the frequency domain: $\phi = \alpha\omega$, where α is the slope constant and ω is the angular frequency, which is the temporal frequency scaled by 2π. The wavelet, identified by an asterisk in Figure 1.1-12, has shifted in time by -0.2 s, but its shape has not changed. *Thus, a linear phase shift is equivalent to a constant time shift. The slope of the line describing the phase spectrum is proportional to the time shift.*

The wavelet can be shifted by any amount of time simply by changing the slope of the line $\phi = \alpha\omega$, that describes the phase spectrum. Starting with the zero-phase wavelet, Figure 1.1-13 shows the effect of increasing amounts of the linear phase shift on a zero-phase wavelet. Although not shown, by changing the sign of the slope in the phase spectrum, the wavelet can be shifted in the opposite time direction.

If a 90-degree phase shift is applied to each of the sinusoids in Figure 1.1-11, as shown in Figure 1.1-14, then the zero crossings are aligned at $t = 0$. The result of this summation yields the antisymmetric wavelet shown on the trace identified by an asterisk. Note that the two wavelets in Figures 1.1-11 and 1.1-14 have the same amplitude spectrum because they have been synthesized from the sinusoidal components with the same peak amplitude and frequency. The difference lies in their phase spectra. The wavelet in Figure 1.1-11 has zero-phase spectrum, while that in Figure 1.1-14 has a constant-phase spectrum (+90 degrees). Therefore, the difference in wavelet shape is a result of the difference in their phase spectra.

Figure 1.1-15 shows the effect of various amounts of constant phase shift on a zero-phase wavelet. The 90-degree phase shift converts the zero-phase wavelet

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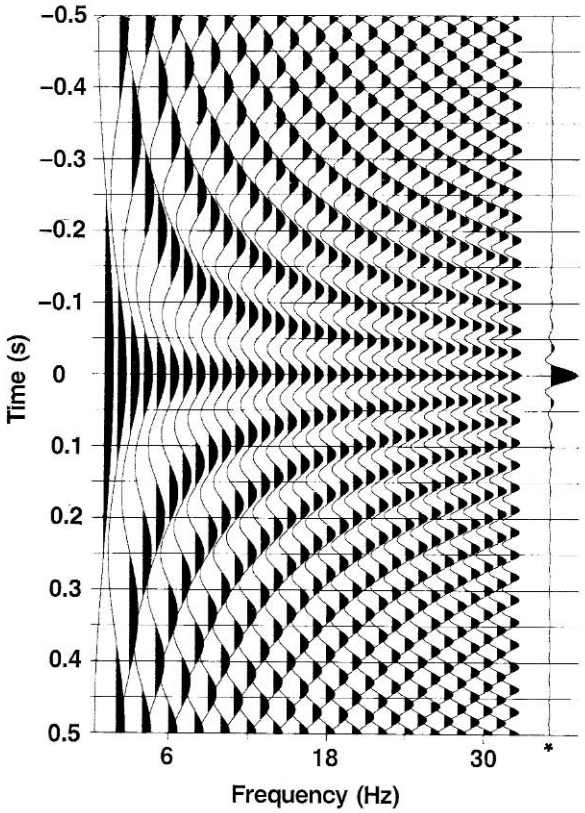

FIG. 1.1-11. Summation of a discrete number of sinusoids with no phase-lag, but with the same peak amplitude, yields a band-limited symmetric wavelet represented by the trace on the right (denoted by an asterisk). This is a zero-phase wavelet.

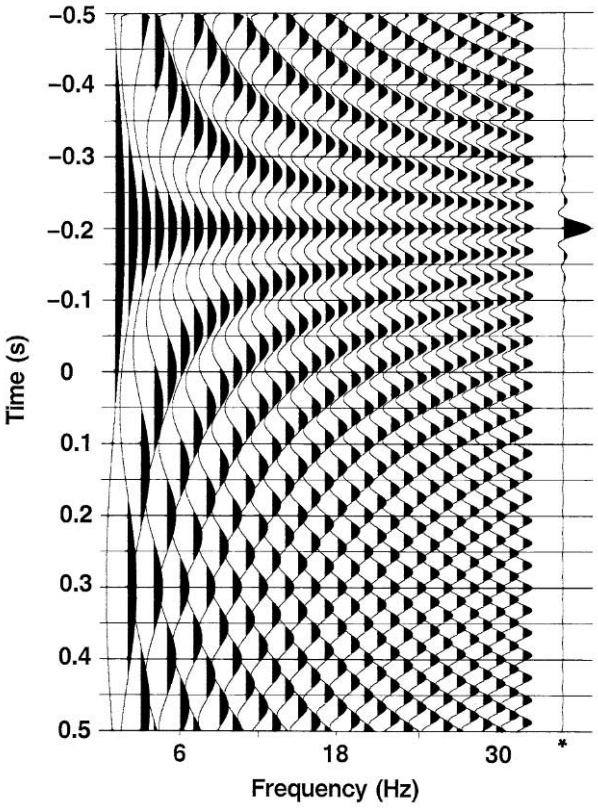

FIG. 1.1-12. The same sinusoidal components as in Figure 1.1-11, but with a −0.2 s constant-time delay. When summed, these sinusoids yield a band-limited symmetric wavelet that is represented by the trace on the right (denoted by an asterisk). This wavelet is the same as that shown in Figure 1.1-11, except that it is shifted in time by −0.2 s. This time shift is related to the linear phase spectrum associated with the summed frequency components.

an antisymmetric wavelet. The 180-degree phase shift changes the polarity of the zero-phase wavelet. The 270-degree phase shift changes the polarity of the zero-phase wavelet, while converting it to an antisymmetric wavelet. Finally, the 360-degree phase shift retains the shape of the original wavelet. *A constant phase shift to changes the shape of a wavelet. In particular, a 90-degree phase shift converts a symmetric wavelet to an antisymmetric wavelet, while a 180-degree phase shift changes its polarity.*

Figure 1.1-16 shows a portion of a seismic section with the application of different degrees of constant phase rotation. Note the change in the wavelet character of the significant reflections. This difference in wavelet character has an impact on picking events for interpretation. When comparing displays of two different sections, which may be associated with two different vintages of processing of the same data or two different lines from the same survey, the wavelet character of the reflection event that is being picked must be consistent from one section to another. A common mistake is displaying two sections to be compared with opposite polarity. The polarity convention set by the Society of Exploration Geophysicists is based on a negative water-bottom reflection coefficient, which corresponds to a positive polarity.

So far, two basic phase spectra have been examined — linear and constant phase shifts. We now examine their combined effect. The phase spectrum is defined by a function $\phi = \phi_0 + \alpha\omega$, where ϕ_0 is the constant phase shift and α is the slope of the linear phase shift. Figure 1.1-17 shows the result of applying a linear phase shift (as in Figure 1.1-12) plus a 90-degree constant phase shift (as in Figure 1.1-14) to the sinusoids in Figure 1.1-11. The zero-phase wavelet with the same amplitude spectrum as that in Figure 1.1-11 was shifted in time by -0.2 s because of the linear phase shift, and converted to an antisymmetric form because of the constant 90-degree phase shift.

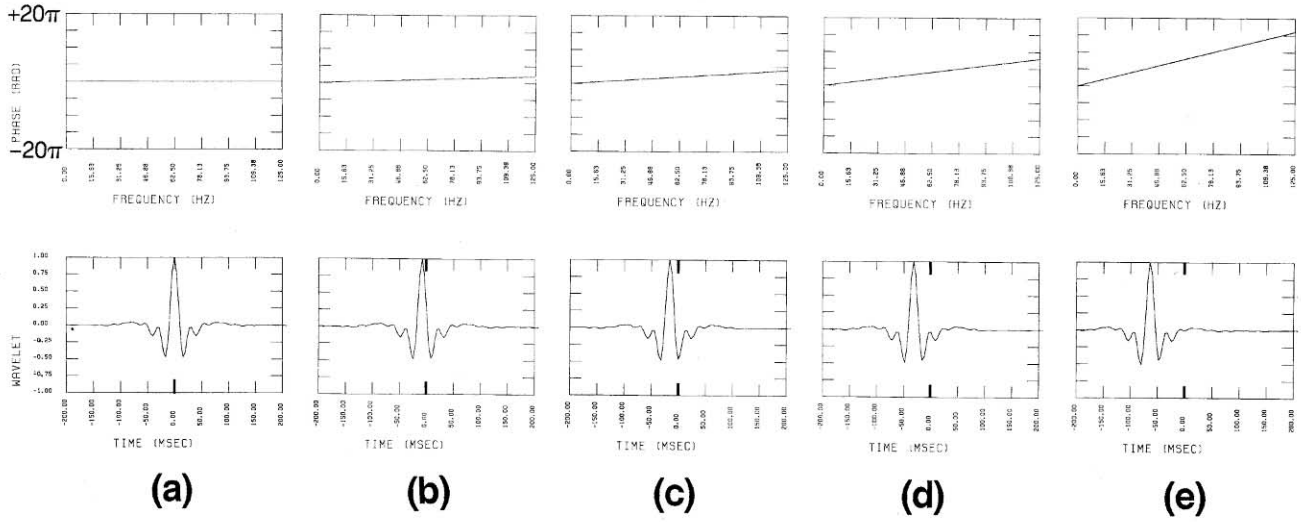

FIG. 1.1-13. Starting with a zero-phase wavelet (a), linear phase shifts are applied to shift the wavelet in time without changing its shape. The slope of the linear phase function is related to the time shift.

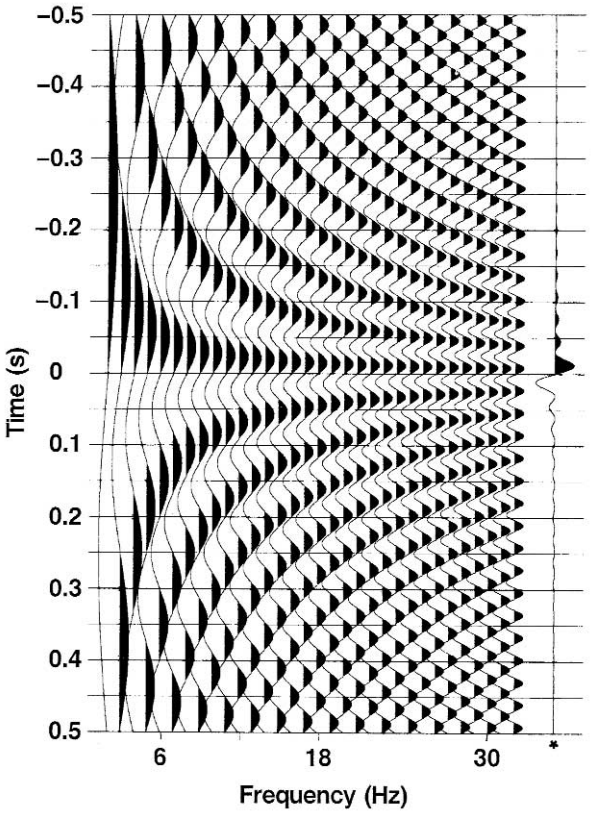

FIG. 1.1-14. The same sinusoidal components as in Figure 1.1-11 but with a constant 90-degree phase shift applied to each. The zero crossings are aligned at $t = 0$. Summation of these sinusoids yields an antisymmetric wavelet that is represented by the trace on the right (denoted by an asterisk).

Other variations in phase spectrum are shown in Figure 1.1-18. The zero-phase wavelet (Figure 1.1-18a) can be modified to different shapes simply by changing the phase spectrum. It can be modified to the extent that it may no longer resemble the original wavelet shape as illustrated by the last example (Figure 1.1-18d). *By keeping the amplitude spectrum unchanged, the wavelet shape can be changed by modifying the phase spectrum.*

Time-Domain Operations

Consider a reflectivity sequence represented by the time series $(1, 0, \frac{1}{2})$. Also consider an impulsive source that causes an explosion at $t = 0$ with an amplitude of 1. The response of the reflectivity sequence to an impulse is called the *impulse response*. This physical process can be described as in Table 1-1.

Table 1-1. Response of the reflectivity sequence $(1, 0, \frac{1}{2})$ to a zero-delay explosive impulse $(1, 0)$.

Time of Onset	Reflectivity Sequence			Source		Response			
0	1	0	$\frac{1}{2}$	1	0	1	0	$\frac{1}{2}$	0

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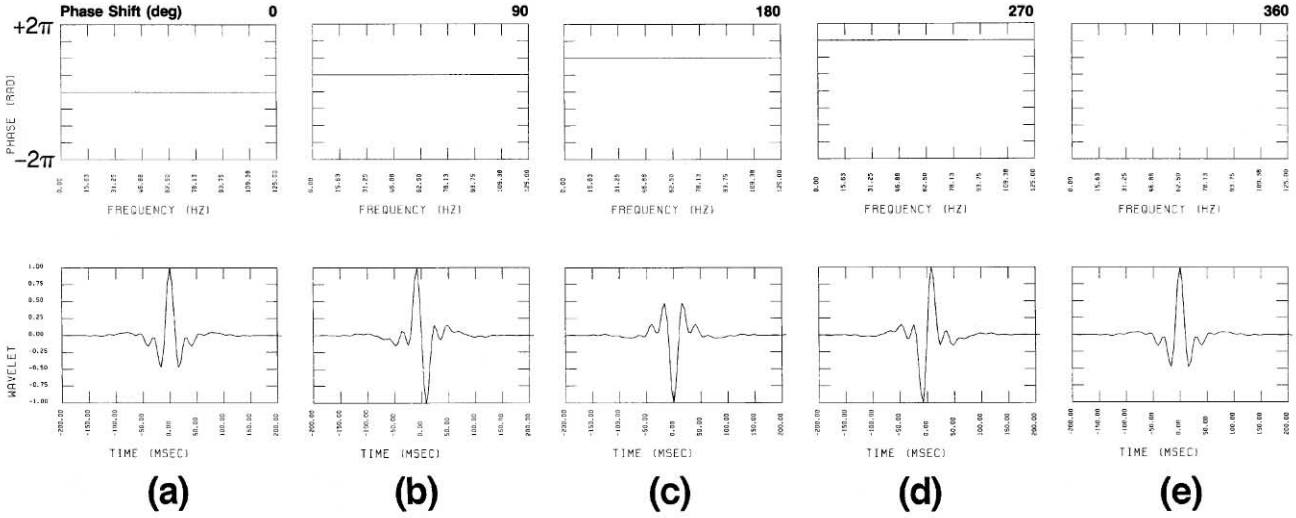

FIG. 1.1-15. Starting with the zero-phase wavelet (a), its shape is changed by applying constant phase shifts. A 90-degree phase shift converts the zero-phase wavelet to an antisymmetric wavelet (b), while a 180-degree phase shift reverses its polarity (c). A 270-degree phase shift reverses the polarity, while making the wavelet antisymmetric (d). Finally, a 360-degree phase shift does not modify the wavelet (e).

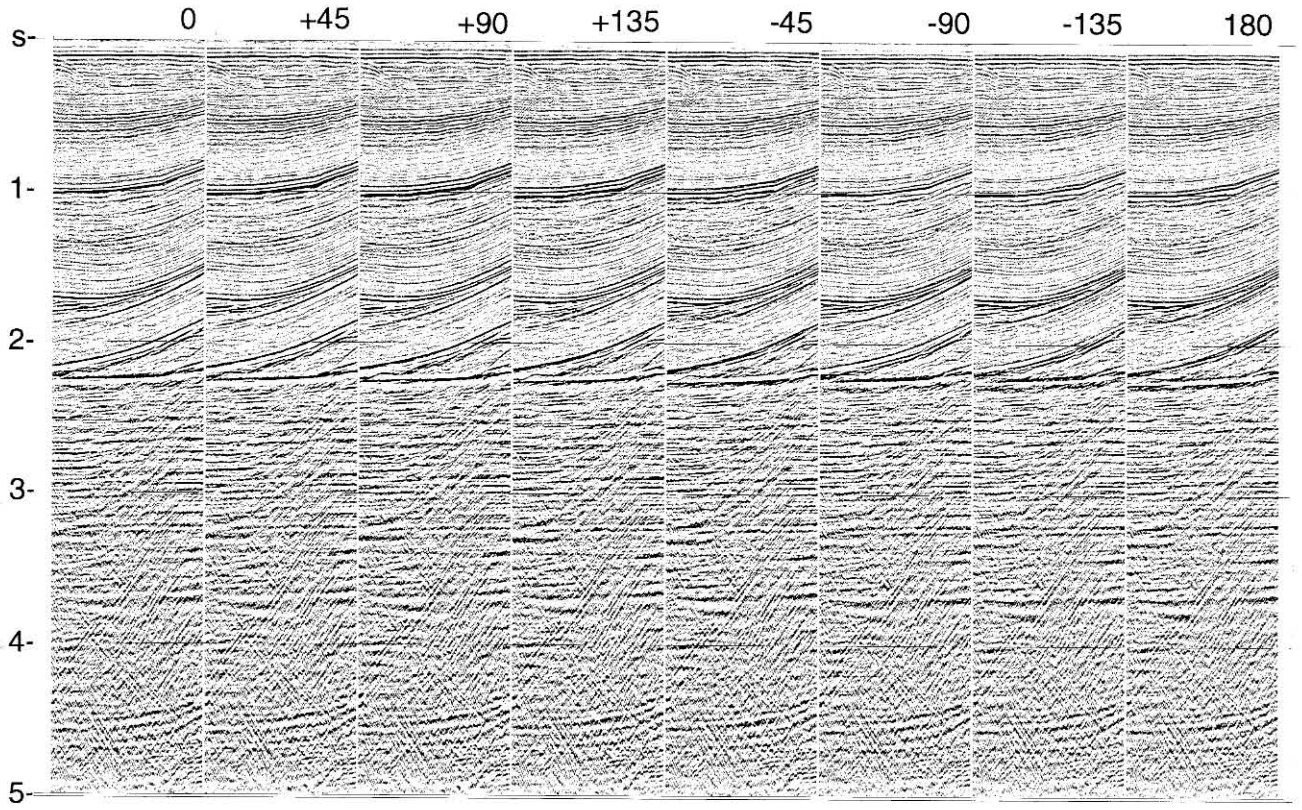

FIG. 1.1-16. A portion of a seismic section with different degrees of constant phase rotations.

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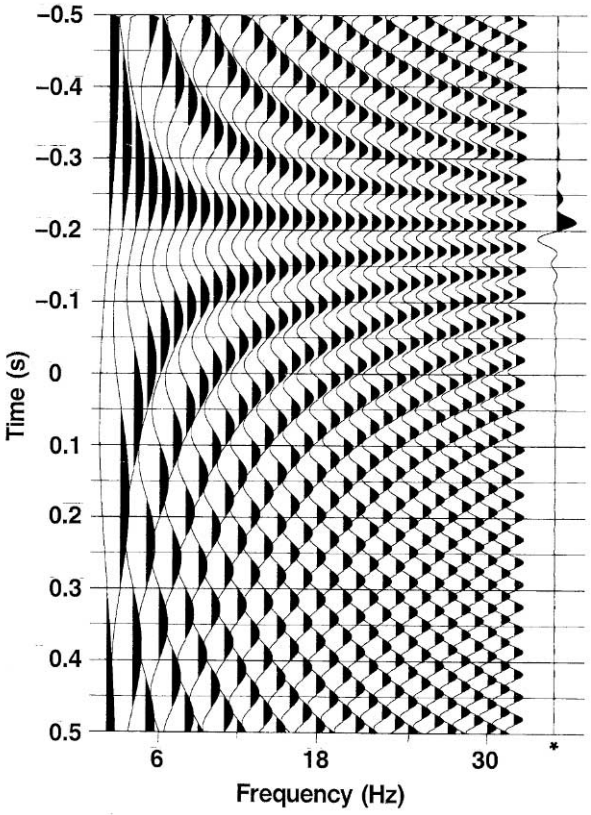

FIG. 1.1-17. A linear (as in Figure 1.1-12) combined with a constant phase shift (as in Figure 1.1-14) results in a time-shifted antisymmetric wavelet. The wavelet is represented by the trace on the right (denoted by an asterisk).

Table 1-2. Response of the reflectivity sequence $(1, 0, \frac{1}{2})$ to a unit-delay implosive impulse $(0, -\frac{1}{2})$.

Time of Onset	Reflectivity Sequence	Source	Response
1	$1 \quad 0 \quad \frac{1}{2}$	$0 \quad -\frac{1}{2}$	$0 \quad -\frac{1}{2} \quad 0 \quad -\frac{1}{4}$

One unit time later, suppose that the impulsive source generates an implosion with an amplitude of $-\frac{1}{2}$. This response is described in Table 1-2.

Note that the response in each case is the reflectivity sequence *scaled* by the impulse strength and *delayed* by the impulse onset. Since a general source function is considered to be a sequence of explosive and implosive impulses, the individual impulse responses are added to obtain the combined response. This process is called *linear superposition* and is described in Table 1-3.

Convolution

In Table 1-3, the asterisk denotes *convolution*. The response of the reflectivity sequence $(1, 0, \frac{1}{2})$ to the source wavelet $(1, -\frac{1}{2})$ was obtained by convolving the two series. This is done computationally as shown in Ta-

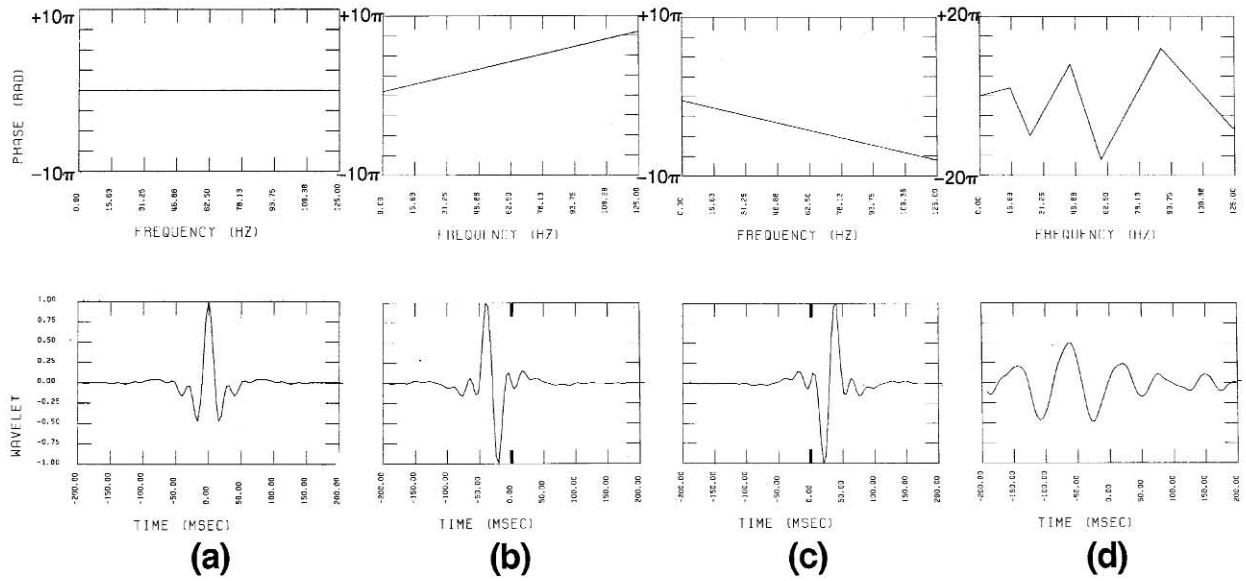

FIG. 1.1-18. The shape of a zero-phase wavelet (a) can be modified by introducing a nonzero-phase spectrum of any form as in (b), (c), and (d).

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Table 1-3. Linear superposition of the two responses described in Tables 1-1 and 1-2.

Time of Onset	Reflectivity Sequence			Source		Response			
0	1	0	$\frac{1}{2}$	1	0	1	0	$\frac{1}{2}$	0
1	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$-\frac{1}{4}$
Superposition:				1	$-\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{4}$

Expressed differently: $(1, 0, \frac{1}{2}) * (1, -\frac{1}{2}) = (1, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{4})$

ble 1-4. A fixed array is set up from the reflectivity sequence. The source wavelet is reversed (folded) and moved (lagged) one sample at a time. At each lag, the elements that align are multiplied and the resulting products are summed.

The mechanics of convolution are described in Table 1-5. The number of elements of output array c_k is given by $m+n-1$, where m and n are the lengths of the operand array a_i and the operator array b_j, respectively.

When the roles of the arrays in Table 1-4 are interchanged, the output array in Table 1-6 results. Note that the output response is identical to that in Table 1-4. Hence, *convolution is commutative — it does not matter which array is fixed and which is moved, the output is the same.*

Crosscorrelation and Autocorrelation

Seismic data processing often requires measurement of the similarity or time alignment of two traces. *Correla-*

Table 1-4. Convolution of the source wavelet $(1, -\frac{1}{2})$ with the reflectivity sequence $(1, 0, \frac{1}{2})$.

Reflectivity Sequence			Output Response
1	0	$\frac{1}{2}$	
$-\frac{1}{2}$	1		1
	$-\frac{1}{2}$	1	$-\frac{1}{2}$
		$-\frac{1}{2}$	$\frac{1}{2}$
			$-\frac{1}{4}$

Table 1-5. Mechanics of the convolutional process.

Fixed Array:
$a_0, a_1, a_2, a_3, a_4, a_5, a_6, a_7$

Moving Array:
b_0, b_1, b_2

Given two arrays, a_i and b_j:

Step 1 : Reverse moving array b_j.
Step 2 : Multiply in the vertical direction.
Step 3 : Add the products and write as output c_k.
Step 4 : Shift array b_j one sample to the right and repeat Steps 2 and 3.

Convolution Table:

a_0	a_1	a_2	a_3	a_4	a_5	a_6	a_7	Output
b_2	b_1	b_0						c_0
	b_2	b_1	b_0					c_1
		b_2	b_1	b_0				c_2
			b_2	b_1	b_0			c_3
				b_2	b_1	b_0		c_4
					b_2	b_1	b_0	c_5
						b_2	b_1	b_0	c_6
							b_2	b_1	b_0	c_7
								b_2	b_1	b_0	c_8
									b_2	b_1	b_0	c_9

where

$$c_k = \sum_{j=0}^{n} a_{k-j} b_j, \quad k = 0, 1, 2, \cdots, m+n-1.$$

Table 1-6. Convolution of the reflectivity sequence $(1, 0, \frac{1}{2})$ with the source wavelet $(1, -\frac{1}{2})$.

Source Wavelet			Output Response
1	$-\frac{1}{2}$		
$\frac{1}{2}$	0	1	1
$\frac{1}{2}$	0	1	$-\frac{1}{2}$
$\frac{1}{2}$	0	1	$\frac{1}{2}$
$\frac{1}{2}$	0	1	$-\frac{1}{4}$

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Table 1-7. Crosscorrelation of wavelet 1 with wavelet 2.

											Output	Lag
			2	1	−1	0	0					
0	0	2	1	−1							−2	−4
	0	0	2	1	−1						1	−3
		0	0	2	1	−1					6	−2
			0	0	2	1	−1				1	−1
				0	0	2	1	−1			−2	0
					0	0	2	1	−1		0	1
						0	0	2	1	−1	0	2
							0	0	2	1 −1	0	3
								0	0	2 1 −1	0	4

tion is another time-domain operation that is used to make such measurements. Consider the following two wavelets:

Wavelet 1: $(2, 1, -1, 0, 0)$
Wavelet 2: $(0, 0, 2, 1, -1)$

Although these wavelets are identical in shape, wavelet 2 is shifted by two samples with respect to wavelet 1. The time lag at which they are most similar can be determined. To do this, perform the operation on wavelet 1 as described in Table 1-5 without reversing wavelet 2 (omit Step 1). This is *crosscorrelation* and the result is shown in Table 1-7. Crosscorrelation measures how much two time series resemble each other.

From Table 1-7, note that maximum correlation occurs at lag −2. This suggests that if wavelet 2 were shifted two samples back in time, then these two wavelets would have maximum similarity.

Table 1-8 shows the crosscorrelation values that result when the arrays are interchanged. This time the maximum correlation occurs at lag 2. Thus, if wavelet

Table 1-8. Crosscorrelation of wavelet 2 with wavelet 1.

											Output	Lag
			0	0	2	1	−1					
2	1	−1	0	0							−0	−4
	2	1	−1	0	0						0	−3
		2	1	−1	0	0					0	−2
			2	1	−1	0	0				0	−1
				2	1	−1	0	0			−2	0
					2	1	−1	0	0		1	1
						2	1	−1	0	0	6	2
							2	1	−1	0 0	1	3
								2	1	−1 0 0	−2	4

Table 1-9. Crosscorrelation of wavelet 1 with wavelet 2 and crosscorrelation of wavelet 2 with wavelet 1.

Wavelet 1 with Wavelet 2	Crosscorrelation Output	Lag	Wavelet 2 with Wavelet 1
−2		−4	0
1		−3	0
6		−2	0
1		−1	0
−2		0	−2
0		1	1
0		2	6
0		3	1
0		4	−2

1 were shifted by two samples forward in time, these two wavelets would have maximum similarity.

Unlike convolution, *crosscorrelation is not commutative — the output depends on which array is fixed and which is moved.* Table 1-9 shows a comparison of the crosscorrelation results listed in Tables 1-7 and 1-8.

Crosscorrelation of a time series with itself is known as *autocorrelation.* Table 1-10 shows the autocorrelation lags of wavelet 1. Note that maximum correlation occurs at zero lag, an important property of autocorrelation. Moreover, *the autocorrelation function is symmetric.* This is a property of a real time series. Therefore, only one side of the autocorrelation needs to be computed.

It is heuristically shown in the next section on frequency filtering that convolution in the time domain is equivalent to multiplication in the frequency domain (Bracewell, 1965). Since correlation is a convolution without reversing the moving array (Table 1-5), a simi-

Table 1-10. Autocorrelation of wavelet 1.

											Output	Lag
			2	1	−1	0	0					
2	1	−1	0	0							−0	−4
	2	1	−1	0	0						0	−3
		2	1	−1	0	0					−2	−2
			2	1	−1	0	0				1	−1
				2	1	−1	0	0			6	0
					2	1	−1	0	0		1	1
						2	1	−1	0	0	−2	2
							2	1	−1	0 0	0	3
								2	1	−1 0 0	0	4

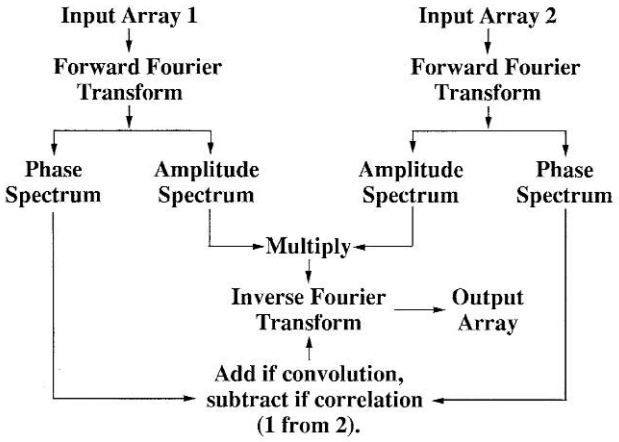

FIG. 1.1-19. The frequency-domain description of convolution and correlation.

lar frequency-domain operation also applies to correlation.

Figure 1.1-19 is a summary of frequency-domain descriptions of convolution and correlation. Note the following properties for convolution and correlation:

(a) Both convolution and correlation produce an output with a spectral bandwidth that is common to both of the input series. The immediate example is the band-pass filtering process.
(b) Phases are additive in case of convolution and subtractive in case of correlation (Bracewell, 1965).
(c) For autocorrelation, this implies that the output series is zero-phase. This fact already was verified by the example in Table 1-10 where it was shown that the autocorrelation is symmetric with respect to zero lag.

As a measure of similarity, crosscorrelation is used widely at various stages of data processing. For instance, traces in a CMP gather are crosscorrelated with a pilot trace to compute residual statics shifts (Section 3.3). The fundamental basis for computing velocity spectra is crosscorrelation (Section 3.2). The constituent elements of the Wiener filter (Section 2.3) are crosscorrelation of the desired output waveform with the input wavelet, and autocorrelation of the input wavelet.

Vibroseis Correlation

One other important process is the vibroseis correlation. This involves crosscorrelation of a *sweep* signal with the recorded vibroseis trace. The sweep is a frequency-modulated vibroseis source signal input to the ground. The convolutional model for vibroseis data is described in Section 2.5.

Figure 1.1-20 shows a vibroseis sweep signal, a recorded common-source gather, and the correlated gather. The sweep length is 10 s with a frequency band of 6 to 60 Hz. The 15-s uncorrelated vibroseis record yields a 5-s correlated record. Note that the early part of the uncorrelated record contains low-frequency energy with increasingly higher frequencies at late times. This is because an *upsweep* (frequency increasing with time) signal was used in this data example.

Frequency Filtering

What happens to a wavelet when its amplitude spectrum is changed while its zero-phase character is preserved? To begin, consider the wavelet in Figure 1.1-21 (summed trace 1) resulting from superposition of two very low-frequency components. Then, add increasingly higher frequency components to the Fourier synthesis (summed traces 2 through 5). Note that the wavelet in the time domain is compressed as the frequency bandwidth (the range of frequencies summed) is increased. Ultimately, if all the frequencies in the inverse Fourier transformation are included, then the resulting wavelet becomes a spike, as seen in Figure 1.1-22 (summed trace 6). Therefore, a spike is characterized as the in-phase synthesis of all frequencies from zero to the Nyquist. For all frequencies, the amplitude spectrum of a spike is unity, while its phase spectrum is zero.

Figure 1.1-23 shows five zero-phase wavelets, synthesized as shown in Figure 1.1-21. Note that all of them have band-limited amplitude spectra. A zero-phase band-limited wavelet can be used to *filter* a seismic trace. The output trace contains only those frequencies that make up the wavelet used in filtering. The time-domain representation of the wavelet is the *filter operator*. The individual time samples of this operator are the *filter coefficients*. The process described here is zero-phase frequency filtering, since it does not modify the phase spectrum of the input trace, but merely band-limits its amplitude spectrum.

Frequency-domain filtering involves multiplying the amplitude spectrum of the input seismic trace by that of the filter operator. The procedure is described in Figure 1.1-24. On the other hand, the filtering process in the time domain involves convolving the filter operator with the input time series. Figure 1.1-25 is a description of the filter design and its time-domain application. The frequency- and time-domain formulations of the filtering process (Figures 1.1-24 and 1.1-25) are based on the following important concept in time series

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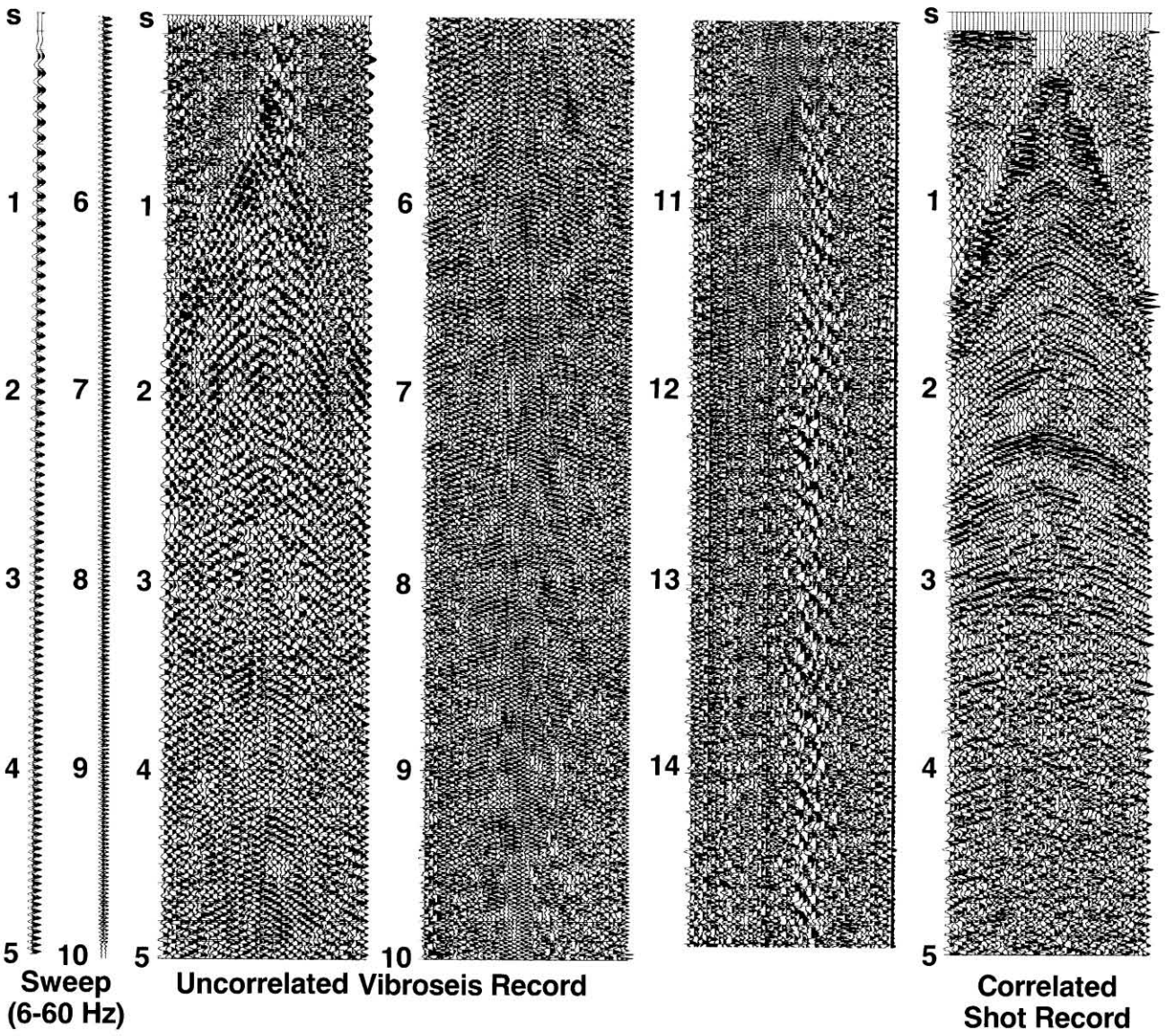

FIG. 1.1-20. Vibroseis correlation: the sweep signal is correlated with the recorded vibroseis record to get correlated field data. A 10-s sweep and 15-s recorded data yield a 5-s correlated record.

analysis (Section A.1): *Convolution in the time domain is equivalent to multiplication in the frequency domain. Similarly, convolution in the frequency domain is equivalent to multiplication in the time domain.*

Frequency filtering can be in the form of band-pass, band-reject, high-pass (low-cut), or low-pass (high-cut) filters. All of these filters are based on the same principle — construction of a zero-phase wavelet with an amplitude spectrum that meets one of the four specifications.

Band-pass filtering is used most commonly, because a seismic trace typically contains some low-frequency noise, such as ground roll, and some high-frequency ambient noise. The usable seismic reflection energy usually is confined to a bandwidth of approximately 10 to 70 Hz, with a dominant frequency around 30 Hz.

Band-pass filtering is performed at various stages in data processing. If necessary, it can be performed before deconvolution to suppress remaining ground-roll energy and high-frequency ambient noise that otherwise would contaminate signal autocorrelation. Narrow band-pass filtering may be necessary before crosscorrelating traces in a CMP gather with a pilot trace for use in estimating residual statics shifts (Section 3.3).

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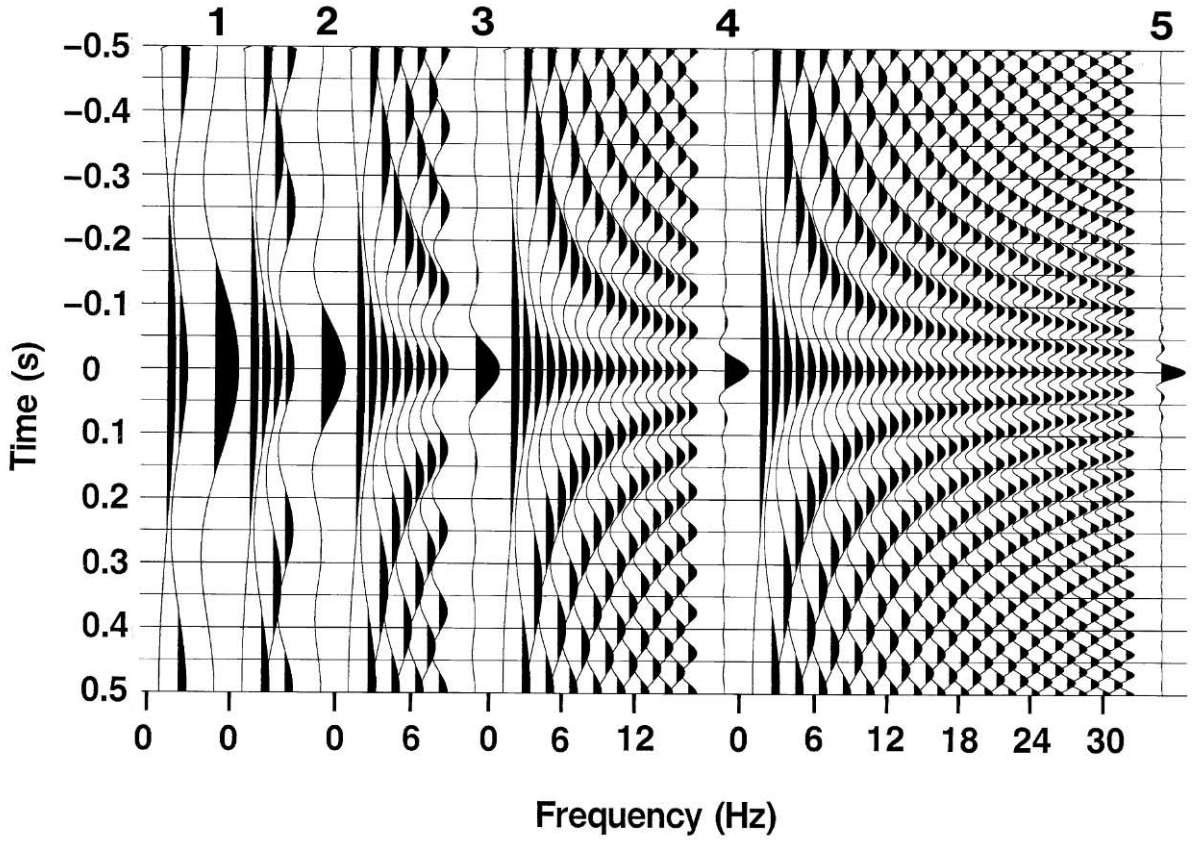

FIG. 1.1-21. The summation of zero-phase sinusoids with identical peak amplitudes. Traces resulting from each summation are numbered from 1 to 5. As the frequency bandwidth is increased, the synthesized zero-phase wavelet is increasingly compressed.

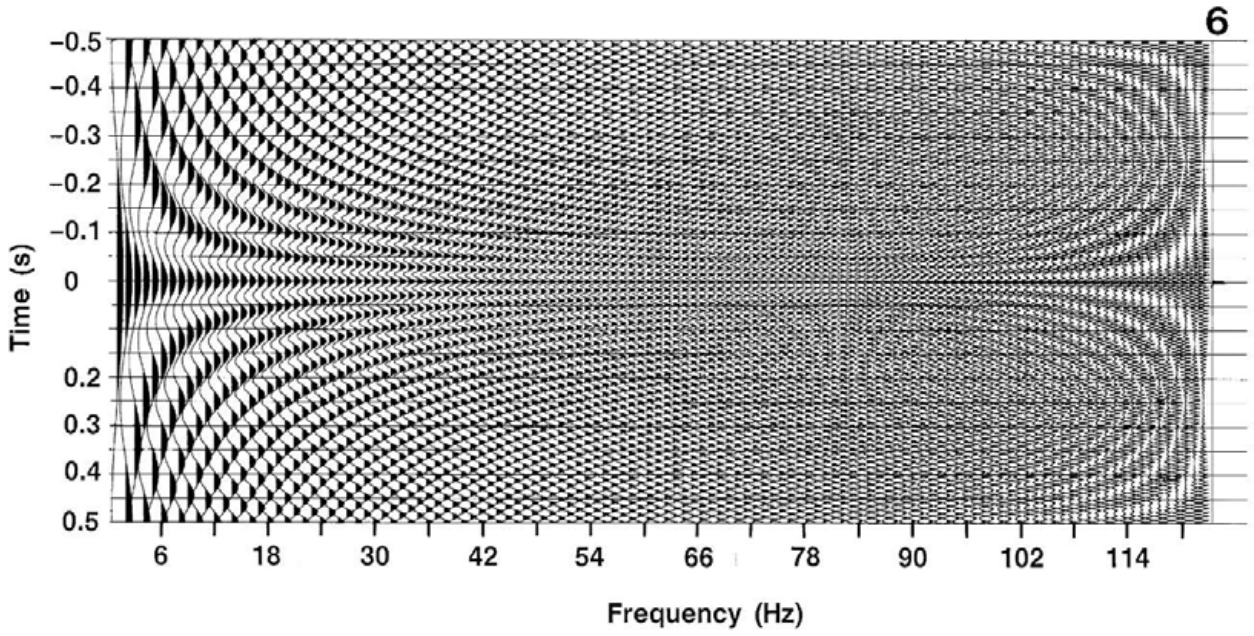

FIG. 1.1-22. The output wavelet becomes a spike (summed trace 6) when the summation includes sinusoids at all frequencies up to the Nyquist frequency. Compare this with output traces 1 through 5 in Figure 1.1-21.

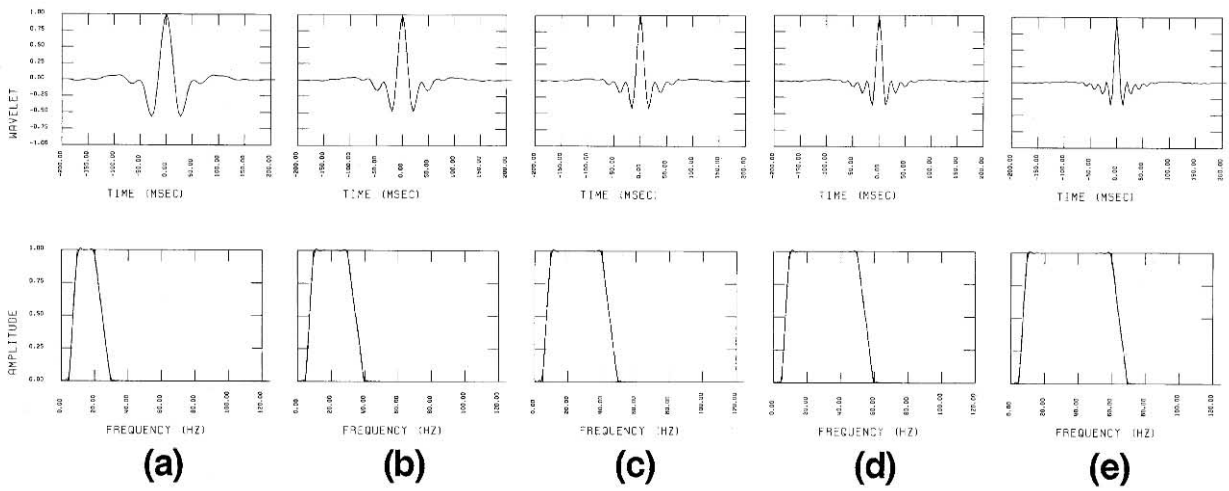

FIG. 1.1-23. A series of zero-phase wavelets (top row) and their respective amplitude spectra (bottom row). As bandwidth is increased, the wavelet is more compressed in time.

Band-pass filtering also can be performed before computing crosscorrelations during construction of the velocity spectrum for improved velocity picking (Section 3.2). Finally, it is a standard practice to apply a time-variant band-pass filter to stacked data (Section 1.5).

Practical Aspects of Frequency Filtering

Application of a filter in the frequency or time domain (Figures 1.1-24 and 1.1-25) yields basically identical results. In practice, the time-domain approach is favored, since convolution involving a short array, such as a filter operator, is more economical than applying Fourier transforms.

From Figure 1.1-23, the fundamental property of frequency filters can be stated as follows: *The broader the bandwidth, the more compressed the filter operator; thus, fewer filter coefficients are required.* This property also follows from the fundamental concept that *the time span of a time series is inversely proportional to its spectral bandwidth.*

In designing a band-pass filter, the goal is to pass a certain bandwidth with little or no modification, and to largely suppress the remaining part of the spectrum as much as practical. At first, it appears that this goal can be met by defining the desired amplitude spectrum for the filter operator as follows:

$$A(f) = \begin{cases} 1, & f_1 < f < f_2; \\ 0, & \text{otherwise}, \end{cases} \qquad (1-3)$$

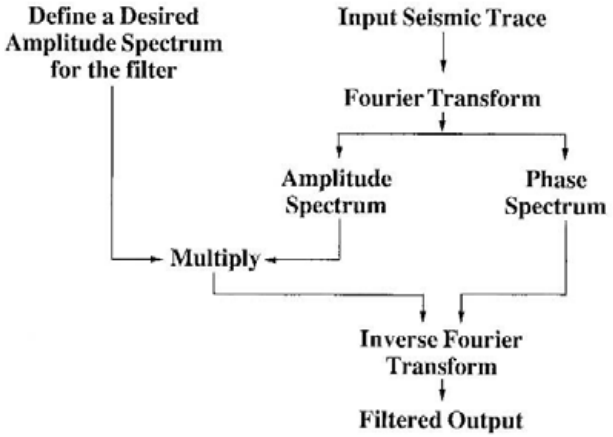

FIG. 1.1-24. Design and application of a zero-phase filter in the frequency domain.

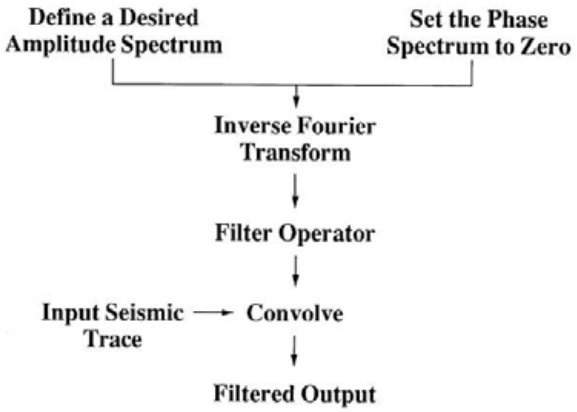

FIG. 1.1-25. Design of a zero-phase frequency filter and its application in the time domain.

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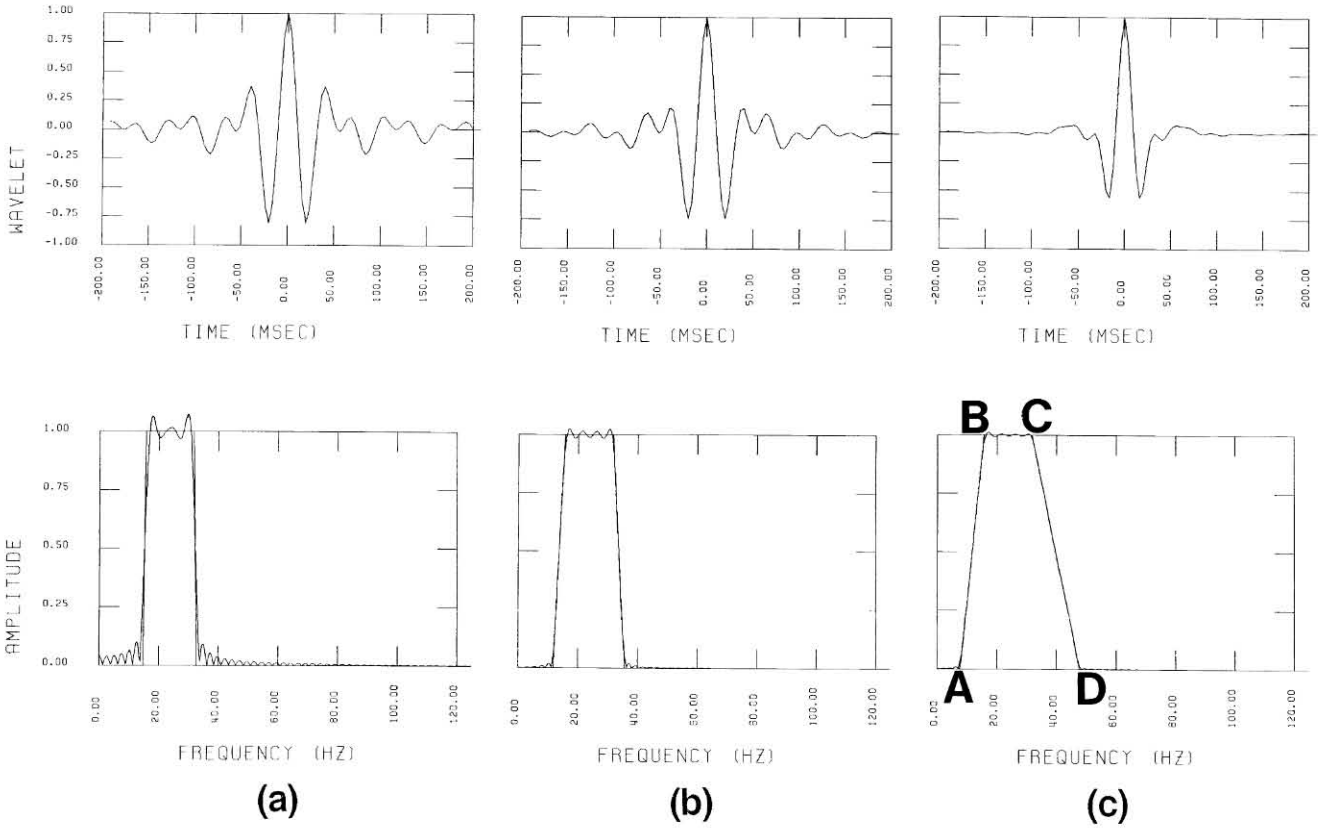

FIG. 1.1-26. Three zero-phase wavelets (top row) and their respective amplitude spectra (bottom row). (a) The steeply defined slopes of the passband cause ripples in the wavelet and the actual amplitude spectrum. (b) A moderate and (c) gentle slope help eliminate the ripples. Refer to the text for a discussion of corner frequencies A, B, C, and D.

where f_1 and f_2 are the cutoff frequencies. This is known as the boxcar amplitude spectrum.

To analyze the properties of such a filter, perform the following sequence of operations:

(a) Define a boxcar amplitude spectrum and zero-phase spectrum.
(b) Apply inverse FFT and obtain a filter operator.
(c) Truncate the operator.
(d) Apply forward FFT and compute the amplitude spectrum of the truncated operator.

Figure 1.1-26a shows the results of this sequence of operations. The operator is at the top, and the actual and desired (boxcar) amplitude spectra are superimposed at the bottom. Note that the actual spectrum has a ringy character. This is known as the Gibbs phenomenon (Bracewell, 1965), and results from representing a boxcar with a finite number of Fourier coefficients. From a practical standpoint, the ringing is undesirable, since some of the frequencies in the passband are amplified, while others are attenuated. Additionally, some of the frequencies in the reject zones on both sides of the boxcar are passed.

How is the Gibbs phenomenon circumvented? Instead of defining the desired passband as a boxcar, assign slopes on both sides as shown in Figure 1.1-26b, and thus define the passband as a trapezoid. Note that the actual and desired amplitude spectra are now closer in agreement and the operator is more compact (it has fewer nonzero coefficients). However, in achieving a more compact operator, the shape of the desired spectrum has been compromised and the passband is broader than intended. The trapezoid slopes must be sufficiently gentle to achieve a satisfactory result as in Figure 1.1-26c, where the actual and desired spectra are approximately equal and the operator is compact. This is most desirable in practice, since it is best to work with operators that are as short as possible. It is recommended that a gentler slope be assigned on the high-frequency side relative to the low-frequency side of the passband. Finally, while defining the passband as a trapezoid, smoothing also must be applied at the

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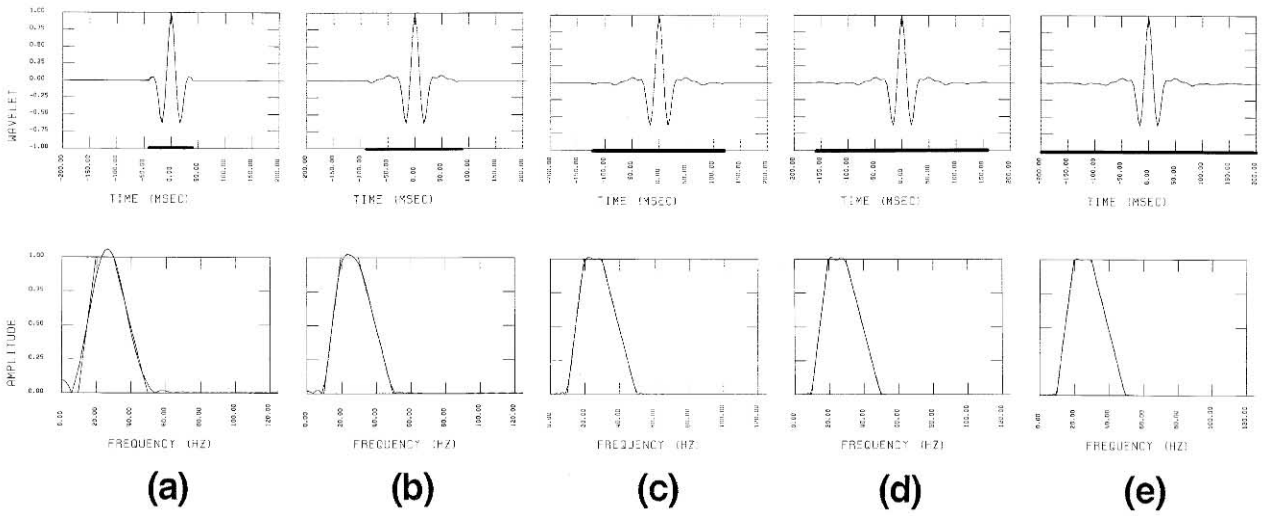

FIG. 1.1-27. Solid bars indicate the live length (nonzero coefficients) of the band-pass filter operator. Severe truncation (a) causes significant departure of the actual amplitude spectrum from the desired (trapezoid) amplitude spectrum, which is the same in all five cases.

corner frequencies (A, B, C, and D, as indicated in Figure 1.1-26c). This must be done because the Fourier transform exists only for continuous functions (Bracewell, 1965).

How short can the operator be? Figure 1.1-27 shows a sequence of increasingly longer operators. Solid bars indicate the operator length of the truncated filters. Note that excessive truncation causes a large deviation from the desired amplitude spectrum even though reasonable slopes were provided to the passband. Extension of the operator length brings the desired and actual spectra closer. However, there is a certain length beyond which nearly zero coefficients are added to the operator. The criterion that is used to define the operator length is that *the frequency bandwidth is inversely proportional to the effective length of the operator.*

Bandwidth and Vertical Resolution

Frequency filtering is intimately tied to vertical (temporal) resolution of seismic data. Consider the filter operators in Figure 1.1-28. Both have the same effective bandwidth — the difference between the high-cut and low-cut frequencies. Therefore, the envelopes of the two operators are identical. The greater ringyness of the second operator (Figure 1.1-28b) results from its lower bandwidth ratio — the ratio of the high-cut to the low-cut frequency.

There is a common misunderstanding that only high frequencies are needed to increase temporal resolution. This is not true. The top frame in Figure 1.1-29 shows a single reflector and three sets of closely situated reflectors with 48-, 24-, and 12-ms time separations. A series of narrow band-pass filters is applied to these data as shown in the lower frames. The reflectors with the 48-ms separation are resolved reasonably well by using the 10-to-20-Hz bandwidth. However, the more closely

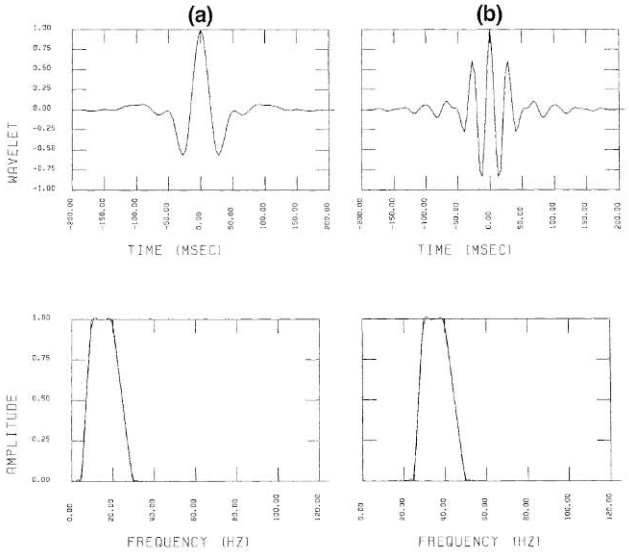

FIG. 1.1-28. Two wavelets (top row) with the same bandwidth (bottom row). The passband of wavelet (a) is centered at 15 Hz, while that of wavelet (b) is centered at 35 Hz. Both wavelets have ripples, although one is low and the other is high frequency in character. Just having low or high frequencies does not suffice; both are needed to increase temporal resolution.

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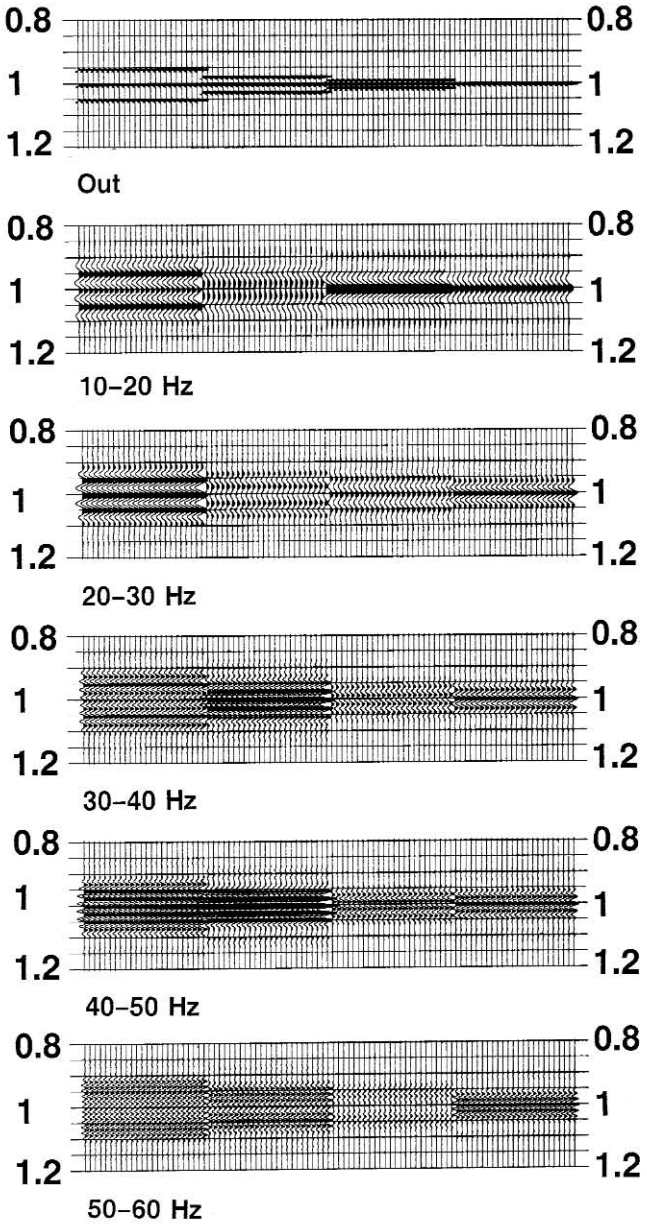

FIG. 1.1-29. The top section is a reflectivity model that consists of, from left to right, three reflectors with 48-ms separation, three reflectors with 24-ms separation, three reflectors with 12-ms separation, and a single reflector — all centered at 1 s. Band-limited responses (the same bandwidth, 10 Hz, centered at different frequencies) do not provide good resolution.

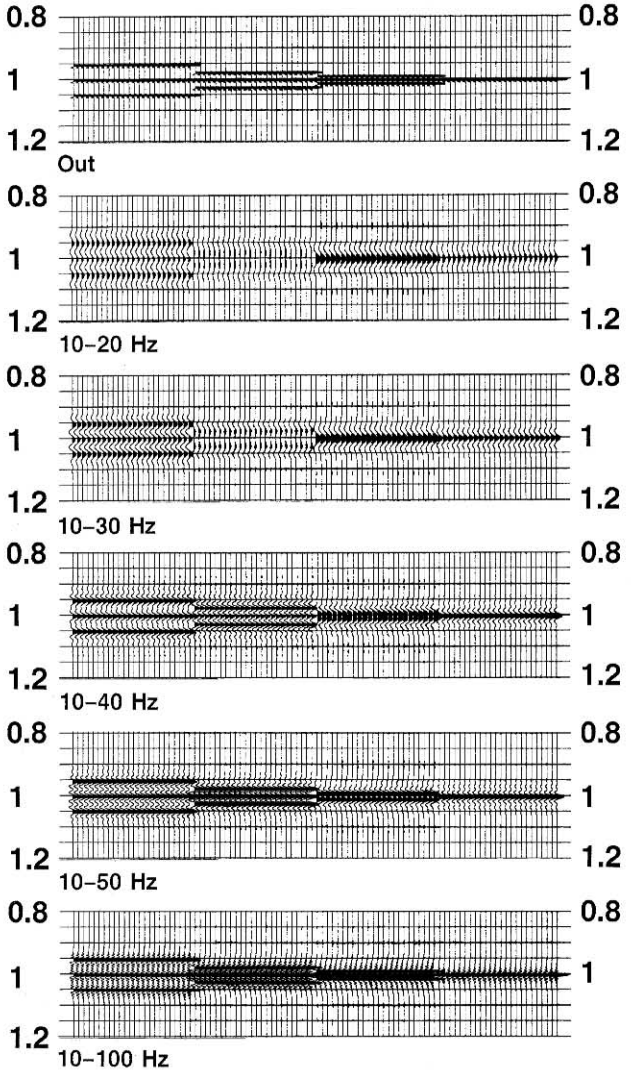

FIG. 1.1-30. The top section is the same reflectivity model as in Figure 1.1-29. Reflectors with large separation (48 ms) are resolvable with a bandwidth as low as 10 to 30 Hz. However, reflectors with smaller separation (24 and 12 ms) require increasingly larger bandwidths for resolution.

situated reflectors cannot be resolved with this filter. For the 20-to-30-Hz bandwidth, again, the 48-ms reflectors are reasonably separated. Nevertheless, none of the narrow band-pass filters provides the resolution needed to distinguish the reflectors situated with smaller separations. *Just having low or high frequencies does not improve temporal resolution. Both low and high frequencies are needed to increase temporal resolution.* This is demonstrated further in Figure 1.1-30. Note that closely situated reflectors can be resolved only with increasingly broader bandwidth. The 10-to-30-Hz bandwidth is sufficient to resolve the reflectors with 48-ms separation. The 10-to-50-Hz bandwidth is sufficient to resolve the reflectors with 24-ms separation. Finally, the 10-to-100-Hz bandwidth is needed to resolve the reflectors that are separated by 12 ms. There is a close relationship between the amount of separation and the desired bandwidth (Section 11.1).

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Time-Variant Filtering

The seismic spectrum, especially the high-frequency end, is subject to absorption along the propagation path because of the intrinsic attenuation of the earth (Section 1.4). Consider the portion of a stacked section and its narrow band-pass filtered panels in Figure 1.1-31. A signal is present from top to bottom within the 10-to-20-, 20-to-30-, 30-to-40-, and 40-to-50-Hz bands. Not much signal is noted below 3.5 s in the 50-to-60-Hz band. Nevertheless, the signal content appears to be retained down to 3.5 s with the 60-to-70-Hz band. Finally, the 70-to-80-Hz band shows signal down to 2.5 s. Higher frequency bands of useful signal are confined to the shallow part of the section. Thus, temporal resolution is reduced greatly in the deeper portion of the section.

From a practical standpoint, the time-variant character of the signal bandwidth requires an application of frequency filters in a time-varying manner. By so doing, the ambient noise, which begins to dominate the signal at late times, is excluded and a section with a higher signal-to-noise ratio is obtained. Table 1-11 lists the time-variant filter (TVF) parameters selected from the panels in Figure 1.1-31. The filtered section is shown on the far right panel of the same figure. In practice, the filters are blended across adjacent time windows to establish a smooth transition of the passband regions.

A second band-pass series of filter scans, which is shown in Figure 1.1-32, allows an assessment of the right choice of the bandwidth for a given time gate. Here, we start with a narrow band-pass filter at the low-frequency end of the spectrum and gradually broaden the passband by including higher frequencies.

For some data, the bandwidth may be kept quite large from top to bottom. The stacked section in Figure 1.1-32 can tolerate wide-band filtering from early to late times. The filter panels for the stacked section in Figure 1.1-33, however, indicate that the signal band rapidly becomes confined to lower frequencies at late times. A signal is present from top to bottom within the frequency bands up to 40 Hz. Noise is noted below 2.5 s in the 40-to-50-Hz band. This noise quickly builds up to shallower times at a higher frequency band.

Time-variant filters typically are applied on stacked data. A uniform bandwidth must be established when filtering two sets of data that may have different vintages, source types, or noise levels. This is especially significant when trying to tie two lines and follow a reflector across them. The interpreter uses the frequency character of a marker horizon as a reference in the tracking procedure. Therefore, two intersecting lines should be filtered so that the reflection character is consistent from one to the other, thus simplifying the interpretation.

Table 1-11. Time-variant filter parameters for the data shown in Figure 1.1-31. The frequency values correspond to the corner frequencies described in Figure 1.1-26.

Time, ms	Filter Band, Hz
0	5, 10–70, 80
2500	5, 10–60, 70
3500	5, 10–50, 60
5000	5, 10–40, 50

1.2 THE 2-D FOURIER TRANSFORM

Multichannel processing operations can be loosely defined as those that must operate on several data traces, simultaneously. Multichannel processes can be useful in discriminating against noise and enhancing signal on the basis of a criterion that can be distinguished from trace to trace, such as dip or moveout. The 2-D Fourier transform is a basis for both analysis and implementation of multichannel processes.

Consider the six zero-offset sections in Figure 1.2-1. The trace spacing is 25 m with 24 traces per section. All have monochromatic events with 12-Hz frequency, but with dips that vary from 0 to 15 ms/trace. From the discussion on the 1-D Fourier transform (Section 1.1), we know about frequency, particularly *temporal frequency*, or the number of cycles per unit time. This is the Fourier dual for the time variable. However, a seismic wavefield is not only a function of time, but also a function of a space variable (offset or midpoint axis). The Fourier dual for the space variable is defined as *spatial frequency*, which is the number of cycles per unit distance, or *wavenumber*. Just as the temporal frequency of a given sinusoid is determined by counting the number of peaks within a unit time, say 1 s, the wavenumber of a dipping event is determined by counting the number of peaks within a unit distance, say 1 km, along the horizontal direction. Just as the temporal Nyquist frequency is defined as in equation (1-1), the Nyquist wavenumber is defined as

$$k_{Nyq} = \frac{1}{2\Delta x}, \qquad (1-4)$$

where Δx is the spatial sampling interval. For all of the sections in Figures 1.2-1 through 1.2-6, the Nyquist wavenumber is 20 cycles/km, since the trace interval is 25 m.

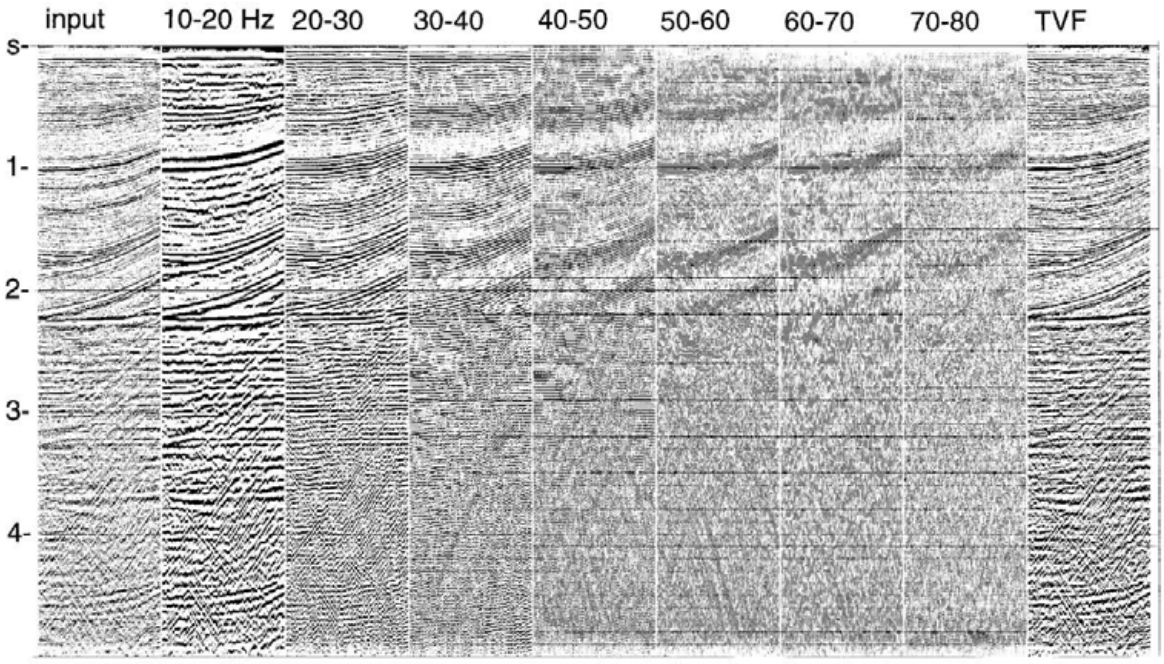

FIG. 1.1-31. The far left panel is a portion of a CMP stack without filtering. The following panels show the same data with different narrow band-pass filters. The frequency bands specified correspond to the corner frequencies B and C in Figure 1.1-26. Appropriate slopes were assigned to both low- and high-frequency ends of each passband. The far right panel is the same section as that in the far left panel after the application of the time-variant filter specified in Table 1-11.

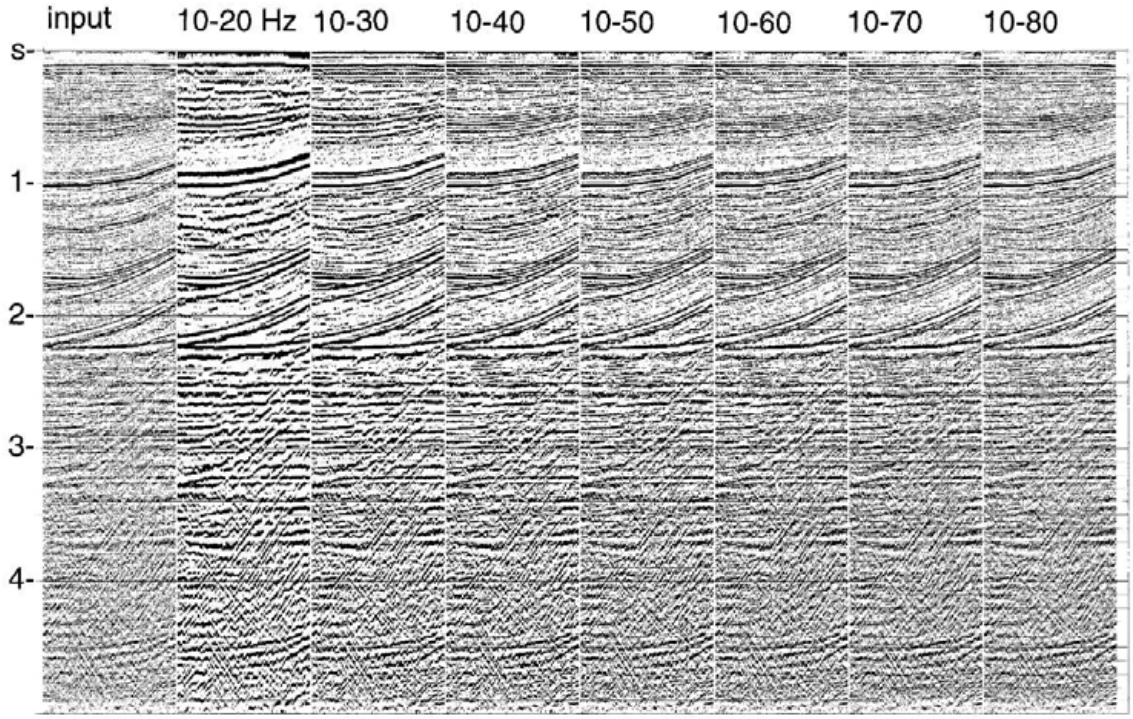

FIG. 1.1-32. The far left panel is a portion of a CMP stack without filtering. The remaining panels show the same data with different band-pass filters which have increasingly wider passbands. The frequency bands specified correspond to the corner frequencies B and C in Figure 1.1-26. Appropriate slopes were assigned to both low- and high-frequency ends of each passband.

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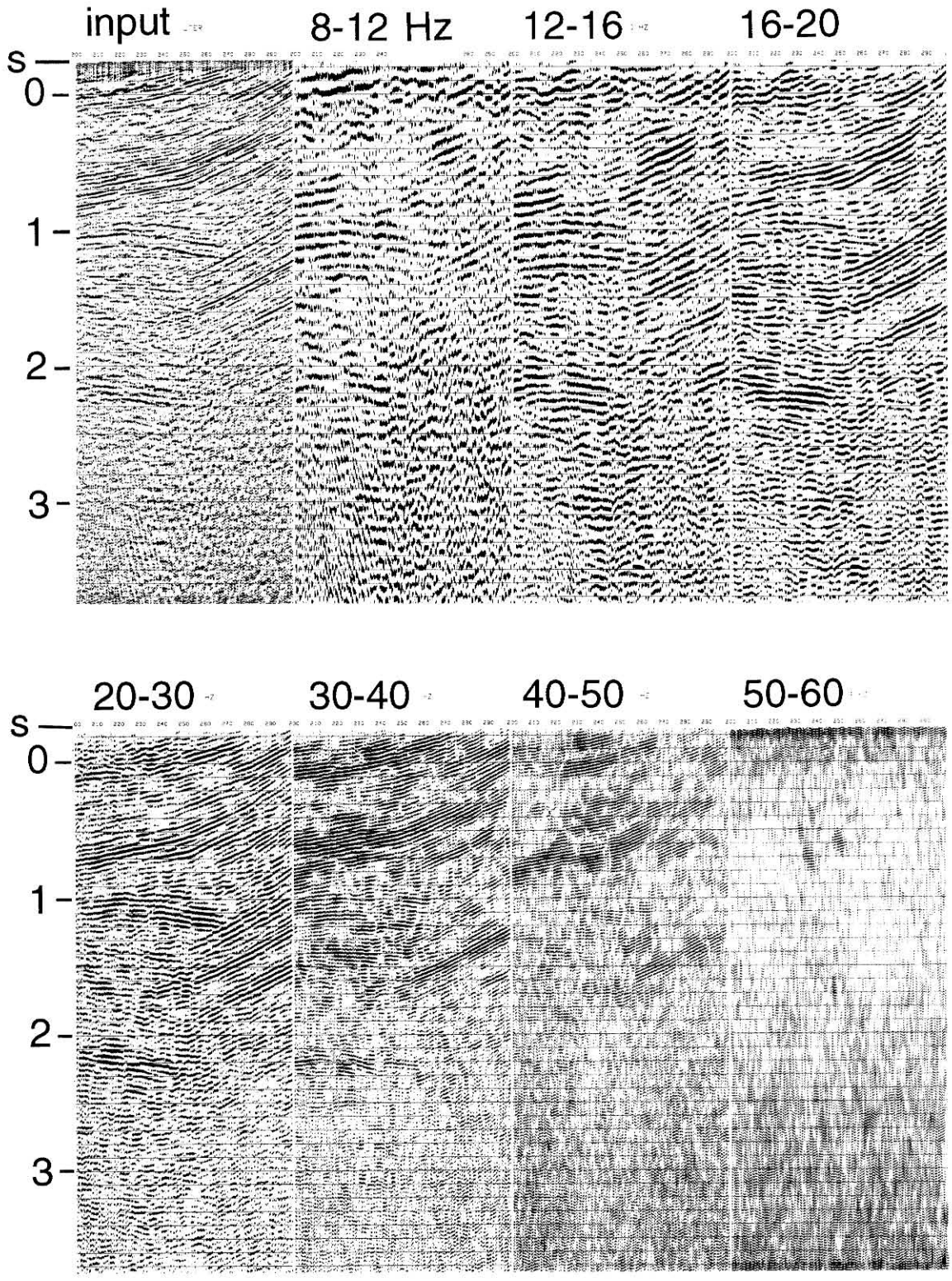

FIG. 1.1-33. The upper far left panel is a portion of a CMP stack without filtering. The remaining panels show the same data with different narrow band-pass filters.

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To compute the wavenumber that is associated with the section corresponding, say, to the 15-ms/trace dip in Figure 1.2-1, follow a peak or trough across the section. First compute the total time difference along the selected peak or trough across the section:

$$(23\ \text{traces})(15\ \text{ms/trace}) = 345\ \text{ms}.$$

Then convert this to number of cycles by dividing by the (temporal) period:

$$\frac{345\ \text{ms}}{(1000\ \text{ms/s})/(12\ \text{cycles/s})} = 4.14\ \text{cycles}.$$

The spatial extent of the section is 575 m; therefore, the wavenumber associated with the 15-ms/trace dip and the 12-Hz frequency is

$$\frac{4.14\ \text{cycles}}{0.575\ \text{km}} = 7.2\ \text{cycles/km}.$$

To continue this discussion, we will map these sections to the plane of temporal frequency versus spatial wavenumber, then look at two quadrants of this plane. The following convention will be used: Events with downdip to the right are assigned positive dip, while events with updip to the right are assigned negative dip. Additionally, positive dips map onto the right quadrant, which corresponds to positive wavenumbers, while negative dips map onto the left quadrant, which corresponds to negative wavenumbers.

The plane of frequency-wavenumber (the $f-k$ plane) appears at the bottom of each section in Figure 1.2-1. The section with zero-dip events maps onto a single point on the frequency axis at 12 Hz. Zero dip is equivalent to zero wavenumber. The magnitude of the spike corresponds to the peak amplitude of the sinusoids that make up the traces in the section. Therefore, the $f-k$ plane actually represents the 2-D amplitude spectrum of the section in the $t-x$ domain. These data have been transferred from the time-space domain to the frequency-wavenumber domain. This process is described mathematically by the 2-D Fourier transform.

There is a practical relationship between the four variables: time-space ($t-x$), and their Fourier duals, frequency-wavenumber ($f-k$). Measure the inverse of the stepout in the 15-ms/trace section in Figure 1.2-1 by following a peak, trough, or zero crossing from trace to trace. Stepout is defined as the slope $\Delta t/\Delta x$. In this case, the inverse of the stepout is

$$\frac{\Delta x}{\Delta t} = \frac{0.575\ \text{km}}{0.345\ \text{s}} = 1.67\ \text{km/s}.$$

Now, compute the ratio:

$$\frac{f}{k} = \frac{12\ \text{cycles/s}}{7.2\ \text{cycles/km}} = 1.67\ \text{km/s}.$$

From this, the inverse of the stepout measured in the $t-x$ domain along a constant phase is equal to the ratio of the frequency to the wavenumber associated with the event

$$\frac{\Delta x}{\Delta t} = \frac{f}{k}. \qquad (1-5)$$

Therefore, *while retaining fixed stepout, doubling the frequency means doubling the wavenumber.*

Note that all sections in Figure 1.2-1 have the same frequency component. However, from 0 to 15 ms/trace, the number of peaks increases horizontally across each section. That is, for a given frequency, higher dips are assigned to higher wavenumbers, as seen on the $f-k$ plots.

Spatial Aliasing

From Figures 1.2-1 through 1.2-6, consider the same dip components, but at different frequencies. Map each individual section to the $f-k$ plane. Nothing unusual happens until the section with 15-ms/trace dip at 36 Hz is reached in Figure 1.2-3. Here there is no positive dip. In fact, as a whole, the section displays a checkerboard character making it difficult to determine whether the dip is positive or negative.

At 48 Hz (Figure 1.2-4), the correct dip direction is observed in the first four sections. However, the fifth section, which corresponds to the 12-ms/trace positive dip, shows a negative dip. Therefore, it is mapped onto the negative quadrant, which is the wrong quadrant for this section. This dip component (12 ms/trace) at this frequency (48 Hz) is *spatially aliased*. In fact, any dip greater than 12 ms/trace is spatially aliased at this frequency.

In the next set of sections in Figure 1.2-5, spatial aliasing occurs at 60 Hz for a 9-ms/trace dip. Spatial aliasing not only causes mapping to the wrong quadrant, but also causes mapping with the wrong dip. One obvious example of this is mapping a 15-ms/trace dip at 60 Hz (Figure 1.2-5). Finally, at 72 Hz (Figure 1.2-6), the 6-ms/trace dip component is on the verge of spatial aliasing. Moreover, the 15-ms/trace dip component is spatially aliased twice; it folds back to the positive-dip quadrant and appears at a lower dip.

This same analysis can be used for the negative-dip components. From Figures 1.2-1 through 1.2-6, note that each section as a whole was mapped onto a single point in the frequency-wavenumber domain. Each section has an associated unique frequency and wavenumber assigned to it. These zero-offset sections can be considered representations of plane waves that propagate at a unique angle from the vertical and carry a monochromatic signal. The wavefront is defined as the line of

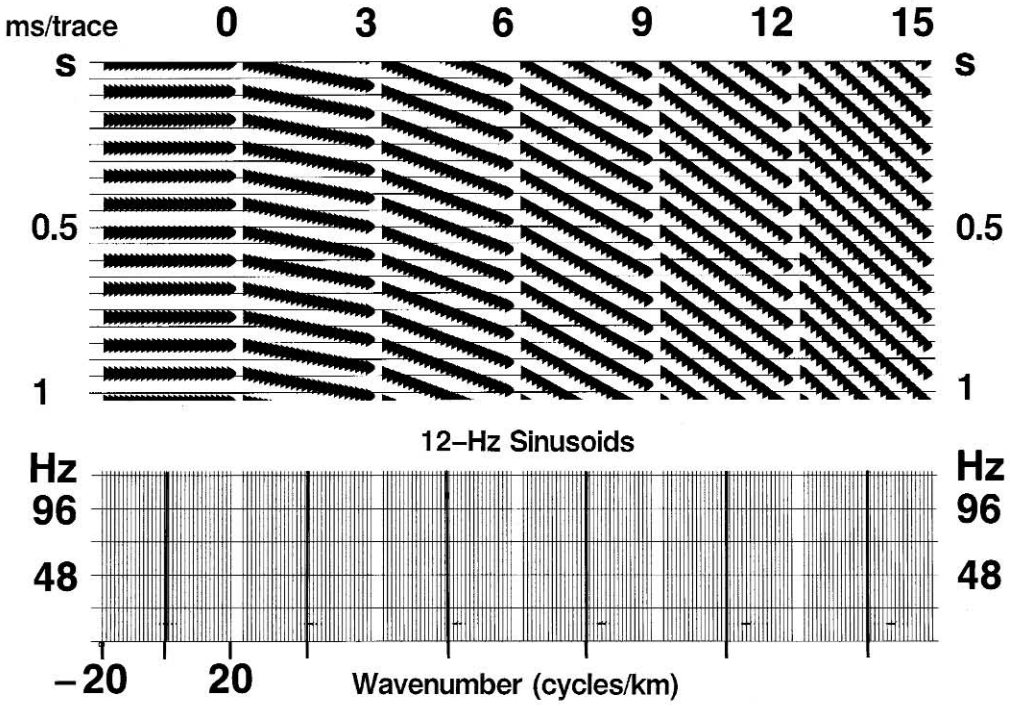

FIG. 1.2-1. Top row: Six gathers, each containing 12-Hz monofrequency events with different dips ranging from 0 to 15 ms/trace. Trace spacing is 25 m. Bottom row: Their respective amplitude spectra. The dots on the spectra represent the mapping of events on the gathers. The solid vertical lines are the frequency axis. If the positive dips are defined as downdip from left to right, then all events map onto the positive quadrant in the frequency-wavenumber $(f-k)$ plane. This is the first in a series of six figures that describes mapping of monofrequency signals in the $(f-k)$ domain (Figures 1.2-1 through 1.2-6).

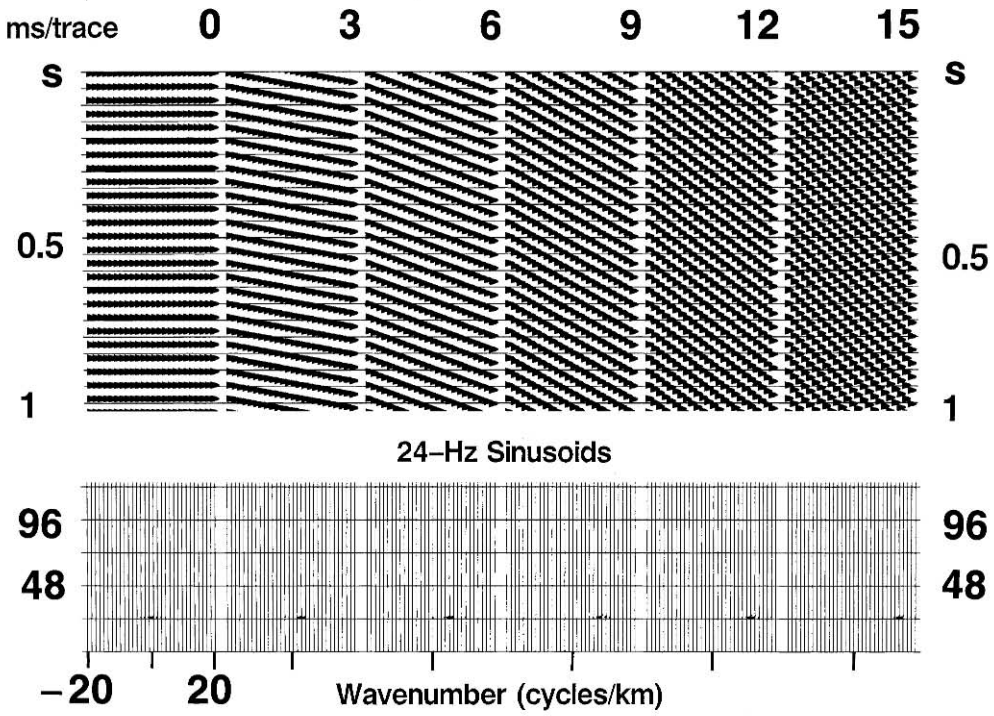

FIG. 1.2-2. The same as Figure 1.2-1, except using 24-Hz monofrequency events.

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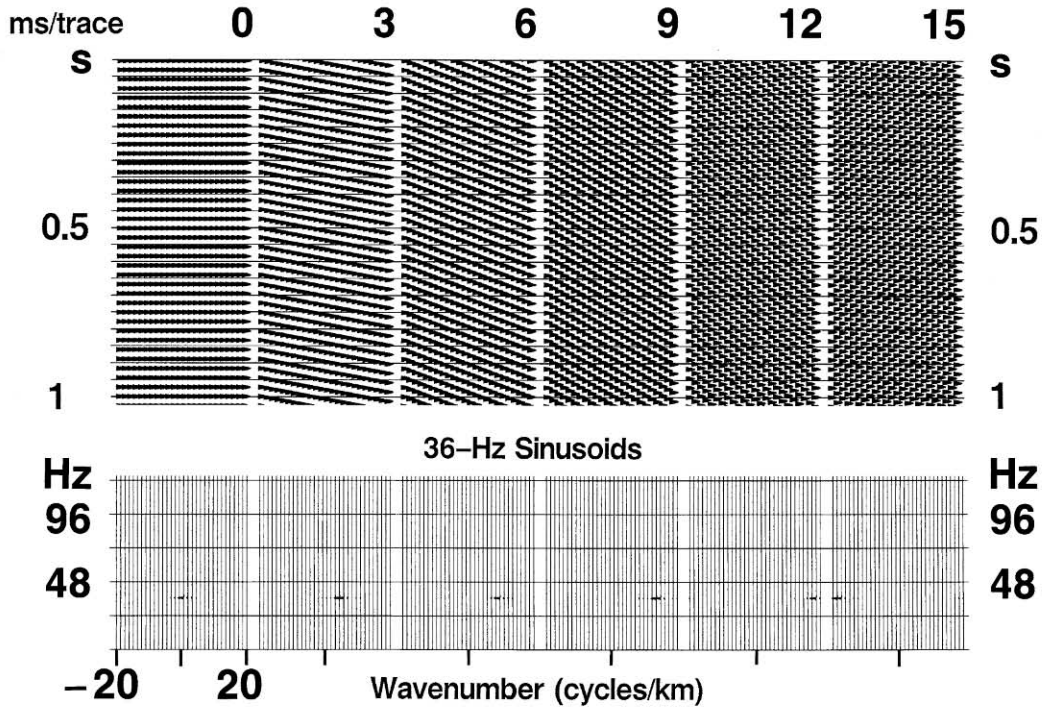

FIG. 1.2-3. The same as Figure 1.2-1, except using 36-Hz monofrequency events.

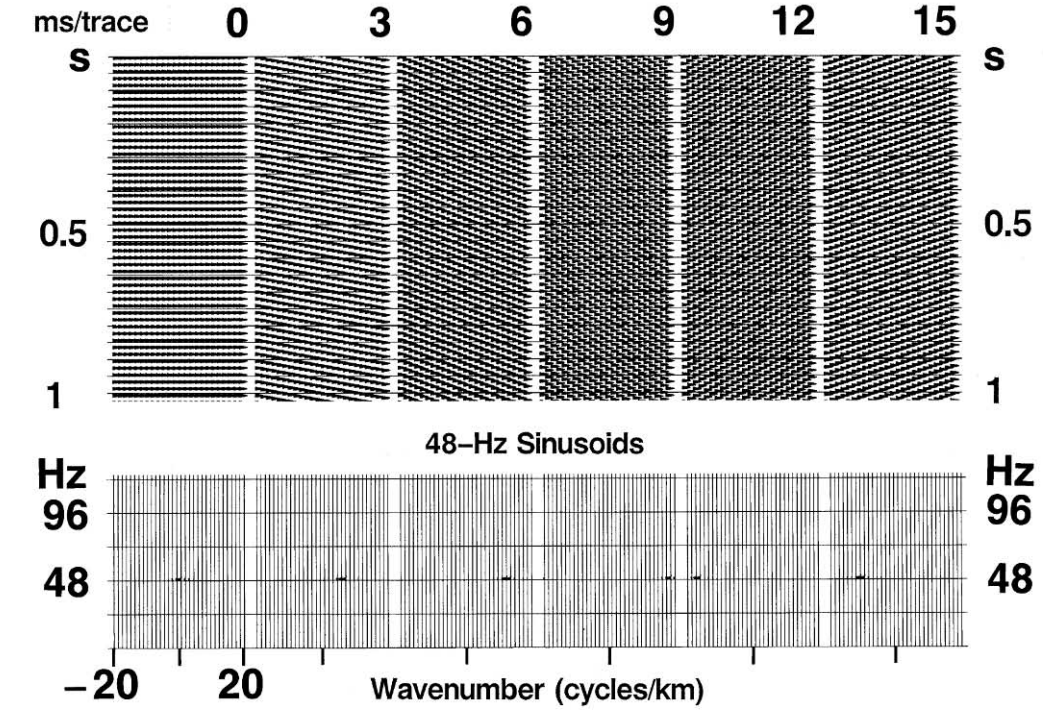

FIG. 1.2-4. The same as Figure 1.2-1, except using 48-Hz monofrequency events.

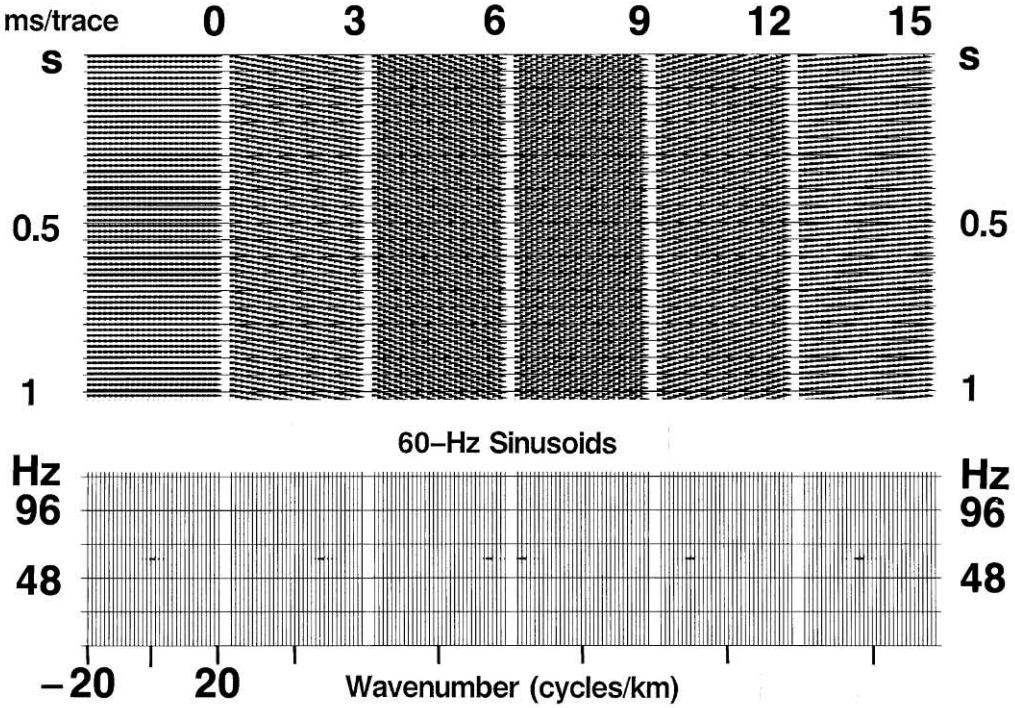

FIG. 1.2-5. The same as Figure 1.2-1, except using 60-Hz monofrequency events.

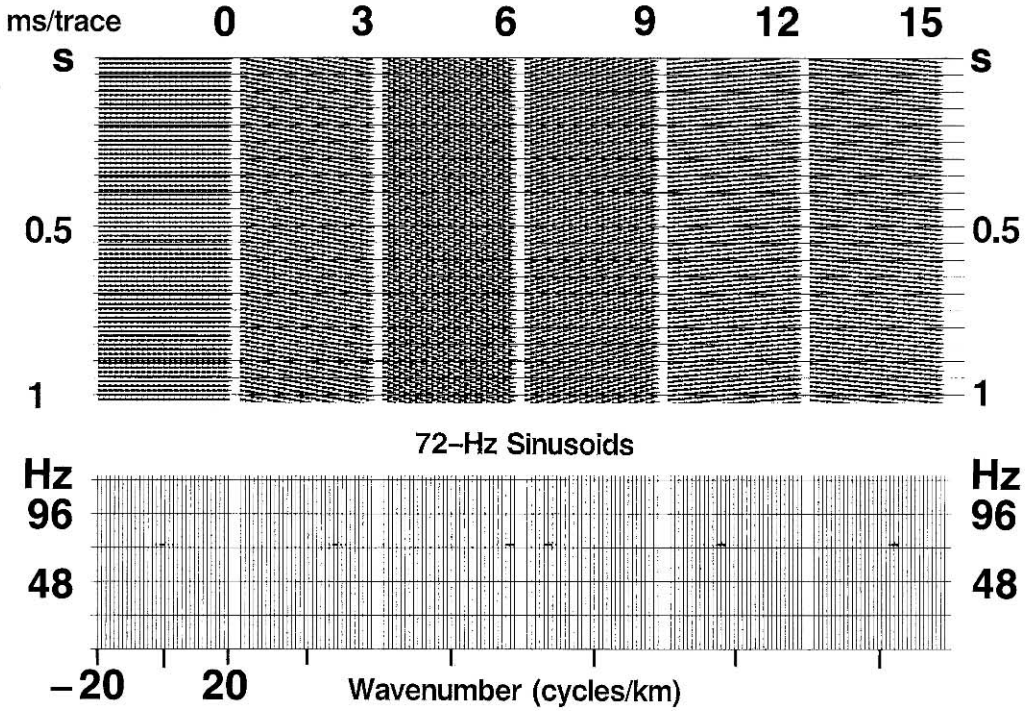

FIG. 1.2-6. The same as Figure 1.2-1, except using 72-Hz monofrequency events.

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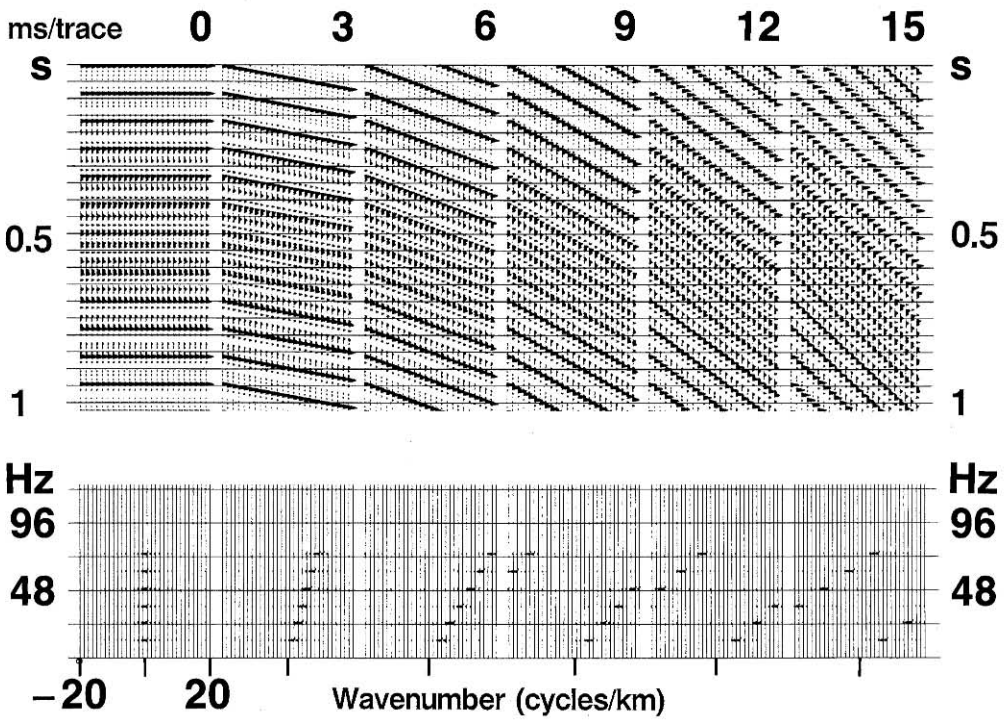

FIG. 1.2-7. Top row: Six gathers, each formed by summing gathers of the like dips in Figures 1.2-1 through 1.2-6. The trace spacing is 25 m. Bottom row: Respective amplitude spectra.

constant phase, while the direction of propagation is perpendicular to the wavefront. Since a seismic wavefield is a superposition of many dips and frequencies, it is equivalent to the *synthesis* of many plane-wave components. In this respect, the physical meaning of the 2-D Fourier transform is important, for it is an *analysis* (decomposition) of a wavefield into its plane-wave components.

A recorded wavefield is a composite of many dip and frequency components, such as those shown in Figures 1.2-1 through 1.2-6. Suppose that sections with the same dip, but with different frequencies, are superimposed. The composite sections are shown in Figure 1.2-7 with the composite amplitude spectra below each section. For a given dip, all frequency components map onto the $f-k$ plane along a straight line that passes through the origin. The higher the dip, the closer the radial line in the $f-k$ domain is to the wavenumber axis. The zero-dip components map along the frequency axis. From the 9-, 12-, and 15-ms/trace dips, note that the spatially aliased frequencies are located along the linear segments that wrap around to the opposite quadrant in the amplitude spectrum. *The steeper the dip, the lower the frequency at which spatial aliasing occurs.*

So far, a discrete number of frequencies was considered. For a continuum of frequency components associated with a single dip, we anticipate that they would map along a straight, continuous line in the $f-k$ domain, as shown in Figure 1.2-8. While the dipping event in Figure 1.2-8 is not aliased, the dipping event in Figure 1.2-9 is spatially aliased beginning at approximately 21 Hz.

Examination of the monochromatic single-dip sections in Figures 1.2-1 through 1.2-6 shows that each section maps onto a single point in the $f-k$ domain. An extension of this observation is made in Figure 1.2-10. Events with the same dip in the $t-x$ domain, regardless of their location, map onto a single radial line in the $f-k$ domain. When events are spatially aliased, the radial line wraps around at the Nyquist wavenumber (Figure 1.2-11). These concepts have important practical implications, for they lead to $f-k$ dip filtering of coherent linear noise (Section 6.2). Events with different dips that may interfere in the $t-x$ domain can be isolated in the $f-k$ domain.

The numerical computation of the 2-D Fourier transform involves two 1-D Fourier transforms. Figure 1.2-12 shows the steps that are involved. A brief mathematical formulation of the 2-D Fourier transform is given in Section A.2.

In summary, 2-D Fourier transformation is a way to decompose a wavefield into its plane-wave components. Each plane wave carries a monochromatic signal that

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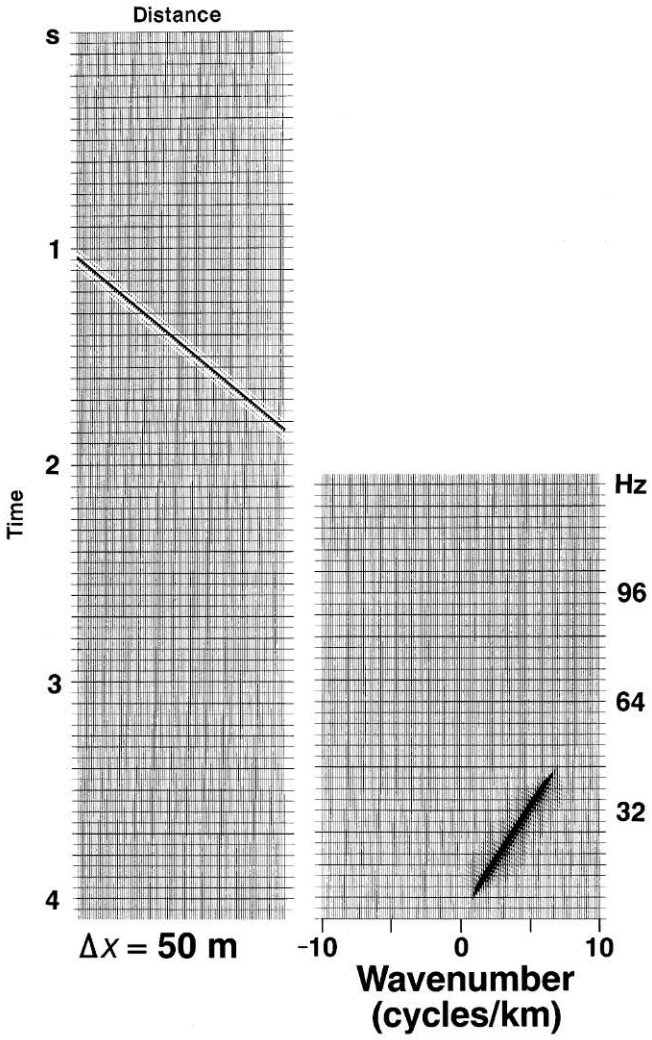

FIG. 1.2-8. A single, isolated dipping event and its 2-D amplitude spectrum. No frequency is spatially aliased. What is the dip of the event in milliseconds per trace?

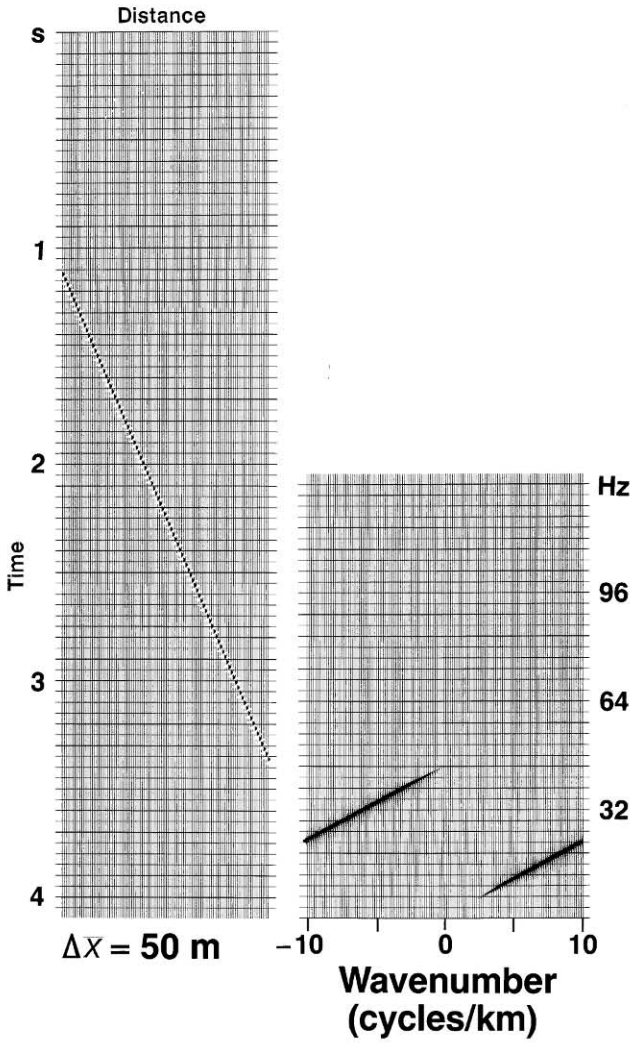

FIG. 1.2-9. A single, isolated dipping event and its 2-D amplitude spectrum. Frequencies beyond 21 Hz are spatially aliased.

propagates at a certain angle from the vertical. Events with the same dip in the $t - x$ domain, regardless of location, are mapped onto a single line in the radial direction in the $f - k$ domain. In Chapter 4, we shall discuss migration methods in the $f - k$ domain, and in Chapter 6, we shall demonstrate use of $f - k$ filtering to remove coherent linear noise and attenuate multiples.

Spatial aliasing has serious effects on the performance of multichannel processes such as $f - k$ filtering (Section 6.2) and migration (Section 4.6). Because of spatial aliasing, these processes can perceive events with steep dips at high frequencies as different from what they actually are and, hence, do not treat them properly. For example, migration moves the spatially aliased frequency components in the wrong direction and generates a dispersive noise that degrades the quality of the migrated section.

How is spatial aliasing avoided? Compare the sections in Figures 1.2-8 and 1.2-9. Both have the same frequency content, 6 to 42 Hz. The data in Figure 1.2-9 are spatially aliased because the dipping event is steeper than in Figure 1.2-8. Some ways to avoid spatial aliasing follow:

(a) Apply time shifts so that the steep events appear to have lower dips. Although this could change the dips that were low to higher dips, making them spatially aliased, it often is a feasible solution for

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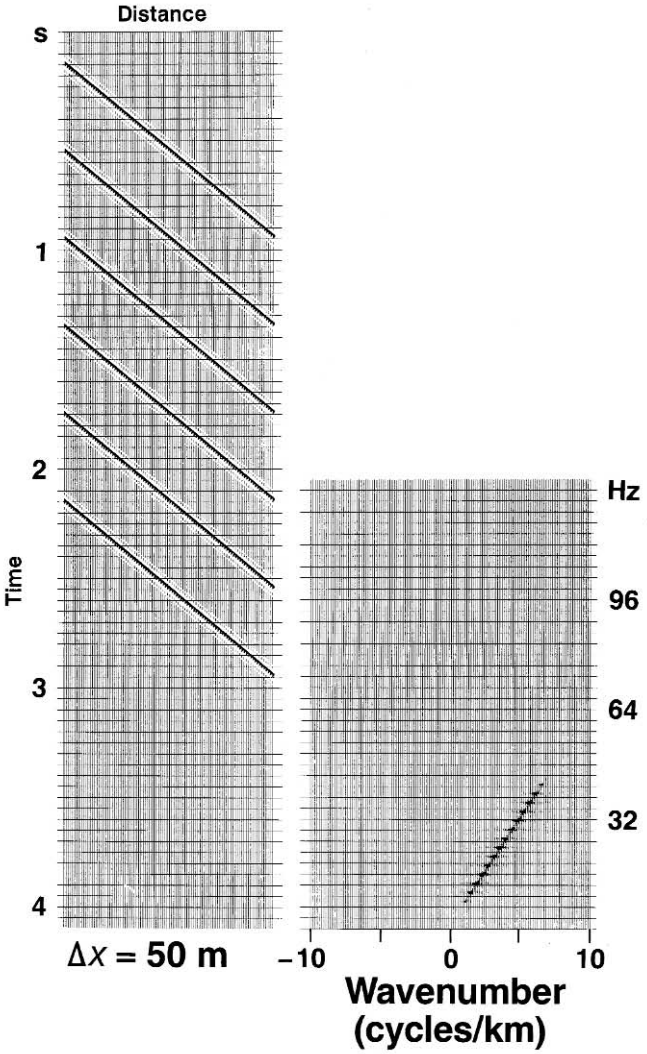

FIG. 1.2-10. Six events with identical dip in the $t-x$ domain map onto the same radial line in the $f-k$ domain. No frequencies are aliased. The dip of these events is the same as that of the single event in Figure 1.2-8. What distinguishes this spectrum from that in Figure 1.2-8?

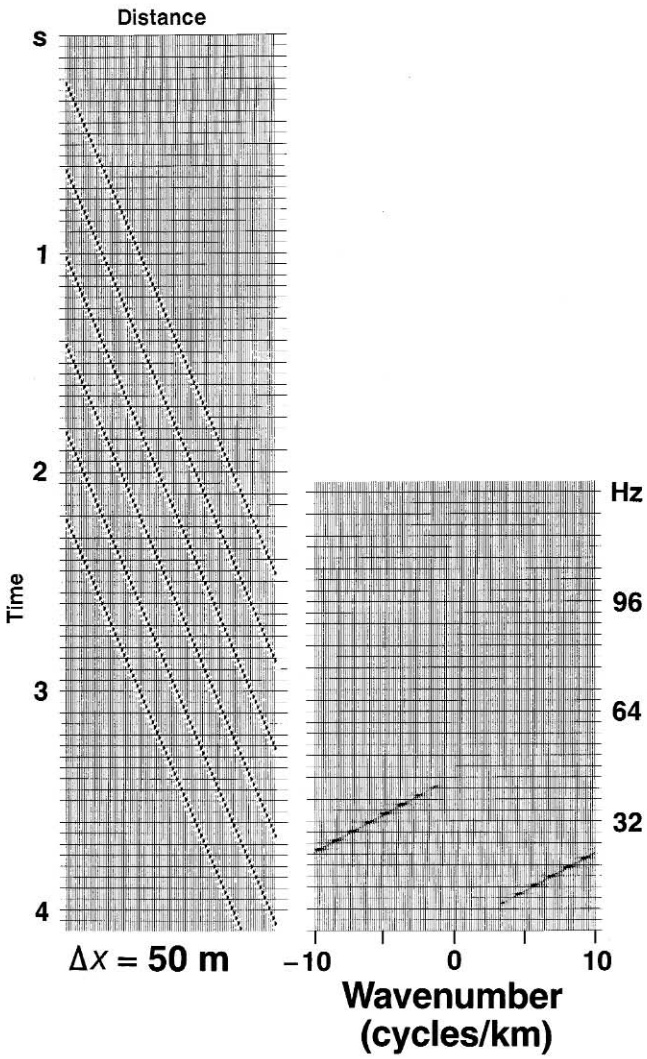

FIG. 1.2-11. Six events with identical dip in the $t-x$ domain map onto the same radial line in the $f-k$ domain. Frequencies beyond 21 Hz are aliased. The dip of these events is the same as that of the single event in Figure 1.2-9.

certain situations. For instance, a linear moveout can be applied to CMP gathers to circumvent spatial aliasing of shallow events in prestack migration (Li et al., 1991). Also, a hyperbolic moveout can be applied to CMP gathers before multiple attenuation in the $f-k$ domain (Section 6.2).

(b) If a low-pass filter were applied to the traces in Figure 1.2-9 so that the frequencies up to 21 Hz were retained, then the segment that is wrapped around to the negative quadrant of the amplitude spectrum is removed. Although spatial aliasing is eliminated, a significant part of the recorded frequency band is lost. This approach is not desirable.

(c) Figure 1.2-13 shows a single dipping event recorded with three different trace spacings. The 2-D amplitude spectra suggest a third approach to solving the spatial aliasing problem. Note that the coarser the trace spacing, the more frequencies are spatially aliased. The same frequency bandwidth is kept in all three cases. The 12.5-m trace spacing provides a frequency band with no spatial aliasing. For a 25-m trace spacing, frequencies beyond 36 Hz are spatially aliased; while for a 50-m trace spacing, frequencies beyond 18 Hz are spatially aliased. For this latter case, spatial aliasing is so severe that the aliased frequencies wrap around the wave-

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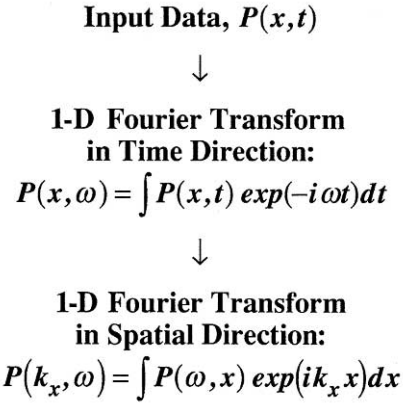

FIG. 1.2-12. Computation of the 2-D Fourier transform.

number axis twice. We see that spatial aliasing can be avoided by selecting a sufficiently small trace spacing. This approach requires either a data-dependent interpolation scheme (Section 7.2) to generate extra traces or modification of the field recording geometry. If the latter approach were taken, more shots and/or more recording channels are needed.

To circumvent spatial aliasing, data often are recorded with twice as many number of channels per shot record as that would be used in processing. Prior to dropping every other trace at the start of a processing sequence, a wavenumber filter is applied to remove the wavenumber components for all frequencies beyond the Nyquist wavenumber that corresponds to the trace spacing of the data after dropping alternating traces from the shot record. This wavenumber filter actually is a spatial high-cut antialias filter akin to the high-cut antialias frequency filter discussed in Section 1.1.

Consider the shot record shown in Figure 1.2-14a recorded with 368 channels at a group interval of 12.5 m. The record which is intended for input to a processing sequence comprises 192 channels with 25-m group interval. Just as dropping every other sample in a time series can result in frequency aliasing of some of the high frequencies, dropping alternating traces could result in spatial aliasing (Figure 1.2-14b). Note that spatial aliasing in the original record (Figure 1.2-14a) occurs starting at about 75 hz. Trace decimation without spatial antialias filter causes spatial aliasing to occur starting at about 40 Hz (Figure 1.2-14b).

To circumvent spatial aliasing as a result of dropping every other trace in the record, a prior application of a spatial high-cut antialias filter is needed (Figure 1.2-15). Apply a wavenumber filter to remove the energy within the $f-k$ region that is between 50 and 100 wavenumber in both quadrants of the $f-k$ plane (Figure 1.2-15a). Since the trace spacing of the original record in Figure 1.2-14a is 12.5 m, the corresponding Nyquist wavenumber is 40 cycles/km. Following trace decimation, the Nyquist wavenumber is 20 cycles/km, corresponding to 25-m trace spacing. Note in the $f-k$ spectrum in Figure 1.2-15a that the wavenumber filter has removed the energy between 20 and 40 cycles/km for all frequencies in both quadrants. Compare the $f-k$ spectra of the decimated data without (Figure 1.2-14b) and with antialias filtering (Figure 1.2-15b) and note that the wavenumber filter has removed the spatially aliased high-frequency components of the steeply dipping events in the original gather.

The spatial antialias filter design and application involves normal-moveout correction of the data (Section 3.1). Consider the 240-channel shot record shown in Figure 1.2-16a with 6.25-m group interval. We want to apply spatial antialias filter and reduce the number of channels by half with 12.5-m group interval. Follow the steps below:

(a) Apply normal-moveout correction to the input gather (Figure 1.2-16a) that needs to be decimated to unalias the aliased energy.
(b) Apply 2-D Fourier transform to map the data onto the $f-k$ plane. Following the normal-moveout correction, the energy maps in the $f-k$ plane closer to the frequency axis, thus alleviating the spatial aliasing (Figure 1.2-17a).
(c) Design a spatial antialias filter by specifying reject zones for all frequencies between 50 and 100 wavenumber in both quadrants of the $f-k$ plane.
(d) Apply the wavenumber filter to the moveout-corrected data in the $f-k$ domain (Figure 1.2-17b).
(e) Apply inverse Fourier transform of the wavenumber-filtered data (Figure 1.2-16b).
(f) Now drop every other trace to obtain the decimated data (Figure 1.2-16c). The Nyquist wavenumber of the gather following the application of the spatial antialias filter is one-half ot the Nyquist wavenumber of the original gather (Figure 1.2-17c).
(g) Apply inverse moveout correction (Figure 1.2-16d). The resulting shot record contains one-half the number of channels as in the original record (Figure 1.2-16a), and its energy is within the $f-k$ region defined by one-half the Nyquist wavenumber associated with the original record (compare Figures 1.2-17a and 1.2-17d).

The original moveout-corrected shot record (Figure 1.2-16a) after dropping every other trace, but without the application of the antialias wavenumber filter, is shown in Figure 1.2-16e with the $f-k$ spectrum of

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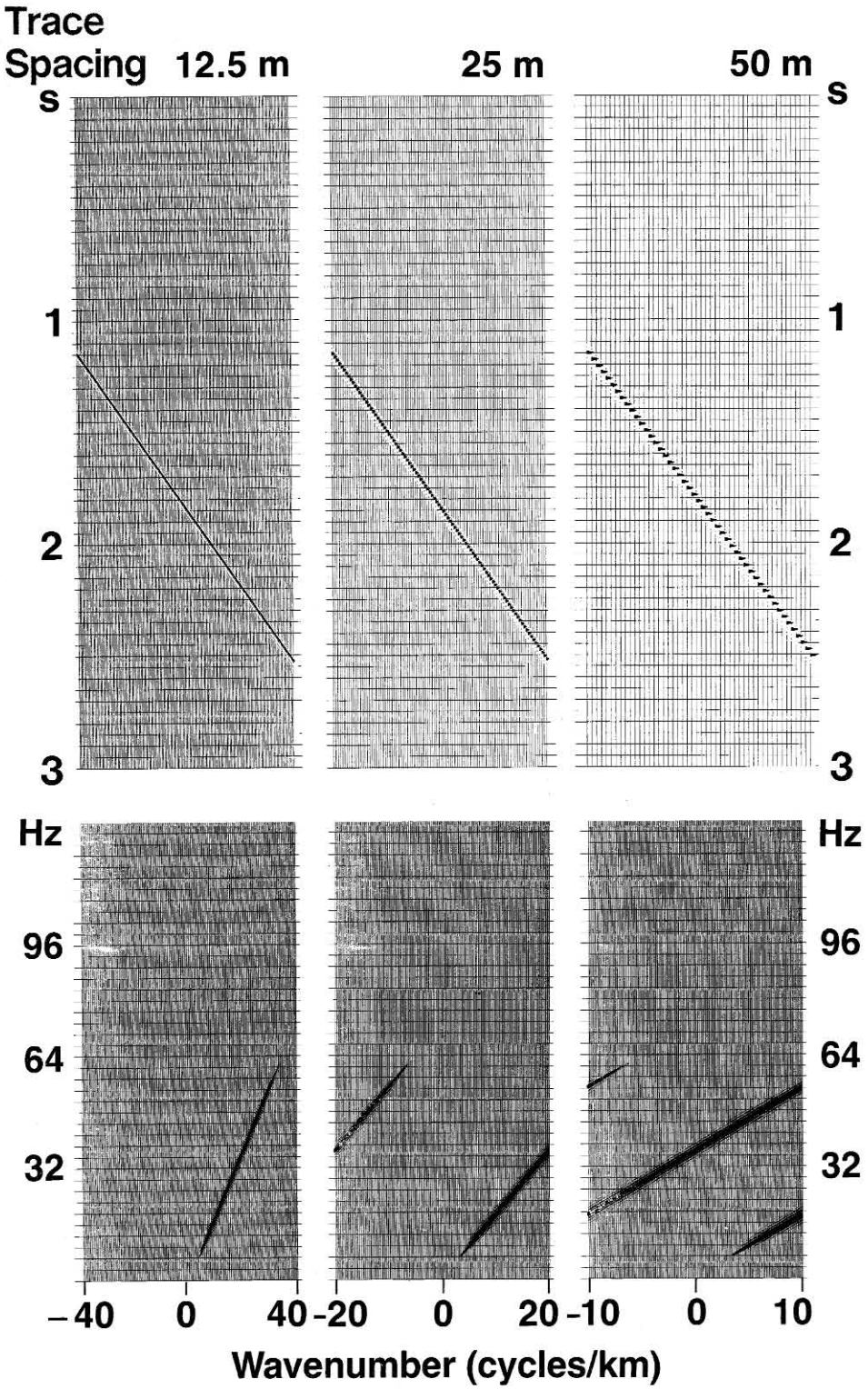

FIG. 1.2-13. A single, isolated dipping event sampled at three different trace spacings with the corresponding $f-k$ spectra. No spatial aliasing occurs with the 12.5-m trace spacing (left). Frequencies beyond 36 Hz are aliased with the 25-m trace spacing (center). Double aliasing occurs with the 50-m trace spacing (right). Although events on the $f-k$ spectra appear to have different dips, all three have the same dip on the $t-x$ gathers (top). This deceptive character is because of the different horizontal scales used in displaying the $f-k$ spectra.

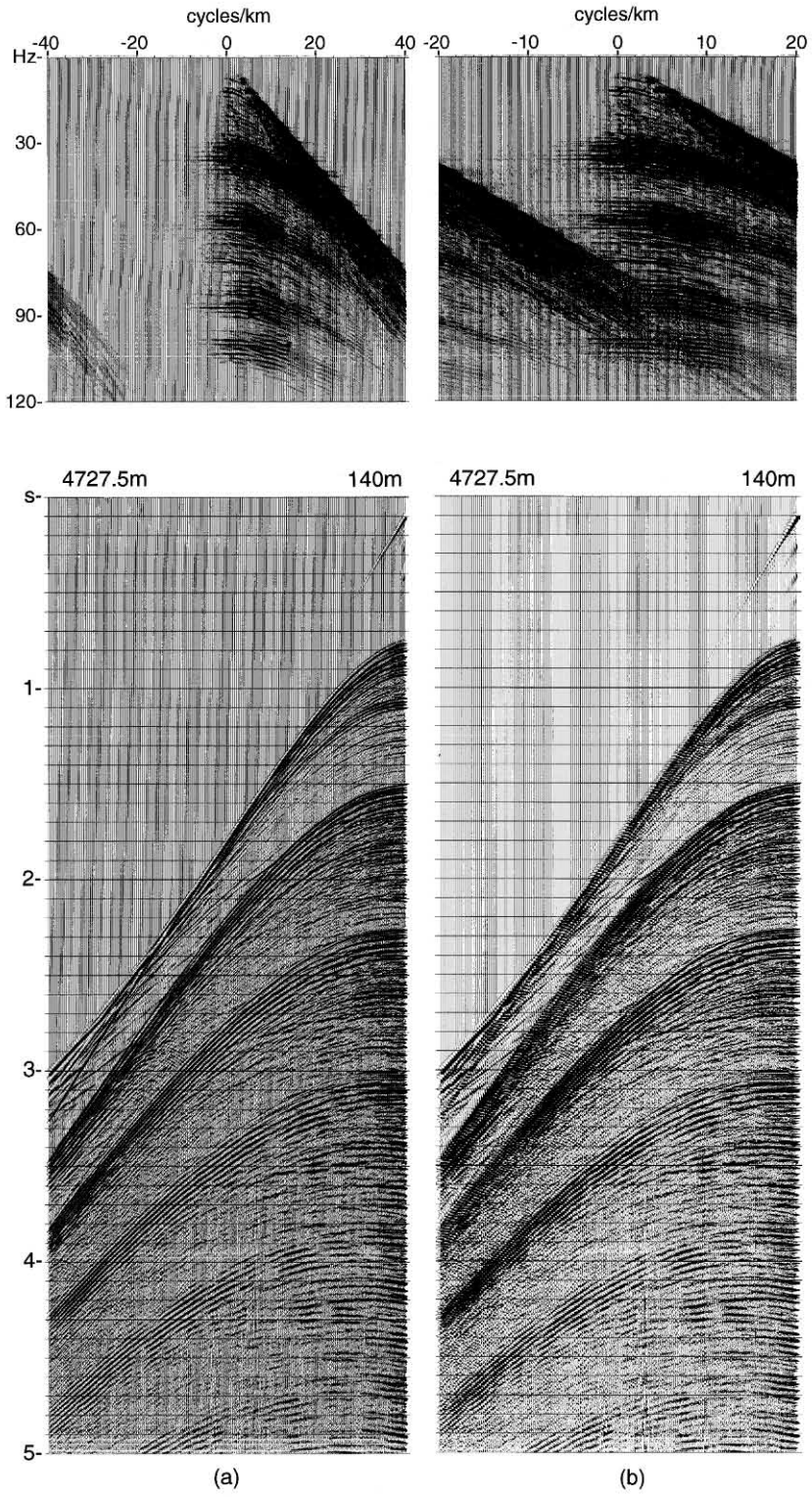

FIG. 1.2-14. (a) A shot record with its 2-D amplitude spectrum displayed on top; trace spacing is 12.5 m and the Nyquist wavenumber is 40 cycles/km. Minimum and maximum offsets are 140 m and 4727.5 m, respectively. (b) The same record as in (a) with every other trace dropped; hence, trace spacing is 25 m and the Nyquist wavenumber is 20 cycles/km. Note that little energy is spatially aliased above 75 Hz in the original record (a), while spatial aliasing occurs beginning at 45 Hz in the record with every other trace dropped (b).

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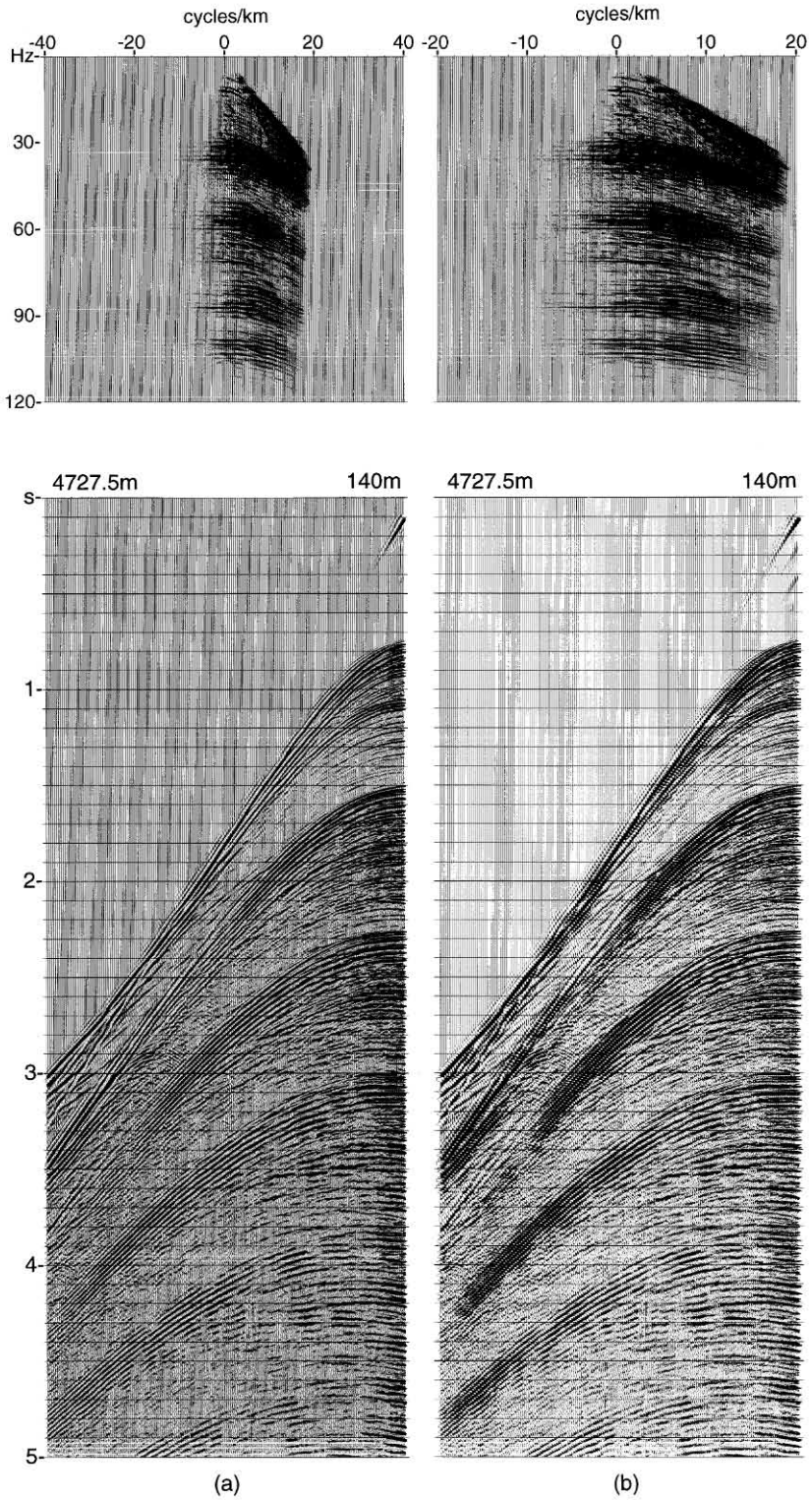

FIG. 1.2-15. (a) The same shot record as in Figure 1.2-14a after the application of a wavenumber filter to remove the energy beyond half the Nyquist wavenumber for all frequencies. (b) The same record as in (a) after dropping every other trace. Note that spatial aliasing is circumvented first by applying a high-cut spatial antialiasing wavenumber filter (a) prior to dropping every other trace. In contrast, dropping every other trace without the prior application of an antialias filter causes spatial aliasing of events with large moveout (Figure 1.2-15b).

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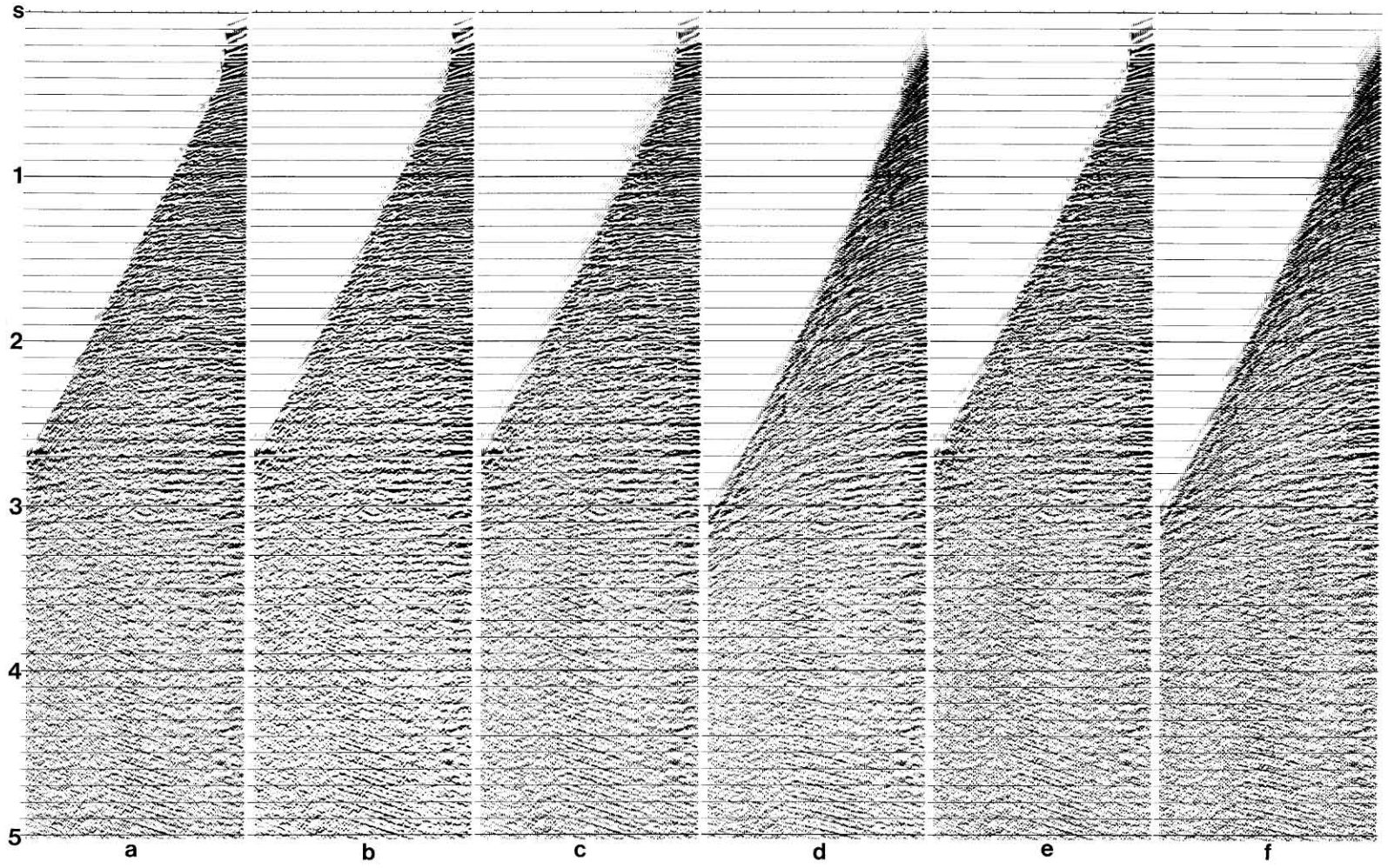

FIG. 1.2-16. (a) A 240-channel shot record with 6.25-m group interval; (b) after normal moveout correction; (b) after the application of wavenumber filter to remove the energy between one-half the Nyquist and the Nyquist wavenumbers for all frequencies; (c) the same record as in (b) after dropping every other trace so that the resulting shot record comprises 120 channels with 12.5-m group interval; (d) the same record as in (c) after inverse moveout correction; (e) the same record as in (b) after dropping every other trace without wavenumber filtering; (f) the same record as in (e) after inverse moveout correction.

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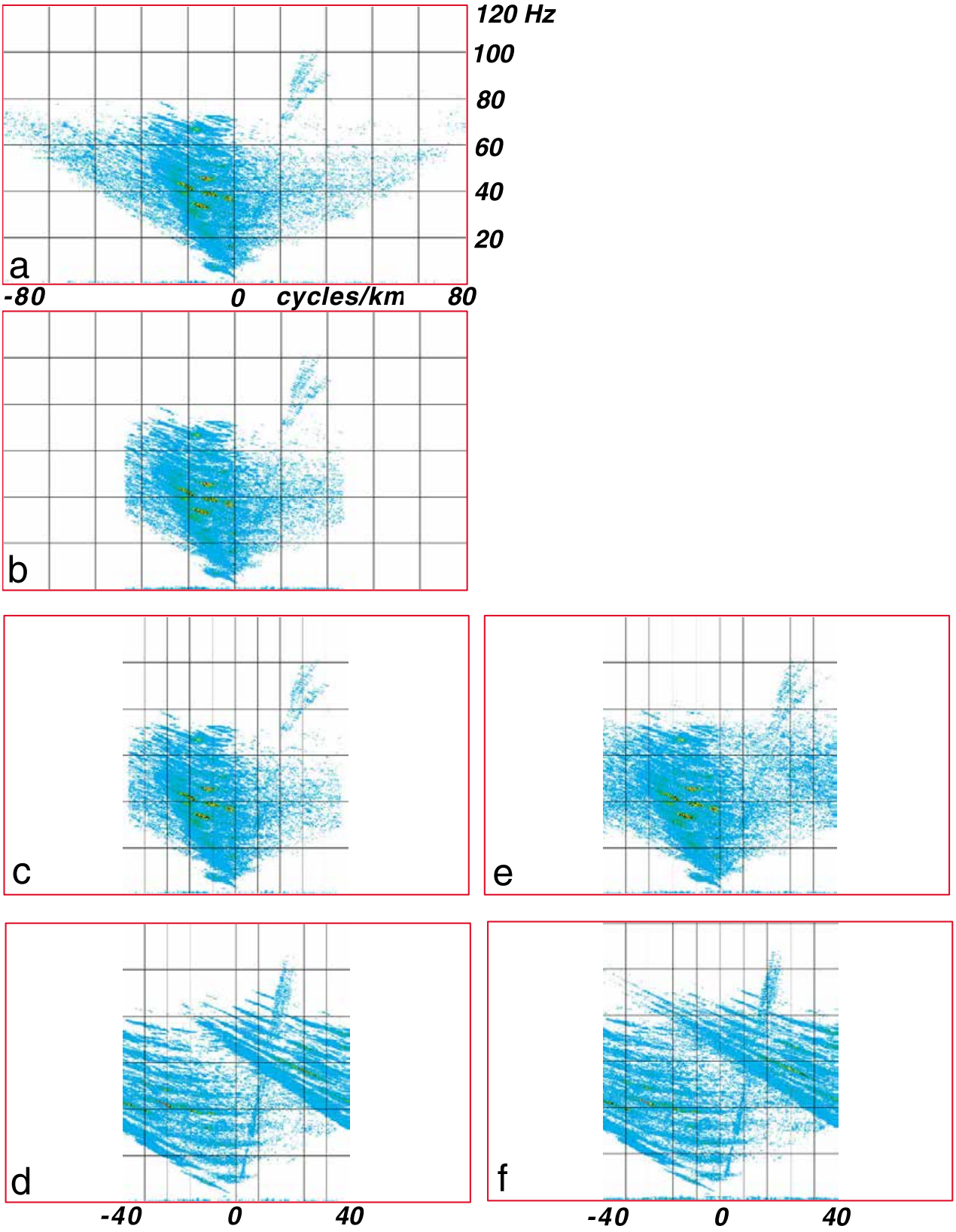

FIG. 1.2-17. The $f - k$ spectra of the shot record at various stages as described in Figure 1.2-16.

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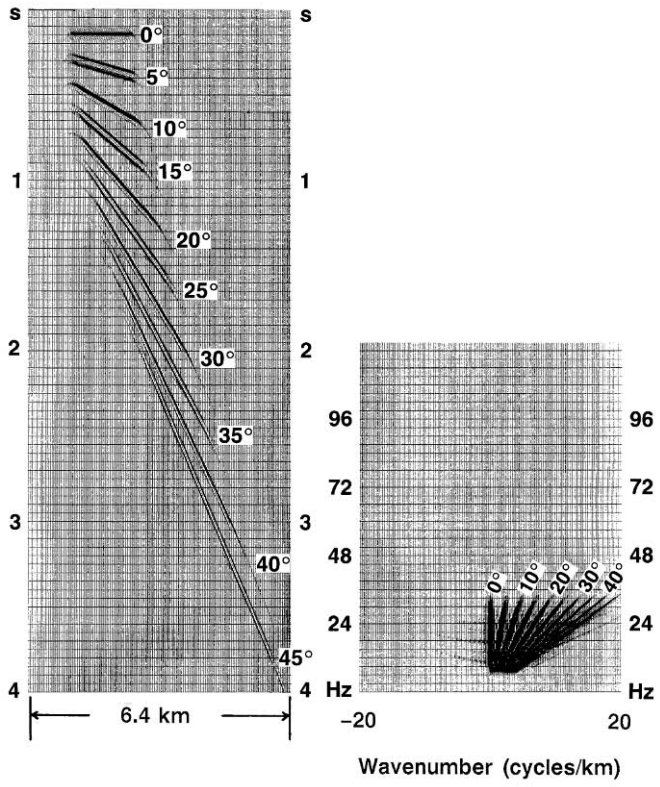

FIG. 1.2-18. A zero-offset section (256 traces with 25-m trace spacing) containing 10 dipping events and its 2-D amplitude spectrum. No frequencies are aliased.

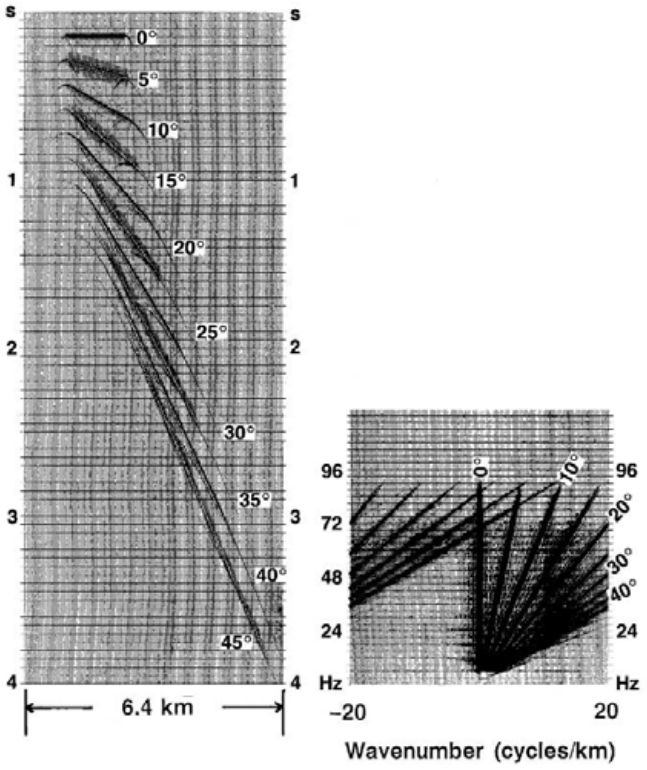

FIG. 1.2-19. A zero-offset section (256 traces with 25-m trace spacing) containing 10 dipping events and its 2-D amplitude spectrum. Steeper dips are aliased at increasingly lower frequencies.

the resulting record shown in Figure 1.2-17e. This decimated shot record (Figure 1.2-16e) should be compared with the antialiased-filtered record (Figure 1.2-16c) after inverse moveout correction (Figures 1.2-16d and f). Examine the $f-k$ spectra of the data after trace decimation with (Figure 1.2-17d) and without antialias filtering (Figure 1.2-17f) and note that, for this data set, wavenumber filtering has not been successful in removing spatially aliased energy. Wavenumber filtering for trace decimation is successful if the aliased energy in the original record is confined to the reject zones for all frequencies between 50 and 100.

So far, only the synthesis of a single dipping event from a discrete number of frequency components has been considered. This analysis now is extended to a range of dips. Figure 1.2-18 shows a section with dips that vary from 0 to 45 degrees and the corresponding 2-D amplitude spectrum. These same dips, but with higher frequency content, also are seen in Figure 1.2-19. Events with 0-, 5-, 10-, and 15-degree dips are not spatially aliased. The 20-degree dip is aliased at nearly 72 Hz, the 30-degree dip at nearly 48 Hz, and the 45-degree dip at nearly 36 Hz. Again, the steeper the dip, the lower the frequency at which spatial aliasing occurs.

Given a dip value, how is the maximum unaliased frequency determined? Consider the 20-degree dipping event in Figure 1.2-19. First, measure the dip in milliseconds per trace. There are 256 traces in the $t-x$ model with 25-m trace spacing. The 20-degree dip is equivalent to 7 ms/trace. Frequency components with periods less than twice the dip are spatially aliased. Thus, given the dip in milliseconds per trace, the threshold frequency at which spatial aliasing begins is 500 per dip. In the present case, the threshold frequency is $500/7 \approx 72$ Hz. This is verified by examining the amplitude spectrum in Figure 1.2-19.

Figure 1.2-20 shows three field records and their 2-D amplitude spectra, known as $f-k$ spectra. By now, it is easy to recognize and relate various events on the shot gathers to those on the $f-k$ spectra. Event A is the high-amplitude dispersive coherent noise with very low group velocity. When the spatial extent of these waves broadens, bandwidth in the wavenumber direction becomes smaller. Conversely, when the spatial extent becomes smaller, the event, such as G, spans a wider wavenumber bandwidth in the $f-k$ spectrum (compare events A, F, and G). Events B and C are parts

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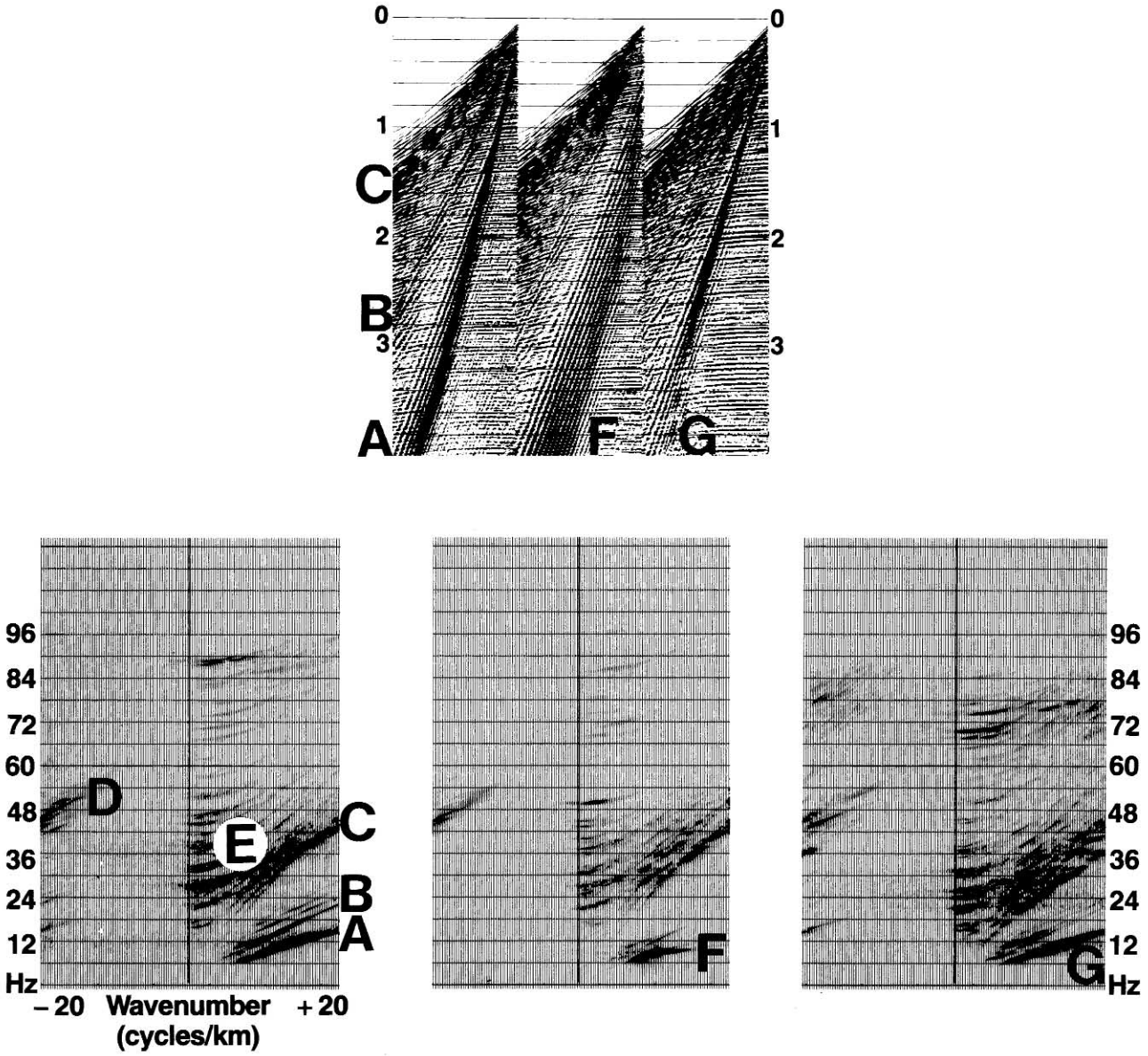

FIG. 1.2-20. Three common-shot gathers (top) and their $f-k$ spectra (bottom). (The marked events are discussed in the text.) Dip convention: An event maps onto a positive-dip quadrant in the $f-k$ spectrum if it dips down moving from near to far offsets. (Data courtesy Deminex Petroleum.)

of the guided wave packet. Event C contains aliased energy above 42 Hz (indicated by D on the $f-k$ spectrum). Primaries and associated multiples are mapped into region E between the frequency axis and event C.

Spatial aliasing not only is a concern in a prestack application of a multichannel filter, such as $f-k$ filtering, but also during poststack processing. Specifically, migration of CMP stacked data suffers from spatial aliasing if the trace interval is too coarse. An optimum CMP trace interval to avoid spatial aliasing can be computed as follows. Consider a dipping reflector with a dip angle of θ (Figure 1.2-21). Also consider a normal-incidence plane wave with a dominant period T recorded at the surface with a trace separation Δx. (This is the zero-offset case where Δx is the CMP trace interval.) From the geometry in Figure 1.2-21, we write

$$\sin\theta = \frac{v\Delta t}{2\Delta x}, \qquad (1-6)$$

where Δt is the two-way time separation between the arrival times of the plane wave at the two receiver locations, A and B, and v is the medium velocity. *Spatial*

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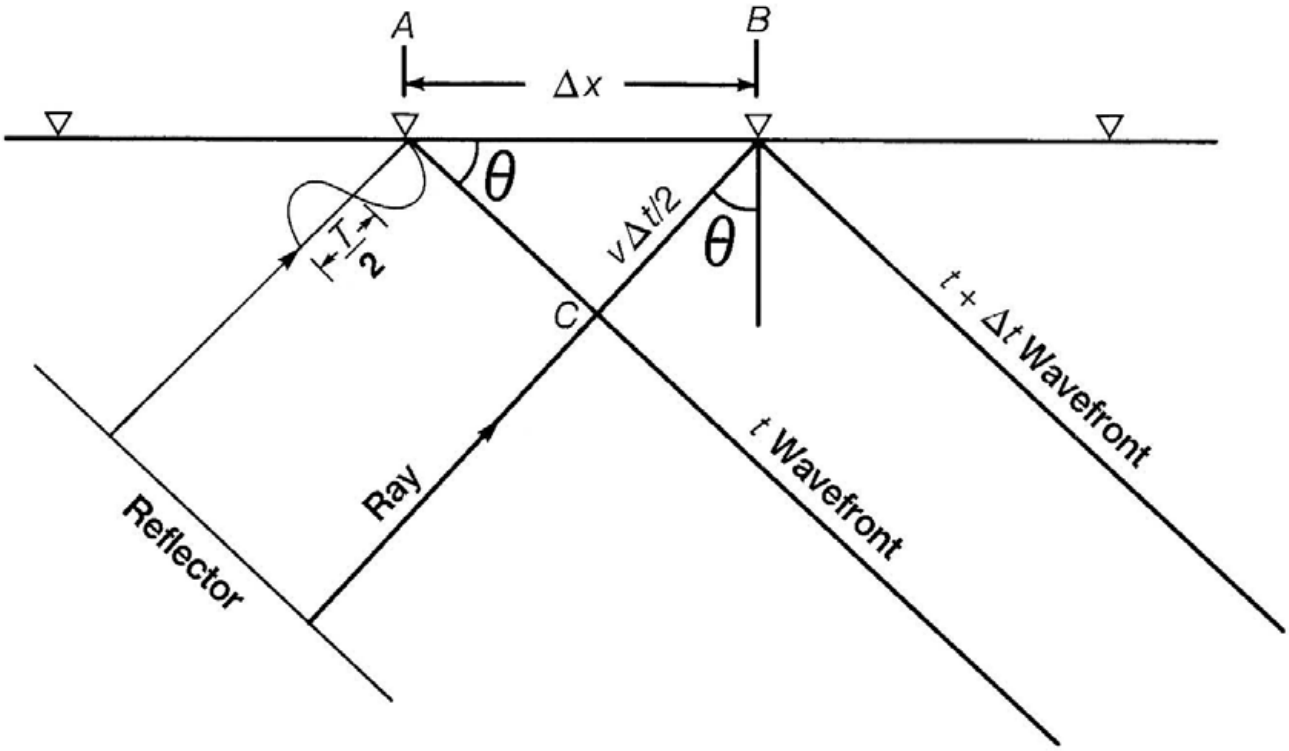

FIG. 1.2-21. A plane wave reflecting at normal incidence from a dipping reflector with a dip angle θ arrives at two consecutive receiver locations A and B at the surface with a separation Δx. Geometry of this plane wave is used to derive equation (1-6).

aliasing occurs when the wavefront separation in time Δt equals half the dominant period T. When this criterion is applied to equation (1-6), we get the following expression for the maximum threshold frequency, $f_{max} = 1/T$, that is not aliased for a given dip, velocity, and CMP trace interval:

$$f_{max} = \frac{v}{4\Delta x \sin\theta}. \qquad (1-7)$$

Table 1-12 shows the evaluation of equation (1-7) for a particular numerical example. Equation (1-7) also can be expressed in terms of receiver group interval $2\Delta x$. Suppose the maximum dip is 30 degrees. If the sampling interval is 4 ms, then the Nyquist frequency is 125 Hz. After antialias filtering, the frequency band extends up to 90 Hz, provided the high-cut filter is at three-quarters of the Nyquist. For a bandwidth without spatial aliasing, we have to select a 12.5-m CMP trace interval.

To circumvent the deleterious effect of spatial aliasing on migrated data, trace interpolation is applied to attain a trace interval appropriate for the bandwidth of the CMP-stacked data. As stated in Section 1.1, 1-D interpolation of a time series, such as a seismic trace, from a coarse to a finer sampling interval does not recover the frequencies lost by the original sampling; but, it only generates extra samples. On the other hand, interpolation of 2-D data, such as a seismic section, from a coarse to a finer trace interval, is possible for all signal frequencies, including those that are aliased. This is because of the fact that one can detect and measure the dip as defined by the stepout of equation (1-5) on a stacked section for the unaliased frequencies and use the dip information to interpolate not only the unaliased but also the aliased frequencies. Modern data acquisition geometries for 2-D data often do not require trace

Table 1-12. Frequency threshold for spatial aliasing. Velocity is 3000 m/s.

	Threshold Frequency (Hz) for CMP Trace Interval (m)			
Dip Angle (deg)	12.5	25	37.5	50
10	346	173	115	86
20	175	88	58	44
30	120	60	40	30
40	93	47	31	23

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interpolation. Nevertheless, trace interpolation usually is required in the crossline direction prior to 3-D migration (Section 7.2).

1.3 WORLDWIDE ASSORTMENT OF SHOT RECORDS

Forty shot records, both land and marine, from North and South America, Europe, the Middle East, North Africa, and the Far East, are presented in Figures 1.3-1 through 1.3-40. Source types are vibroseis, Geoflex, dynamite, air gun, Maxipulse, Aquapulse, and Aquaseis. The recording parameters, including the number of traces, number of samples per trace, sampling interval, trace interval, and inner offset, are indicated in Table 1-13. Study the field records to learn how to recognize different types of waves. For display purposes, an instantaneous type of gain (AGC) (Section 1.4) was applied to all 40 records. These records will be referred to by their record numbers in the following discussions.

The main goal in processing reflection seismic data is to enhance genuine reflection signal by suppressing unwanted energy in the form of coherent and random ambient noise. In the following paragraphs, shot gathers are examined to point out the different types of seismic energy.

Record 1 is a correlated vibroseis data set. (For vibroseis correlation, refer to Section 1.1.) A number of reflections is present in this record with reasonably good signal-to-noise ratio. A genuine reflection is recognized on common-shot gathers by its hyperbolic nature. Reflections behave the same way on CMP gathers. A flat horizon with no dip yields a symmetric hyperbola on both common-shot and common-midpoint gathers recorded using split-spread geometry. (In split-spread geometry, the source is located somewhere in the middle of the receiver cable, usually at the center.) A dipping horizon yields a skewed hyperbola on a common-shot gather, while still yielding a symmetric hyperbola on a common-midpoint gather. Reciprocity of sources and receivers provides this symmetry. From the reflection hyperbolas in Record 1, note that the subsurface is made up of nearly horizontally flat layers. Any irregularity in the shape of the moveout hyperbola can be attributed to near-surface effects and/or lateral variations in velocity.

Record 2 is an asymmetric shot gather. Note the reflection energy between 1 and 2 s, with rather irregular moveout.

Record 3, which was obtained by using dynamite, contains a series of reflections with nearly perfect hyperbolic moveout, especially between 1 and 3 s. This record is from the times of analog recording. It is not uncommon to digitize old analog data and process it with modern techniques.

Record 4 contains events with complex moveout between 2.5 and 3.5 s. Events A, B, and C have skewed hyperbolic moveouts, which, in this case, suggests that they are dipping up toward the left. Also note the traveltime distortions along the moveouts caused, most likely, by near-surface irregularities.

Record 5 has some ground-roll energy, which is characterized by its low-frequency, high-amplitude appearance, particularly on short-offset traces. This kind of energy typically is suppressed in the field by using a proper receiver array.

Record 6 contains weak and strong, nearly flat reflectors (A and B). The break in the reflection hyperbola (C1-C2) suggests the presence of a fault C1 on the upthrown side and C2 on the downthrown side). Again, note the ground-roll energy with its dispersed low-frequency character on inside traces (event D).

Record 7 contains three interesting events. Event A is a skewed hyperbola, which suggests that it is dipping up toward the left, while event B is nearly symmetric, which suggests a flat dip. Event C shows a discontinuity F along its moveout curve, indicating the presence of a fault.

Record 8 shows a record with excellent signal quality. This dynamite record has a number of reflections and associated interbed reverberations. Note the progressive decrease in the signal-to-noise ratio at late times. This is true for almost all seismic data. Event A has large moveout because it is shallow, while event B has small moveout because it is deep. (Linear energy C is referred to in Exercise 1-3.)

Record 9, which is a correlated vibroseis data set, has a series of reflections and ground roll. Unlike data from impulsive sources such as dynamite, first breaks in vibroseis data may not be distinguishable (compare, for example, Records 8 and 9). This is because the correlated vibroseis record contains some of the side lobes of the sweep signal autocorrelation. Note the increase in random noise in the later part of the record below 3 s.

Record 10 contains two strong shallow reflectors, A and B, in addition to ground-roll energy C. Also, a bundle of energy with extremely large moveout is noted between 2.5 and 5 s (D1-D2). This coherent noise may be attributed to side-scattered energy, which is caused by inhomogeneities in the subsurface (particularly at the water bottom) that behave as point sources.

Record 11 contains four prominent reflections. This record is from Alaska, where the thickness of the permafrost layer can be irregular. Such near-surface irreg-

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Table 1-13. Parameter index of a worldwide assortment of common-shot gathers.

Record Number	Area	Number of Samples per Trace	Number of Traces	Sampling Interval, ms	Trace Interval, ft or m	Inner Offset, ft or m	Source
1	South Texas	1,275	48	4	330 ft	990 ft	V
2	West Texas	1,025	120	4	100 ft	400 ft	V
∗3	Louisiana	1,500	24	4	340 ft	340 ft	D
4	Turkey	1,275	48	4	100 m	250 m	V
5	South America	3,000	48	2	100 m	200 m	D
6	Far East	1,250	48	4	100 m	150 m	D
7	South America	2,600	48	2	100 m	300 m	V
8	Central America	1,300	96	4	50 m	100 m	D
9	Alaska	1,000	96	4	220 ft	990 ft	V
10	North Africa	1,325	120	4	25 m	300 m	V
11	Alaska	1,000	96	4	220 ft	990 ft	V
12	Mississippi	1,275	48	4	330 ft	990 ft	V
13	Offshore Texas	2,025	48	4	220 ft	875 ft	A
14	Offshore Texas	1,525	48	4	220 ft	690 ft	P
15	Offshore Canada	2,500	48	2	25 m	360 m	A
16	South America	1,275	48	4	25 m	233 m	A
17	South America	2,000	48	4	50 m	250 m	A
18	Offshore Louisiana	1,500	120	4	82 ft	716 ft	A
19	Turkey	1,250	216	4	10 m	50 m	D
20	South Aleutians	2,025	120	4	82 ft	921 ft	A
21	Denver Basin	1,550	48	2	220 ft	220 ft	V
22	Williston Basin	1,550	48	2	110 ft	110 ft	V
23	San Juaquin Basin	1,550	48	2	220 ft	220 ft	V
24	Arctic	3,000	48	2	220 ft	220 ft	S
25	Alberta	2,000	96	2	50 m	50 m	D
26	Alberta	1,500	48	2	67 m	67 m	D
27	Canada	1,791	92	4	50 m	200 m	A
28	Canada	2,500	48	2	25 m	300 m	A
29	Offshore Spain	2,000	48	4	50 m	250 m	M
30	Offshore Crete	2,125	96	4	25 m	230 m	A
31	North Sea	1,550	96	4	25 m	228 m	A
32	North Sea	1,550	96	4	25 m	178 m	A
33	North Sea	1,625	96	4	25 m	200 m	A
34	Celtic Sea	1,500	60	4	50 m	253 m	A
35	Denmark	2,500	52	2	100 m	100 m	D
36	Middle East	1,024	48	4	50 m	250 m	V
37	Turkey	1,000	48	4	75 m	187 m	V
38	North Africa	2,500	60	2	100 m	100 m	V
39	Middle East	2,500	60	2	50 m	100 m	G
40	West Africa	2,600	96	2	30 m	120 m	D

∗ Analog recording. V: vibroseis, D: dynamite, A: Air gun, P: Aquapulse, M: Maxipulse, G: Geoflex, S: Aquaseis. All vibroseis records have been correlated. Aquapulse and Maxipulse are registered trademarks of Western Geophysical Company of America. Aquaseis and Geoflex are registered trademarks of Imperial Chemical Industries.

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ularities can have lateral dimensions that range from less than a group interval to wavelengths that are several times a cable length. As seen on the right flank of the hyperbolas (events A, B, C, and D), these irregularities cause substantial time shifts in reflection arrivals. Such distortions in moveout could be dynamic (time-dependent) or static (time-independent). They should be corrected before stacking. Except for these distortions, all events seem to have symmetric, hyperbolic moveouts that indicate nearly horizontally layered substrata.

Record 12 is a field record with a low signal-to-noise ratio. A complex subsurface structure is implied between 2 and 3.5 s.

From Record 13, note the high-frequency hyperbolic energy S that is associated with a side scatterer, possibly at the water bottom.

Record 14 has three identifiable reflections: A, B, and C. Reverberations and multiples also make up a significant portion of the data.

Record 15 is a marine record. The hard water bottom causes refraction arrival A. This shot gather primarily contains guided waves, which are manifested as linear trends such as B, C, and D. The genuine reflection E has little moveout.

Guided waves are trapped within a water layer and travel in the horizontal direction. They are dispersive — each frequency component travels at a different speed, which is called *horizontal phase velocity.* Their behavior is variable, primarily dictated by water bottom conditions and the thickness of the water layer. They are an important source of coherent noise and are confined mainly to the supercritical region of propagation, where no transmission occurs into the substratum. The nature of guided waves is analyzed in Section F.1.

Wave packet A in Record 16 is made up entirely of guided waves. Direct arrivals B carry the highest frequency components, while lower frequencies C arrive earlier. Moderate frequencies D make up the later portion of the dispersive wave packet. This record has a reflection E and long-period multiples M1 through M4. The reflection and its multiples also have an accompanying reverberating wavetrain that is nearly 300 ms long.

Record 17 is longer than the common length (4 to 6 s) used in seismic data acquisition. There is no apparent signal after 4 s; nevertheless, very weak signals can sometimes be uncovered by stacking.

Record 18 has some events worth mentioning. Dispersive waves, which include the head wave and direct arrivals A make up the early portion of the record. Some reflections, B, C, and D, are followed by short-period reverberations. In the deeper part of the record, note the events with extremely large moveout E, which is unusual for deep data. These events represent the side-scattered coherent noise.

Record 19 is a walk-away noise test. It is actually a composite of six shot records. The receiver cable was held constant while the shots were moved away without overlap. The receiver group interval is 10 m. The receivers in each group were bundled together without array forming. This allowed all signal and noise wavelengths to be recorded. Horizontal wavelength is determined by measuring the dominant frequency (the reciprocal of the time between successive peaks or troughs) and the horizontal phase velocity (the reciprocal of stepout $\Delta t/\Delta x$) of the unwanted ground roll. The horizontal wavelength then is used to design the receiver array length that is needed to suppress this energy (see Exercise 1-4). Wave packet A1-A2 (between 1.7 and 4.6 s on the far-left trace) is an excellent example of ground roll. The linear coherent energy with opposite dip A1-A2 is the backscattered component. Reflection C is being disrupted by ground-roll energy.

Ground roll is different from guided waves, although both are dispersive. Ground roll is one type of Rayleigh wave that arises because of the coupling of compressional waves (P) and the vertical component of shear waves (SV) that propagate along the free surface (Grant and West, 1965). On the other hand, guided waves are one kind of compressional wave that travels within a layer just as sound waves travel in an organ pipe.

Record 20 (marine) shows a variety of wave types. Direct arrivals A are significantly suppressed by receiver arrays in the field. We can see the water bottom reflection B on short-offset traces. Note a shallow reflector C and associated refraction arrival D. At 1 s, another reflector E is seen. Much of the energy between 1 and 3 s most likely is multiples associated with B, C, and E. Linear noise (possibly cable noise) F and the lower-frequency propeller noise G appear in the deeper portion of the record after 4 s.

Record 21 (vibroseis) has a weak A and a strong B shallow reflector. Below 2 s, ambient noise dominates the record.

Record 22 is another vibroseis record. Note reflection arrivals A, B, and C. Although reasonably hyperbolic, there are some fluctuations in traveltimes that may be attributed to near-surface complexity (as inferred by the first breaks). Record 23 shows similar characteristics.

Record 24 is a marine shot gather that was acquired with an Aquaseis source. Direct arrivals A, water-bottom reflection B, and first-order multiples M1, M2, are recognized easily. A primary reflection P and its peg-leg multiple PL also are distinguished.

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Record 25 (land) has a very good signal quality. In addition to the several primary reflections, note ground-roll energy A. It is predominantly low frequency and travels with low group velocity (the speed with which the energy in a wave packet travels). Also note the near-surface effects that cause traveltime distortions along the right flanks of reflections B, C, D, and E.

Record 26 (dynamite) does not have a well-developed ground-roll energy; however, it still is recognizable from its low-frequency character A. Traveltime paths that correspond to reflections (for example, B, C, and D) have been disrupted by ground roll and possibly distorted by irregularities in the near surface.

Record 27 (marine) is interesting. Note the change in the cable geometry A (Exercise 1-6). There is a well-developed dispersive wave packet B that spans between 1.9 and 2.9 s at the far-offset trace. This includes the head wave and direct arrivals. Also note the distinct moveout difference between events C and D. Event C, with a larger moveout, belongs to the short-period multiple wavetrain that is associated with the water-bottom reflection. Event D, with a small moveout, is a primary reflection with its own peg-leg series F.

The air-gun data in Record 28 contain high-velocity reflections with little moveout. Note the predominant guided wave packet C that spans between 0.7 and 1.9 s at the far-offset trace. It results from the strong water-bottom refractor D.

Record 29 is puzzling (see Exercise 1-7). The skewness of the reflection hyperbolas (B, C, D, and E) increases in depth.

Record 30 is a deep-water shot record. The direct arrival A, water-bottom reflection B, and shallow reflection C, can be identified easily. First-order water-bottom multiples M and peg-leg multiples PL, which are associated with the shallow reflector C, also are prominent in this record.

Record 31 is a shot record primarily made up of guided waves. The following wave types are identified: A is the refraction arrival, B is its multiples, C is the direct arrival, D is the dispersive medium-frequency components of guided waves between 1.8 and 3 s at the far-offset trace, and E is the backscattered energy associated with side scatterers at the water bottom.

Record 32 is another marine record that contains strong guided wave energy. A refractor A, direct arrival B, and dispersed wave packet C span between 1.2 and 4 s at the far-offset trace. Events D and E represent subcritical reflection energy, most of which is reverberation.

The events on Records 33 and 34 are referred to in Exercises 1-9 and 1-10, respectively.

The four far left traces in Record 35 are associated with the channels that are used to record auxiliary information.

Record 36 seems to have no events with hyperbolic moveout. Record 37 has a few reflections, AD, BE, CF; however, they are buried in strong ambient noise. Record 38 contains virtually no reflections. A strong dispersive wave (ground roll) makes up the early part of the record, while the remaining part contains primarily random noise.

Record 39 is a Geoflex record containing strong ground-roll energy A. Additionally, the record from top to bottom contains a strong high-frequency reverberating wavetrain and short-period multiples that are associated with the water bottom and, perhaps, a few shallow reflectors.

Record 40 has a small usable segment — the longer offset traces on the left side between 1 and 4 s. The remaining part of the record contains strong random noise and transient noise, A, B, C, D, and E, which are attributed to electronic instrument noise (possibly resulting from weather conditions).

Wave Types

To summarize, field records contain (a) reflections, (b) coherent noise, and (c) random ambient noise. One important aspect of data processing is to uncover genuine reflections by suppressing noise of various types. *Processing, however, cannot yield signal from field data without signal.* At best, it suppresses whatever noise is in the field data and enhances the reflection energy that is buried in the noise. Seismic data must not be acquired with the attitude, *"Don't worry, processing will bring out the signal."*

Reflections on shot records are recognized by their hyperbolic traveltimes. If the reflecting interface is horizontal, then the apex of the reflection hyperbola is situated at zero offset. On the other hand, if it is a dipping interface, then the reflection hyperbola is skewed in the updip direction.

There are several wave types under the coherent noise category:

(a) *Ground roll* is recognized by low frequency, strong amplitude, and low group velocity. It is the vertical component of dispersive surface waves. In the field, receiver arrays are used to eliminate ground roll. Ground roll can have strong backscattered components because of lateral inhomogeneities in the near-surface layer.

(b) *Guided waves* are persistent, especially in shallow marine records in areas with hard water bottom. The water layer makes a strong velocity contrast with the substratum, which causes most of the energy to be trapped within and guided laterally

(text continues on p. 81)

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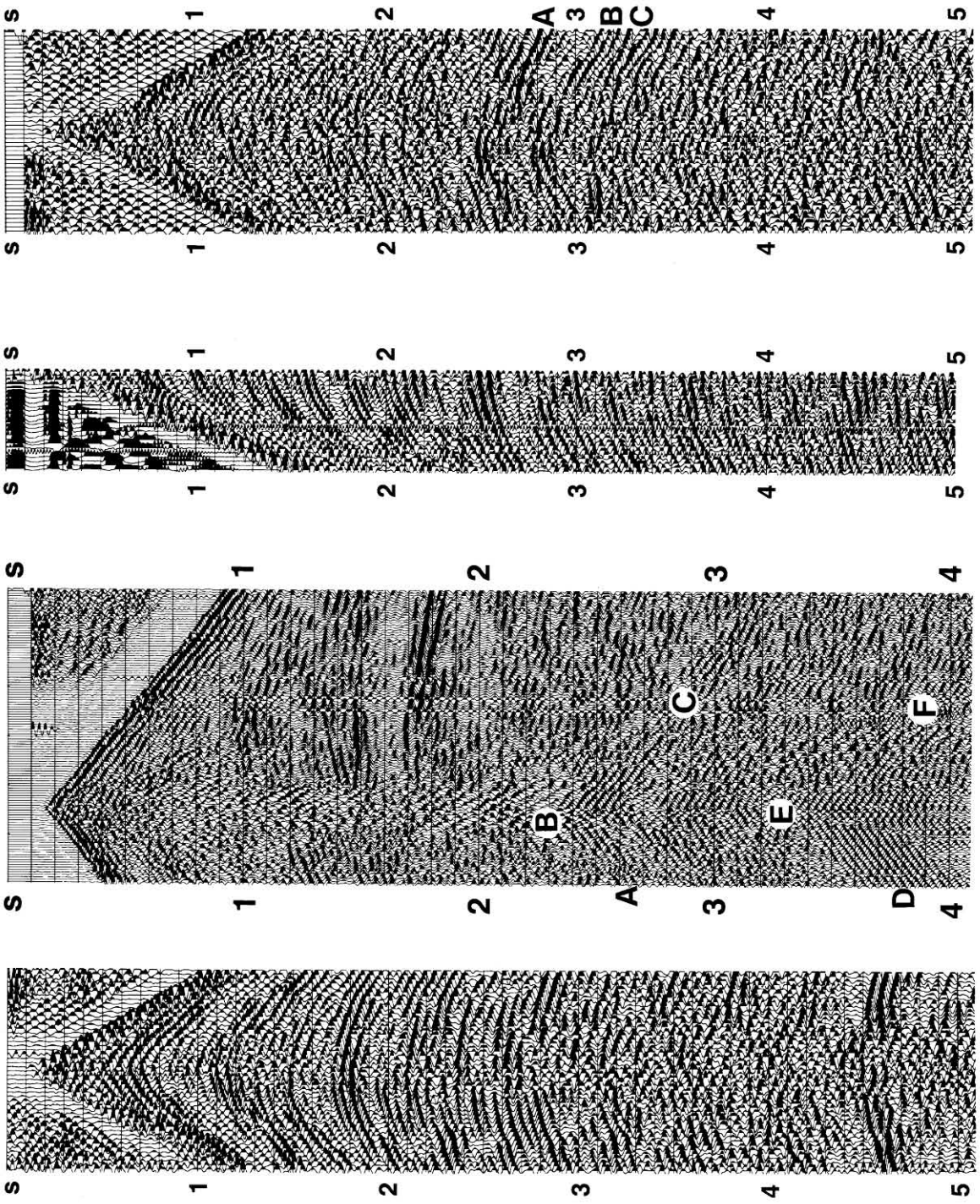

FIG. 1.3-1. Worldwide assortment record 1. For this and the following records, refer to Table 1-12 for recording parameters.
FIG. 1.3-2. Worldwide assortment record 2. Identify events ABC and DEF.
FIG. 1.3-3. Worldwide assortment record 3.
FIG. 1.3-4. Worldwide assortment record 4. What is the cause of the energy preceding the first breaks?

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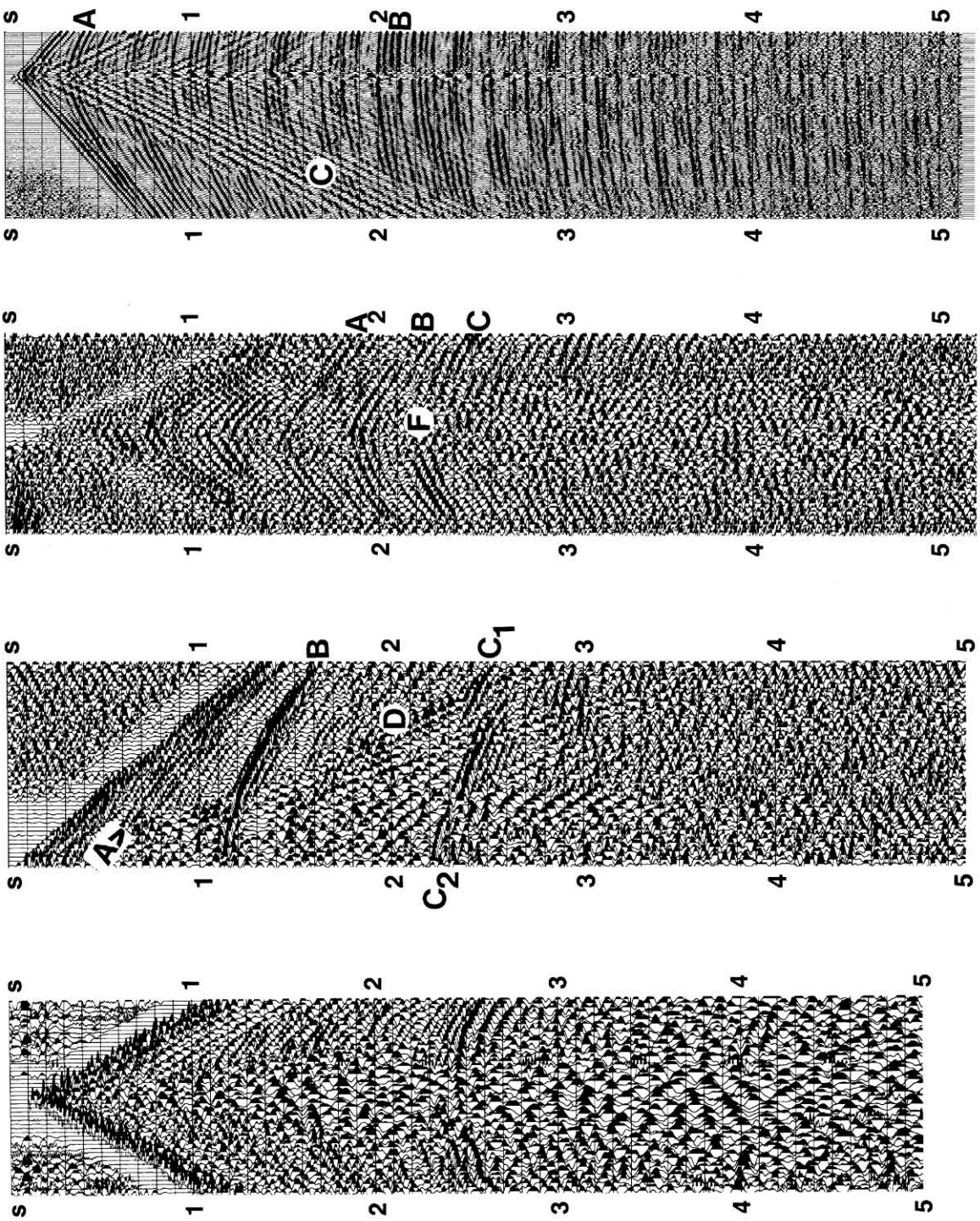

FIG. 1.3-5. Worldwide assortment record 5.
FIG. 1.3-6. Worldwide assortment record 6.
FIG. 1.3-7. Worldwide assortment record 7.
FIG. 1.3-8. Worldwide assortment record 8.

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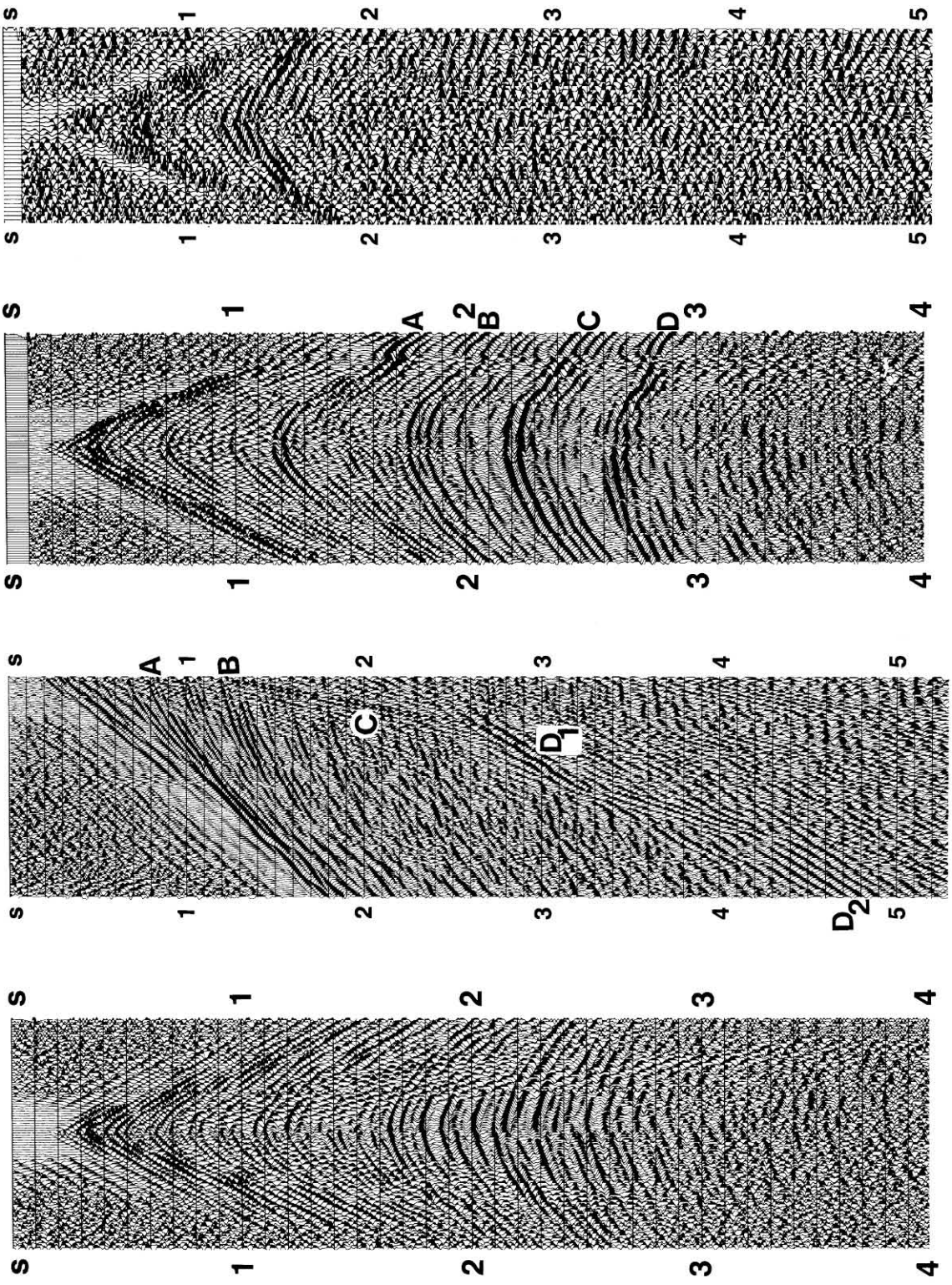

FIG. 1.3-9. Worldwide assortment record 9.
FIG. 1.3-10. Worldwide assortment record 10.
FIG. 1.3-11. Worldwide assortment record 11.
FIG. 1.3-12. Worldwide assortment record 12.

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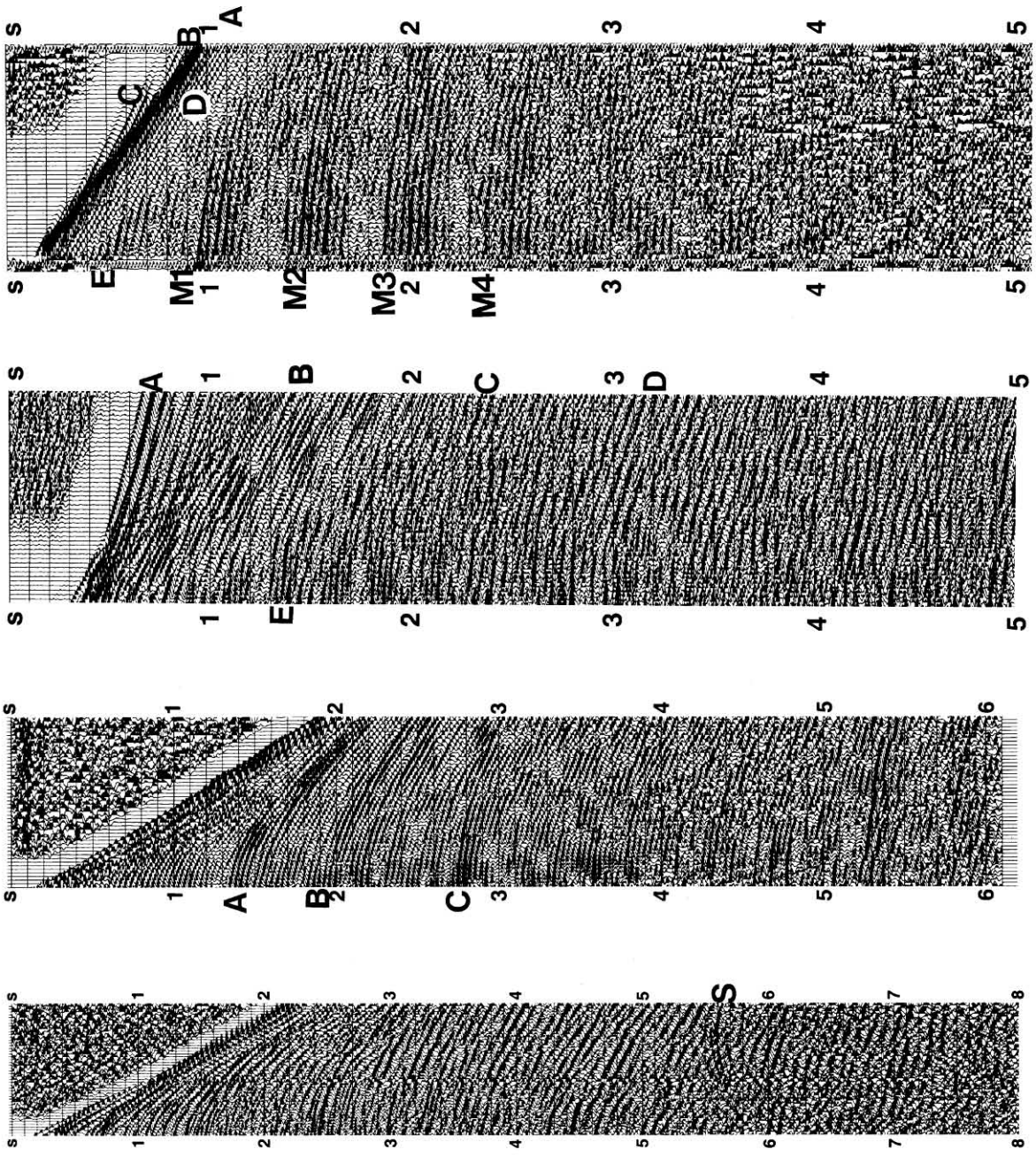

FIG. 1.3-13. Worldwide assortment record 13.
FIG. 1.3-14. Worldwide assortment record 14.
FIG. 1.3-15. Worldwide assortment record 15.
FIG. 1.3-16. Worldwide assortment record 16.

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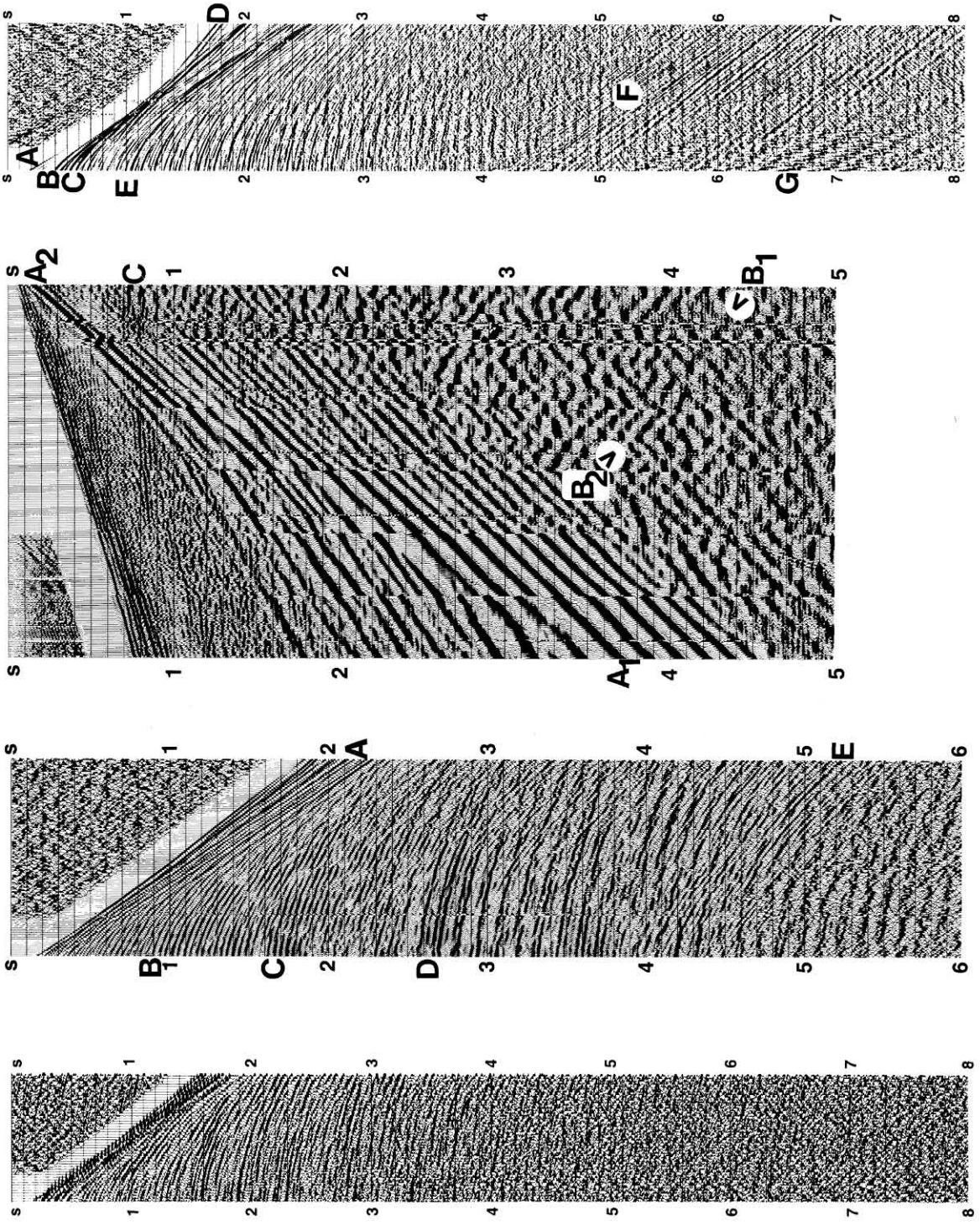

FIG. 1.3-17. Worldwide assortment record 17.
FIG. 1.3-18. Worldwide assortment record 18. Identify the direct wave on this record.
FIG. 1.3-19. Worldwide assortment record 19.
FIG. 1.3-20. Worldwide assortment record 20.

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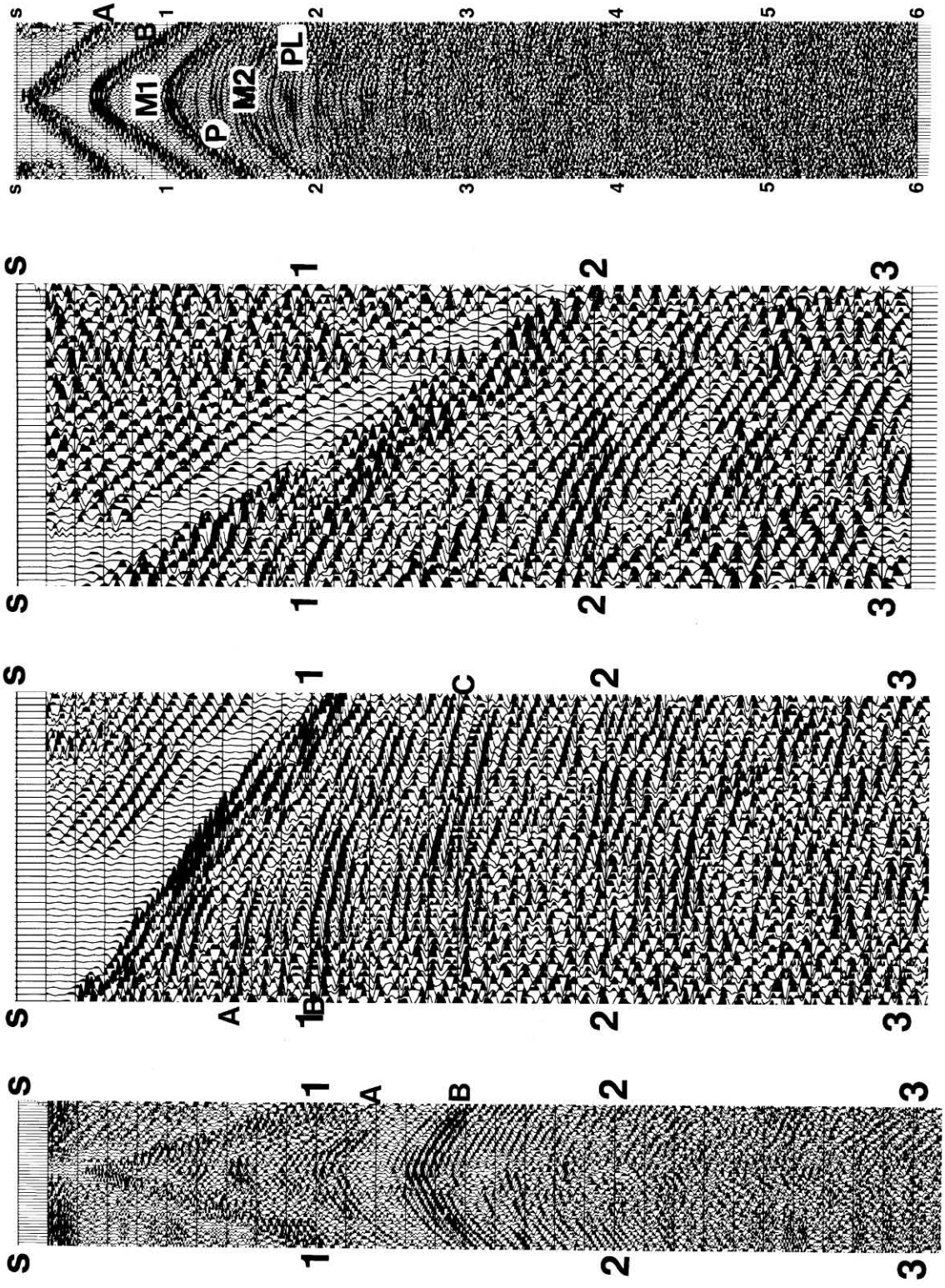

FIG. 1.3-21. Worldwide assortment record 21.
FIG. 1.3-22. Worldwide assortment record 22.
FIG. 1.3-23. Worldwide assortment record 23.
FIG. 1.3-24. Worldwide assortment record 24.

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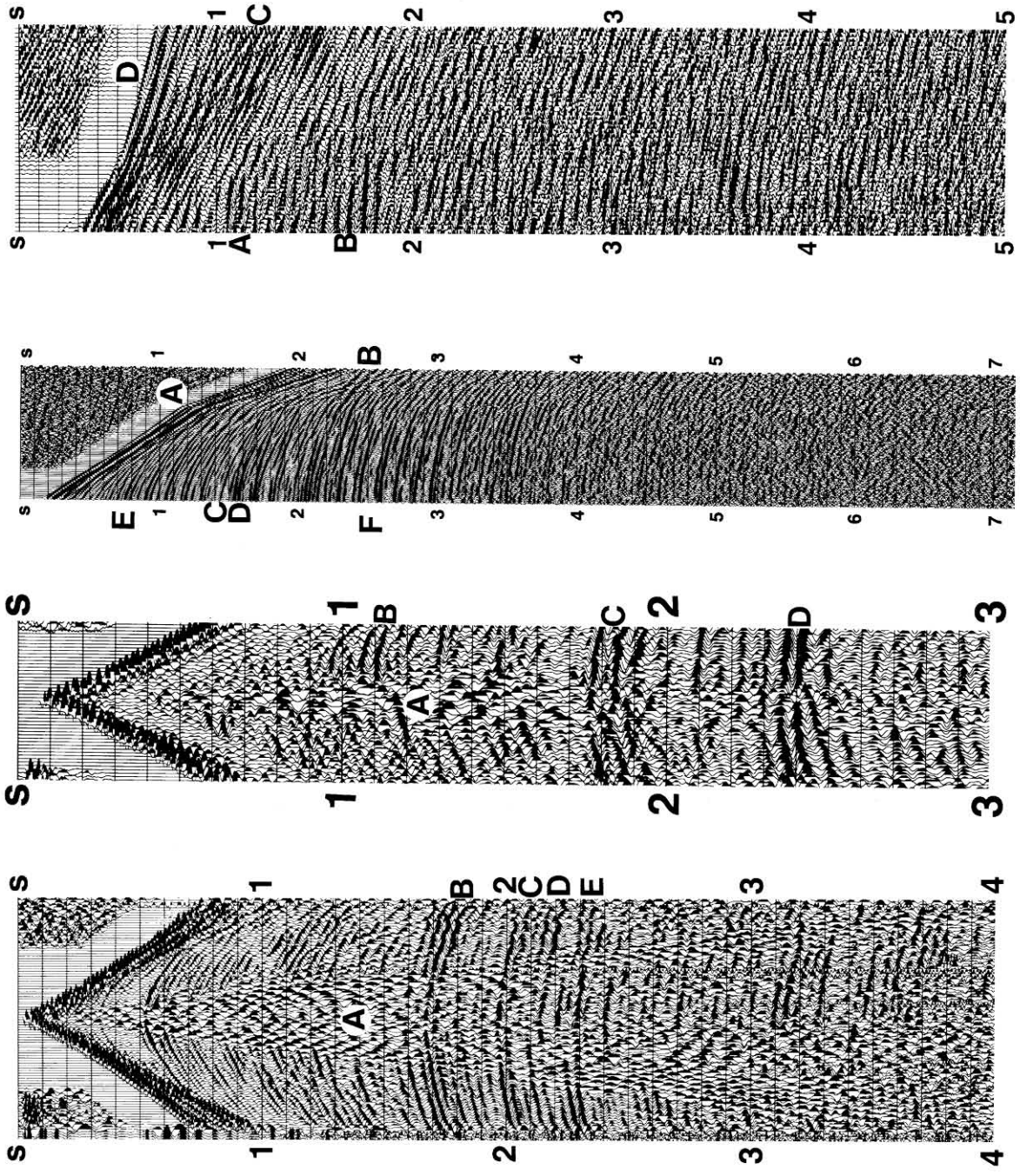

FIG. 1.3-25. Worldwide assortment record 25.
FIG. 1.3-26. Worldwide assortment record 26.
FIG. 1.3-27. Worldwide assortment record 27. Identify the direct wave on this record.
FIG. 1.3-28. Worldwide assortment record 28.

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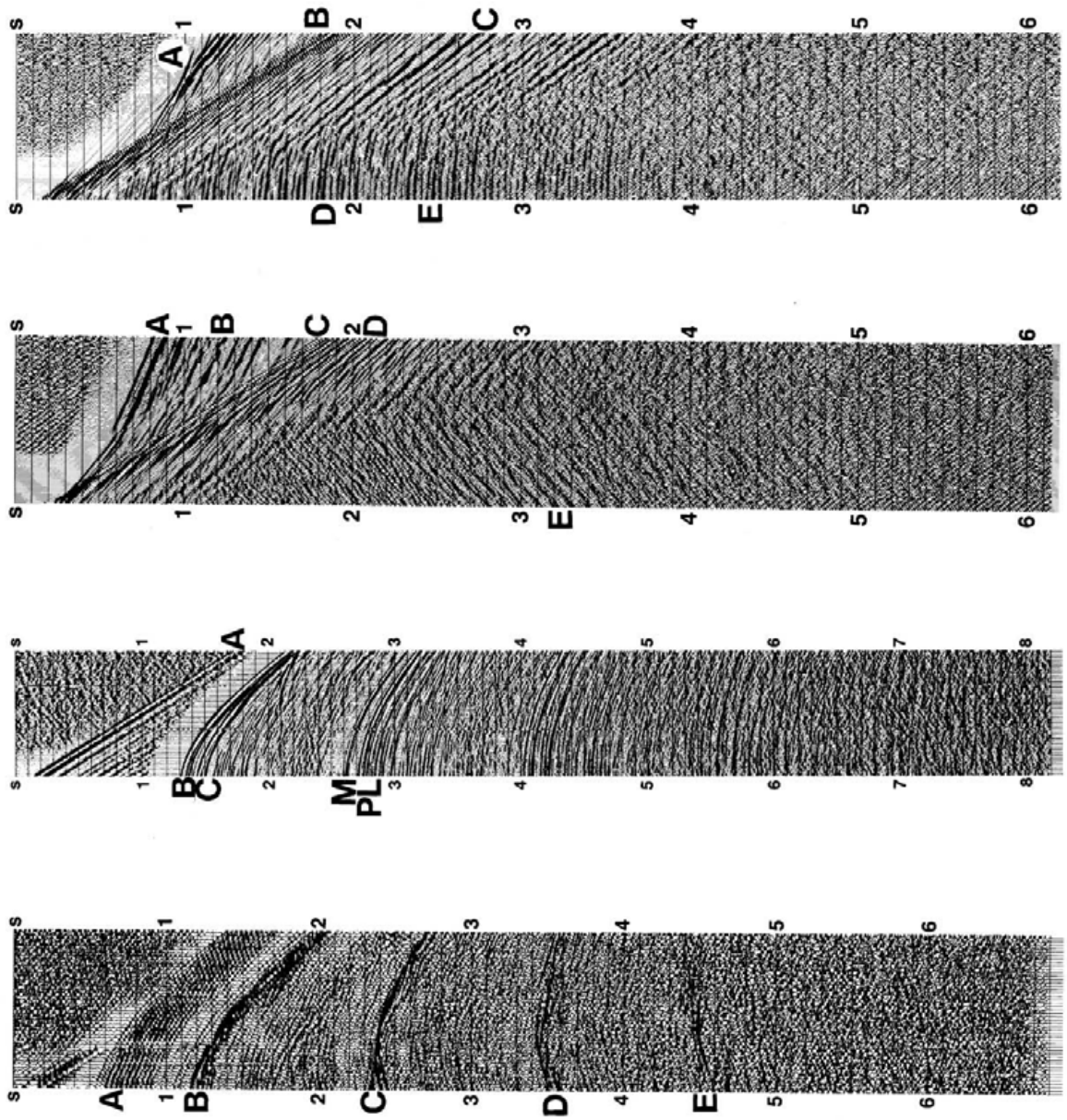

FIG. 1.3-29. Worldwide assortment record 29.
FIG. 1.3-30. Worldwide assortment record 30.
FIG. 1.3-31. Worldwide assortment record 31.
FIG. 1.3-32. Worldwide assortment record 32.

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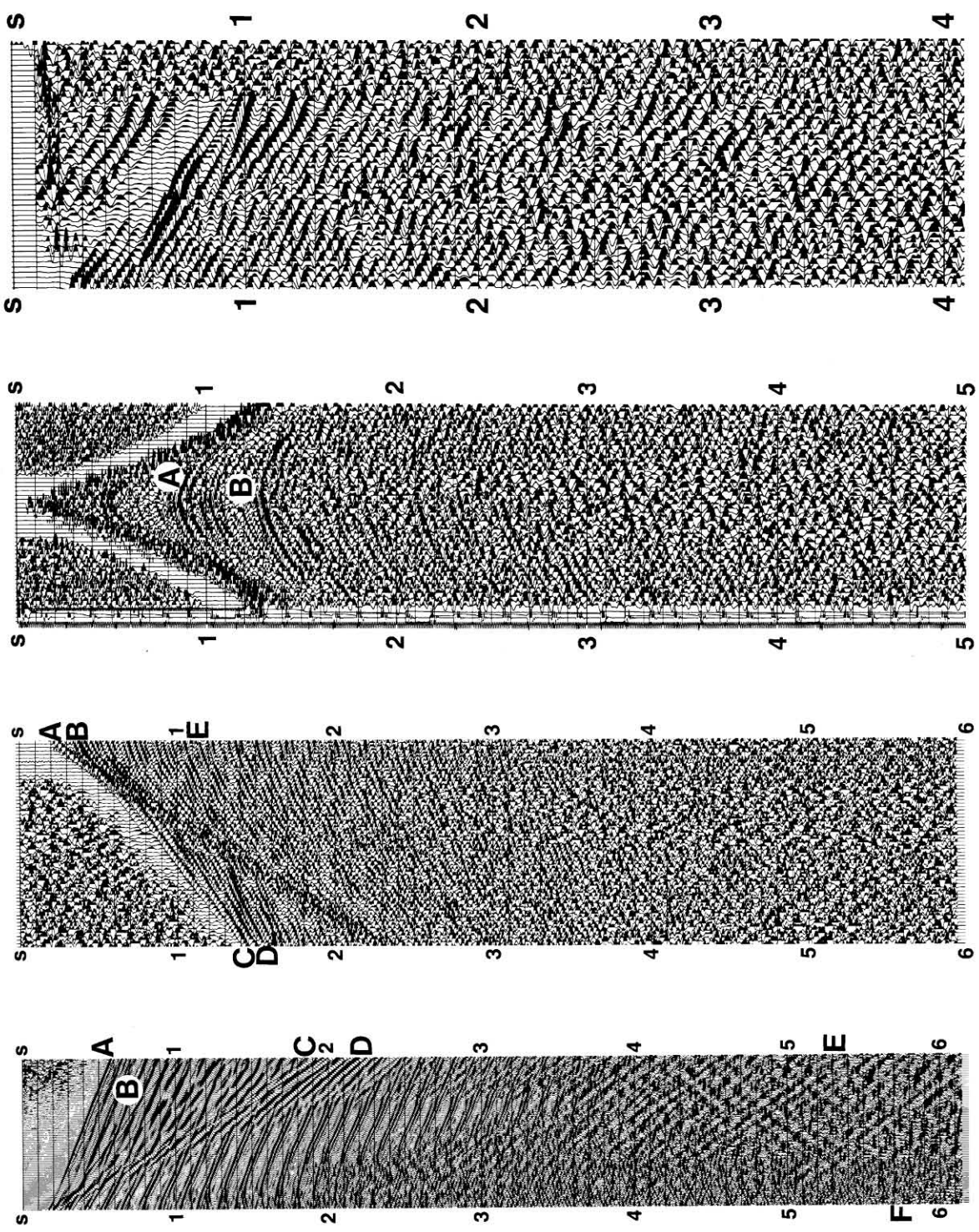

FIG. 1.3-33. Worldwide assortment record 33.
FIG. 1.3-34. Worldwide assortment record 34.
FIG. 1.3-35. Worldwide assortment record 35.
FIG. 1.3-36. Worldwide assortment record 36.

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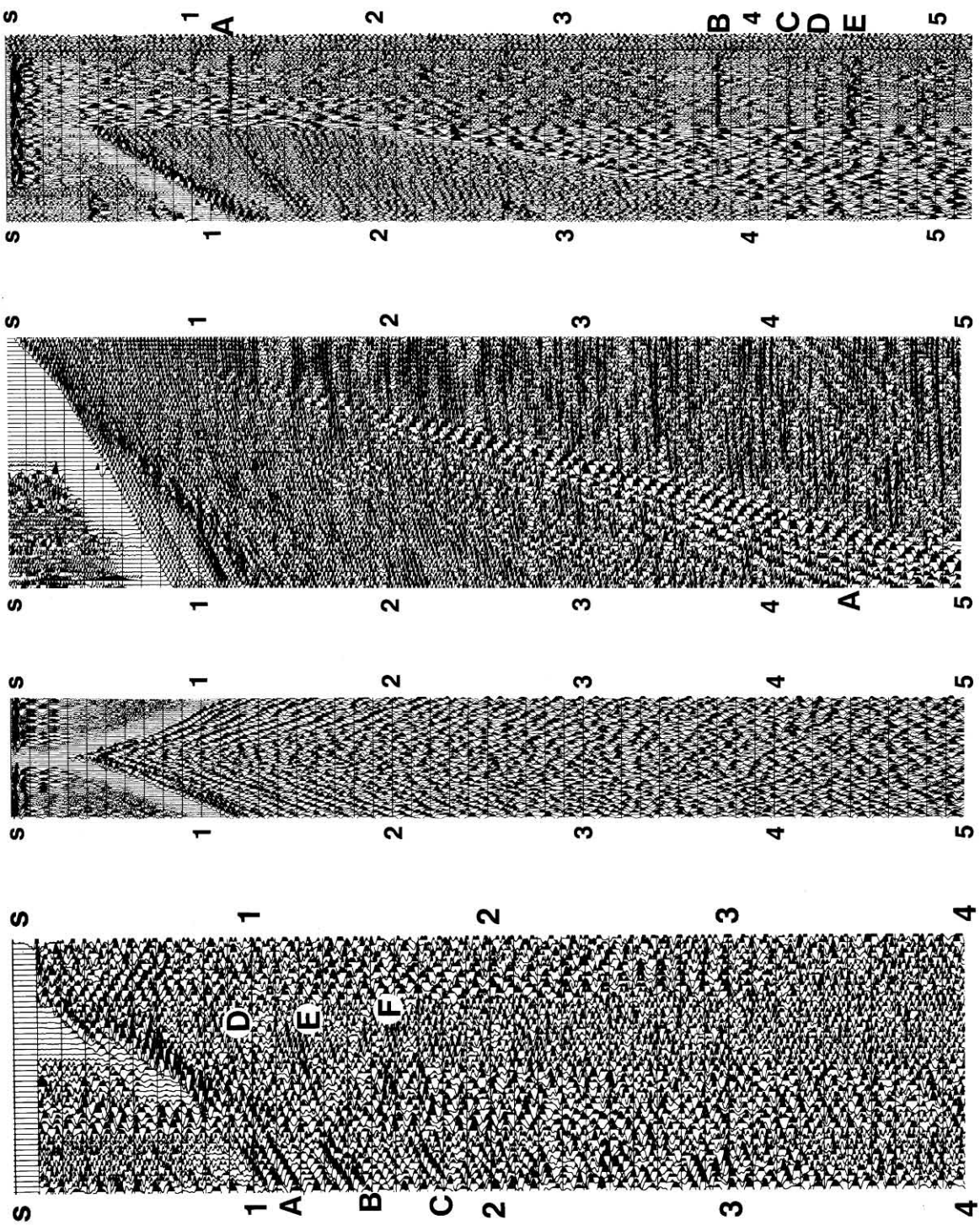

FIG. 1.3-37. Worldwide assortment record 37.
FIG. 1.3-38. Worldwide assortment record 38.
FIG. 1.3-39. Worldwide assortment record 39.
FIG. 1.3-40. Worldwide assortment record 40.

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through the water layer. The dispersive nature of these waves makes them easy to recognize on shot records. Guided waves also make up the early arrivals. The stronger the velocity contrast between the water layer and the substratum, the smaller the critical angle; thus, more guided-wave energy is trapped in the supercritical region. When there is a strong velocity contrast, refraction energy propagates in the form of a head wave. Guided waves also are found in land records. These waves are largely attenuated by CMP stacking. Because of their prominently linear moveout, in principle they also can be suppressed by dip filtering techniques. One such filtering technique is based on 2-D Fourier transformation of the shot record. This is discussed in Section 6.2. Another approach is based on slant stacking, which is described in Section 6.3.

(c) *Side-scattered noise* commonly occurs at the water bottom, where there is no flat, smooth topography. Irregularities of varying size act as point scatterers, which cause diffraction arrivals with table-top trajectories. They can be on or off the vertical plane of the recording cable. These arrivals typically exhibit a large range of moveouts, depending on the spatial position of the scatterers in the subsurface.

(d) *Cable noise* is linear and low in amplitude and frequency. It primarily appears on shot records as late arrivals.

(e) *The air wave* with a 300 m/s velocity can be a serious problem when shooting with surface charges such as Geoflex, Poulter, or land air gun. Perhaps the only effective way to remove air waves is to zero out the data on shot gathers along a narrow corridor containing this energy (notch muting). It often is impossible to recover any data arriving after the air wave on Poulter data.

(f) *Power lines* also cause noisy traces in the form of a monofrequency wave. A monofrequency wave may be 50 or 60 Hz, depending on where the field survey was conducted. Notch filters often are used in the field to suppress such energy.

(g) *Multiples* are secondary reflections with interbed or intrabed raypaths. Guided waves include supercritical multiple energy. Multiples are attacked by methods, which are based on moveout discrimination, and prediction theory, which uses the periodic behavior of multiples. The most effective moveout-based suppression technique often is CMP stack with inside-trace mute (Section 6.1). Prediction theory should be particularly effective, at least in theory, in the slant-stack domain (Section 6.3).

Random noise has various sources. A poorly planted geophone, wind motion, transient movements in the vicinity of the recording cable, wave motion in the water that causes the cable to vibrate, and electrical noise from the recording instrument all can cause ambient noise. The net result of scattered noise from many scatterers in the subsurface also contributes to random noise (Larner et al., 1983).

In Section 1.4, it is noted that energy propagating within the earth is subject to a decay in amplitude because of wavefront divergence and frequency-dependent absorption from the intrinsic attenuation of rocks. Signal strength therefore decreases in time, while random noise persists and eventually dominates. Unfortunately, gain corrections to restore signal strength at later times boost random noise in the process. Fortunately, CMP stack suppresses a significant part of the random noise uncorrelated from trace to trace.

1.4 GAIN APPLICATIONS

Gain is a *time-variant scaling* in which the scaling function is based on a desired criterion. For instance, geometric spreading correction is applied to compensate for wavefront divergence early in processing, before deconvolution. Also before deconvolution, an exponential gain may be applied to compensate for attenuation losses. Often, gain is applied to seismic data for display. An automatic gain control (AGC) is applied to seismic data to bring up weak signals. Gain must be used with care, since it can destroy signal character. For example, an instantaneous AGC with a very small time gate makes strong reflections indistinguishable from weak reflections.

Unlike a gain function, trace balancing is a *time-invariant scaling* of amplitudes. Trace balancing usually is based on rms-amplitude criterion. Specifically, each trace in a group of traces is scaled so that they all have the same desired rms amplitude level.

Geometric Spreading Correction

A field record represents a wavefield that is generated by a single shot. Conceptually, a single shot is thought of as a point source that generates a spherical wavefield. The earth has two effects on a propagating wavefield:

(a) In a homogeneous medium, energy density decays proportionately to $1/r^2$, where r is the radius of the wavefront. Wave amplitude is proportional to the square root of energy density; it decays as $1/r$.

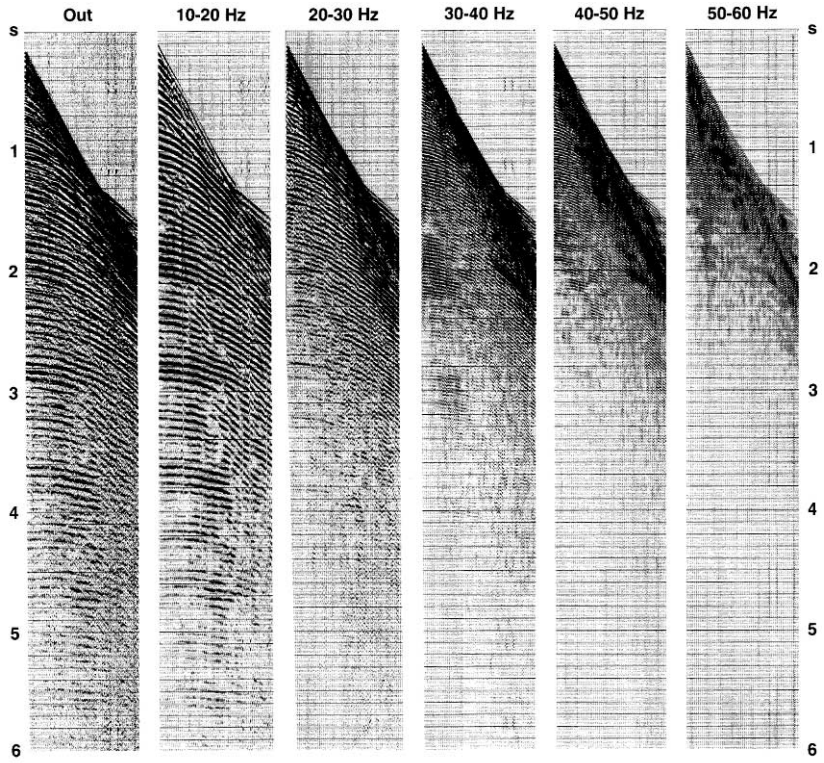

FIG. 1.4-1. A raw field record with no geometric spreading correction (far left panel) and its band-pass filtered versions. Note that the larger reflection amplitudes are confined to shallower times at increasingly higher frequency bands.

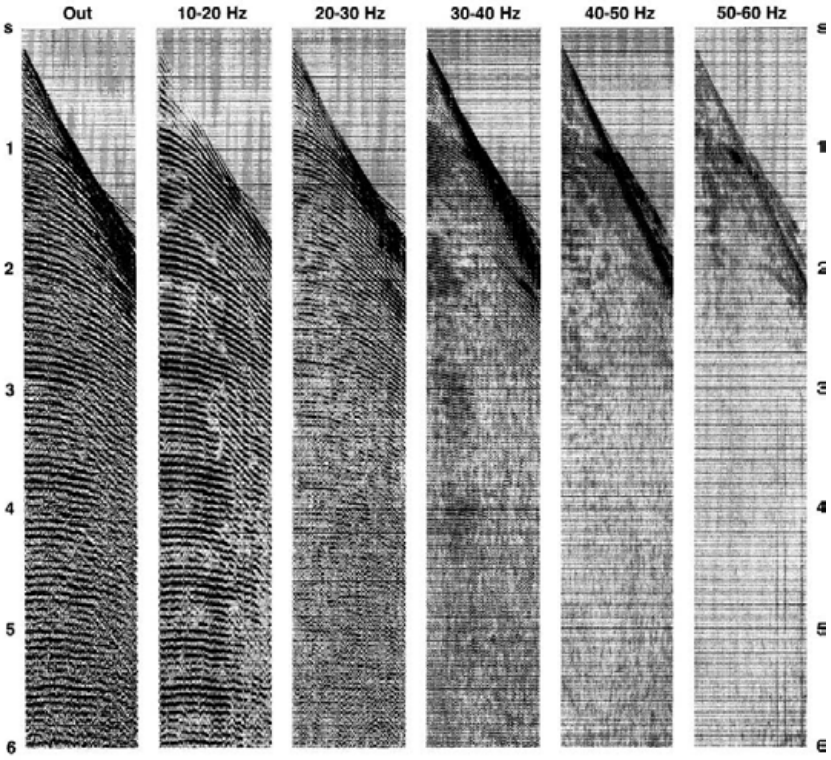

FIG. 1.4-2. The same field record as in Figure 1.4-1 (far left panel) after correcting for geometric spreading. Amplitudes are restored but frequency absorption remains.

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In practice, velocity usually increases with depth, which causes further divergence of the wavefront and a more rapid decay in amplitudes with distance.

(b) The frequency content of the initial source signal changes in a time-variant manner as it propagates. In particular, high frequencies are absorbed more rapidly than low frequencies. This is because of the intrinsic attenuation in rocks.

Attenuation mechanisms still are the subject of extensive research. However, one plausible mechanism for attenuation is related to pore fluids. As the wavefront passes through rocks, the fluids that are present in the pores are disturbed. This disturbance is greater in partially saturated rocks than fully saturated rocks. Pore fluids consume part of the energy of the propagating wavefield, which causes a frequency-dependent decay.

From Figure 1.4-1, note the wavefront divergence and frequency absorption on the field record. The first panel represents field data without any gain recovery function applied. Note the gradual decay in amplitude at later times. This record was filtered with a series of 10-Hz-wide band-pass filters. The signal in the 10-to-20-Hz panel exists down to about 6 s. On the 20-to-30-Hz panel, however, signal is visible only down to about 4 s. Moving to the higher frequency panels, note that the signal level is confined mainly to increasingly shallower times. Now apply the geometric spreading correction to the original field record in the far left panel of Figure 1.4-1. The result is shown in the far left panel of Figure 1.4-2. The amplitude level has been restored at late traveltimes. Filter panels of this record also are shown in Figure 1.4-2. When the filter panels in Figures 1.4-1 and 1.4-2 are compared with the same pass band, we see that the geometric spreading correction brought up some of the signal level at late times. However, note that the geometric spreading correction did not restore the amplitudes of the high frequencies as much as it restored the low frequencies, since the high frequencies were subject to stronger attenuation.

Figure 1.4-3 shows a portion of a CMP-stacked section and its filtered versions using narrow band-pass filters. Note that signal bandwidth only up to 20 Hz is observed down to 5 s, and frequencies up to 36 Hz are observed down to 3.5 s. The erosional unconformity just below 3.5 s constitutes a boundary with large frequency absorption. This results in attenuation of much of the high-frequency signal below this boundary. Frequencies above 36 Hz are confined to the shallow portion of the stacked section down to 2.5 s.

The effect of attenuation must be removed by modifying the amplitude spectrum of the signal, thereby making it broader. Deconvolution is one process that

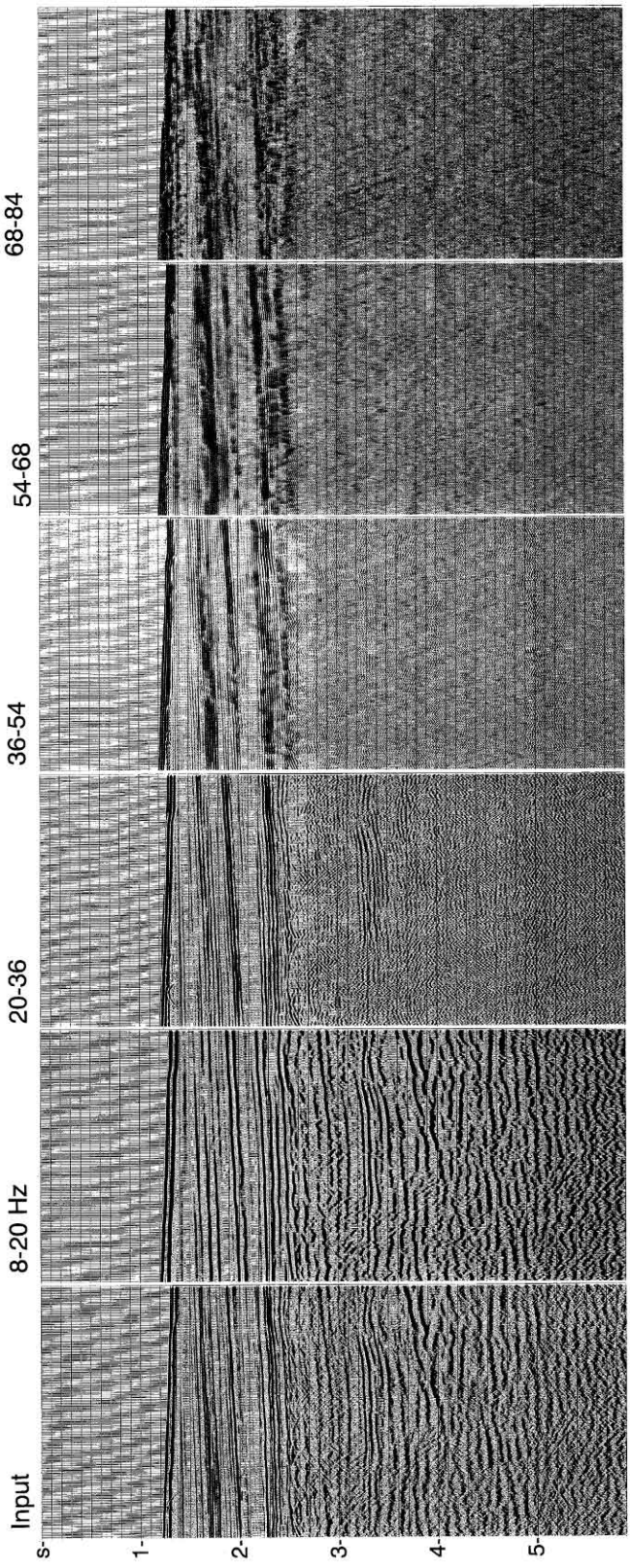

FIG. 1.4-3. A portion of a CMP-stacked section and its filtered versions. Note the attenuation of high frequencies below 2.5 s.

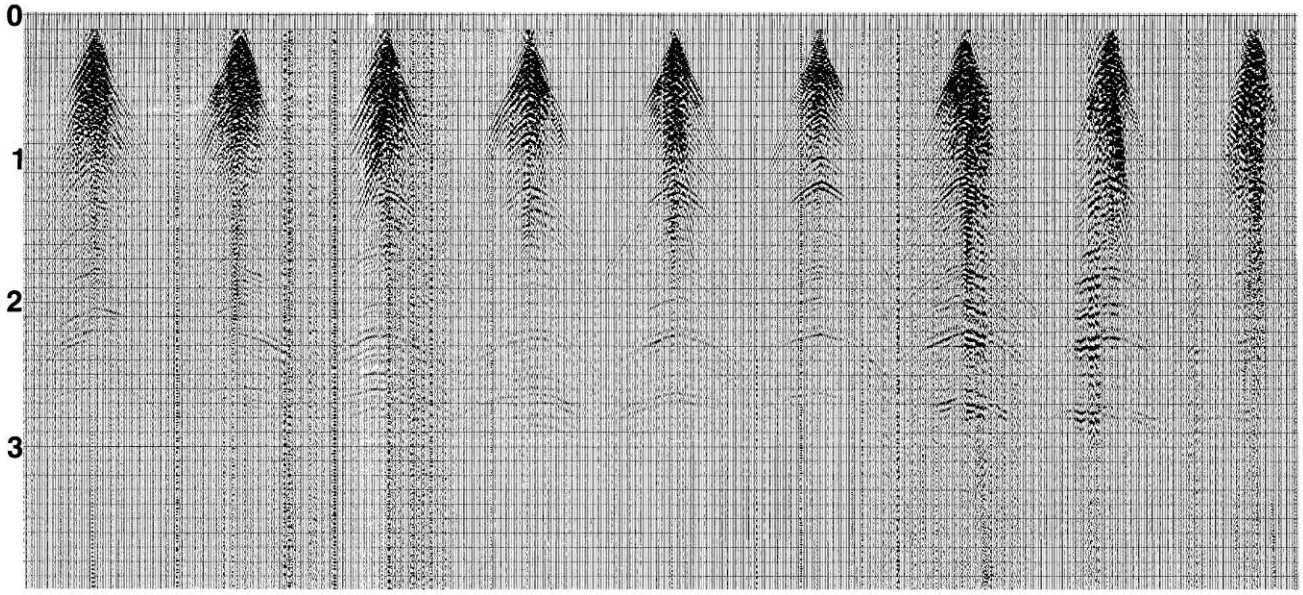

FIG. 1.4-4. Raw field records from a land survey. Note the rapid decay in amplitudes at late times.

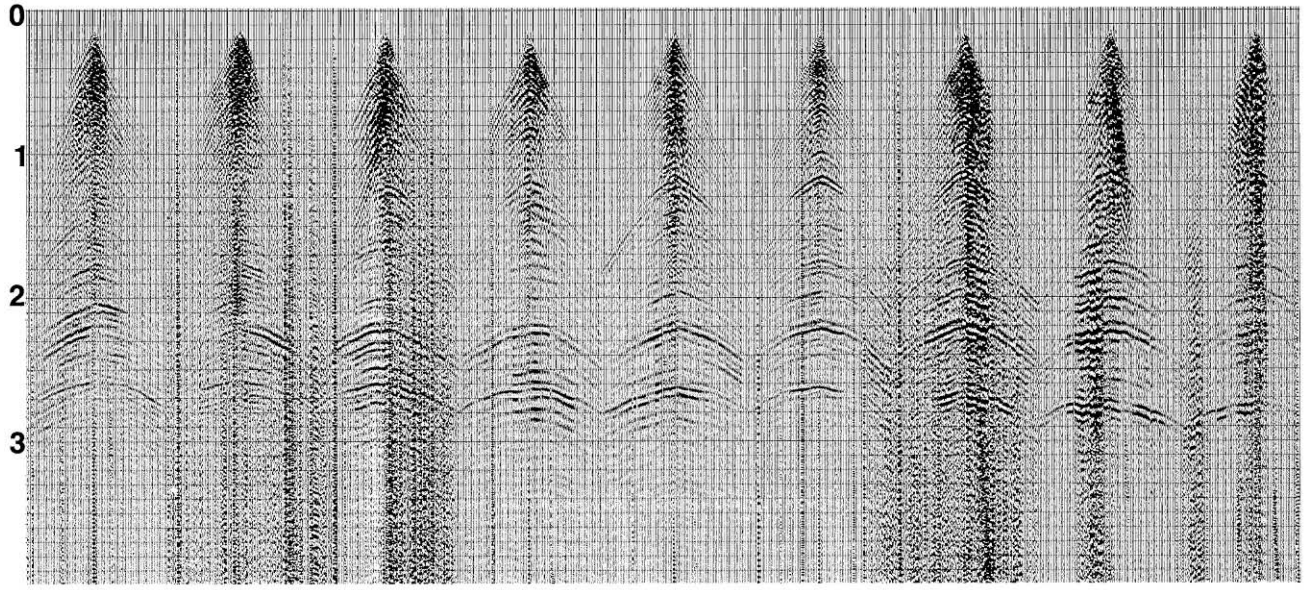

FIG. 1.4-5. The same field records as in Figure 1.4-4 after correcting for geometric spreading. The amplitudes have been restored at late times. Unfortunately, ambient noise also has been strengthened.

is used to achieve this goal. Alternative methods to compensate for frequency attenuation are time-variant spectral whitening and inverse-Q filtering. All three processes are described in Chapter 2.

The factor $1/r$ that describes the decay of wave amplitudes as a function of the radius of the spherical wavefront is valid for a homogeneous medium without attenuation. For a layered earth, amplitude decay can be described approximately by $1/\left[v^2(t)\,t\right]$ (Newman, 1973). Here, t is the two-way traveltime and $v(t)$ is the root-mean-squared (rms) velocity (Section 3.1) of the primary reflections (those reflected only once) averaged over a survey area. Therefore, the gain function for geometric spreading compensation is defined by

$$g(t) = \frac{v^2(t)\,t}{v_0^2 t_0}, \qquad (1-8a)$$

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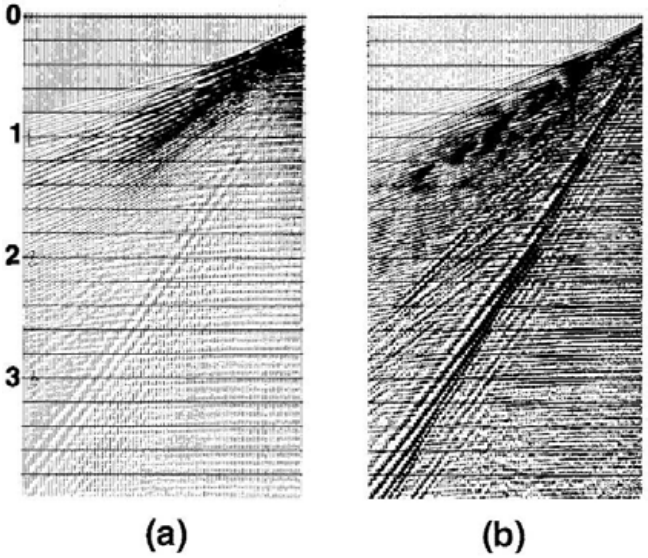

FIG. 1.4-6. (a) A raw field record from a marine survey. Before correcting for geometric spreading, refraction and guided wave energy dominate the record. (b) After the geometric spreading correction, while reflection amplitudes have been restored, multiples and coherent noise also have been boosted.

where v_0 is the reference velocity at specified time t_0. A more rigorous offset-dependent and time-dependent description of the geometric spreading correction function also can be used.

Signal-level decay is evident in the field records in Figure 1.4-4. Note the weak appearance of reflections, particularly below 1 s. This does not mean that there are no strong reflections below this time. Because of the amplitude decay resulting from wavefront divergence, no signal is seen at late times. As stated previously, this earth effect must be removed to bring up any signal that may be present in the deep portion of the record.

The same shot records after geometric spreading correction are shown in Figure 1.4-5. While reflections have been brought up in strength, noise components in the data also have been boosted. This is one undesirable aspect of any type of gain application.

Besides ambient noise, coherent noise in the data may be boosted as shown in Figure 1.4-6. By using the primary velocity function in correcting for geometric spreading, the amplitudes of the dispersive coherent noise and multiples have been overcorrected. Another example of overcorrected multiples is shown in Figure 1.4-2. (Compare the far left panel with its equivalent in Figure 1.4-1.)

To prevent overcorrection of amplitudes of multiple reflections, a velocity-independent scaling function, such as

$$g(t) = t^{\alpha}, \qquad (1-8b)$$

where α usually is set to 2, can be used for geometric spreading correction (Claerbout, 1985). Figure 1.4-7 shows a marine record after muting the guided waves and applying $t-$squared scaling. Note from the amplitude spectrum averaged over the shot record that geometric spreading correction does not restore frequency components of the wavefield which are subject to absorption effects in rocks. Nevertheless, by correcting for amplitude decay caused by wavefront divergence, the autocorrelogram better describes the reverberations across all offsets. Actually, $t-$squared scaling now is a commonly used scaling function for geometric spreading correction.

Programmed Gain Control

Various types of gain criteria are used in practice. Based on a desired criterion, a gain function $g(t)$ is derived from the data and multiplied with trace amplitudes at each time sample. This is illustrated in Figure 1.4-8. The gain function is specified or estimated at the time samples indicated by the dots and interpolated between these samples.

Programmed gain control (PGC) is the simplest type of gain. Referring to part of a stacked section in Figure 1.4-9, a gain function can be defined by interpolating between some scalar values specified at particular time samples. Larger scalar values naturally would be assigned at late times. In Figure 1.4-9, the applied PGC factors are indicated by the pairs of numbers corresponding to 0 and 6 s. While the input panel indicates strong amplitudes at shallow times, the section scaled by the PGC factors (1,4) indicates a balanced amplitude level from top to bottom.

Rather than picking the scalars in a qualitative manner, the envelope of the ungained trace can be computed and smoothed. The envelope, which is the curve drawn by smoothly connecting the adjacent peaks (or troughs) along the trace, is a reliable attribute that describes amplitude decay rate. The PGC function then is the inverse of the trace envelope. A single PGC function is applied to all traces in a gather or stacked section to preserve the relative amplitude variations in the lateral direction.

RMS Amplitude AGC

The rms amplitude AGC gain function is based on the rms amplitude within a specified time gate on an input trace. This gain function is computed as follows. The input trace is subdivided into fixed time gates. First, the amplitude of each sample in a gate is squared. Second,

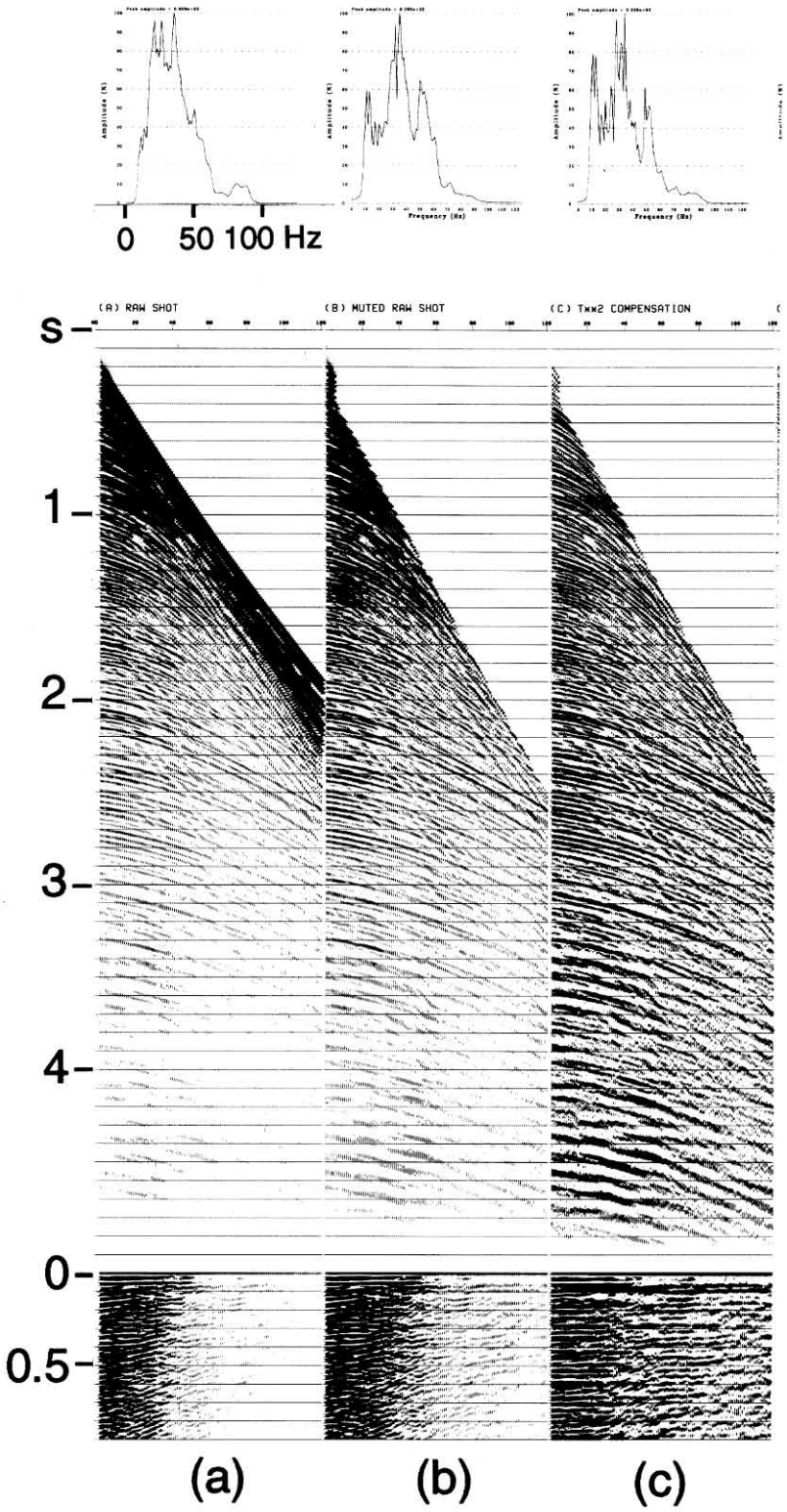

FIG. 1.4-7. (a) A field record from a marine survey, (b) after muting guided waves, and (c) after the geometric spreading correction using t^2 as the scaling function. Shown on top are the amplitude spectra averaged over the shot record, and at the bottom are the autocrrelograms.

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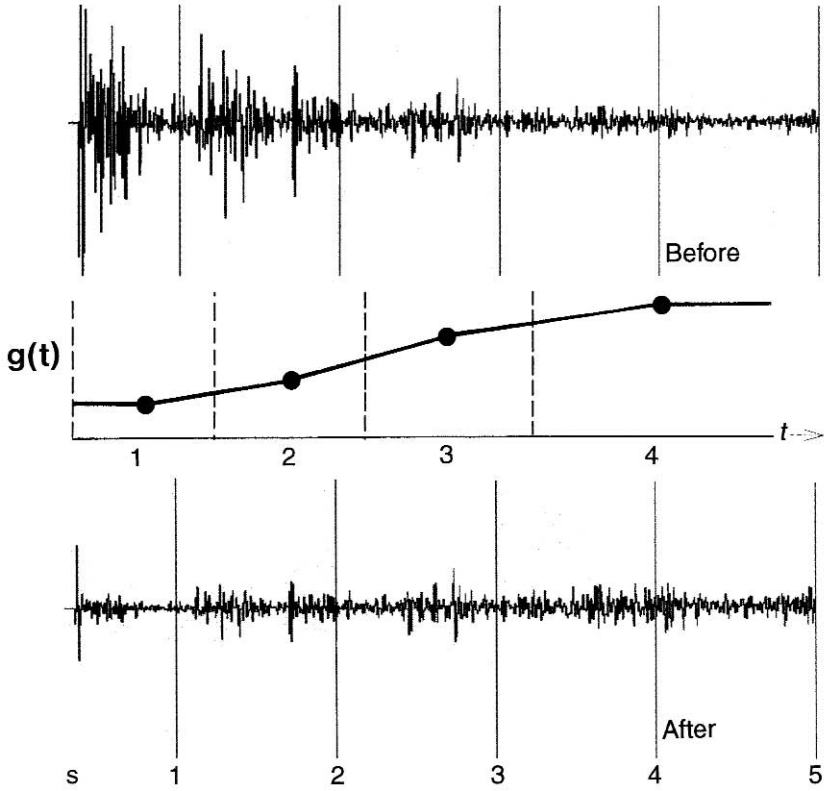

FIG. 1.4-8. Gain is a time-variant scaling defined by a function, $g(t)$. Based on some criteria, this function is defined at the time samples (shown by solid circles) that are usually at the center of specified time gates along the trace as indicated by 1, 2, 3, and 4. Gain application simply involves multiplying $g(t)$ by the input trace amplitudes.

the mean of these values is computed and its square root is taken. This is the rms amplitude over that gate. The ratio of a desired rms amplitude (say 2000) to the actual rms value is assigned as the value of the gain function at the center of the gate. Hence, the scaling function $g(t)$ at the gate center is given by

$$g(t) = \frac{\text{desired rms}}{\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}}, \qquad (1-9a)$$

where x_i is the trace amplitude and N is the number of samples within the gate.

Typically, we start out with a certain gate length at the shallow part of the trace. Gate length can be kept either constant or it can be increased systematically down the trace. At each gate center, the value of the gain function is computed as described above. Function $g(t)$ then is interpolated between the gate centers. Note that the specified time gates are stationary — they do not slide down the trace.

Figure 1.4-10 shows the ungained data and two rms-gained sections. The gate lengths are indicated at the top of each panel. When the gate used in the computation is kept small, say 64 ms, then strong reflections become less distinct.

Instantaneous AGC

Instantaneous AGC is one of the most common gain types used. This gain function is computed as follows. First, the mean absolute value of trace amplitudes is computed within a specified time gate. Second, the ratio of the desired rms level to this mean value is assigned as the value of the gain function. Unlike the rms amplitude AGC, this value is assigned to any desired time sample of the gain function within the time gate, say the nth sample of the trace, rather than to the sample at the center of the gate. The next step is to move the time gate one sample down the trace and compute the value of the gain function for the $(n+1)$th time sample, and so on. No interpolation is therefore required to define this gain function. Hence, the scaling function $g(t)$ at the gate center is given by

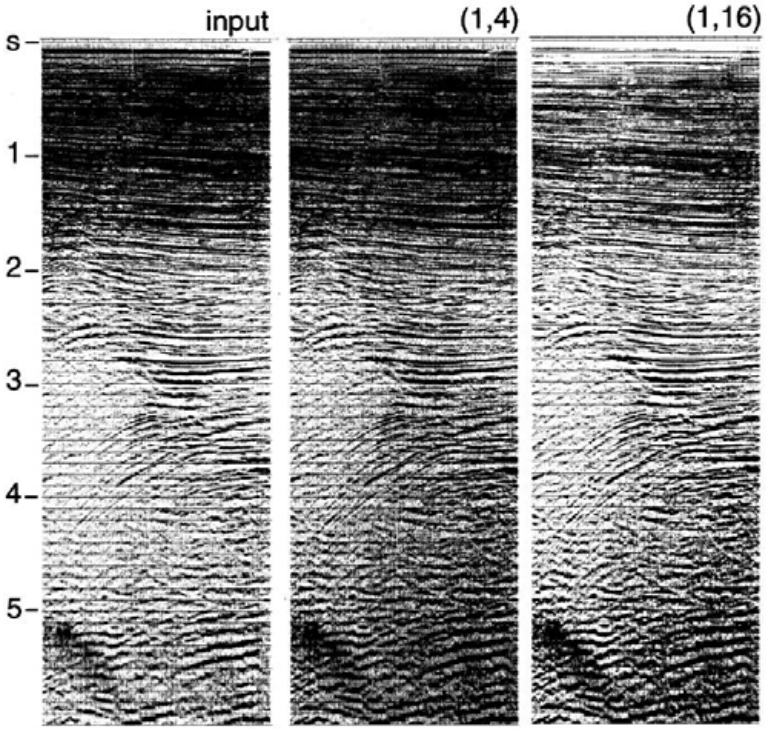

FIG. 1.4-9. A portion of a CMP stack before and after application of two different PGC functions. The scale factors used in constructing the gain functions are indicated by the pairs of numbers on top of the gained sections; the first scalar in the pair corresponds to time $t = 0$, and the second scalar corresponds to time $t = 6$ s.

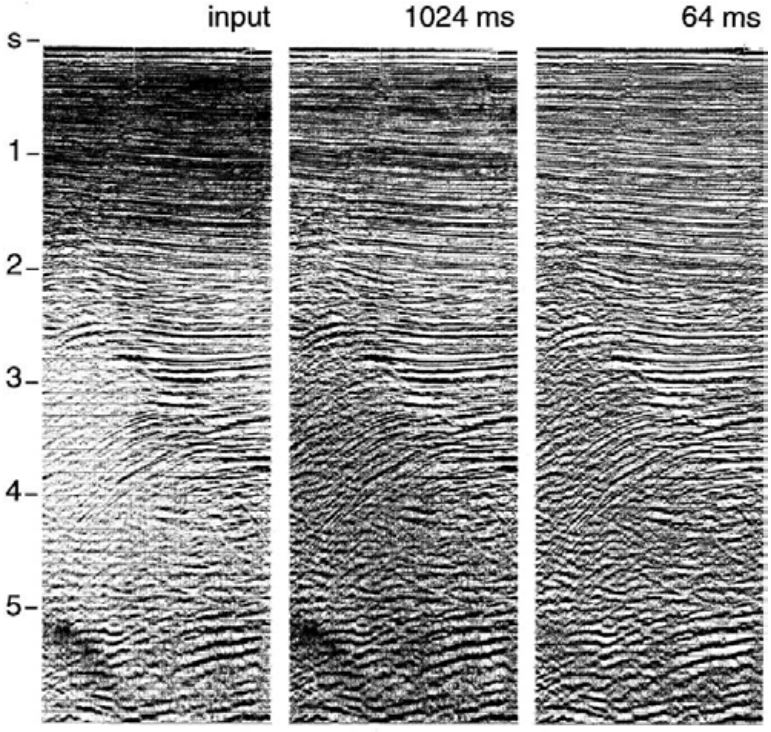

FIG. 1.4-10. A portion of a CMP stack before and after application of two different rms AGC functions. Numbers on the top indicate the window sizes in milliseconds used in computing the AGC gain function described by equation (1-9a).

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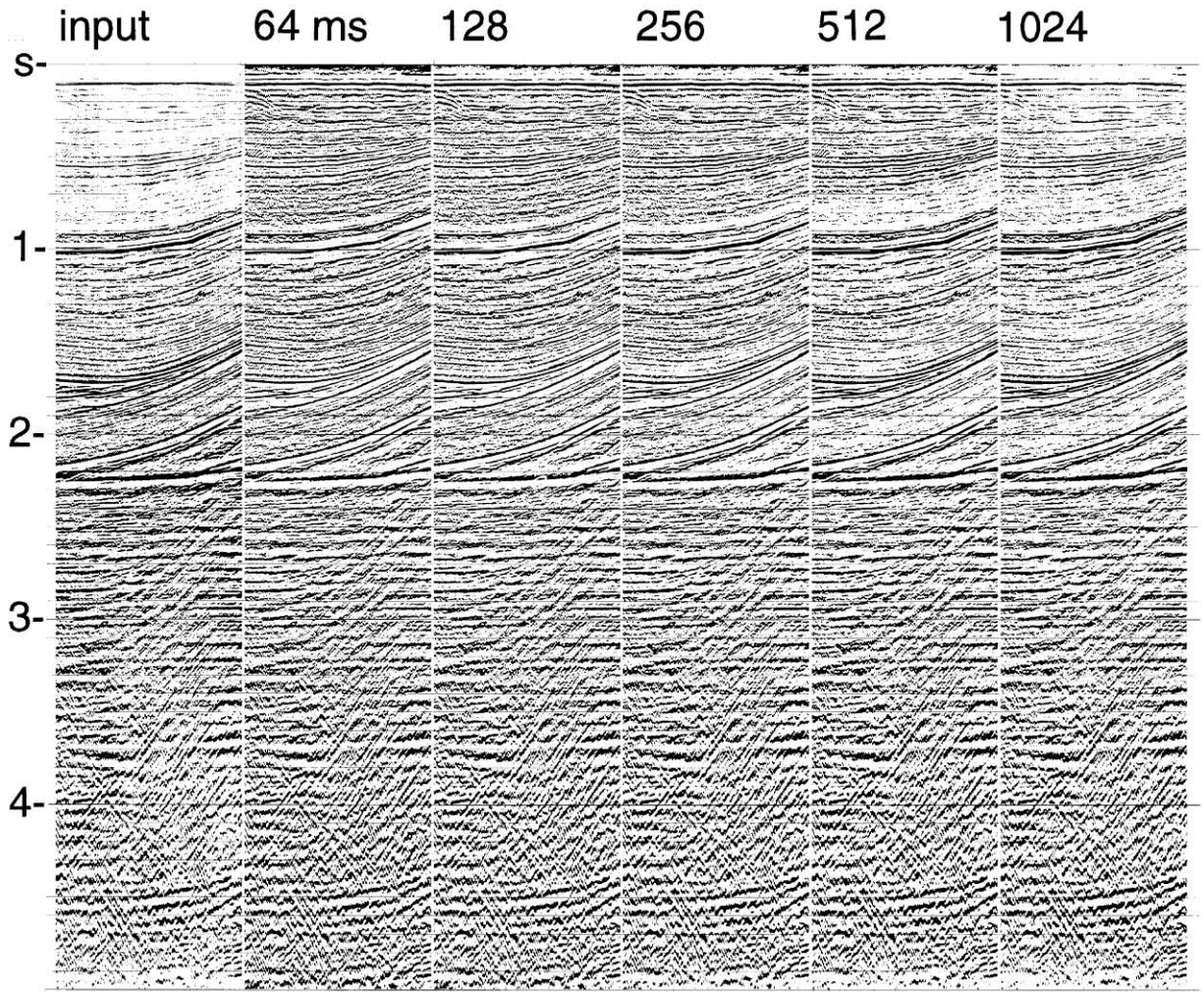

FIG. 1.4-11. A portion of a CMP stack before and after application of five different instantaneous AGC functions. The numbers on top indicate gain window sizes in milliseconds used in computing the AGC gain function described by equation (1-9b).

$$g(t) = \frac{\text{desired rms}}{\frac{1}{N}\sum_{i=1}^{N} |x_i|}, \qquad (1-9b)$$

where x_i is the trace amplitude and N is the number of samples within the gate.

Figure 1.4-11 shows the ungained data and four instantaneous AGC-gained sections. Gate lengths are indicated on top of each panel. Very small time gates cause a significant can loss of signal character by boosting zones that contain small amplitudes. This occurs with the 64-ms AGC output. In processing, this is called a fast AGC. In the other extreme, if a large time gate is selected, then the effectiveness of the AGC process is lessened. In practice, AGC time gates commonly are specified between 200 and 500 ms.

Relative Trace Balancing

All of the gain applications described in this section modify the trace amplitudes by function $g(t)$ in a time-varying manner (equations 1-4 and 1-5). In true amplitude processing, it is necessary to display the data without applying a time-varying data-dependent gain function. However, some amplitude scaling is always necessary for display, since plotters require input data amplitudes to fall in a specific range. Trace balancing (trace equalization) schemes are used for this type of scaling. The balance factor is defined as the ratio of the desired rms to the rms amplitude that is computed from a specified time window. A separate balance factor

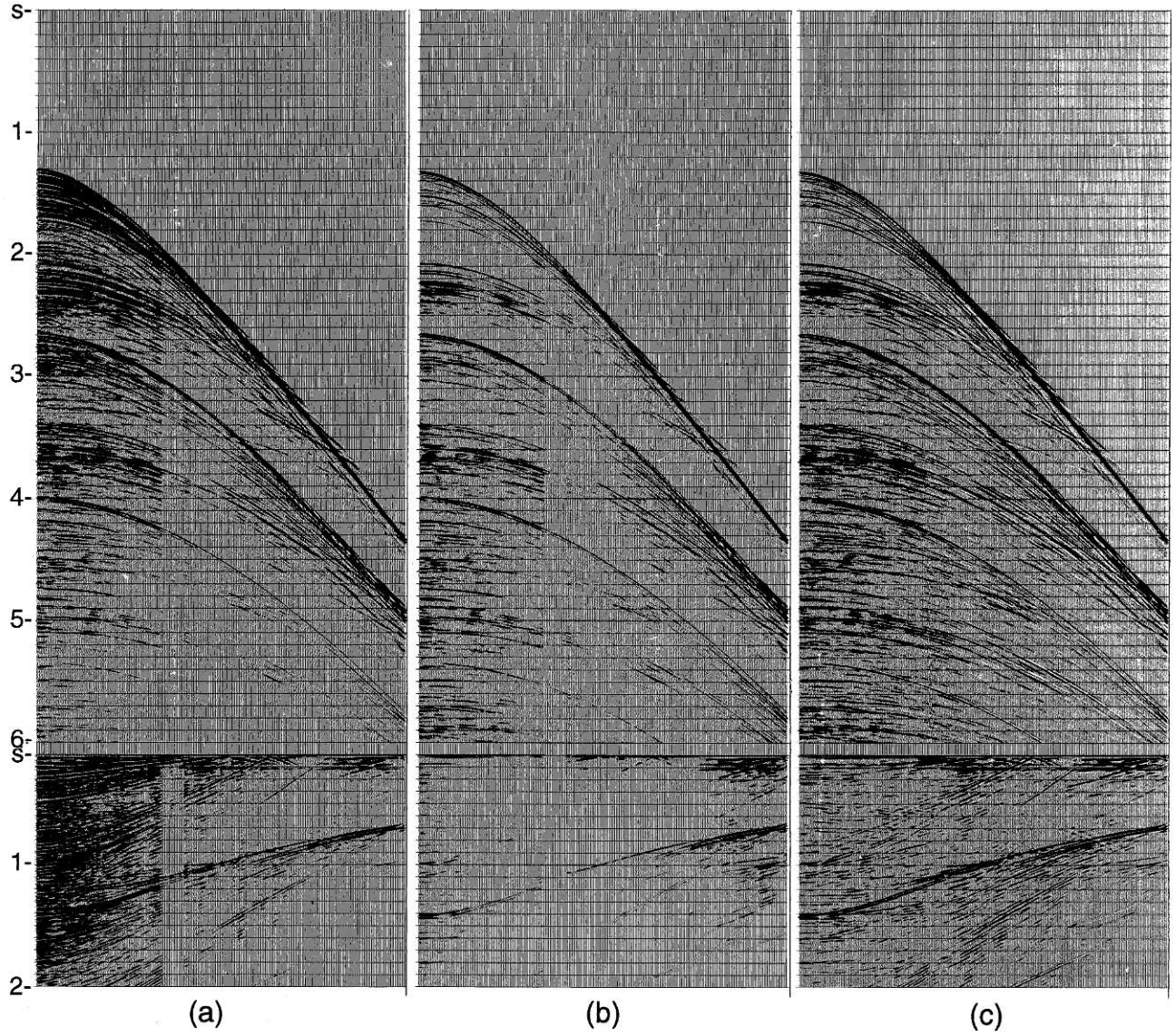

FIG. 1.4-12. (a) A field record, (b) after geometric spreading correction which shows differences in amplitude levels of the near- and far-offset channels caused by differences in gain settings during recording. (c) Following the application of trace balancing, these differences in amplitudes are removed. Displayed at the bottom are the autocorrelograms. Note in (a) the effect of different gain settings on the amplitude level of the autocorrelogram from trace to trace.

is computed for and applied to each trace, individually. Alternatively, a single balance factor based on a selected trace within a group of traces can be applied to the entire group. This is called *relative trace balancing.* Note that trace balancing amounts to scaling the trace by using a single factor that is time-invariant (equivalent to a single-window rms AGC). Figure 1.4-12 shows rms trace balancing of field data to correct for the differences in gain settings between the recording channels. Trace balancing commonly is applied immediately after deconvolution, and on final stacks using large gates.

1.5 BASIC DATA PROCESSING SEQUENCE

Since the introduction of digital recording, a routine sequence in seismic data processing has evolved. This basic sequence now is described to gain an overall understanding of each step. There are three primary steps in processing seismic data — deconvolution, stacking, and migration, in their usual order of application. Figure 1.5-1 represents the seismic data volume in processing coordinates — midpoint, offset, and time. Deconvolu-

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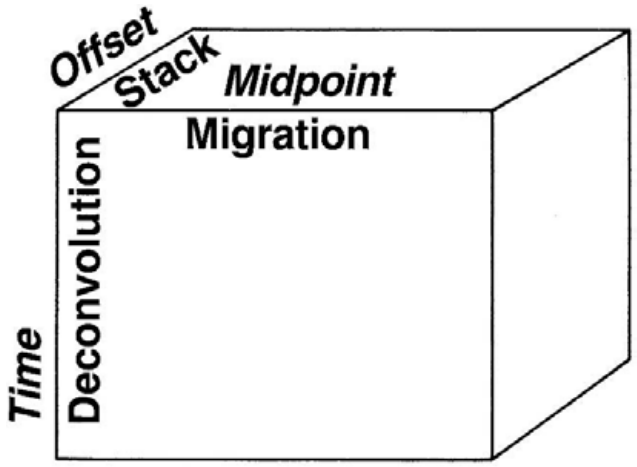

FIG. 1.5-1. Seismic data volume represented in processing coordinates — midpoint-offset-time. Deconvolution acts on the data along the time axis and increases temporal resolution. Stacking compresses the data volume in the offset direction and yields the plane of stacked section (the frontal face of the prism). Migration then moves dipping events to their true subsurface positions and collapses diffractions, and thus increases lateral resolution.

tion acts along the time axis. It removes the basic seismic wavelet (the source time function modified by various effects of the earth and recording system) from the recorded seismic trace and thereby increases temporal resolution. Deconvolution achieves this goal by compressing the wavelet (Chapter 2). Stacking also is a process of compression (Chapter 3). In particular, the data volume in Figure 1.5-1 is reduced to a plane of midpoint-time at zero offset (the frontal face of the prism) first by applying normal moveout correction to traces from each CMP gather (Section 3.1), then by summing them along the offset axis. The result is a *stacked section*. (The terms *stacked section*, *CMP stack*, and *stack* often are used synonymously.) Finally, migration commonly is applied to stacked data. It is a process that collapses diffractions and maps dipping events on a stacked section to their supposedly true subsurface locations (Chapter 4). In this respect, migration is a spatial deconvolution process that improves spatial resolution.

All other processing techniques may be considered secondary in that they help improve the effectiveness of the primary processes. For example, dip filtering may need to be applied before deconvolution to remove coherent noise so that the autocorrelation estimate is based on reflection energy that is free from such noise. Wide band-pass filtering also may be needed to remove very low- and high-frequency noise. Before deconvolution, correction for geometric spreading is necessary to compensate for the loss of amplitude caused by wavefront divergence. Velocity analysis, which is an essential step for stacking, is improved by multiple attenuation and residual statics corrections.

Many of the secondary processes are designed to make data compatible with the assumptions of the three primary processes. Deconvolution assumes a stationary, vertically incident, minimum-phase source wavelet and white reflectivity series that is free of noise. Stacking assumes hyperbolic moveout, while migration is based on a zero-offset (primaries only) wavefield assumption. A pessimist could claim that none of these assumptions is valid. However, when applied to field data, these techniques do provide results that are close to the true subsurface image. This is because these three processes are robust and their performance is not very sensitive to the underlying assumptions in their theoretical development.

Keep in mind that the success of a process depends not only on the proper choice of parameters pertinent to that particular process, but also on the effectiveness of the previous processing steps.

We shall use a 2-D seismic line from the Caspian Sea to demonstrate the basic processing sequence. Table 1-14 provides the processing parameters for the line. The water depth at one end of the line is approximately 750 m and decreases along the line traverse to approximately 200 m at the other end.

Preprocessing

Field data are recorded in a multiplexed mode using a certain type of format. The data first are demultiplexed as described in Figure 1.5-2. Mathematically, demulti-

Table 1-14. Processing parameters for the Caspian line used to describe the basic processing sequence in this section.

Shot interval in m	25
Group interval in m	25
Number of receiver groups	180
Minimum offset in m	175
Maximum offset in m	4,650
CMP interval in m	12.5
Fold of coverage	90
Number of CMPs	6,212
Line length in km	77.64
Sampling interval in ms	4
Maximum time in ms	8,000
Data volume in gigabytes	4.5

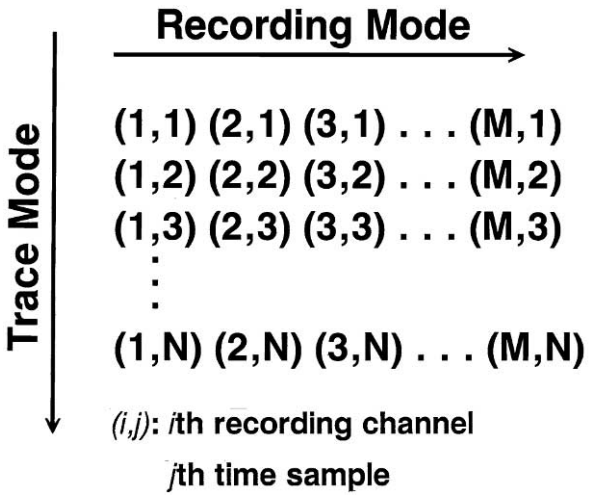

FIG. 1.5-2. Seismic data are recorded in rows of samples — samples at the same time at consecutive channels. Demultiplexing involves sorting the data into columns of samples — all the time samples in one channel followed by those in the next channels.

plexing is seen as transposing a big matrix so that the columns of the resulting matrix can be read as seismic traces recorded at different offsets with a common shot point. At this stage, the data are converted to a convenient format that is used throughout processing. This format is determined by the type of processing system and the individual company. A common format used in the seismic industry for data exchange is SEG-Y, established by the Society of Exploration Geophysicists.

Figure 1.5-3 shows selected shot records along the Caspian line under consideration. Note the strong amplitudes at the early part and the relatively weaker energy at the deeper part of the records. Such decay in amplitude primarily is caused by wavefront divergence. The dispersive nature of the guided waves resulting from normal-mode propagation within the water layer appears to vary from record to record. This results from a combination of varying water depth, depth of the source array, and water-bottom conditions (Section F.1).

Preprocessing also involves trace editing. Noisy traces, traces with transient glitches (see Figure 1.3-40), or monofrequency signals (see Figure 1.3-3) are deleted; polarity reversals (see Figure 1.3-2) are corrected. In case of very shallow marine data, guided waves are muted since they travel horizontally within the water layer and do not contain reflections from the substratum.

As seen in Figure 1.5-3, most marine data are contaminated by swell noise and cable noise. These types of noise carry very low-frequency energy but can be high in amplitudes. They can be recognized by their distinctive linear pattern and vertical streaks. The swell noise and cable noise are removed from shot records by a low-cut filtering as shown in Figure 1.5-4. Attenuation of coherent linear noise associated with side scatterers and ground roll may require techniques based on dip filtering (Chapter 6).

Following the trace editing and prefiltering, a gain recovery function is applied to the data to correct for the amplitude effects of spherical wavefront divergence. This amounts to applying a geometric spreading function, which depends on traveltime (Section 1.4). Optionally, this amplitude correction is made dependent on a spatially averaged velocity function, which is associated with primary reflections in a particular survey area. Additionally, an exponential gain function may be used to compensate for attenuation losses.

The data in Figure 1.5-5 have been corrected for geometric spreading using a t^2 scaling function. While primary reflection amplitudes are corrected for wavefront divergence, energy associated with multiple reflections, coherent linear noise generated by water-bottom point scatterers and the recording cable, and random noise also is inevitably boosted by geometric spreading correction.

Finally, field geometry is merged with the seismic data. This precedes any gain correction that is offset-dependent. Based on survey information for land data or navigation information for marine data, coordinates of shot and receiver locations for all traces are stored on trace headers. Changes in shot and receiver locations are handled properly based on the information available in the observer's log. Many types of processing problems arise from setting up the field geometry, incorrectly. As a result, the quality of a stacked section can be degraded severely.

For land data, elevation statics are applied at this stage to reduce traveltimes to a common datum level. This level may be flat or vary (floating datum) along the line. Reduction of traveltimes to a datum usually requires correction for the near-surface weathering layer in addition to differences in elevation of source and receiver stations. Estimation and correction for the near-surface effects usually are performed using refracted arrivals associated with the base of the weathering layer (Section 3.4).

Deconvolution

Typically, prestack deconvolution is aimed at improving temporal resolution by compressing the effective source wavelet contained in the seismic trace to a spike (spiking deconvolution). Predictive deconvolution (Sections 2.3 and 2.4) with a prediction lag (commonly termed

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gap) that is equal to the first or second zero crossing of the autocorrelation function also is used commonly. Although deconvolution usually is applied to prestack data trace by trace, it is not uncommon to design a single deconvolution operator and apply it to all the traces on a shot record. Deconvolution techniques used in conventional processing are based on optimum Wiener filtering (Section 2.3).

Figure 1.5-6 shows the common-shot gathers after spiking deconvolution. By examining some of the individual reflections and comparing them with those in Figure 1.5-5, note how the wavelet associated with the significant reflections is compressed and reverberatory energy that trails behind each reflection is largely attenuated by deconvolution. Because both low- and high-frequency noise and signal are boosted, the data often need filtering with a wide band-pass filter after deconvolution. In addition, some kind of trace balancing (Section 1.4) often is applied after deconvolution to bring the data to a common root-mean-squared (rms) level (Figure 1.5-7).

CMP Sorting

Seismic data acquisition with multifold coverage is done in shot-receiver (s, g) coordinates. Figure 1.5-8a is a schematic depiction of the recording geometry and ray paths associated with a flat reflector. Seismic data processing, on the other hand, conventionally is done in midpoint-offset (y, h) coordinates. The required coordinate transformation is achieved by sorting the data into CMP gathers. Based on the field geometry information, each individual trace is assigned to the midpoint between the shot and receiver locations associated with that trace. Those traces with the same midpoint location are grouped together, making up a CMP gather. Albeit incorrectly, the term *common depth point* (CDP) and *common midpoint* (CMP) often are used interchangeably.

Figure 1.5-8b depicts the geometry of a CMP gather and raypaths associated with a flat reflector. Note that CDP gather is equivalent to a CMP gather only when reflectors are horizontal and velocities do not vary horizontally. However, when there are dipping reflectors in the subsurface, these two gathers are not equivalent and only the term CMP gather should be used. Selected CMP gathers obtained from sorting the deconvolved shot gathers (Figure 1.5-7) are shown in Figure 1.5-9.

Figure 1.5-10 shows the superposition of shot-receiver (s, g) and midpoint-offset (y, h) coordinates, and raypath geometries for various gather types. The (y, h) coordinates have been rotated 45 degrees relative to the (s, g) coordinates. The dotted area represents the coverage used in recording the seismic profile along the midpoint axis, Oy. Each dot represents a seismic trace with the time axis perpendicular to the plane of paper. The following gather types are identified in Figure 1.5-10:

(1) Common-shot gather (shot record, field record),
(2) Common-receiver gather,
(3) Common-midpoint gather (CMP gather, CDP gather),
(4) Common-offset section (constant-offset section),
(5) CMP-stacked section (zero-offset section).

The recording cable length is FG and the line length is AD. The number of dots along the offset axis (cross-section 3) is equal to the *CMP fold.* The fold tapers off at the ends of the profile (segments AB and CD). Full-fold coverage along the line is at midpoints over segment BC. The diagram in Figure 1.5-10 is known as a *stacking chart* and is useful when setting up the geometry of a line for preprocessing. If there is a missing shot or a bad receiver, the affected midpoints are identified easily (Exercise 1-15).

For most recording geometries, the fold of coverage n_f for CMP stacking is given by

$$n_f = \frac{n_g \Delta g}{2 \Delta s}, \qquad (1-10)$$

where Δg and Δs are the receiver-group and shot intervals, respectively, and n_g is the number of recording channels. By using this relationship, the following rules can be established:

(a) The fold does not change when alternating traces in each shot record are dropped.
(b) The fold is halved when every other shot record is skipped, whether or not alternating traces in each record are dropped.

Velocity Analysis

In addition to providing an improved signal-to-noise ratio, multifold coverage with nonzero-offset recording yields velocity information about the subsurface (Chapter 3). Velocity analysis is performed on selected CMP gathers or groups of gathers. The output from one type of velocity analysis is a table of numbers as a function of velocity versus two-way zero-offset time (velocity spectrum). These numbers represent some measure of signal coherency along the hyperbolic trajectories governed by velocity, offset, and traveltime.

Figure 1.5-11 shows the velocity spectra derived from the CMP gathers as in Figure 1.5-9. The horizontal

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axis in each spectrum represents the scanned normal-moveout velocity with a range of 1000 to 5000 m/s, and the vertical axis represents the two-way zero-offset time from 0 to 8 s. Red indicates the maximum coherency measure. The curve in each spectrum represents the velocity function based on the picked maximum coherency values associated with primary reflections. The pairs of numbers along each curve denote the time-velocity values for each pick. Velocity-time pairs are picked from these spectra based on maximum coherency peaks to form velocity functions at analysis locations.

The velocity functions picked at analysis locations then are spatially interpolated between the analysis locations to create a velocity field as shown in Figure 1.5-12. Red in the shallow portion and blue in the deep portion of the section correspond to low and high velocities, respectively. This velocity field is used to supply a velocity function for each CMP gather along the profile.

In areas with complex structure, velocity spectra often fail to provide sufficient accuracy in velocity picks. When this is the case, the data are stacked with a range of constant velocities, and the constant-velocity stacks themselves are used in picking velocities.

Normal-Moveout Correction

The velocity field (Figure 1.5-12) is used in normal moveout (NMO) correction of CMP gathers (Section 3.1). Based on the assumption that, in a CMP gather, reflection traveltimes as a function of offset follow hyperbolic trajectories, the process of NMO correction removes the moveout effect on traveltimes. Figure 1.5-13 shows the CMP gathers in Figure 1.5-9 after moveout correction. Note that events are mostly flattened across the offset range — the offset effect has been removed from traveltimes. Traces in each CMP gather are then summed to form a stacked trace at each midpoint location. The stacked section comprises the stacked traces at all midpoint locations along the line traverse.

As a result of moveout correction, traces are stretched in a time-varying manner, causing their frequency content to shift toward the low end of the spectrum. Frequency distortion increases at shallow times and large offsets (Figure 1.5-13). To prevent the degradation of especially shallow events, the amplitudes in the distorted zone are zeroed out (muted) before stacking (Figure 1.5-14).

The CMP recording technique, which was invented in the 1950s and published later (Mayne, 1962), uses redundant recording to improve the signal-to-noise ratio during stacking. To achieve redundancy, multiple sources per trace n_s, multiple receivers per trace n_r, and multiple offset coverage of the same subsurface point n_f, are used in the field. Given the total number of elements in the recording system, $N = n_s \times n_r \times n_f$, the signal amplitude-to-rms noise ratio theoretically is improved by a factor of $\sqrt{N}$. This improvement factor is based on the assumptions that the reflection signal on traces of a CMP gather is identical and the random noise is mutually uncorrelated from trace to trace (Sengbush, 1983). Because these assumptions do not strictly hold in practice, the signal-to-noise ratio improvement gained by stacking is somewhat less than $\sqrt{N}$. Common-midpoint stacking also attenuates coherent noise such as multiples, guided waves, and ground roll. This is because reflected signal and coherent noise usually have different stacking velocities.

In areas with complex overburden structure that gives rise to strong lateral velocity variations, the hyperbolic moveout assumption associated with reflection traveltimes in CMP gathers is no longer valid. As a result, hyperbolic moveout correction and CMP stacking do not always yield a stacked section in which reflections from the underlying strata are faithfully preserved. In such circumstances, imaging in depth and before stack becomes imperative.

Multiple Attenuation

Multiple reflections and reverberations are attenuated using techniques based on their periodicity or differences in moveout velocity between multiples and primaries. These techniques are applied to data in various domains, including the CMP domain, to best exploit the periodicity and velocity discrimination criteria (Chapter 6).

Deconvolution is one method of multiple attenuation that exploits the periodicity criterion. Often, however, the power of conventional deconvolution in attenuating multiples is underestimated. As for the Caspian data example in this section, despite theoretical limitations, deconvolution can remove a significant portion of the energy associated with short-period multiples and reverberations. It also can attenuate long-period multiples if it is applied in data domains in which periodicity is preserved (Chapter 6).

Dip-Moveout Correction

The normal-moveout correction in Figure 1.5-14 was applied to the CMP gathers using the velocity field of Figure 1.5-12 that is optimum for flat events. Stacking velocities, however, are dip-dependent. Dip-moveout correction (DMO) is needed to correct for the dip effect on stacking velocities and thus preserve events with

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conflicting dips during CMP stacking (Chapter 5). Dip-moveout correction has been an integral part of a conventional processing sequence for 2-D and 3-D seismic data since 1985.

Dip-moveout correction is applied to data following the normal-moveout correction using flat-event velocities (Figure 1.5-15). This then is followed by inverse moveout correction (Figure 1.5-16) and subsequent velocity analysis at closely spaced intervals. Figure 1.5-17 shows the velocity spectra asociated with a subset of the analysis locations which correspond to those of Figure 1.5-11. As for the velocity spectra in Figure 1.5-11, the velocity range is 1000-5000 m/s and the maximum time is 8 s. Also, red indicates the maximum coherency measure.

CMP Stacking

A new velocity field as shown in Figure 1.5-18 is derived from the velocity functions picked from the velocity spectra after DMO correction. As for the velocity field in Figure 1.5-12, red in the shallow portion and blue in the deep portion of the section correspond to low and high velocities, respectively. This new velocity field is used to apply NMO correction to the CMP gathers (Figure 1.5-19). Finally, a CMP stack is obtained (Figure 1.5-20) by summing over the offset axis. The stack is the frontal face of the data volume shown in Figure 1.5-1.

Poststack Processing

A typical poststack processing sequence includes the following steps:

(a) Deconvolution after stack (Section 2.5) is usually applied to restore high frequencies attenuated by CMP stacking. It also is often effective in suppressing reverberations and short-period multiples. Figure 1.5-21 shows the CMP stack as in Figure 1.5-20 after spiking deconvolution.
(b) Although not included in the processing sequence for the Caspian data example in this section, often, time-variant spectral whitening (Section 2.6) is used to further flatten the spectrum and accounts for the time-variant character of the source waveform.
(c) Time-variant band-pass filtering (Section 1.1) is then used to remove noise at the high- and low-frequency end of the signal spectrum (Figure 1.5-22).
(d) The basic processing sequence sometimes includes a step for attenuation of random noise uncorrelated from trace to trace (Chapter 6).
(e) Finally, some type of display gain (Section 1.4) is applied to the stacked data (Figure 1.5-23). For true amplitude preservation, time-variant scaling of stacked amplitudes is avoided; instead, a relative amplitude compensation function that is constant from trace to trace is applied (Section 1.4). This is a slow time-varying gain function that amplifies weak late reflections without destroying the amplitude relationships from trace to trace that may be caused by subsurface reflectivity.

Migration

Dipping events are then moved to their supposedly true subsurface positions, and diffractions are collapsed by migrating the stacked section prior to amplitude scaling (Chapter 4). Figure 1.5-24 shows the CMP stack as in Figure 1.5-22 after migration. As for the unmigrated stack, the migrated section also is displayed with the scaled amplitudes (Figure 1.5-25). Although the output of migration is intended to represent the geological cross-section along the line traverse, it often is displayed in time as for the input stacked section. The provided lateral velocity variations are mild to moderate, *time migration* often is acceptable; otherwise, *depth migration* is imperative (Chapter 4).

The structural highs below midpoints 4200, 6800, and 8200 in Figure 1.5-25 are associated with mud diapirism which is prominent in the Caspian basin. Structural complexity caused by faulting and folding generally introduces problems in stacking and imaging the subsurface in three respects:

(a) Steeply dipping reflections associated with fault planes and salt flanks often conflict during stacking with gently dipping or near-flat reflections associated with the less undisturbed strata. The remedy for this problem is prestack time migration for which the robust alternative is dip-moveout correction combined with poststack time migration.
(b) Nonhyperbolic moveout caused by strong lateral velocity variations associated with complex overburden structures involving salt tectonics and overthurst tectonics yields traveltime and amplitude distortions during stacking based on the hyperbolic moveout assumption. The remedy for this problem is prestack depth migration.

(text continues on p. 122)

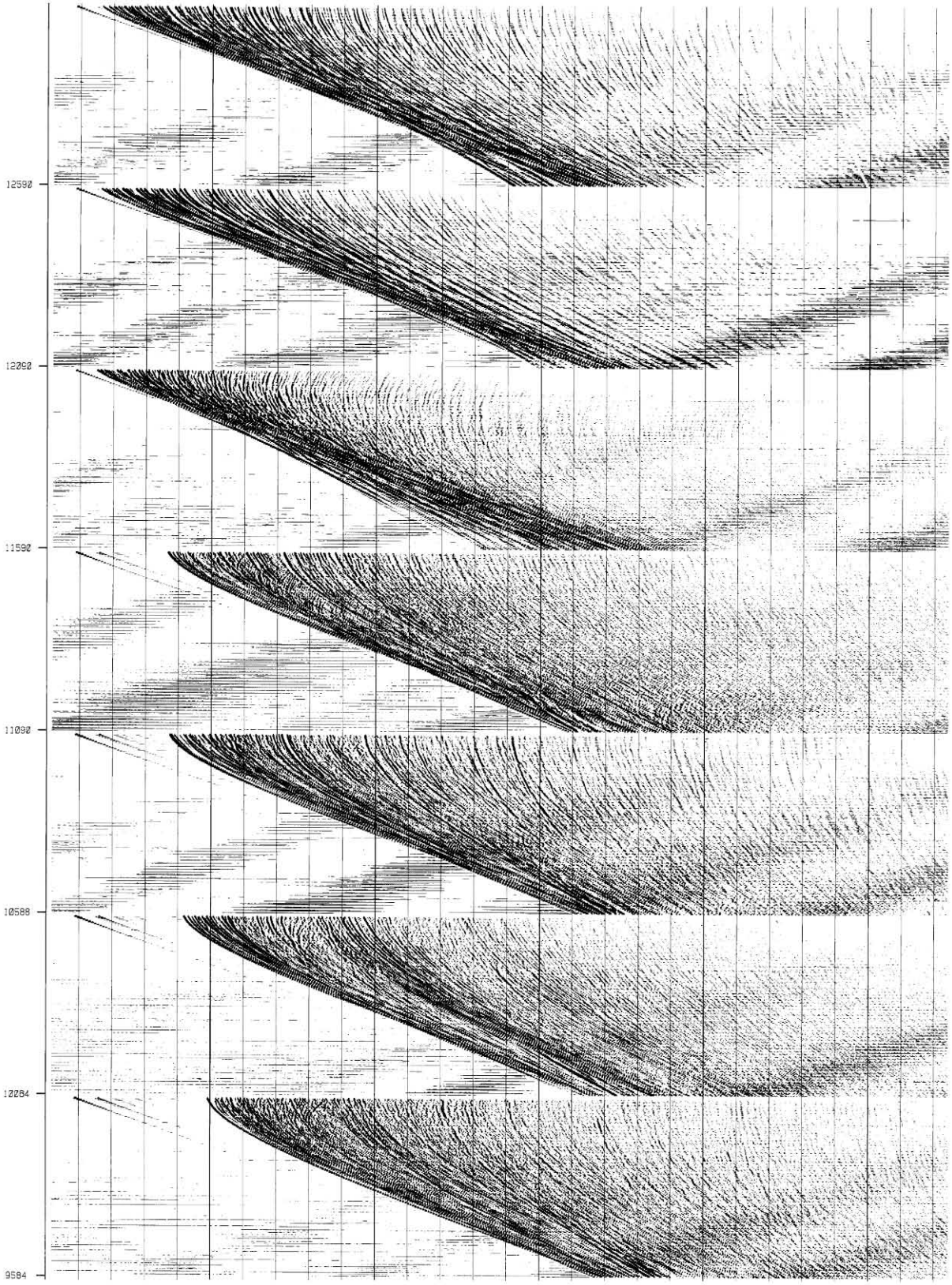

FIG. 1.5-3. Selected common-shot gathers from an offshore survey just after demultiplexing. (Data courtesy Davud Babayev, Kaspmorneftegeofizika.)

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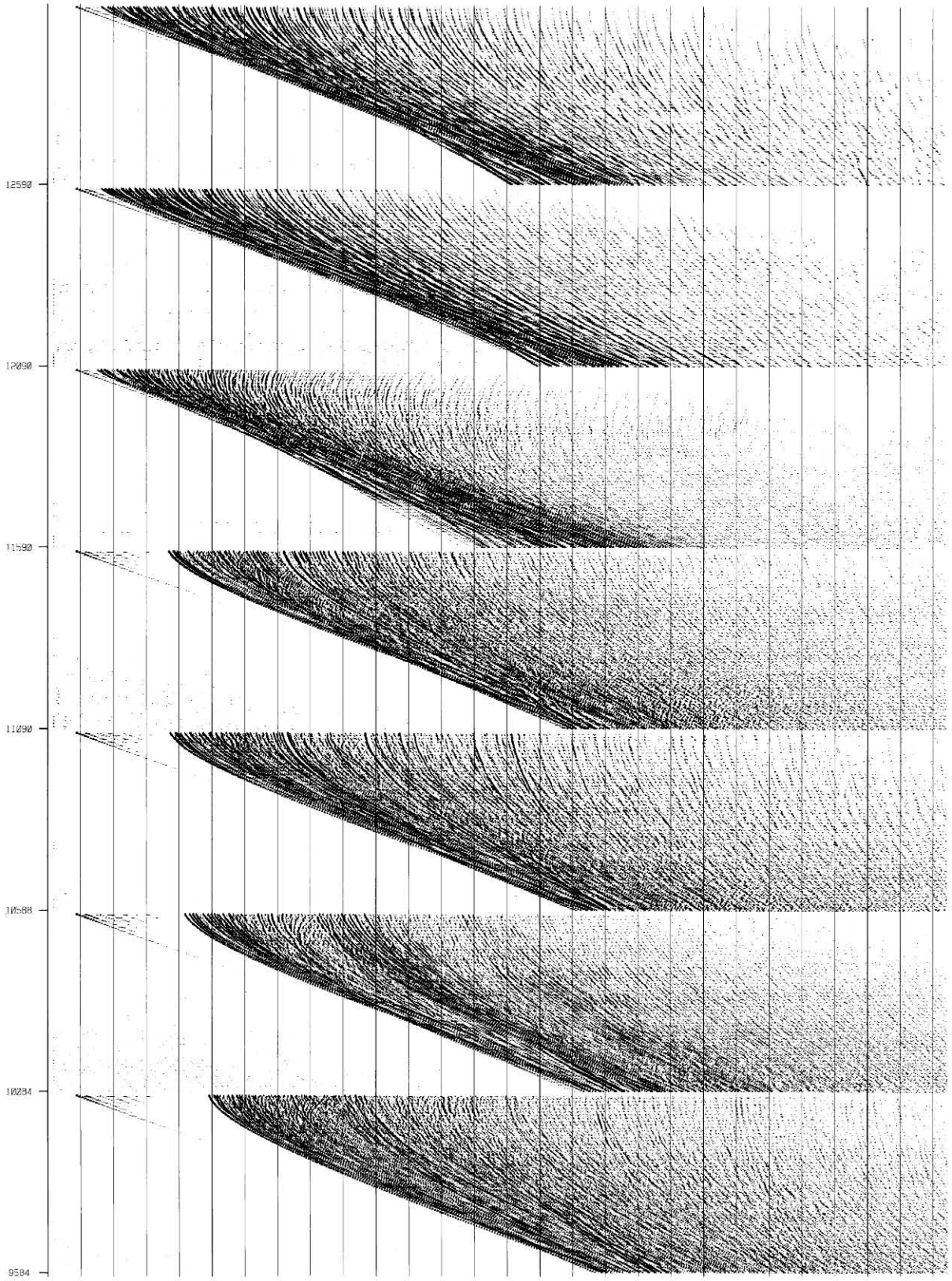

FIG. 1.5-4. The common-shot gathers as in Figure 1.5-3 after removal of the swell and cable noise by low-cut filtering.

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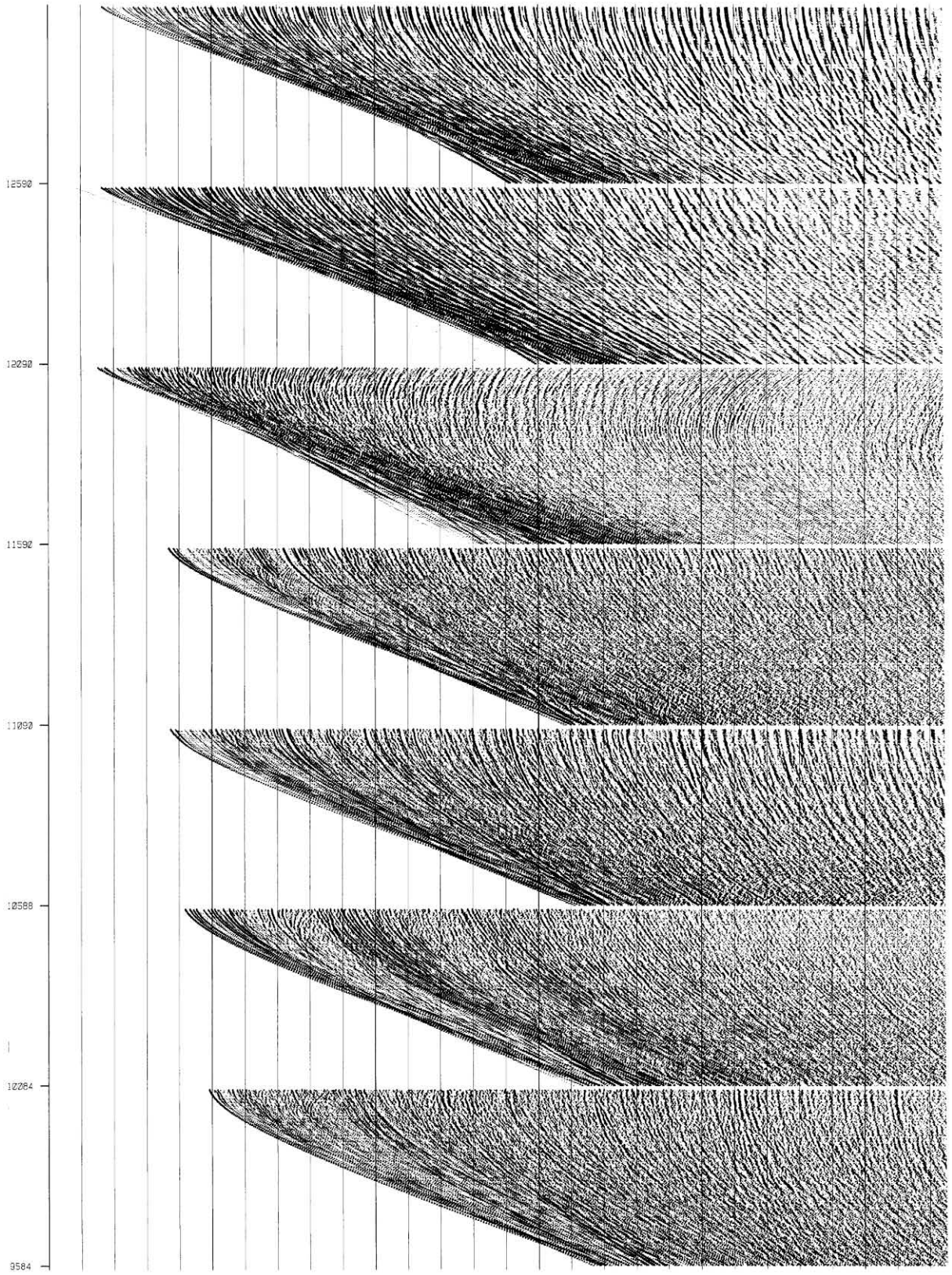

FIG. 1.5-5. The common-shot gathers as in Figure 1.5-4 after correcting for the amplitude effects of wavefront divergence. Note the restored amplitudes at late times on the records.

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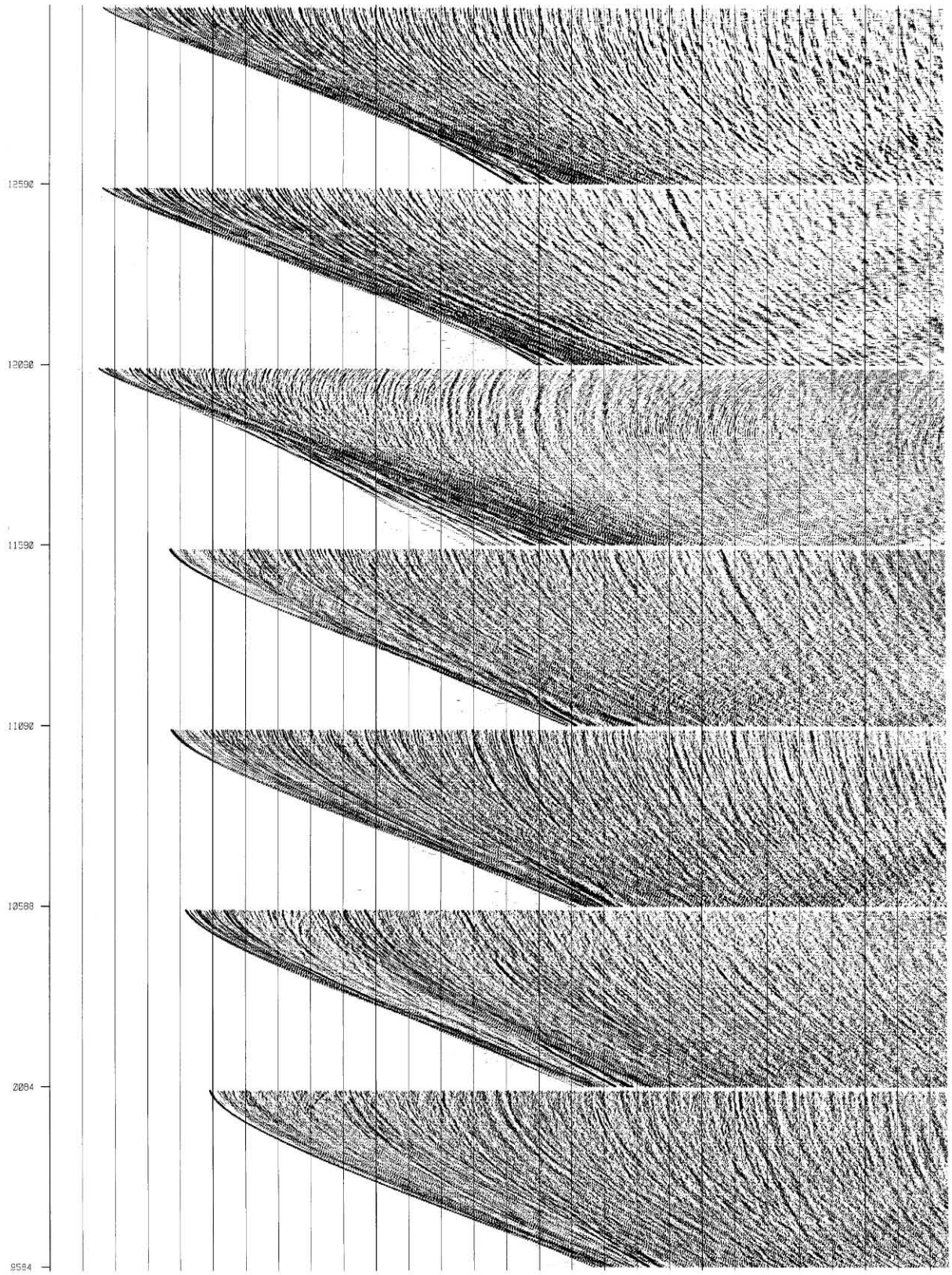

FIG. 1.5-6 The common-shot gathers as in Figure 1.5-5 after spiking deconvolution.

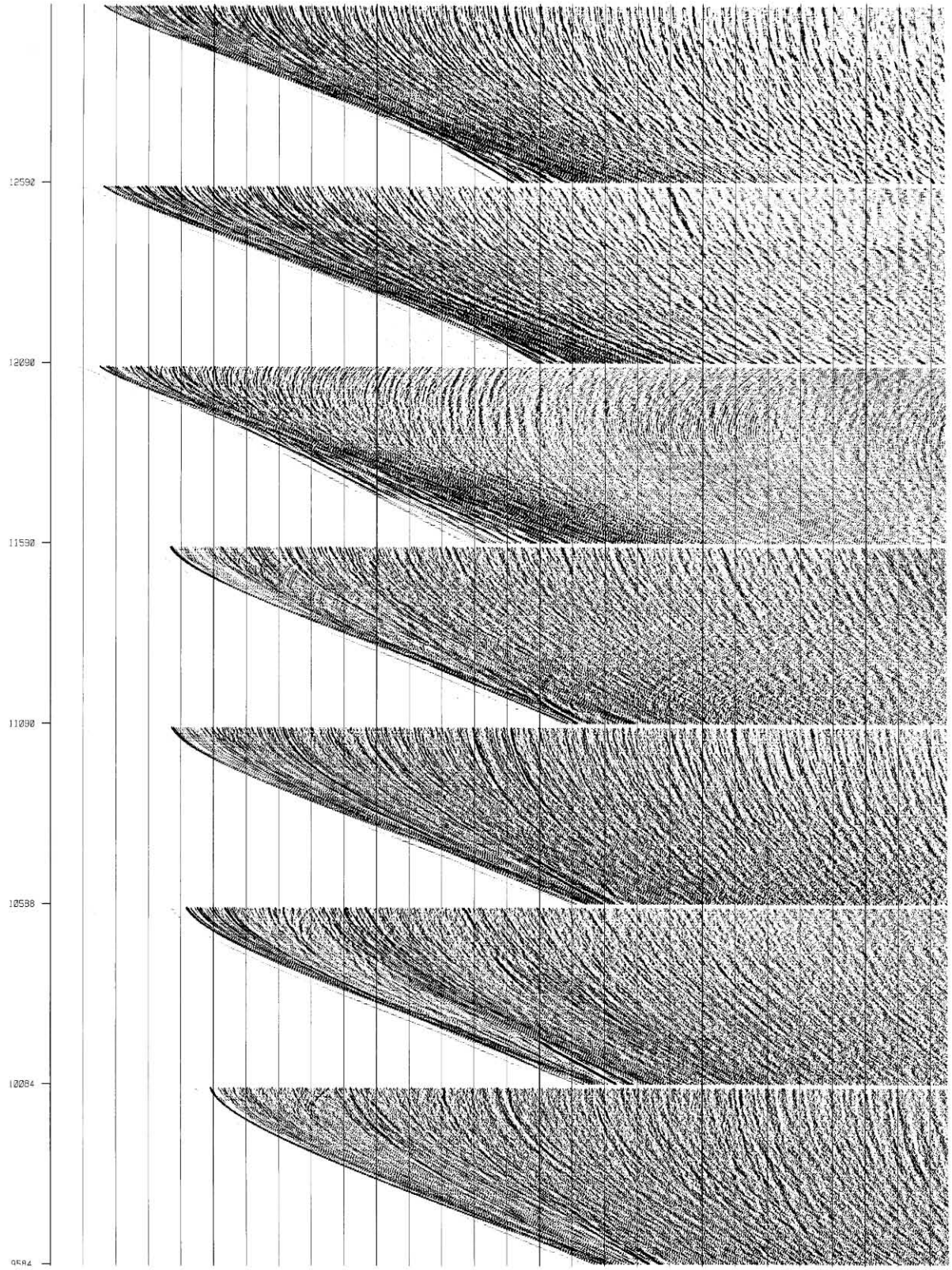

FIG. 1.5-7. The common-shot gathers as in Figure 1.5-6 after a wide band-pass filter to remove very low- and very high-frequency noise, followed by trace balancing. Balancing is a time-invariant scaling of amplitudes to a common rms level for all traces.

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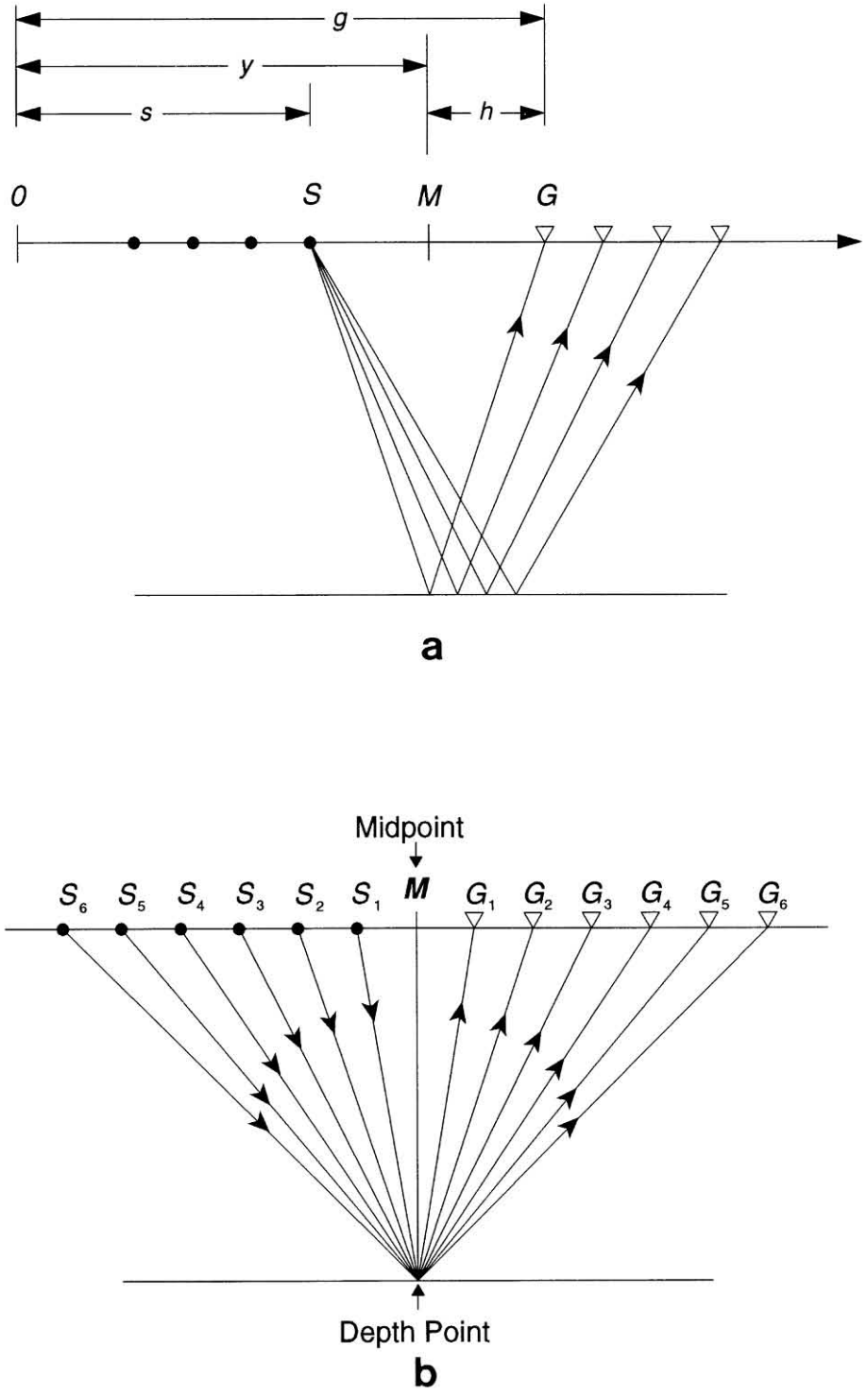

FIG. 1.5-8. (a) Seismic data acquisition is done in shot-receiver (s, g) coordinates. The raypaths are associated with a planar horizontal reflector from a shot point (indicated by the solid circles) to several receiver locations (indicated by the triangles). The processing coordinates, midpoint-(half) offset, (y, h) are defined in terms of (s, g): $y = (g + s)/2$, $h = (g - s)/2$. The shot axis here points opposite the profiling direction, which is to the left. On a flat reflector, the subsurface is sampled by reflection points which span a length that is equal to half the cable length. (b) Seismic data processing is done in midpoint-offset (y, h) coordinates. The raypaths are associated with a single CMP gather at midpoint location M. A CMP gather is identical to a CDP gather if the depth point were on a horizontally flat reflector and if the medium above were horizontally layered.

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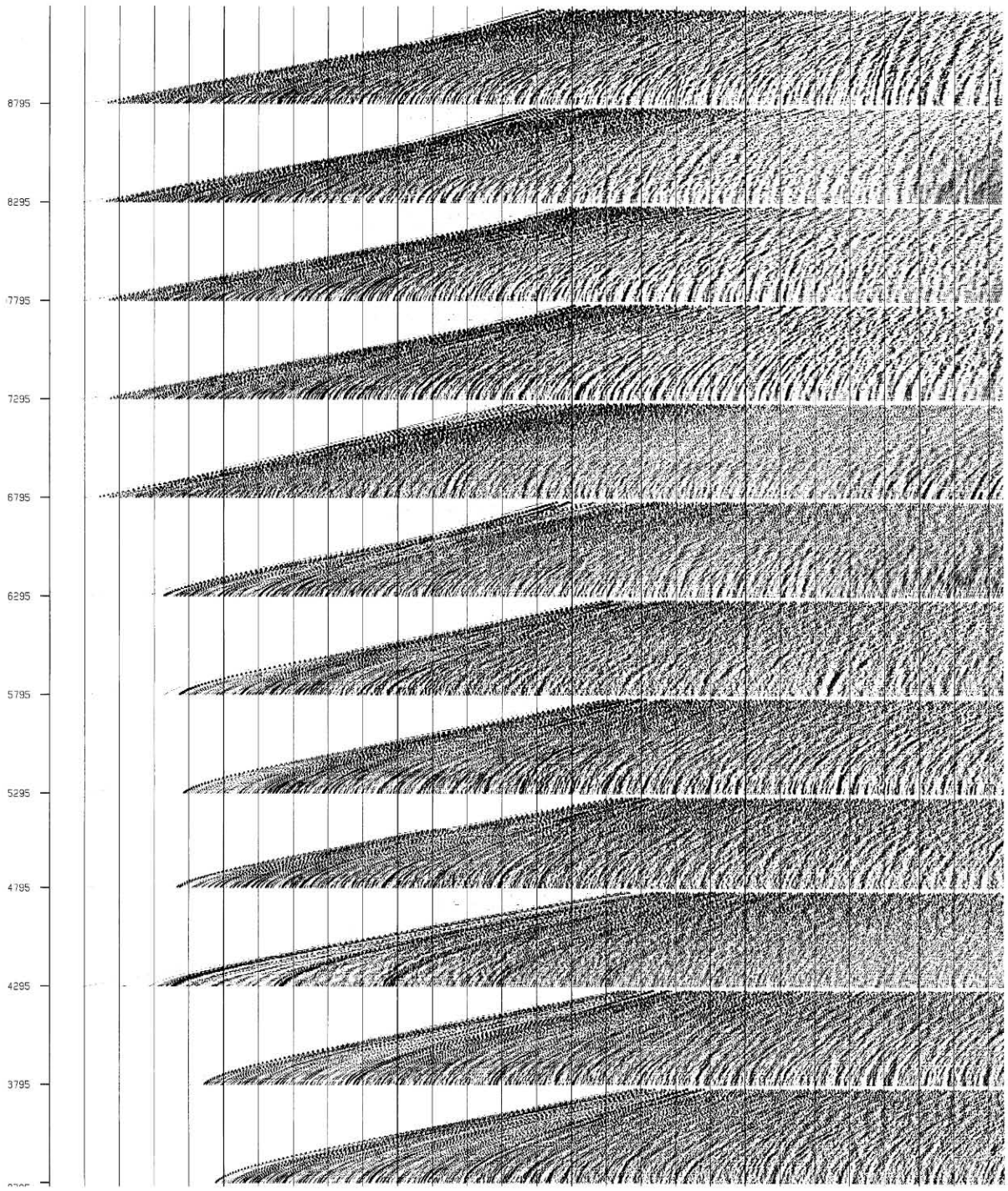

FIG. 1.5-9. Selected CMP gathers corresponding to the same data as in Figure 1.5-7.

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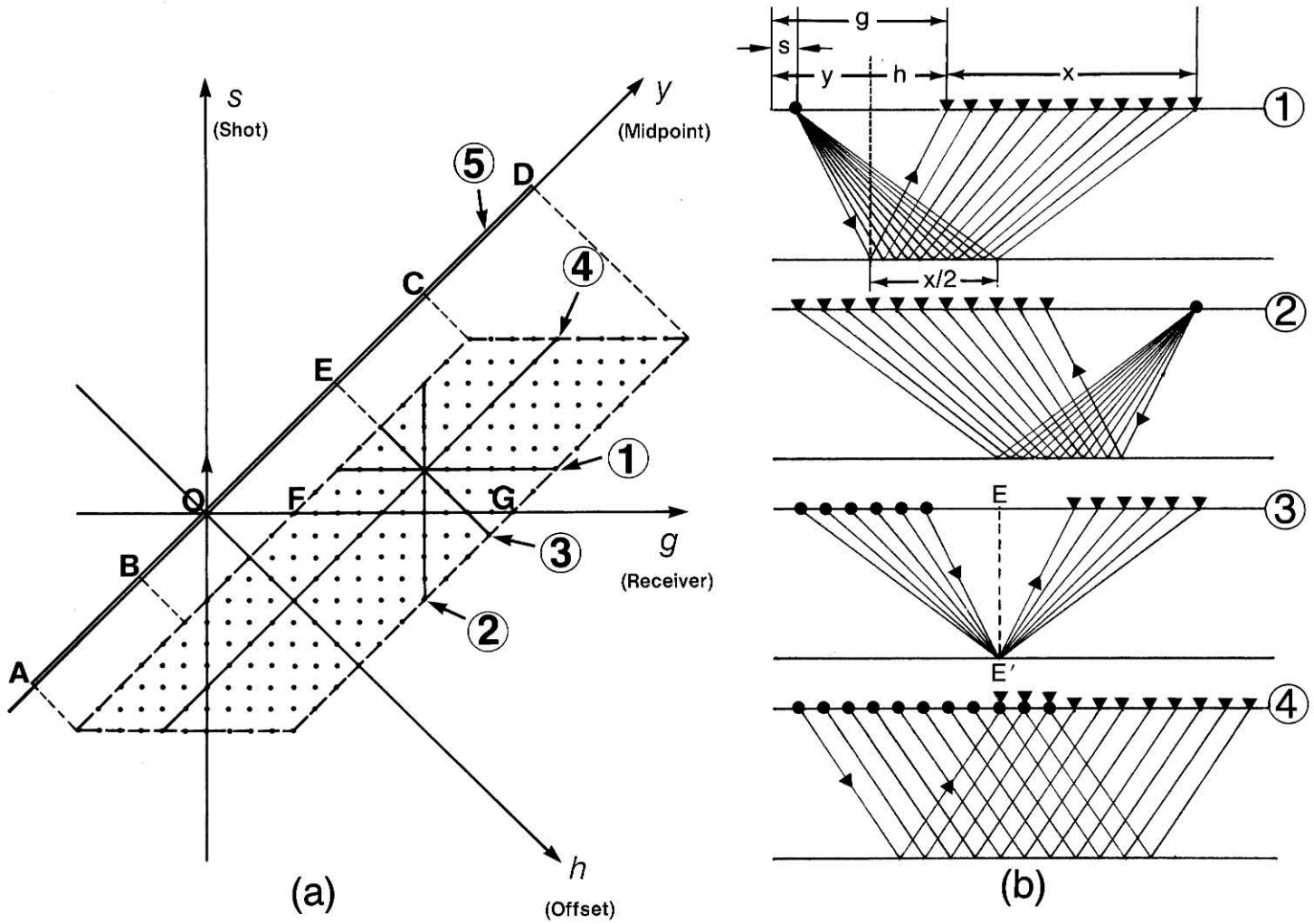

FIG. 1.5-10. (a) A hypothetical stacking chart (adapted from Claerbout, 1976). Each dot represents a single trace with the time axis perpendicular to the plane of the page. Shot-geophone (s, g), and midpoint-offset (y, h) coordinates are superimposed with the (y, h) plane rotated 45 degrees with respect to the (s, g) plane. Here, (1) is a common-shot gather, (2) is a common-receiver gather, (3) is a CMP gather, (4) is a common-offset section, and (5) is a CMP-stacked section. The remaining notation is defined in the text. (b) Raypaths associated with (1) a common-shot gather, (2) a common-receiver gather, (3) a CMP gather, and (4) a common-offset section. Solid triangles denote receiver locations and solid circles denote shot locations, x is the effective cable length — the difference between the maximum and minimum offsets, E denotes a midpoint and E' denotes a depth point on a flat reflector.

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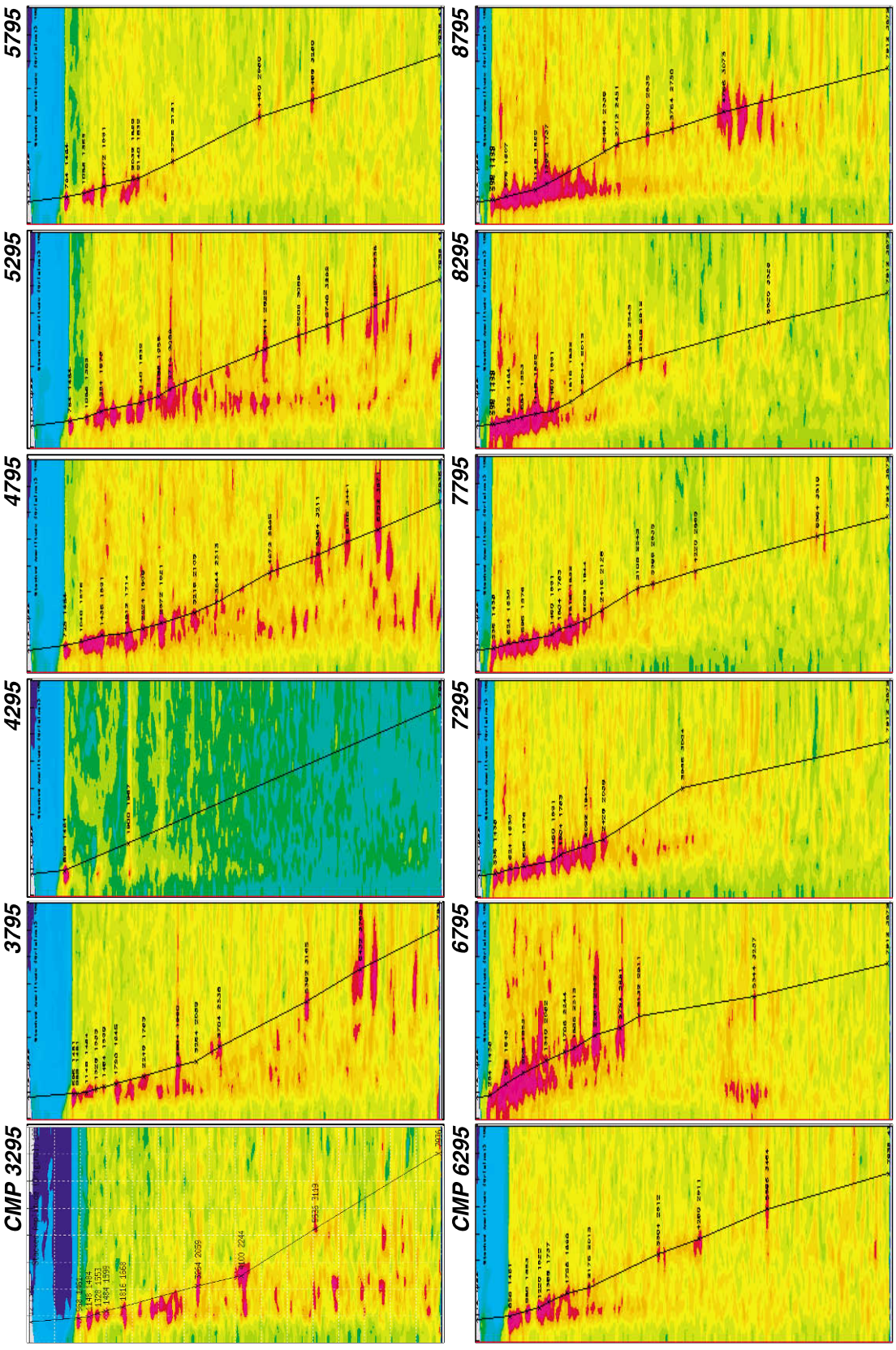

FIG. 1.5-11. Velocity spectra derived from the CMP gathers as in Figure 1.5-9.

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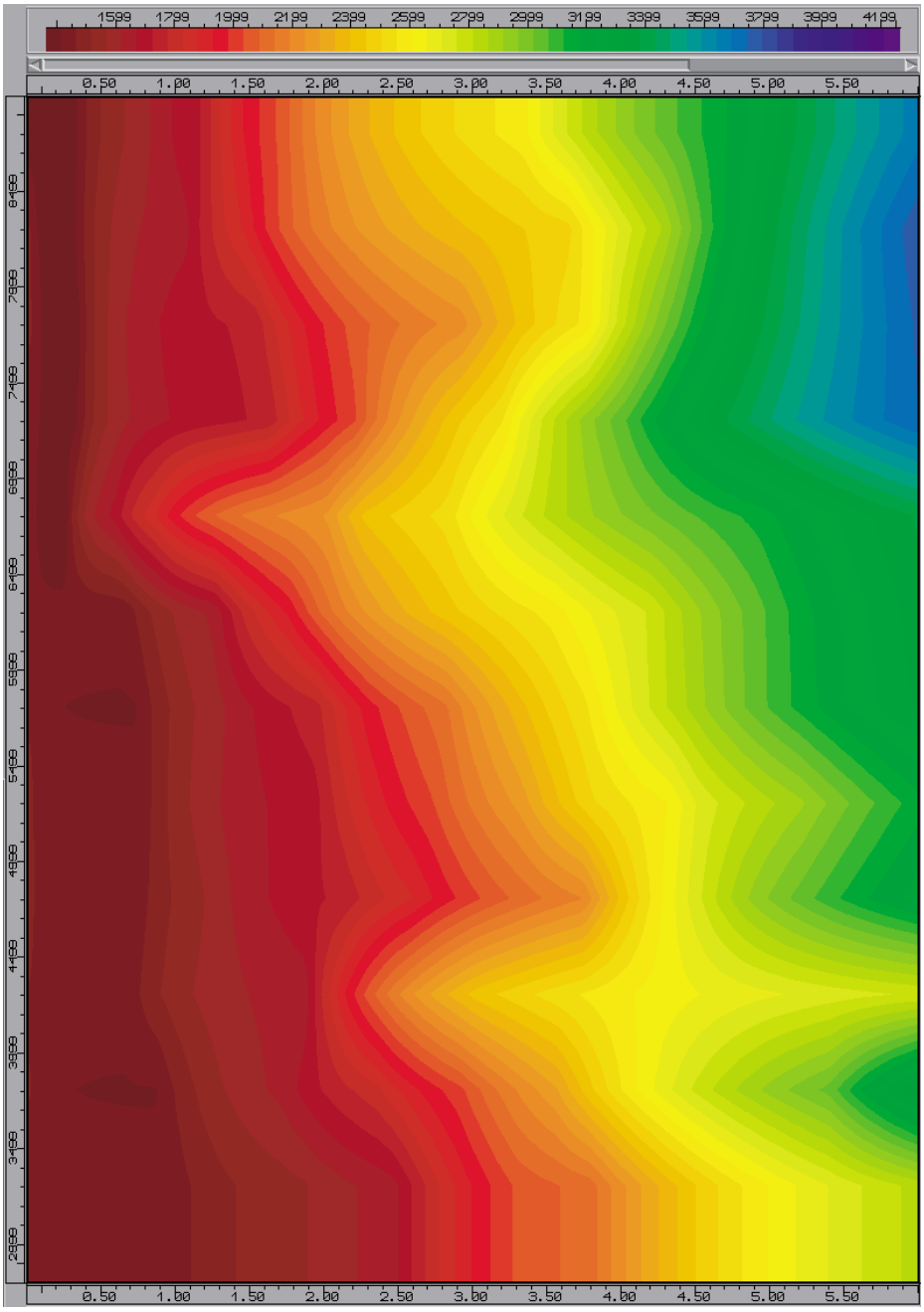

FIG. 1.5-12. Velocity field over the length of the seismic line under consideration. This isovelocity contour map was derived using the velocity picks from the spectra in Figure 1.5-11.

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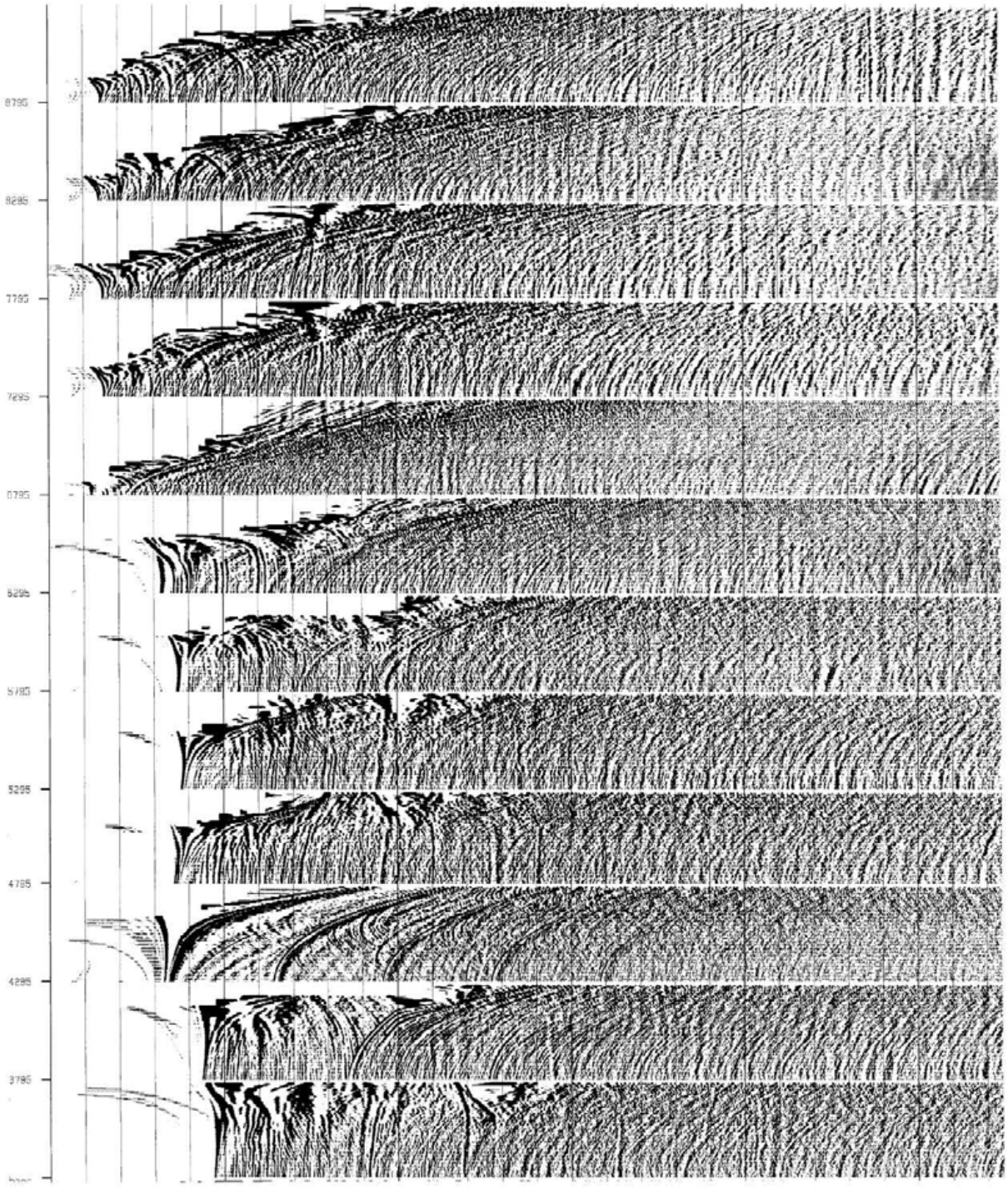

FIG. 1.5-13. The CMP gathers as in Figure 1.5-9 after NMO correction using the velocity field shown in Figure 1.5-12.

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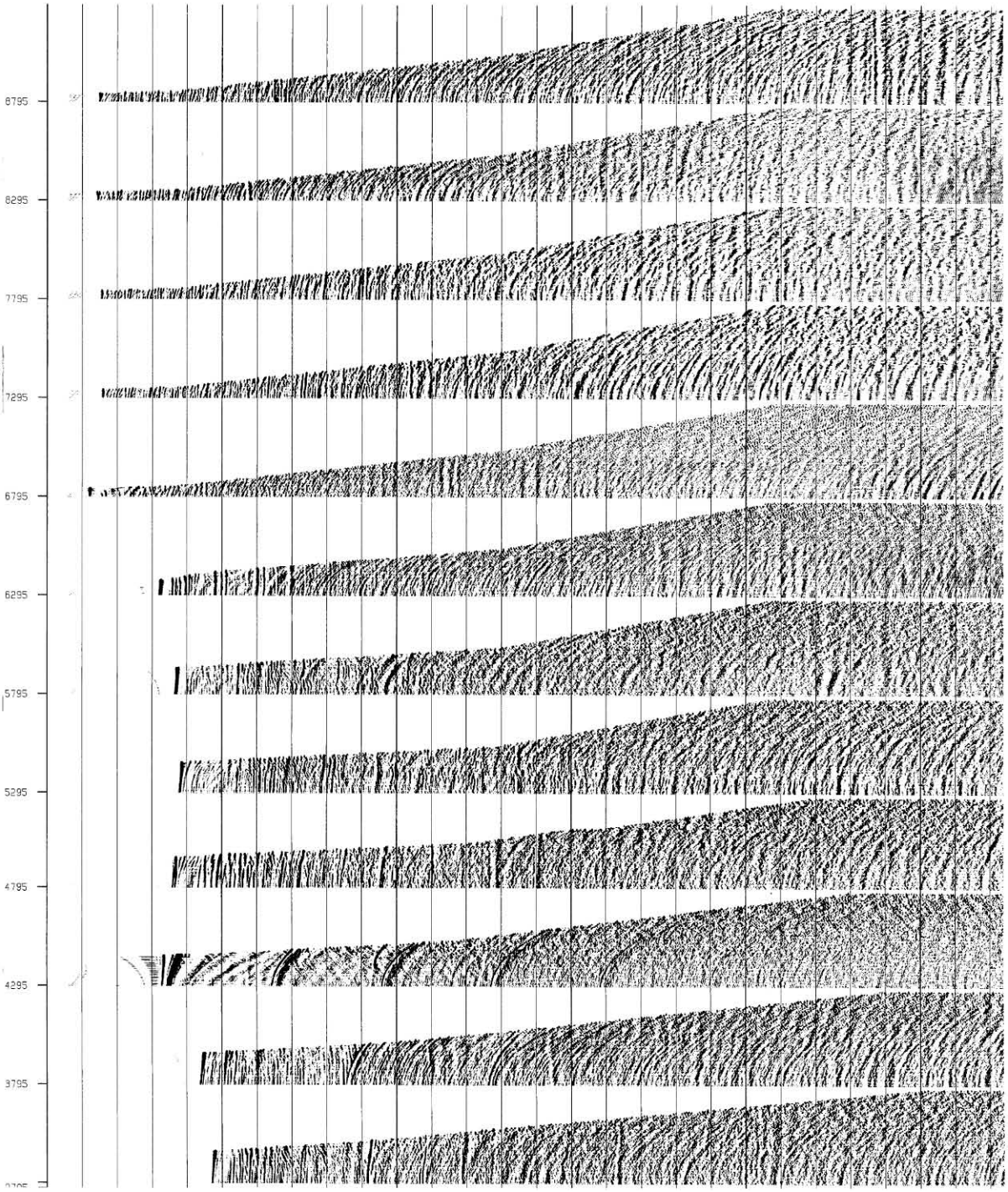

FIG. 1.5-14. The CMP gathers as in Figure 1.5-13 after muting to remove frequency distortion at shallow portions and large offsets resulting from moveout correction.

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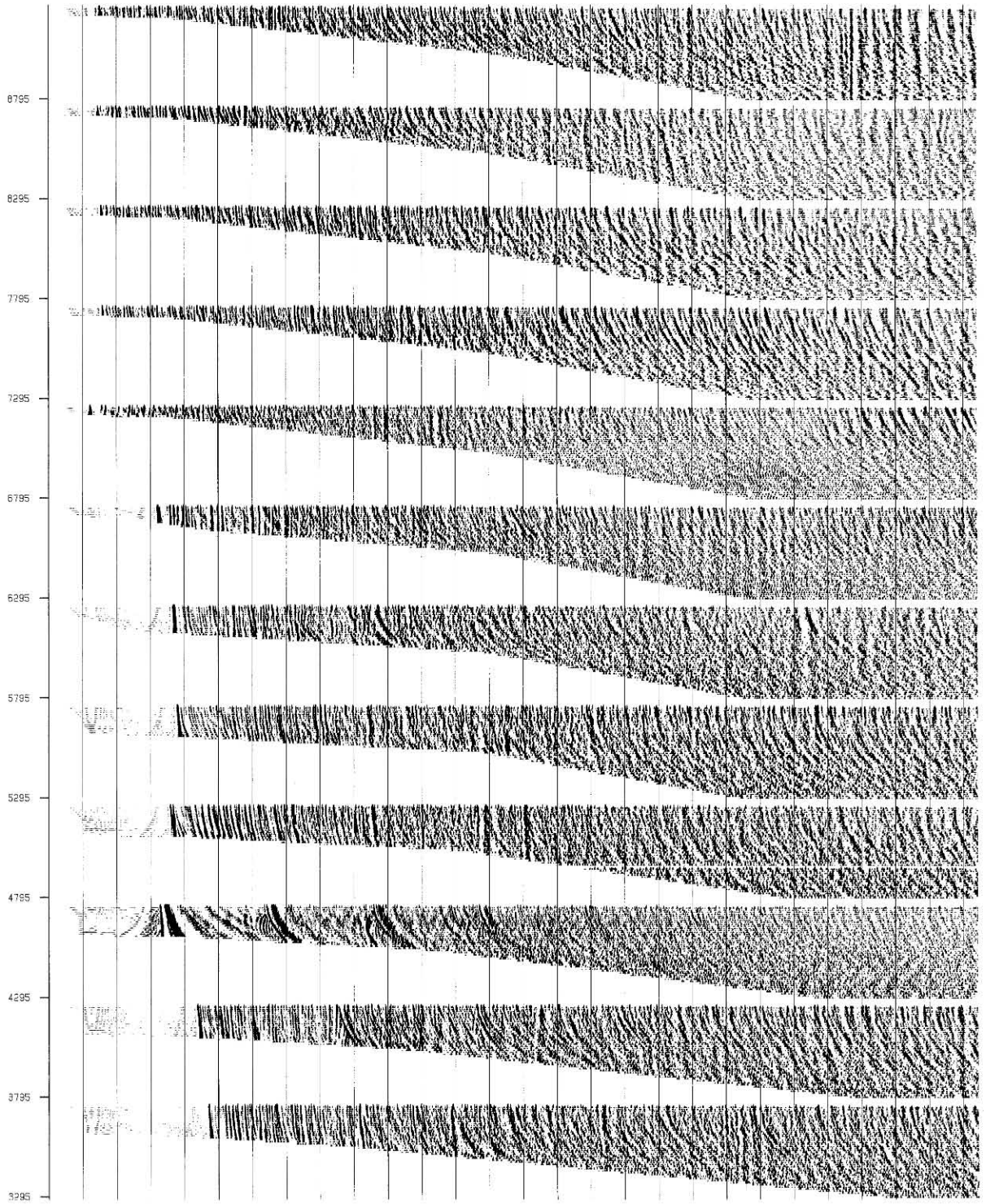

FIG. 1.5-15. The CMP gathers as in Figure 1.5-14 after dip-moveout (DMO) correction.

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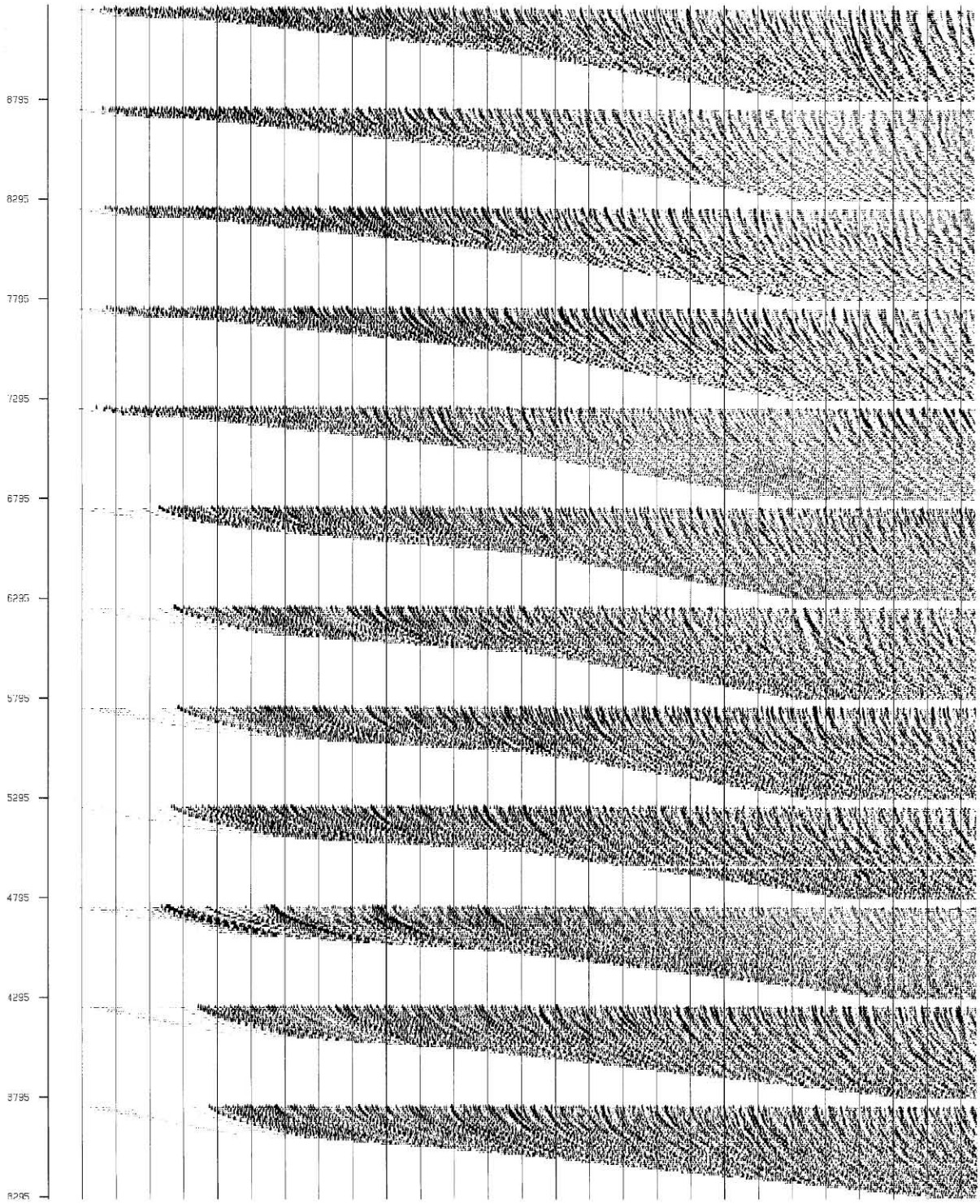

FIG. 1.5-16. The CMP gathers as in Figure 1.5-15 after inverse moveout correction using the velocity field in Figure 1.5-12.

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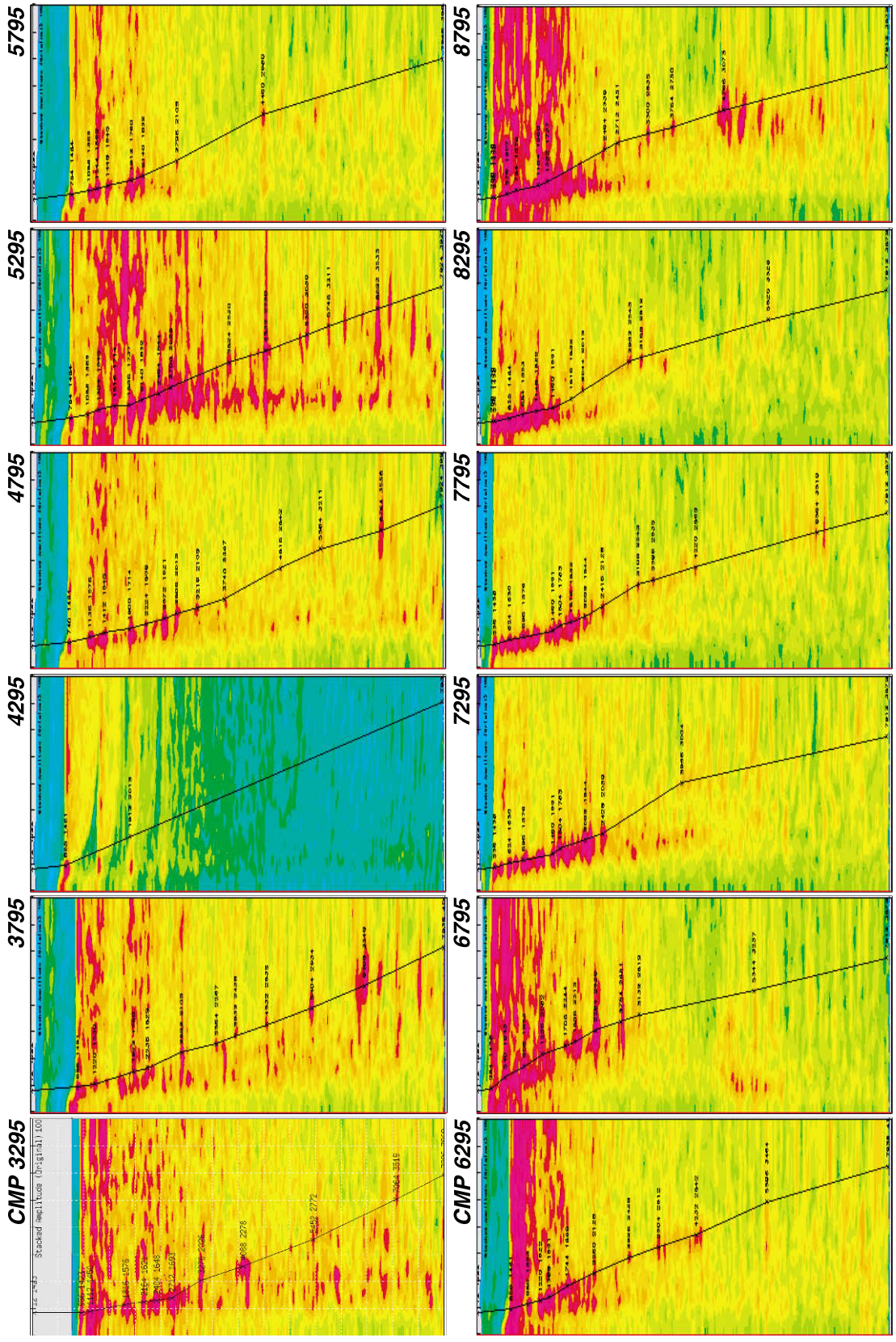

FIG. 1.5-17. Velocity spectra derived from NMO- and DMO-corrected gathers as in Figure 1.5-16.

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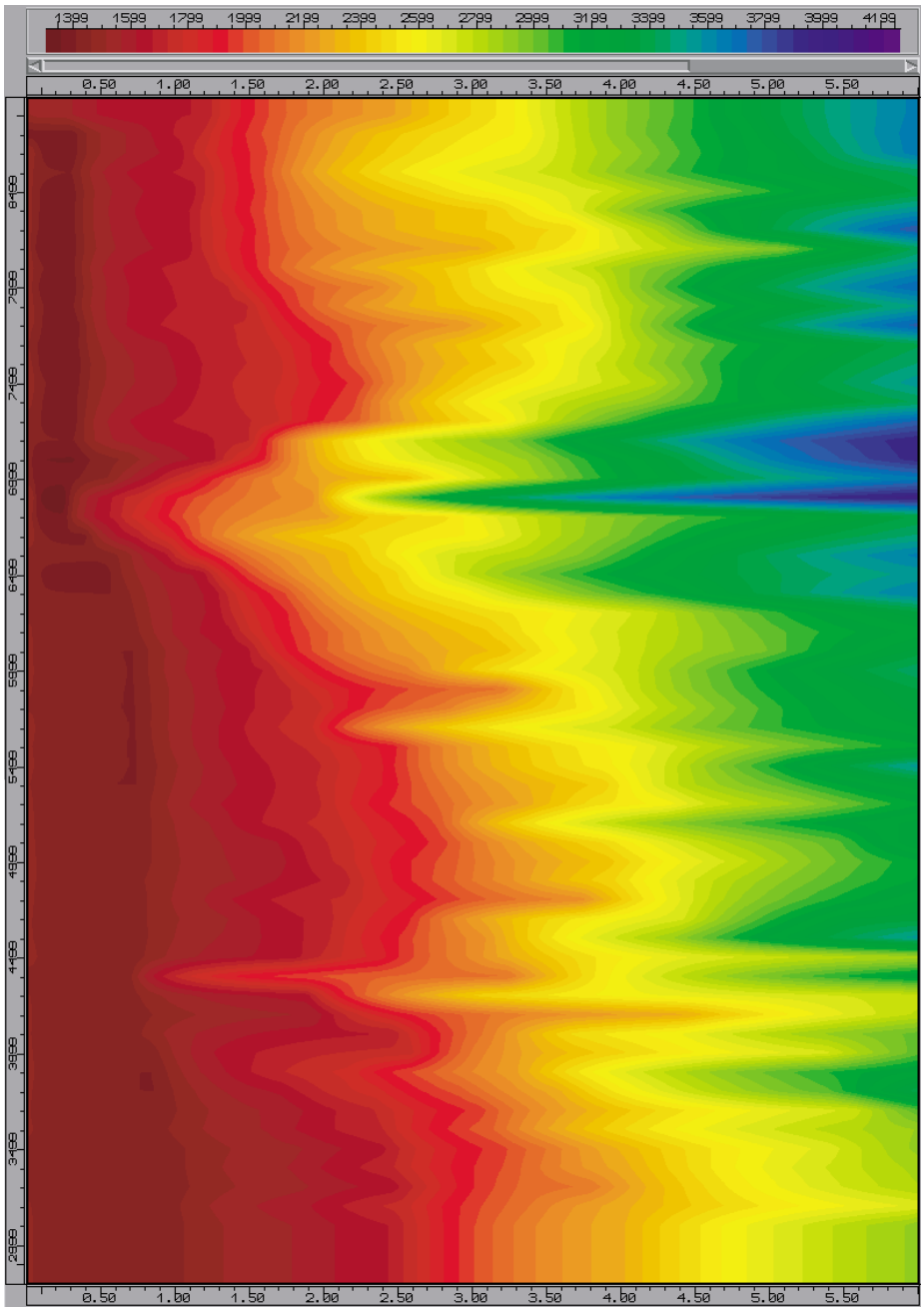

FIG. 1.5-18. Stacking velocity field over the length of the seismic line under consideration. This isovelocity contour map was derived using the velocity picks from the spectra in Figure 1.5-17.

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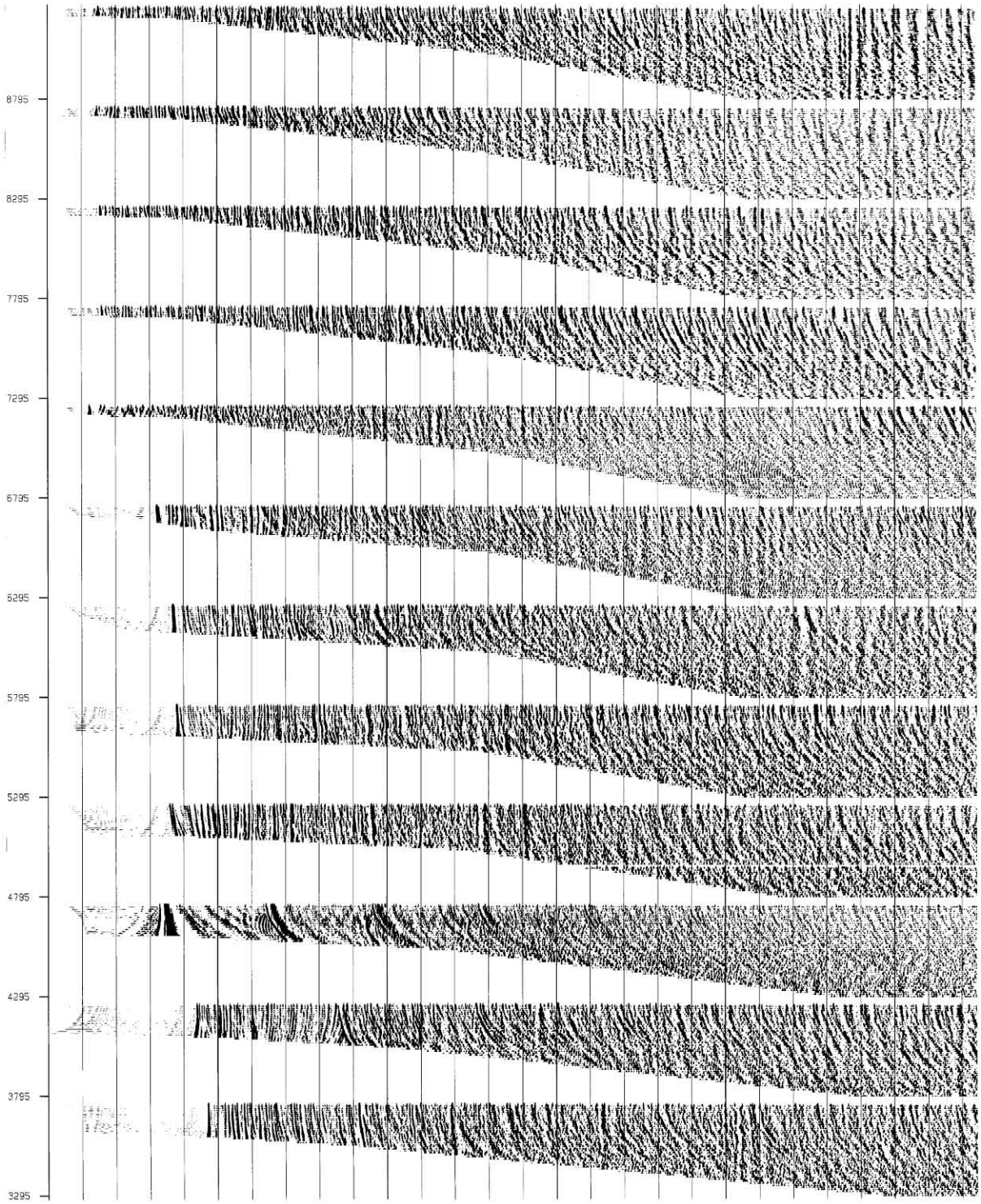

FIG. 1.5-19. The CMP gathers as in Figure 1.5-16 after NMO correction and muting using the stacking velocity field in Figure 1.5-18.

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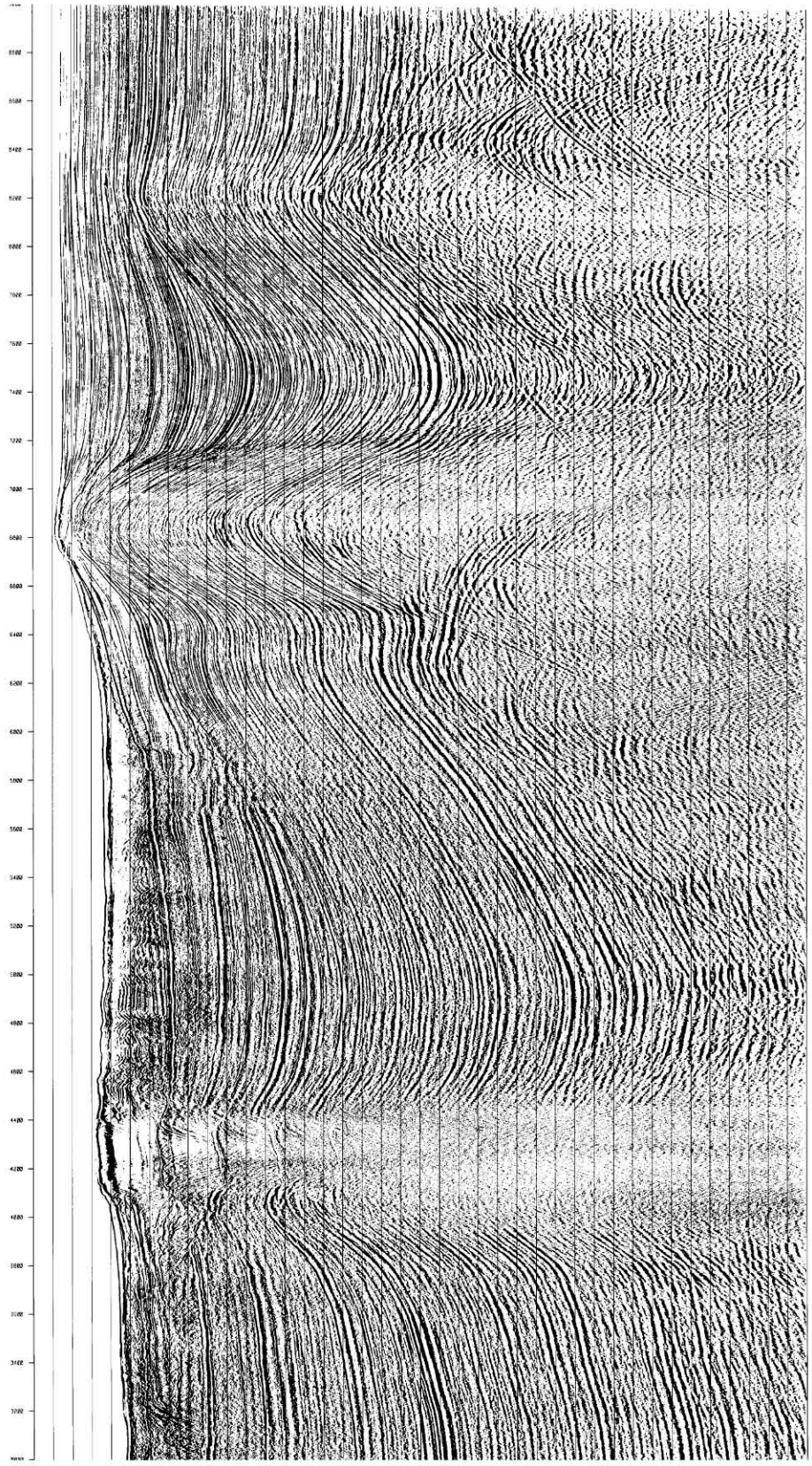

FIG. 1.5-20. The CMP stack associated with the gathers in Figure 1.5-19.

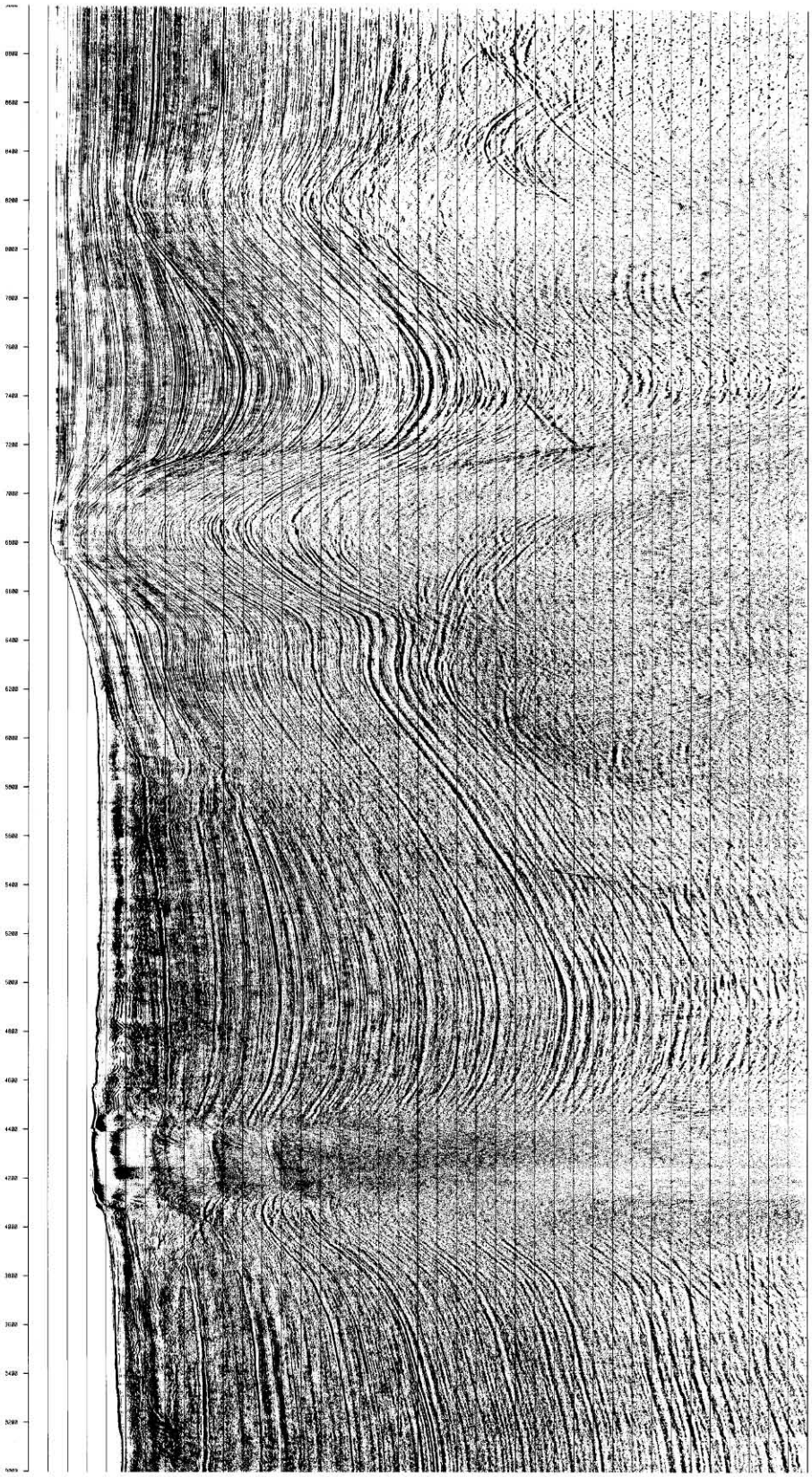

FIG. 1.5-21. The CMP stack as in Figure 1.5-20 after poststack spiking deconvolution.

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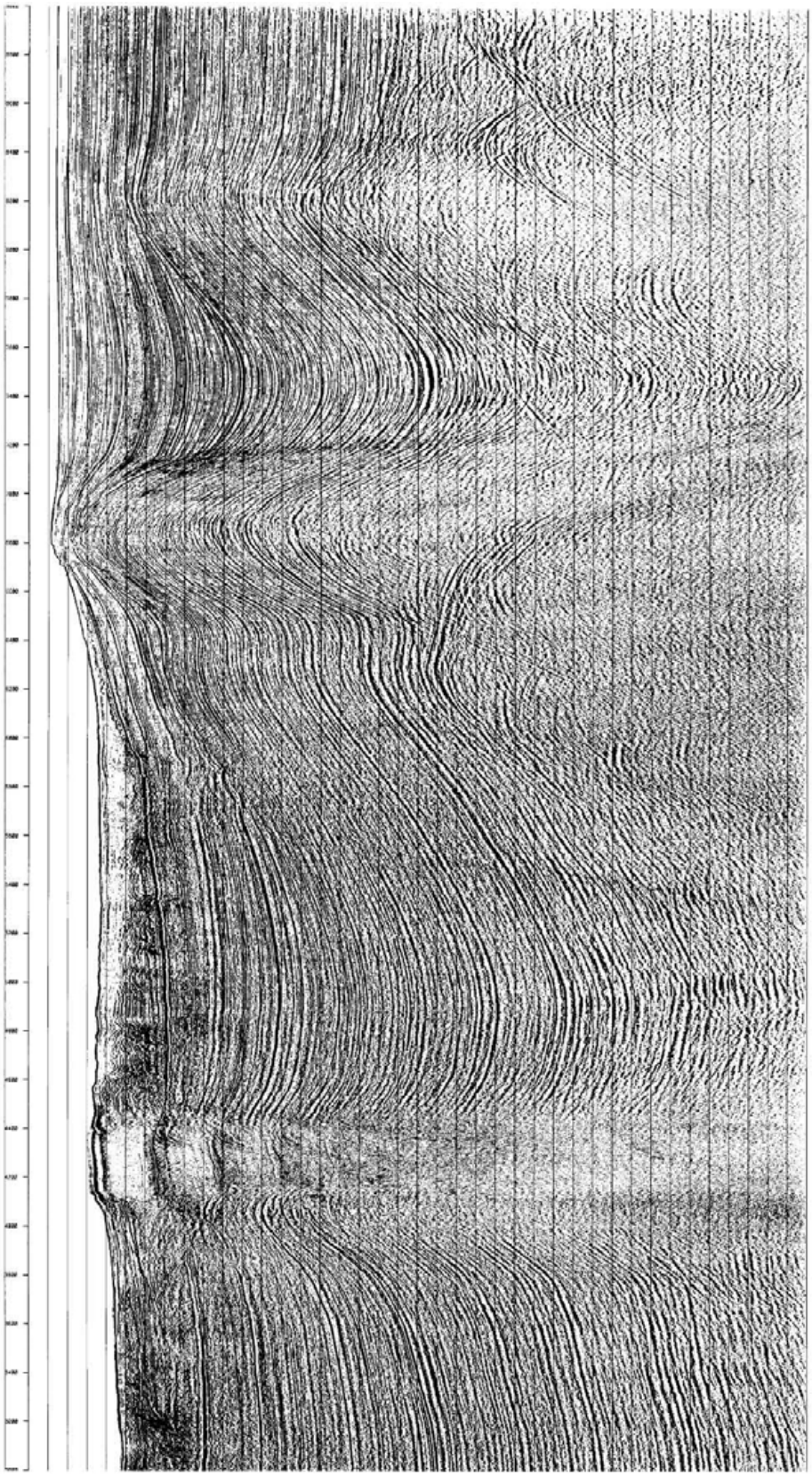

FIG. 1.5-22. The CMP stack as in Figure 1.5-21 after time-variant filtering.

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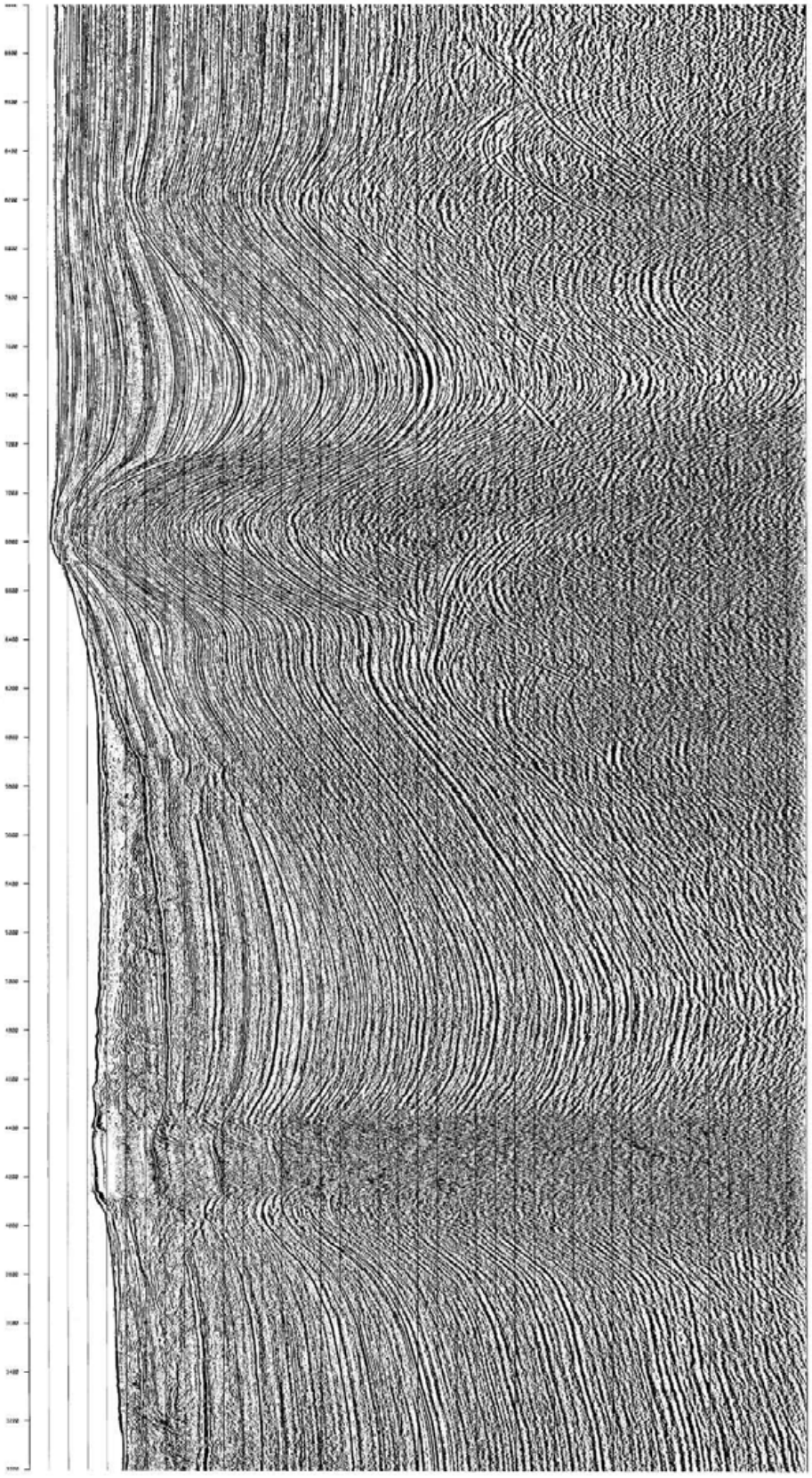

FIG. 1.5-23. The CMP stack as in Figure 1.5-22 after instantaneous AGC scaling.

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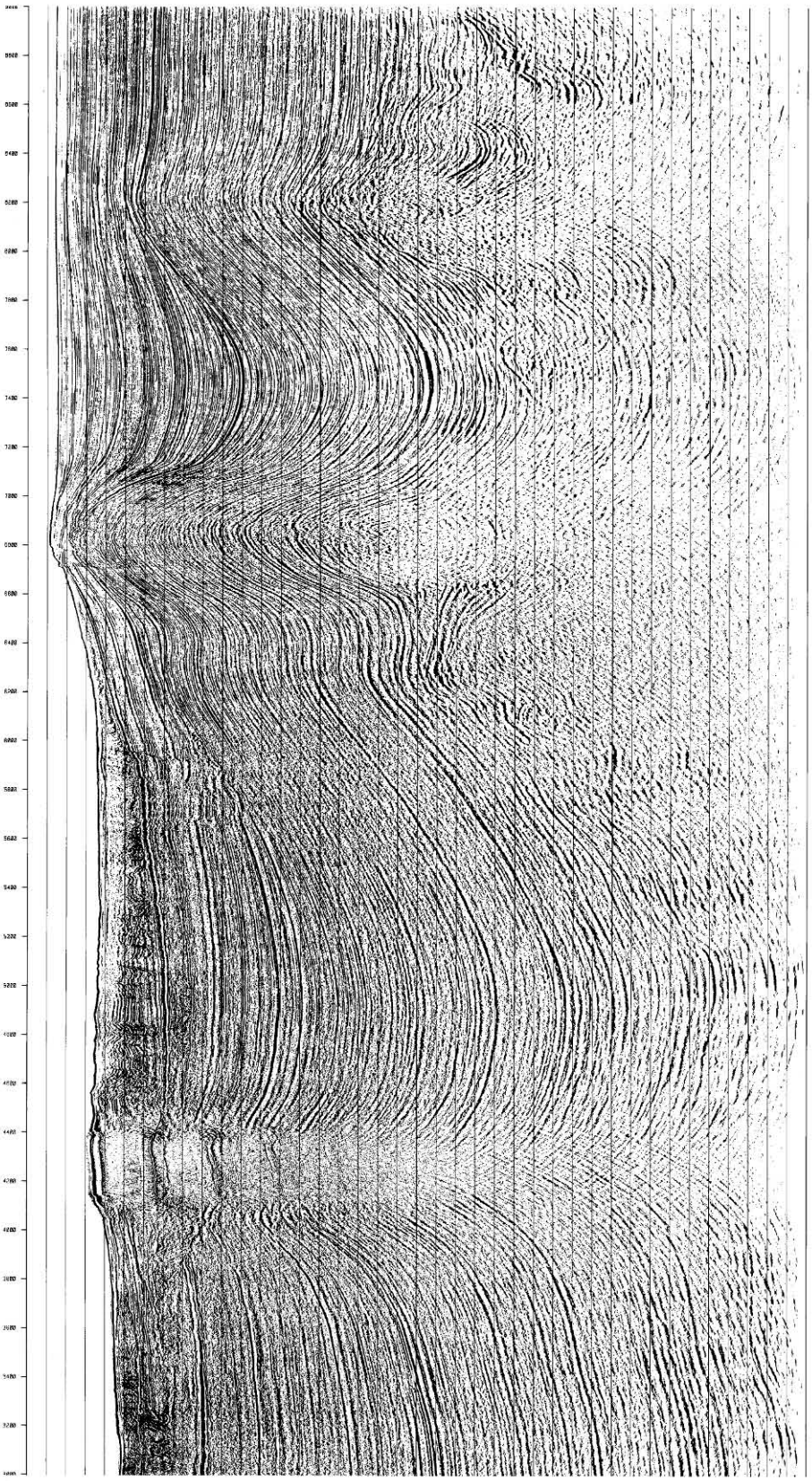

FIG. 1.5-24. The CMP stack shown in Figure 1.5-22 after migration.

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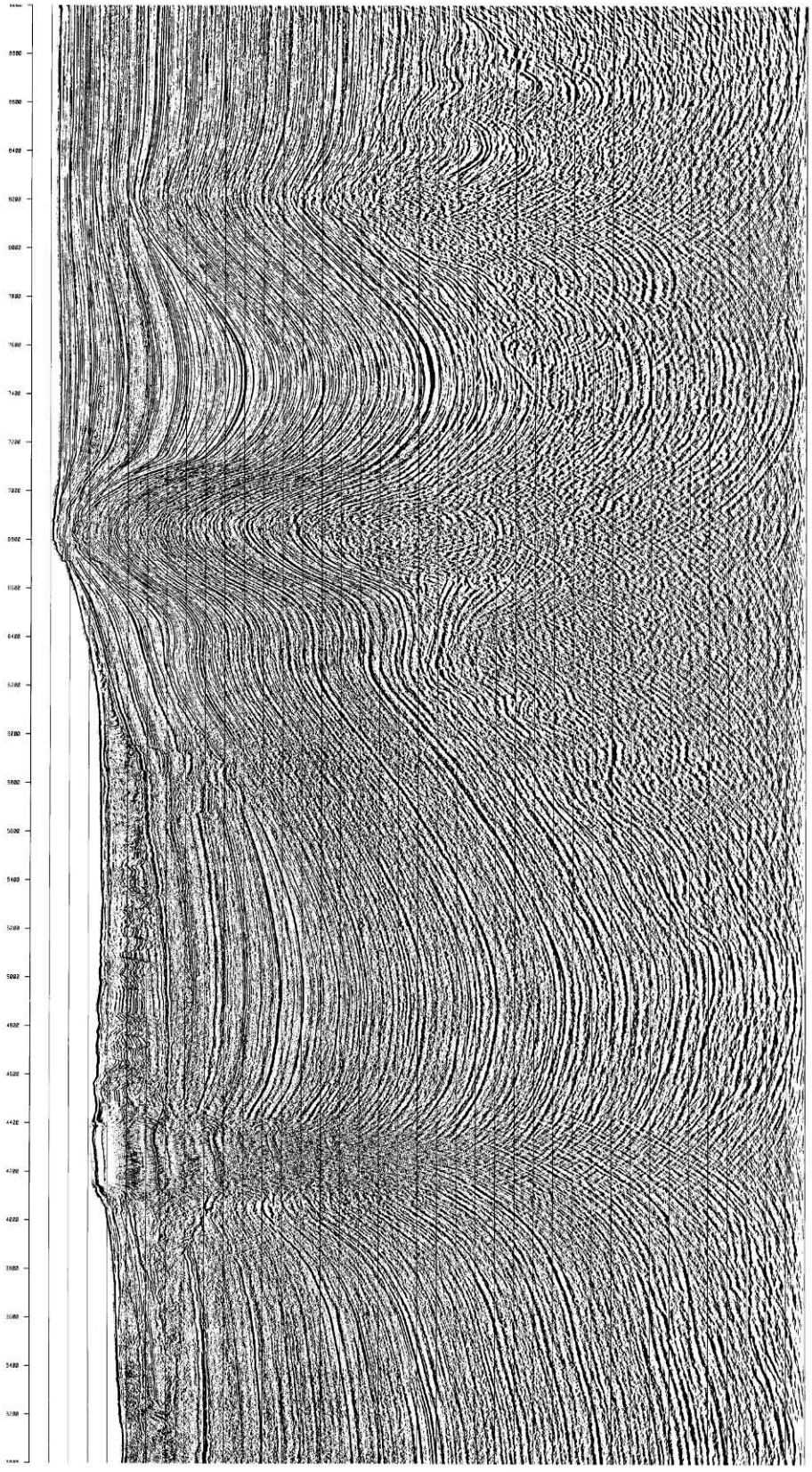

FIG. 1.5-25. The migrated section shown in Figure 1.5-24 after instantaneous AGC scaling.

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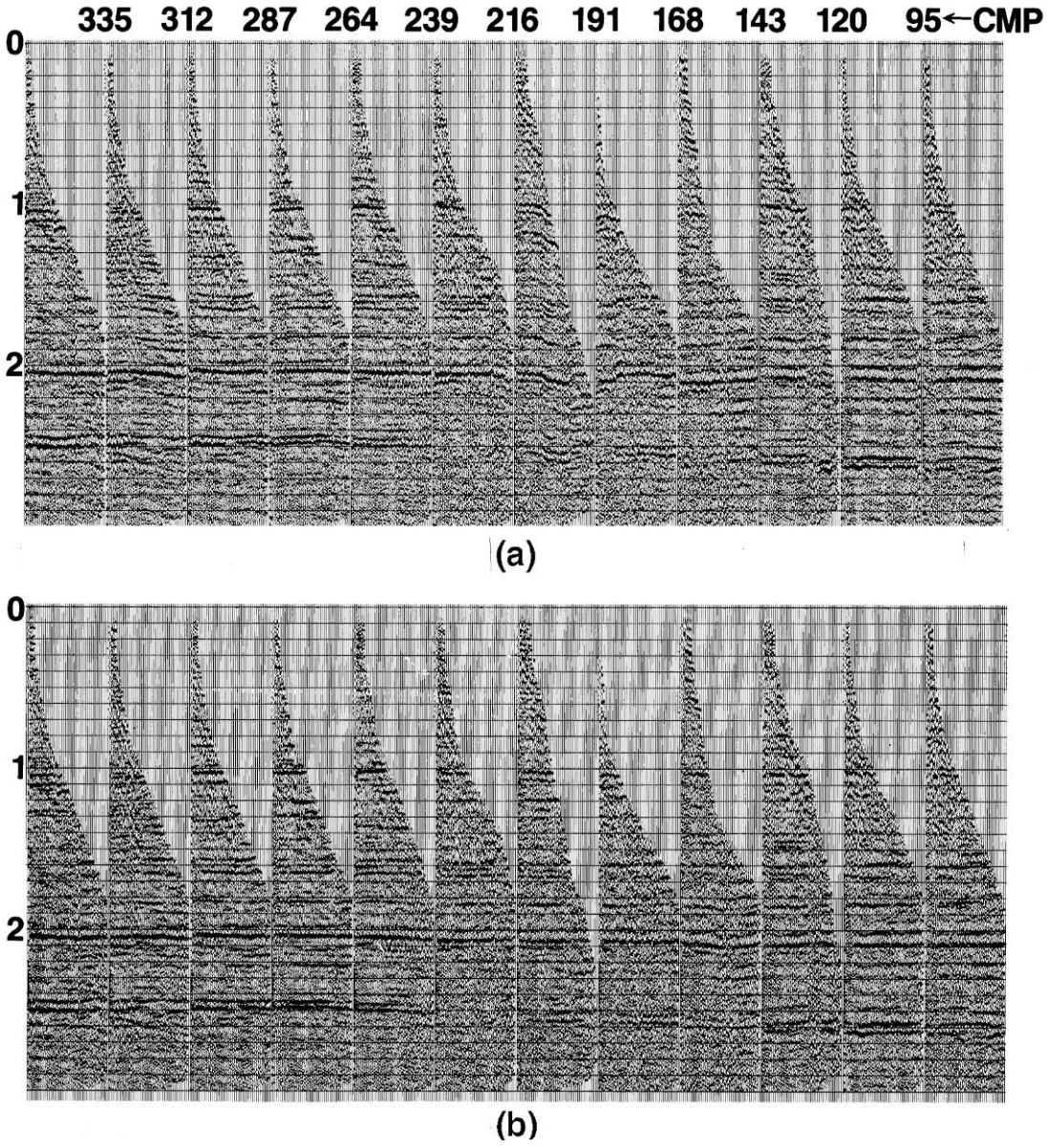

FIG. 1.5-26. The NMO-corrected CMP gathers from a land seismic line (a) before and (b) after residual statics corrections. Note that the distorted events (CMP 191, 216) have nearly been flattened (b).

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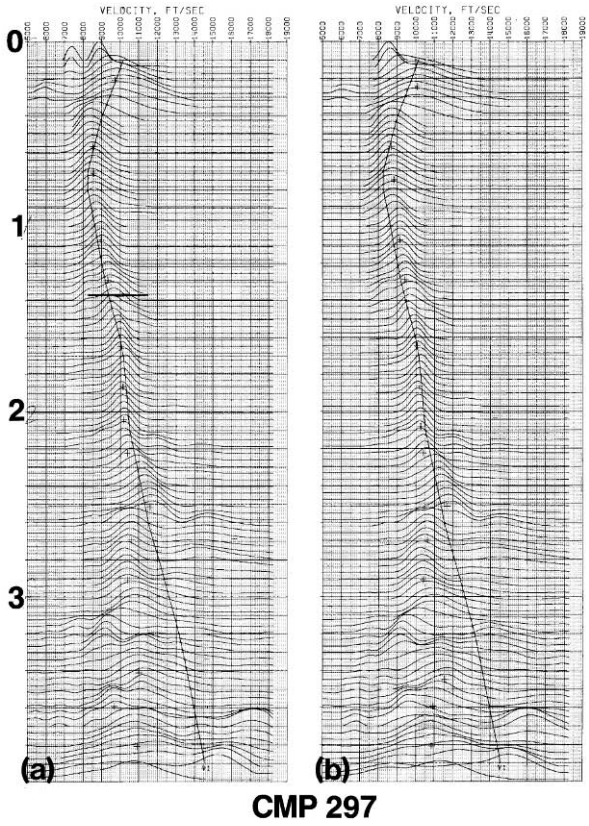

FIG. 1.5-27. Velocity spectra derived from the same data as in Figure 1.5-26, (a) before and (b) after residual statics corrections. Note that no significant difference exists between spectra derived from CMP gather 297 with and without the application of residual statics corrections. Reflection times in this gather did not have significant residual static shifts.

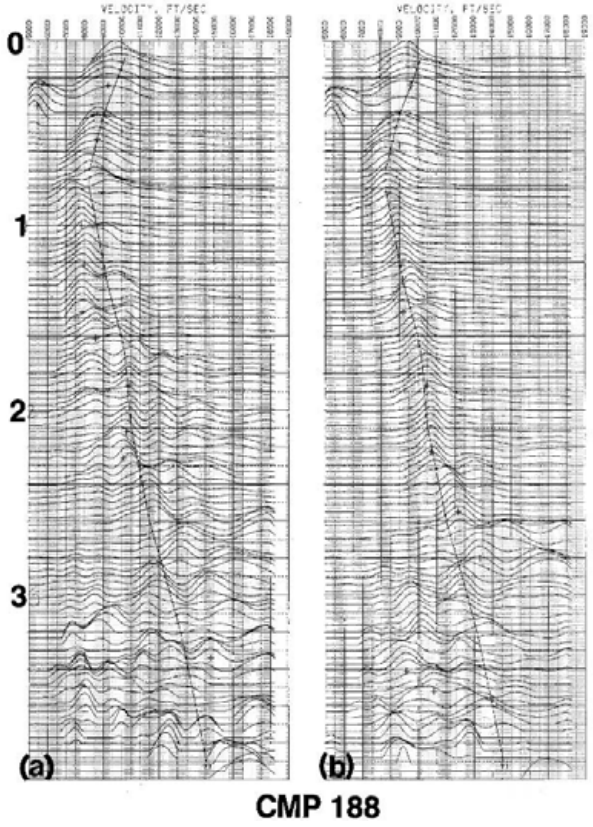

FIG. 1.5-28. Velocity spectra derived from the same data as in Figure 1.5-26, (a) before and (b) after residual statics corrections. Note the improvement after corrections down to 2.6 s.

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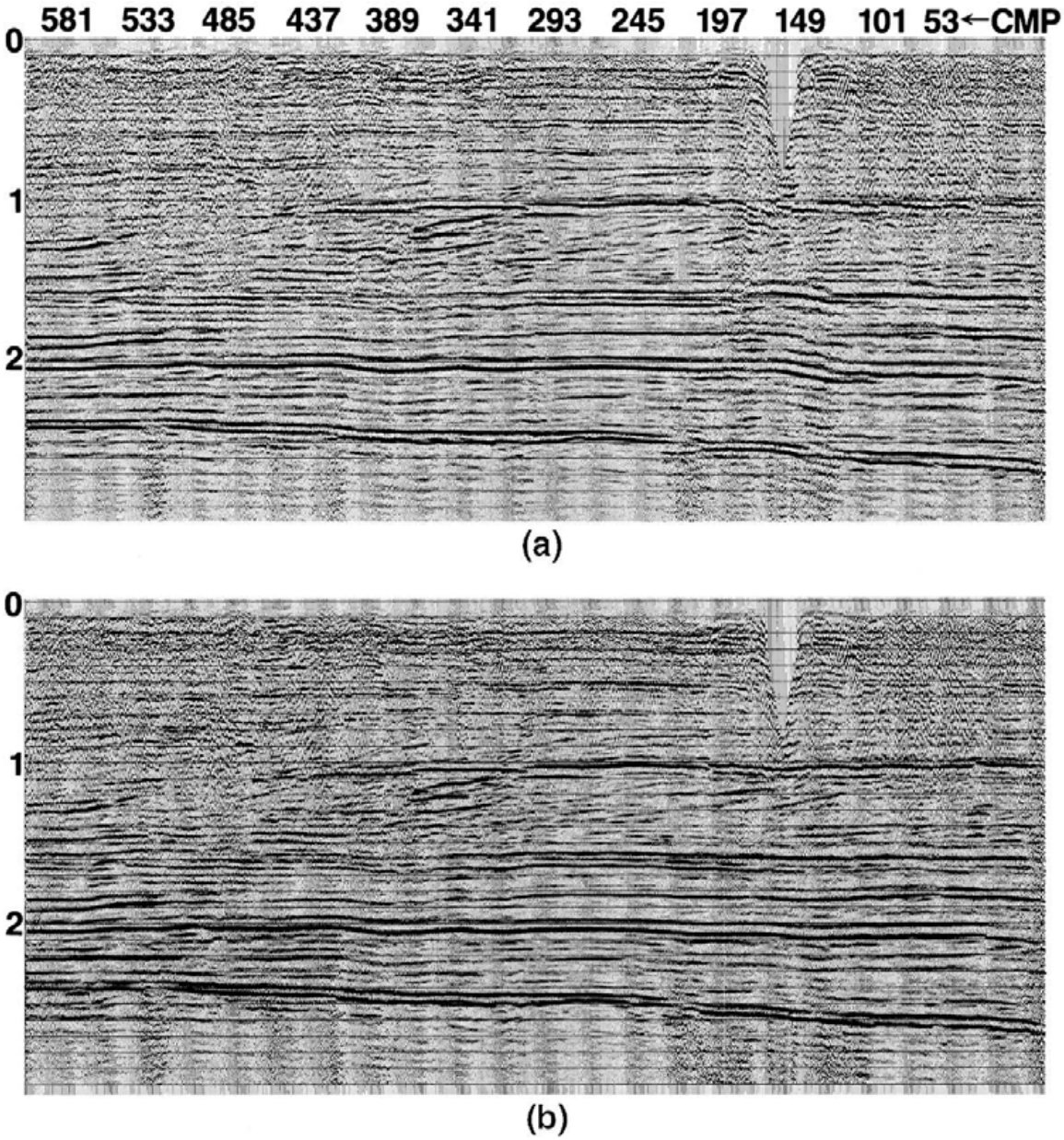

FIG. 1.5-29. The CMP stacks derived from the gathers in Figure 1.5-26. The stack (a) without residual statics corrections shows false structure and poor coherence in the vicinity of CMP 149-197. Both are eliminated by correcting for residual statics (b).

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(c) Any of the two cases described in (a) and (b) often manifest themselves as 3-D problems in nature. The remedy for the 3-D effects, of course, is 3-D migration.

The migrated section in Figure 1.5-25 must be evaluated within the above limitations in stacking and imaging the subsurface.

Residual Statics Corrections

There is one additional step in conventional processing of land and shallow-water seismic data before stacking — residual statics corrections. From the NMO-corrected gathers in Figure 1.5-26a, note that the events in CMP 216 are not as flat as they are in the other gathers. The moveout in CMP gathers does not always conform to a perfect hyperbolic trajectory. This often is because of near-surface velocity irregularities that cause a static or dynamic distortion problem. Lateral velocity variations caused by a complex overburden can cause moveouts that could be negative — a reflection event arrives on long-offset traces before it arrives on short-offset traces. Close examination of the velocity spectra indicates that some are easier to pick (Figure 1.5-27a) than others (Figure 1.5-28a). The velocity spectrum that corresponds to CMP 297 has sharp coherency peaks that are associated with a distinctive velocity trend. However, the velocity spectrum that corresponds to CMP 188 does not yield a distinctive trend, thus making it relatively difficult to pick (Figure 1.5-28a).

To improve stacking quality, residual statics corrections (Section 3.3) are performed on the moveout-corrected CMP gathers. This is done in a surface-consistent manner; that is, time shifts are dependent only on shot and receiver locations, not on the ray paths from shots to receivers. The estimated residual corrections are applied to the original CMP gathers with no NMO correction. Velocity analyses then are often repeated to improve the velocity picks (Figures 1.5-27b and 1.5-28b). With the improved velocity field, the CMP gathers are NMO-corrected (Figure 1.5-26b). Finally, the gathers are stacked as shown in Figure 1.5-29b. For comparison, the stack without the residual statics corrections is shown in Figure 1.5-29a. Reflection continuity over the problem zone between midpoints 53-245 has been improved.

Quality Control in Processing

The conventional processing sequence is outlined in Figure 1.5-30. Each of the processes described above is presented in detail in subsequent chapters. In a seismic data processing sequence, the step that is most vulnerable to human errors is defining the geometry for the survey under consideration and merging it with the seismic data. This involves correctly assigning sources and receivers to their respective surface locations and correctly specifying the source-receiver separation and azimuth for each recorded trace in the survey.

To demonstrate just how important it is to correctly specify the geometry of a survey, consider the impact of a deliberately incorrect geometry assignment on velocity estimation and normal-moveout correction. Figure 1.5-31 shows CMP gathers before and after moveout correction and velocity spectra at three analysis locations along a seismic traverse. The case shown in Figure 1.5-31a does not appear to exhibit any abnormal moveout behavior. The velocity spectrum yields a fairly unambiguous primary velocity function, and primary events on the moveout-corrected gather are nearly flat. The case shown in Figure 1.5-31b, however, begins to show signs of something being wrong with the data.

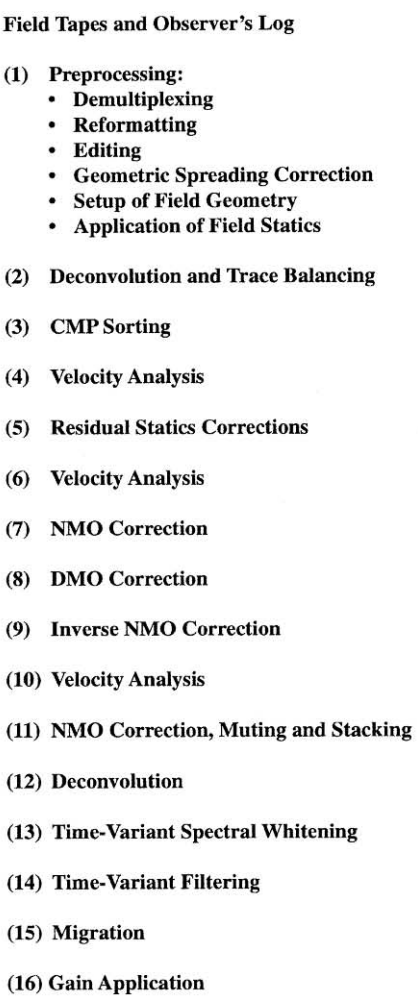

FIG. 1.5-30. A conventional processing flowchart.

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Although the velocity spectrum, again, yields a fairly unambiguous primary velocity function, note that the events associated with the major primary reflections in the CMP gather do not submit themselves to flattening properly after normal-moveout correction. Such behavior in the moveout may be attributed to some physical phenomenon, for instance, anisotropy or nonhyperbolic moveout caused by lateral velocity variations. Nevertheless, it is caused in this case by incorrect geometry specification related to wrong offset assignment to the traces in the gather. The abnormal moveout behavior is strikingly more obvious in the case shown in Figure 1.5-31c. Note the ambiguous semblance peaks in the velocity spectrum, which cause failure in normal-moveout correction to properly flatten the primary events in the gather. Note the differences in the degree of abnormal behavior in event moveout from one location to another (Figures 1.5-31a,b,c); the simpler and the flatter the subsurface structure, the less obvious the adverse impact of incorrect geometry on the moveout.

The care required for correct assignment of the geometry of a survey, of course, does not undermine the care required for proper specification of the parameters associated with any other step in a processing sequence. Specifically, each step must be executed with the necessary quality control. Displays of appropriate data attributes, such as amplitude spectrum and autocorrelogram, help the analyst understand signal and noise chracteristics of the recorded data and the effect of a step included in the processing sequence on the data, thus facilitating appropriate specification of parameters associated with that step. Figures 1.5-32 through 1.5-41 show quality control panels that are examples of recommended standard displays for parameter selection at various stages in the analysis. All displays include the amplitude spectrum on the top row averaged over the gather, if it is a prestack test panel, and averaged over the portion of the stack, if it is a poststack test panel, and autocorrelogram of the respective data type on bottom row.

Figure 1.5-32 is the quality control panel for prestack signal processing. Shown from left to right are: (a) a CMP gather which exhibits strong, low-frequency swell noise; (b) low-cut filtering to remove the swell noise; (c) t^2 scaling to correct for geometric spreading (Section 1.4); (d) prestack spiking deconvolution (Sections 2.3, 2.4, and 2.5); (e) and wide bandpass filtering to remove the high-frequency noise boosted by spiking deconvolution. Note that the autocorrelogram better exhibits over the entire cable length the characteristics of the source waveform and reverberations and multiples after t^2 scaling. Also note that spiking deconvolution has removed much of the energy associated with the reverberations and multiples. The broadening and flattening of the amplitude spectrum after spiking deconvolution are indicative of the increase of vertical resolution.

Figure 1.5-33 shows the spectra which are associated with the gathers from left to right in Figure 1.5-32. The horizontal axis is frequency in Hz and the vertical axis is two-way traveltime in s. Note from (a) that the swell noise at very low frequencies occupies the spectrum along the entire time axis. Note also that the energy in the gather is largely confined to shallow times within a bandlimited region of the spectrum. Following the low-cut filtering (b), note the elimination of the swell noise energy. The t^2 scaling (c) has restored the energy at late times, and deconvolution (d) has broadened the spectrum. Following the wide bandpass filtering (e), note that the signal bandwidth has been preserved [compare with (a)], and the spectrum has been flattened within the passband.

Figures 1.5-34 and 1.5-35 show two standard test panels for determining prestack deconvolution parameters. With the help of the amplitude spectrum and autocorrelogram, the analyst chooses an optimum operator length and prediction lag. Figure 1.5-34 shows the test panel for prestack spiking deconvolution (Sections 2.3, 2.4, and 2.5). Shown from left to right are: the input gather after low-cut filtering and t^2 scaling as in Figure 1.5-32, followed by deconvolution using operator lengths of 120 ms, 160 ms, 240 ms, 360 ms, and 480 ms. Note that deconvolution using an operator length of 480 ms best flattens the spectrum within the signal passband. Failure of deconvolution in flattening the spectrum at very high frequencies is most likely due to nonstationarity of the signal. This effect usually is accounted for by time-variant spectral whitening after stack. Since autocorrelation of input data is used in designing a deconvolution operator, it is appropriate to examine the autocorrelation before and after deconvolution. Note from the autocorrelograms in Figure 1.5-34 that operator length dictates the ability of deconvolution in removing reverberations and short-period multiples.

Figure 1.5-35 shows the test panel for prestack predictive deconvolution (Sections 2.3, 2.4, and 2.5). Shown from left to right are: the input gather after low-cut filtering and t^2 scaling as in Figure 1.5-32, followed by deconvolution using prediction lags of 2 ms (unit prediction lag), 8 ms, 16 ms, 24 ms, and 32 ms, with the same operator length of 480 ms. Note that the unit-prediction lag yields a flat spectrum across the passband, while increasing the prediction lag results in departure from a flat spectrum. Prediction lag controls the ability of deconvolution to increase the vertical resolution (Section 2.4).

Figures 1.5-36 and 1.5-37 show two standard test panels for determining poststack deconvolution parameters. Note from the average amplitude spectrum of the

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section on the left-hand side of each test panel that CMP stacking inherently attenuates high frequencies which need to be restored by poststack deconvolution. Figure 1.5-36 shows the test panel for poststack spiking deconvolution. Shown from left to right are: the input stack, followed by deconvolution using operator lengths of 120 ms, 160 ms, 240 ms, 360 ms and 480 ms, and high-cut filtering to retain the acceptable signal band and remove the high-frequency noise.

Figure 1.5-37 shows the test panel for poststack predictive deconvolution. Shown from left to right are: the input stack, followed by deconvolution using prediction lags of 2 ms (unit prediction lag), 8 ms, 16 ms, 24 ms and 32 ms, using the same operator length of 480 ms, and high-cut filtering to retain the acceptable signal band and remove the high-frequency noise. Again, note that the unit-prediction lag yields a flat spectrum across the passband, while increasing the prediction lag results in departure from a flat spectrum.

Figure 1.5-38 shows the standard quality control panel for poststack signal processing. Shown from left to right are: a portion of the stacked section with prestack processing as described by Figure 1.5-32; spiking deconvolution (Section 2.5) to restore the high frequencies attenuated by the stacking process; time-variant spectral whitening to account for nonstationarity and to further flatten the spectrum (Section 2.6) — all three steps followed by high-cut filtering; bandpass filtering to retain the acceptable signal band and remove the high-frequency noise; instantaneous AGC scaling and rms amplitude AGC scaling.

Figures 1.5-39 and 1.5-40 show the test panels for defining the parameters for time-variant filtering (Section 1.1). A portion of the stacked section is bandpass filtered using a 10-Hz bandwidth that slides from low to high-frequency end of the spectrum. Note that the coherent signal at high-frequency bands is confined to shallow times. Nevertheless, these filter panels indicate that signal up to 90 Hz is present in the data down to 2.2 s, and the signal up to 100 Hz is present down to 1.4 s.

Finally, Figure 1.5-41 shows the test panel for poststack noise attenuation using $f-x$ deconvolution (Section 6.5). A parameter that needs to be tested for $f-x$ deconvolution is the percent add-back of the estimated noise to circumvent the smeared appearance of events following noise attenuation. Shown from left to right are: a portion of the stacked section with poststack deconvolution, time-variant spectral whitening and bandpass filtering; noise attenuation with 80, 60, 40, 20, and 0 percent add-back. Note that without any add-back, the amplitude spectrum of the section after noise attenuation indicates dampening of high-frequency energy that may be attributed to the random noise uncorrelated from trace to trace.

The test panels for quality control in processing of seismic data are not limited to those presented in Figures 1.5-32 through 1.5-41. Additional panels with appropriate and convenient format may be constructed to test parameters associated with refraction and residual statics corrections, multiple attenuation, dip-moveout correction, and migration. Powerful interactive tools, including 3-D visualization techniques, facilitate efficient parameter testing and quality control in processing.

Parsimony in Processing

The primary objective in data processing is to enhance the signal-to-noise ratio while preserving the useful signal bandwidth associated with the recorded data at all stages in the analysis. The principle of parsimony in processing is the basis to achieve this objective. Specifically, a processing sequence should be optimally lean and not include any step that may do more harm than the intended action by that process. A further compelling reason for parsimony is preserving relative amplitudes for amplitude-driven exploration objectives associated with stratigraphic plays.

Figures 1.5-42 through 1.5-53 show the step-by-step appearance of a portion of a stacked section based on a very basic processing sequence intended to minimize amplitude distortions while largely attenuating reverberations, multiples, and random noise and ultimately increasing vertical and lateral resolution.

The stacked sections in Figures 1.5-42 through 1.5-47 were created based on the following prestack processing sequence:

(a) Figure 1.5-42: stack based on unprocessed data that contain low-frequency swell noise.
(b) Figure 1.5-43: stack using CMP gathers with low-cut filtering applied to remove the swell noise.
(c) Figure 1.5-44: stack as in (b) with the additional step for t^2 scaling to compensate for wavefront divergence; note the restoration of amplitudes at late times.
(d) Figure 1.5-45: stack as in (c) with the additional step for prestack spiking deconvolution; note the attenuation of reverberations.
(e) Figure 1.5-46: stack as in (d) with the additional step for wide bandpass filtering to improve velocity analysis.
(f) Figure 1.5-47: stack as in (e) with the additional step for dip-moveout correction; note the preservation of diffractions that interfere with the nearly flat reflections.

(text continues on p. 150)

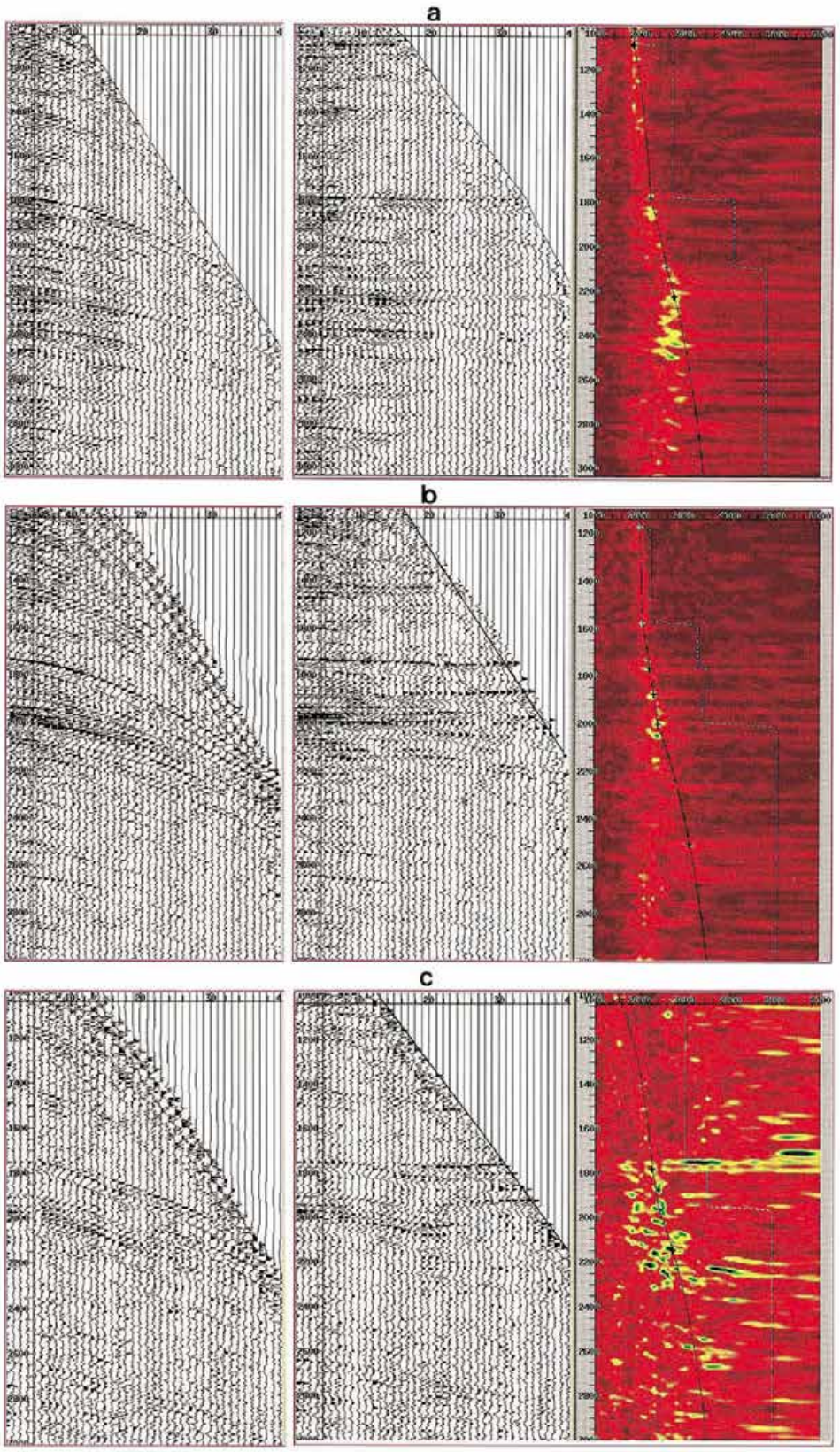

FIG. 1.5-31. Anomalous moveout behavior on CMP gathers caused by incorrect geometry specification. See text for details.

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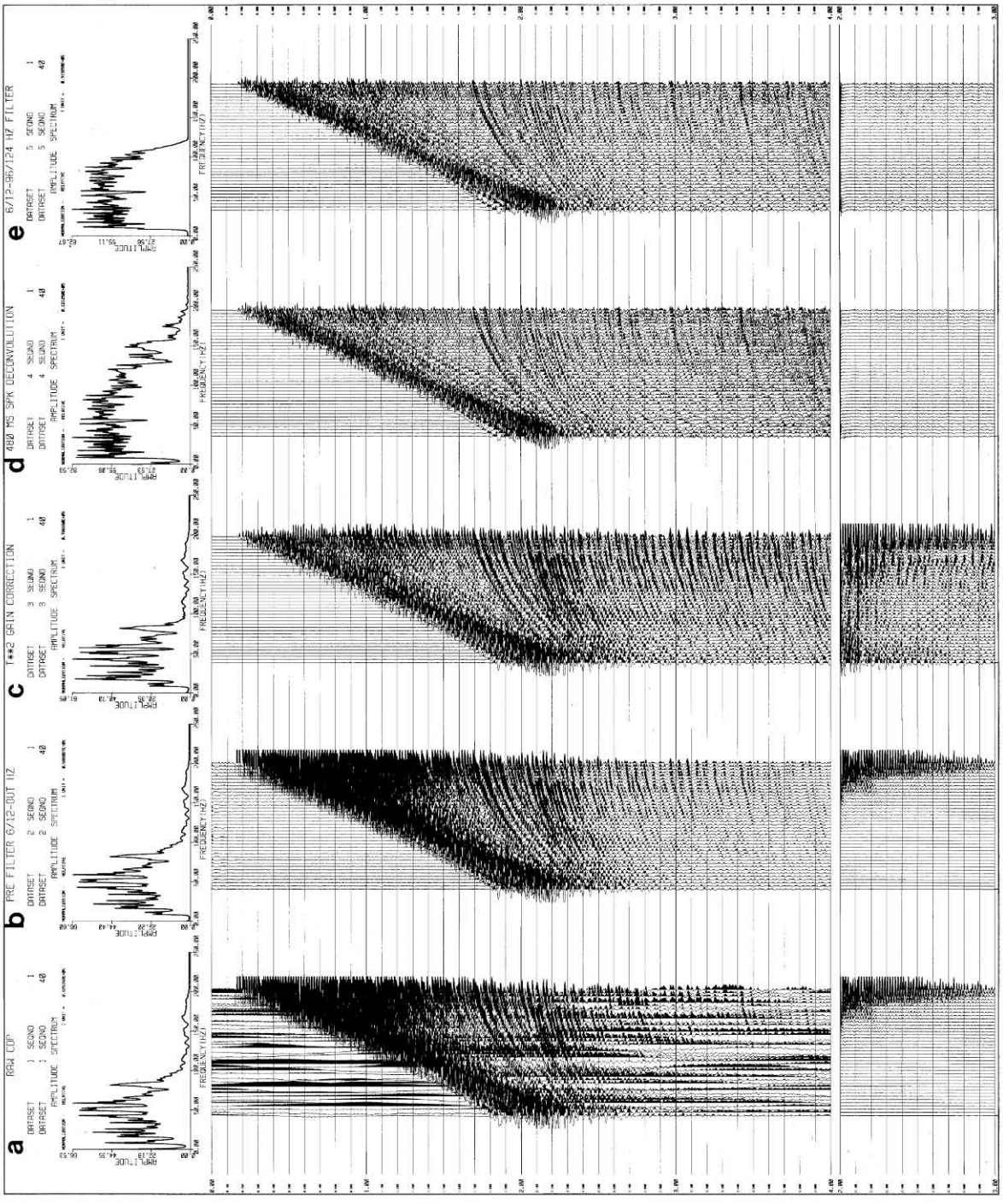

FIG. 1.5-32. Test panel for prestack signal processing. See text for details. (Data courtesy ONGC.)

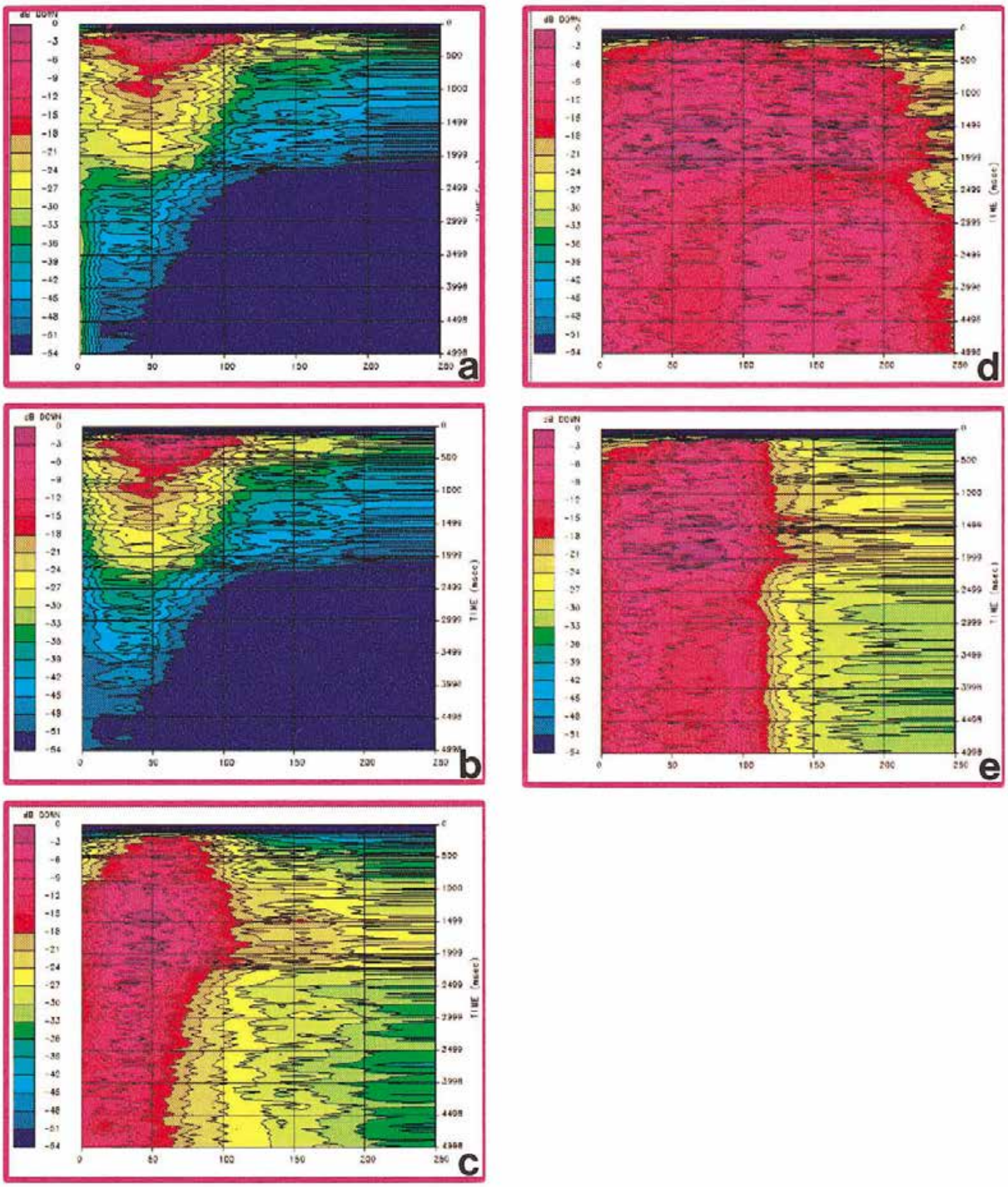

FIG. 1.5-33. Spectral analysis associated with the test panel in Figure 1.5-32. See text for details.

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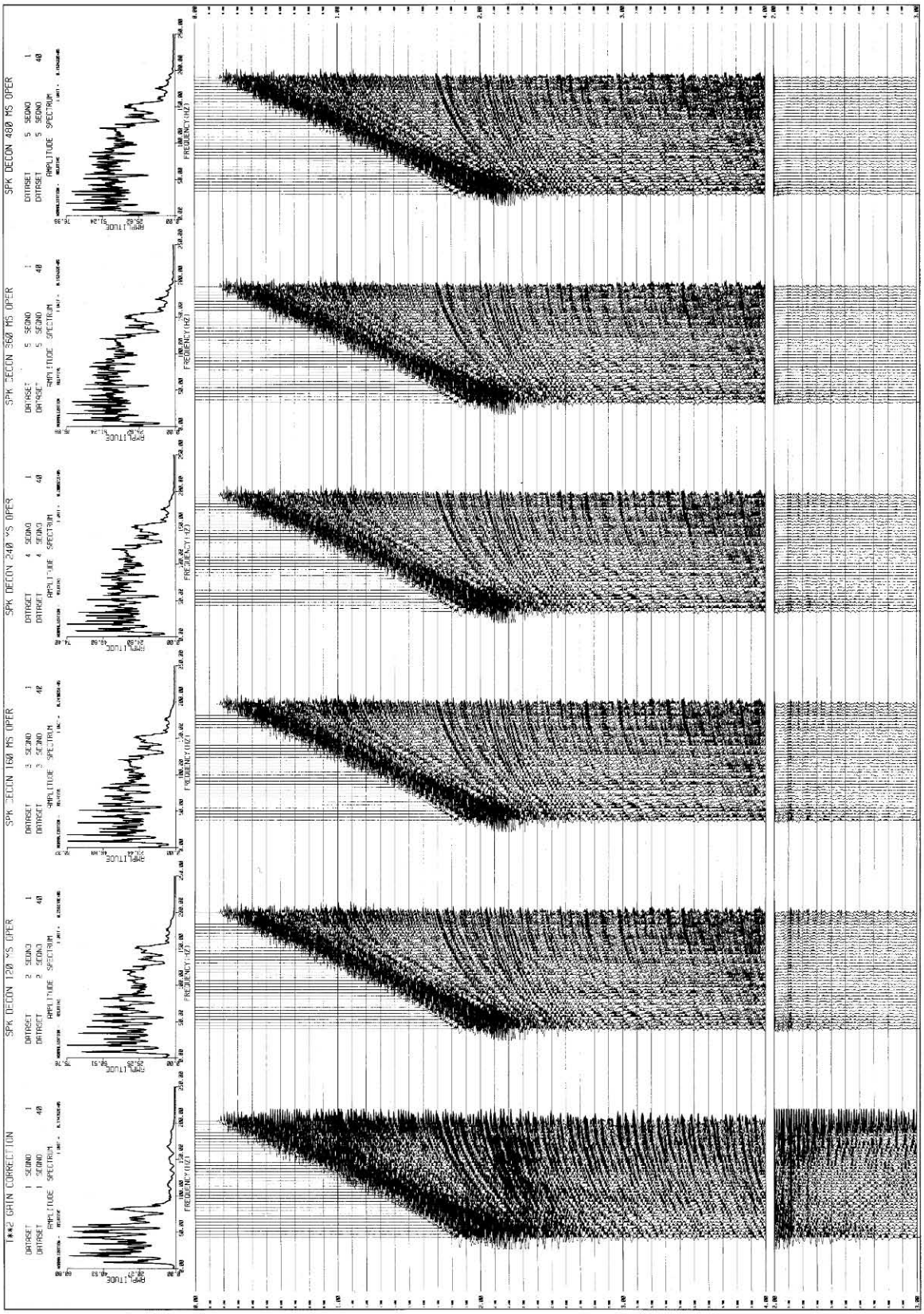

FIG. 1.5-34. Test panel for prestack spiking deconvolution. See text for details.

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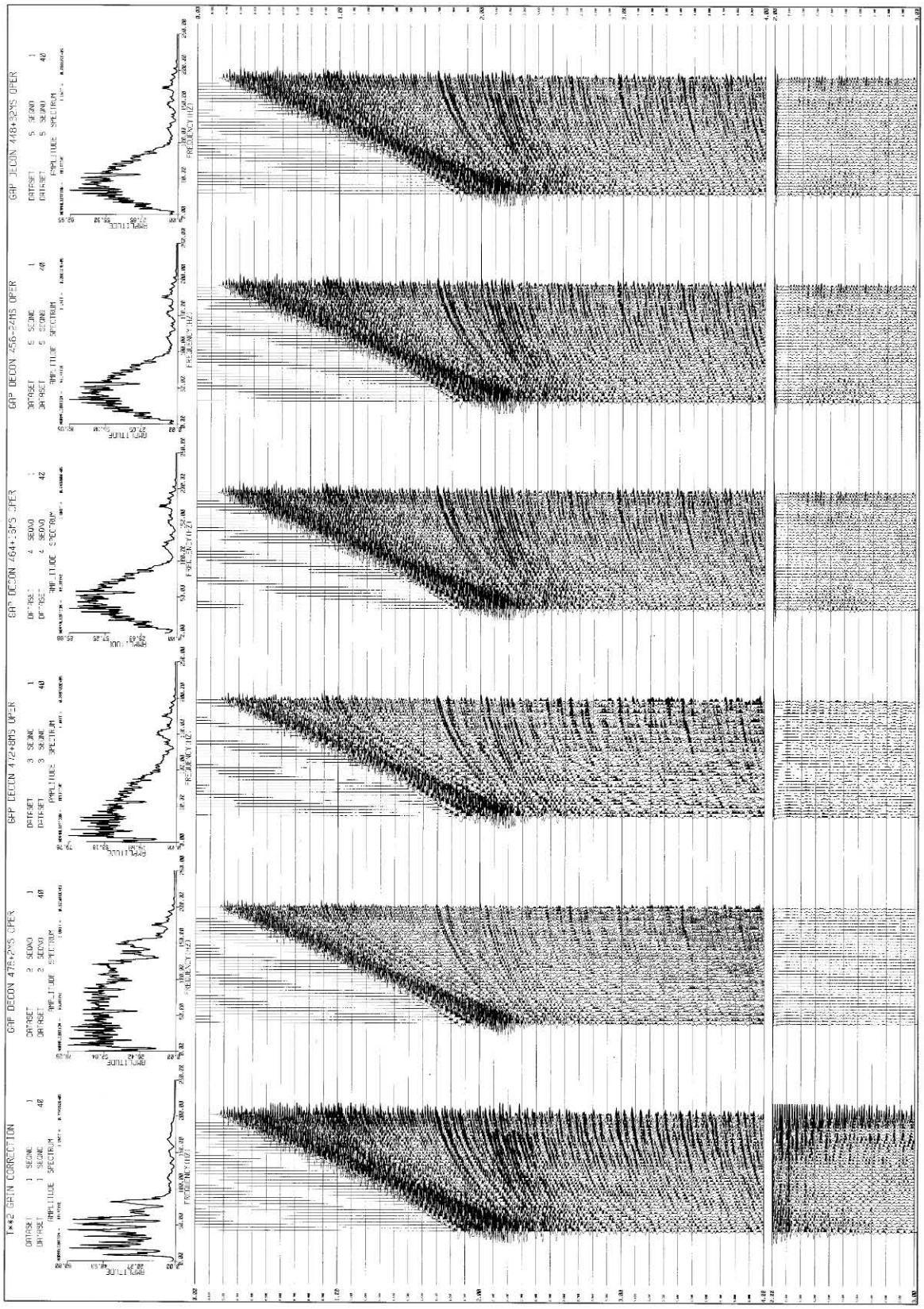

FIG. 1.5-35. Test panel for prestack predictive deconvolution. See text for details.

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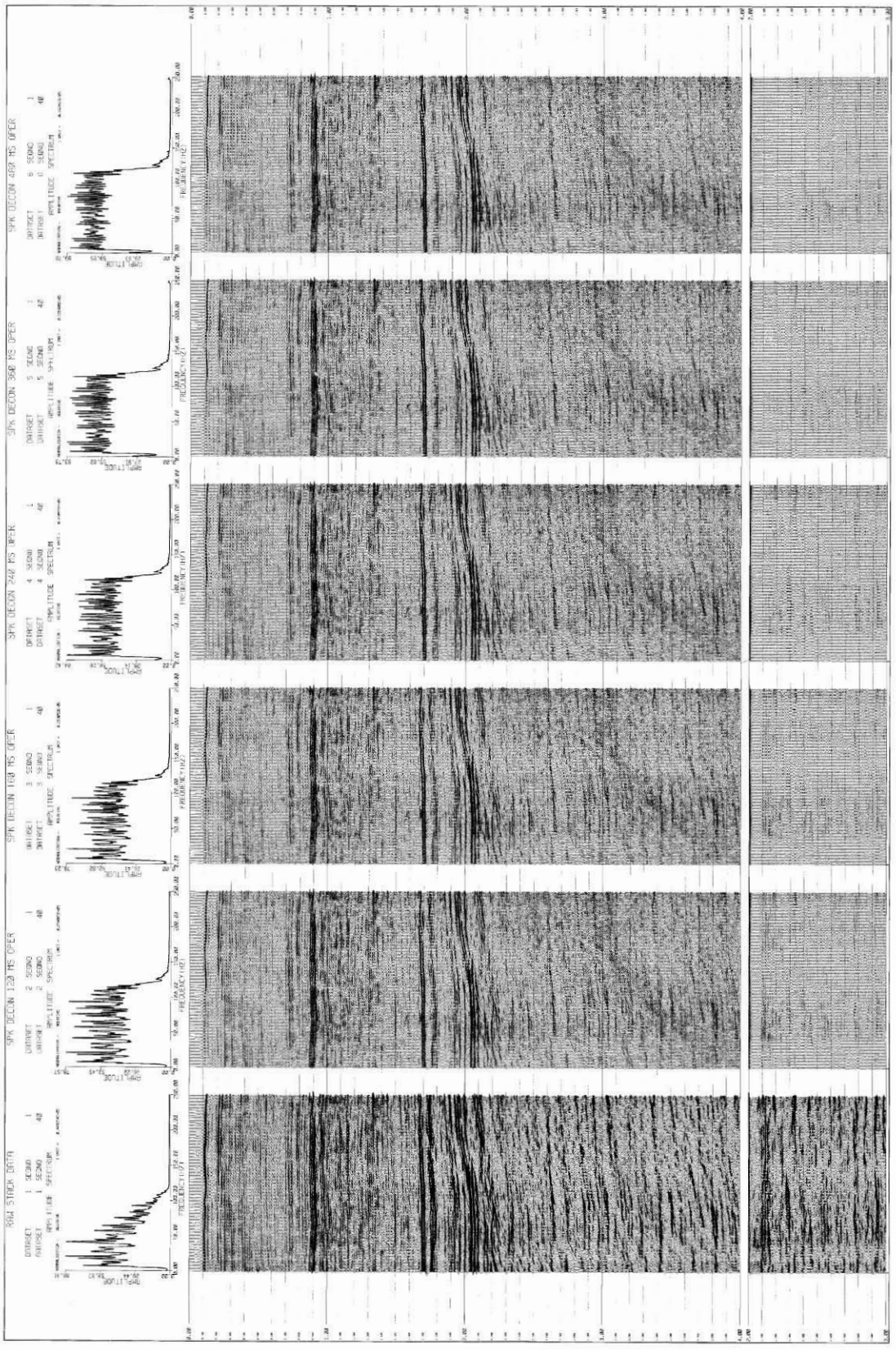

FIG. 1.5-36. Test panel for poststack spiking deconvolution. See text for details.

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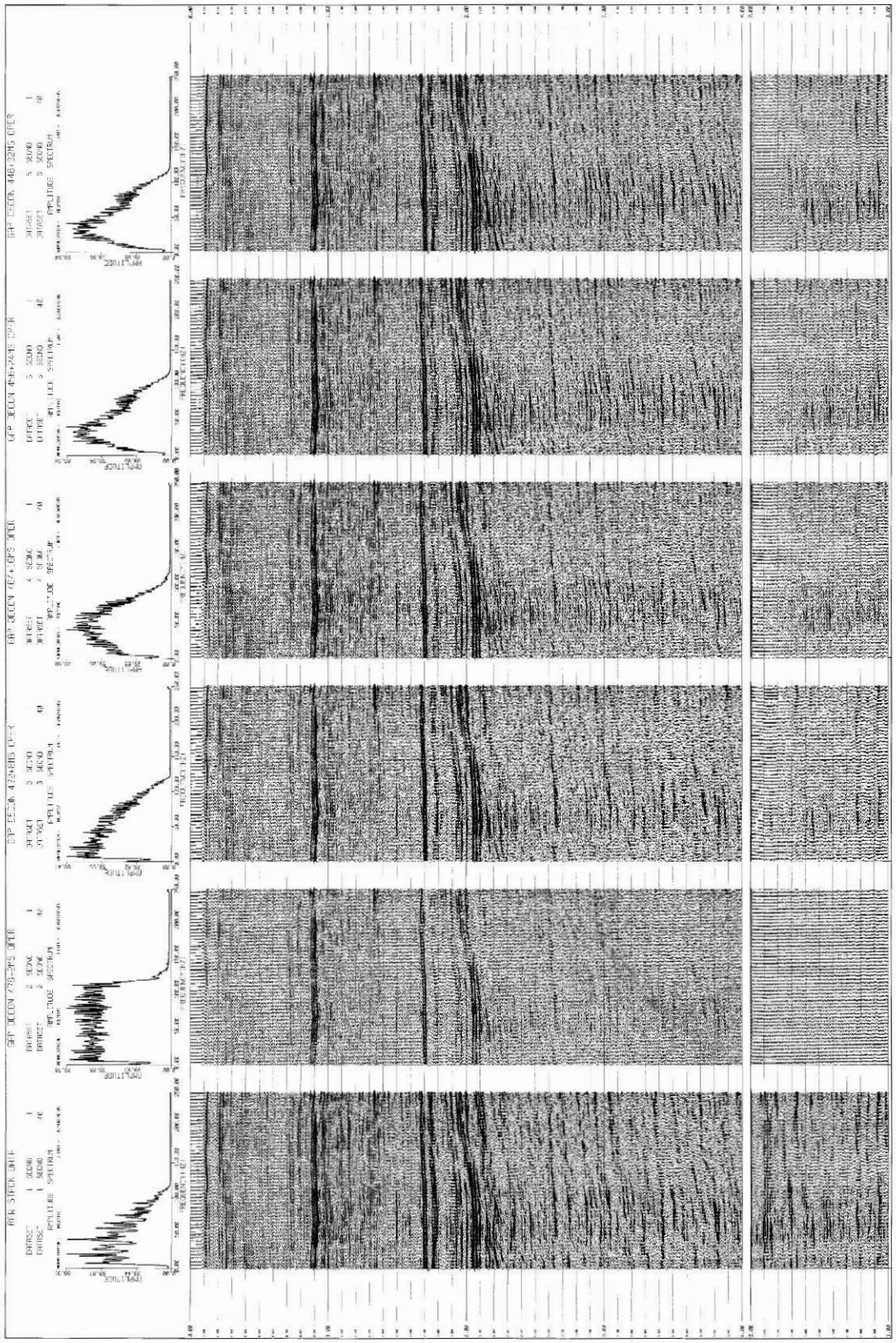

FIG. 1.5-37. Test panel for poststack predictive deconvolution. See text for details.

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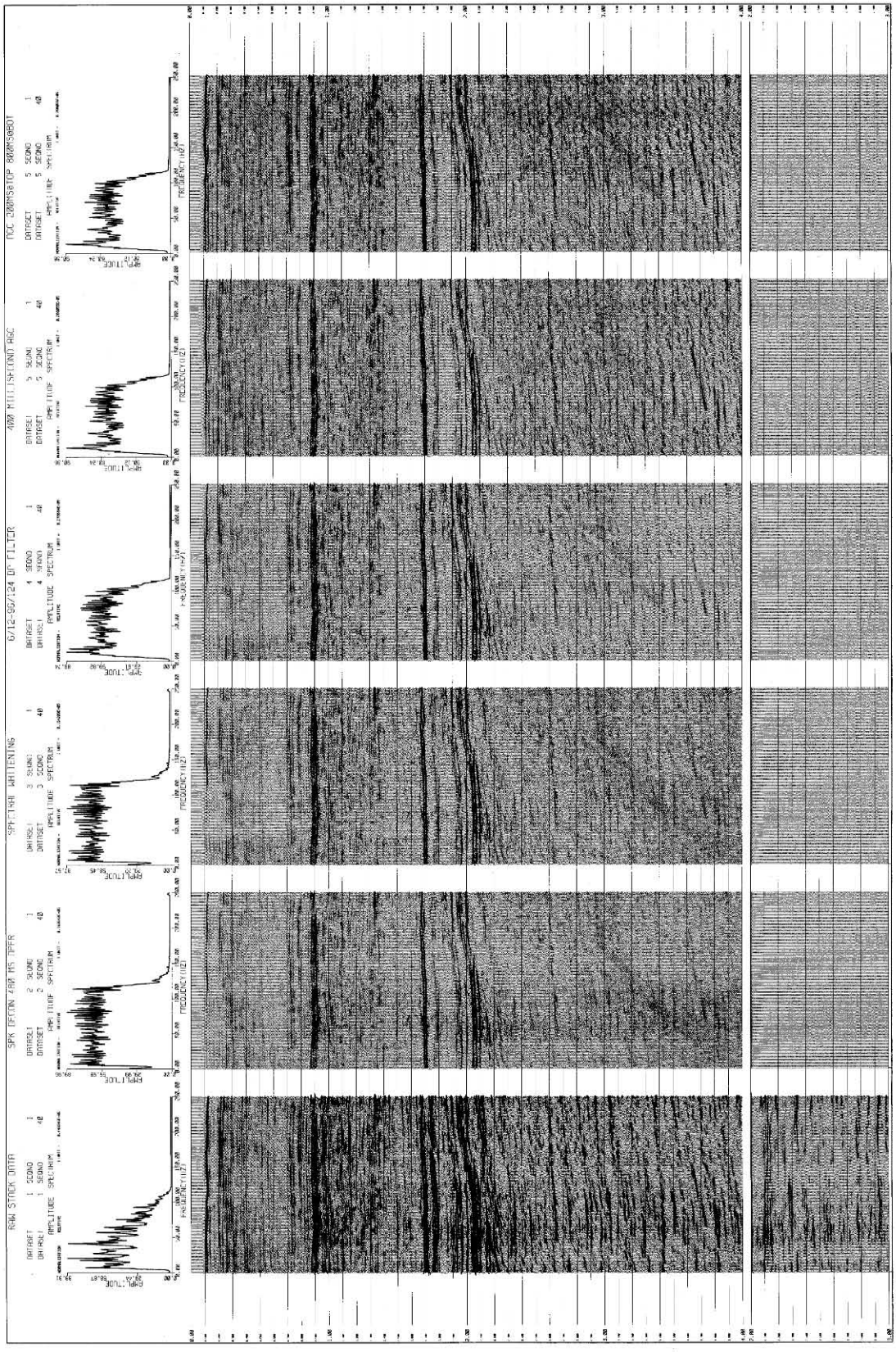

FIG. 1.5-38. Test panel for poststack signal processing. See text for details.

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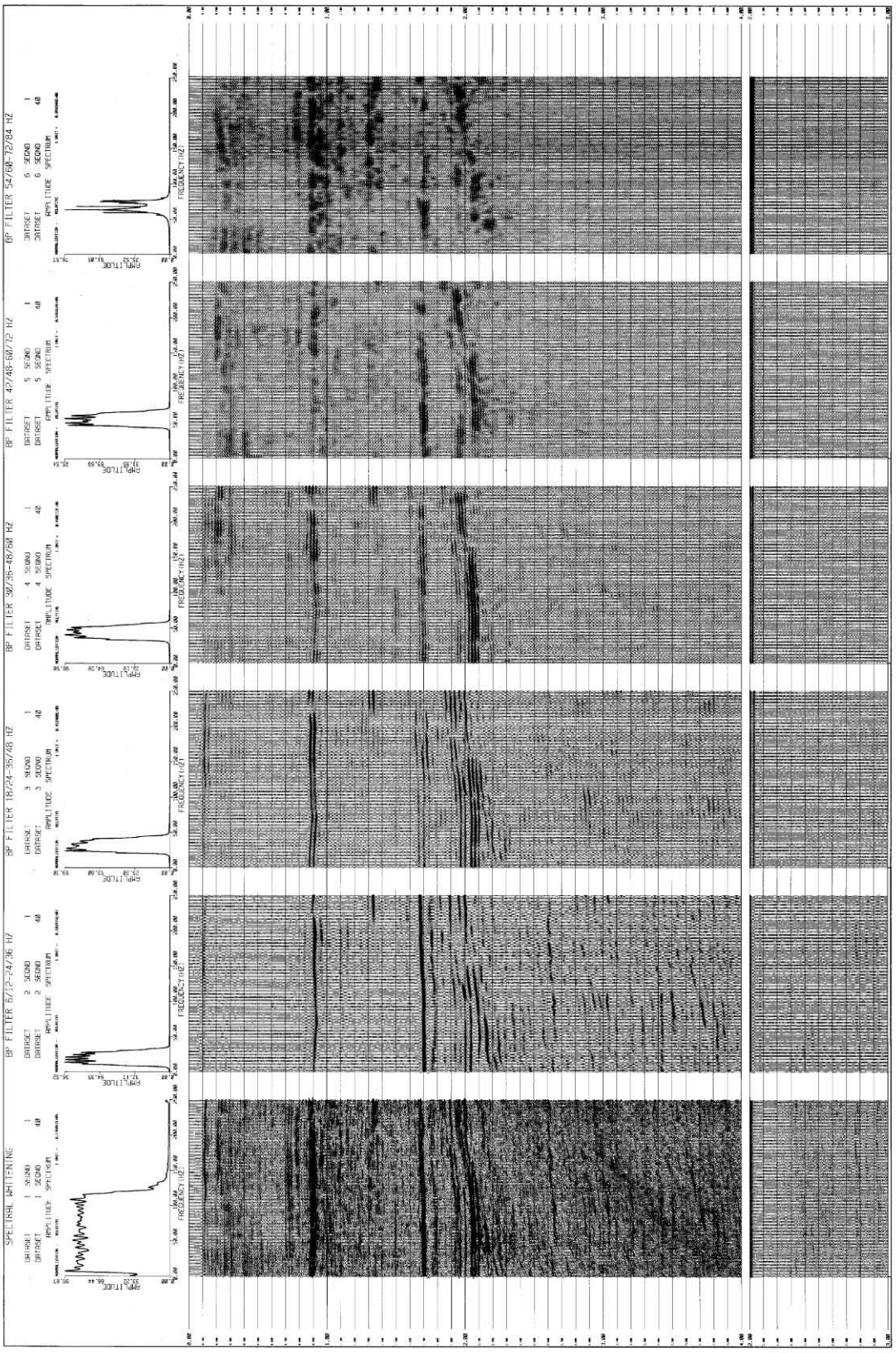

FIG. 1.5-39. Filter test panel (first half). See text for details.

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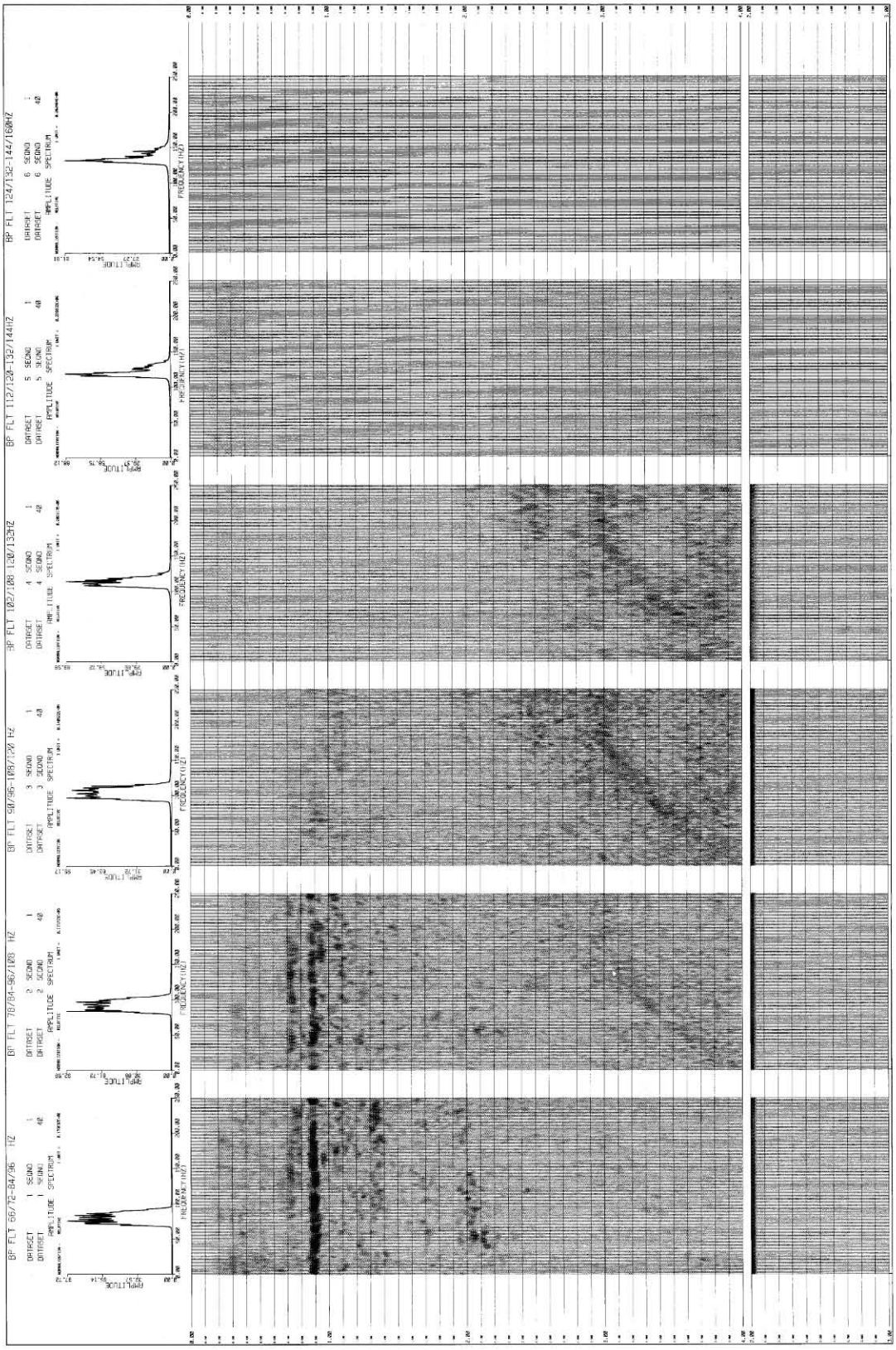

FIG. 1.5-40. Filter test panel (second half). See text for details.

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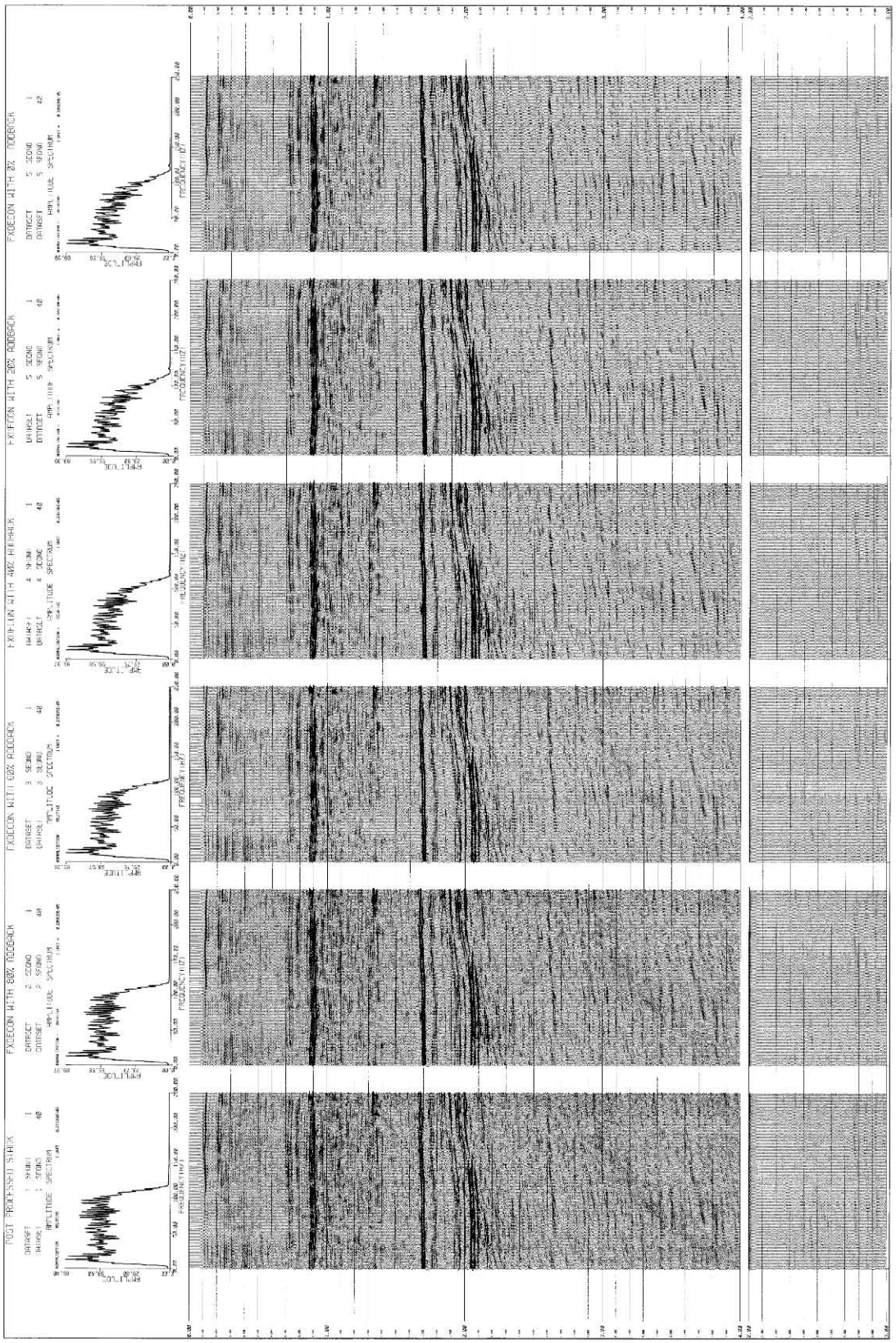

FIG. 1.5-41. Random noise attenuation test panel. See text for details.

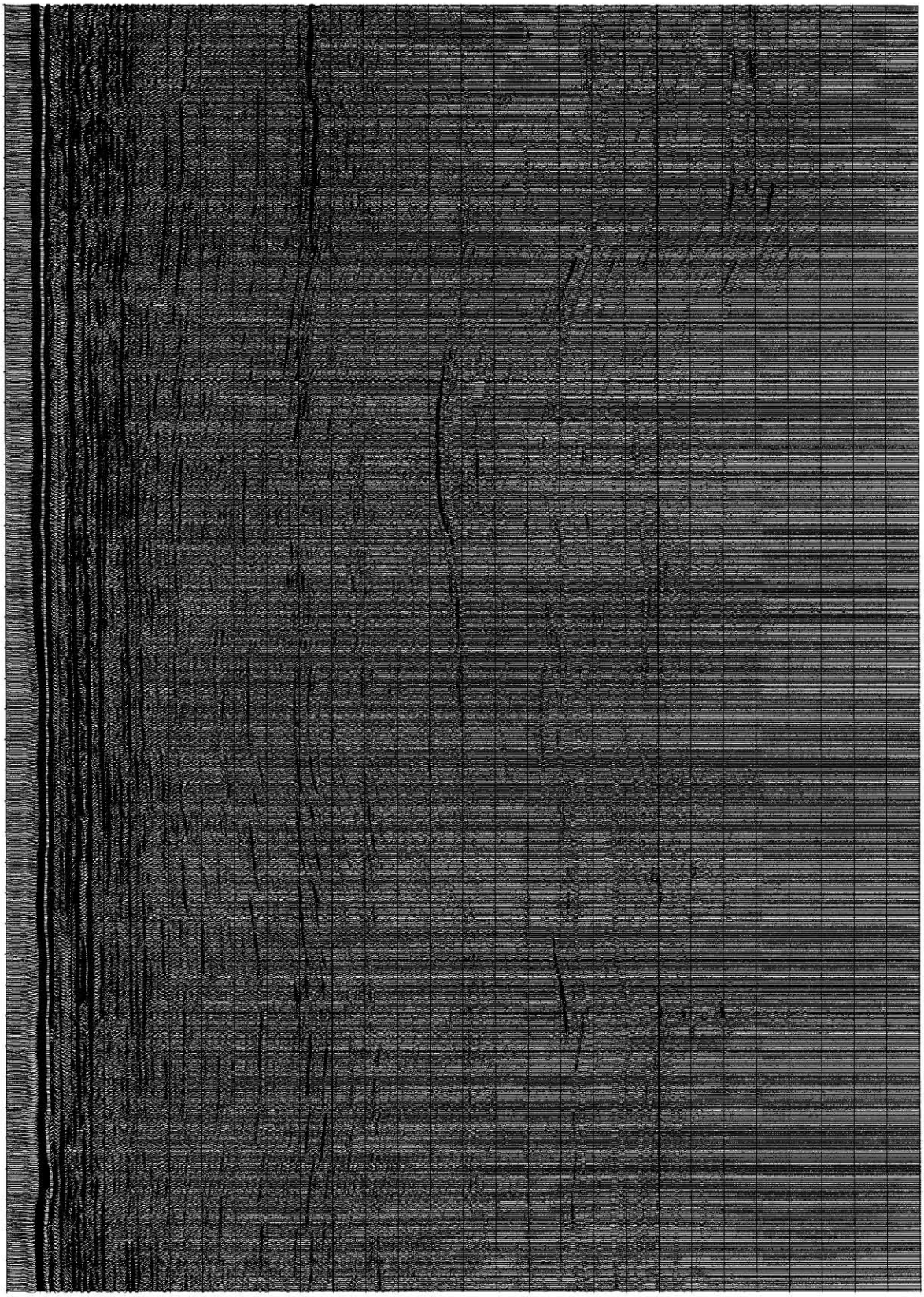

FIG. 1.5-42. CMP stack based on recorded data without any prestack and poststack processing.

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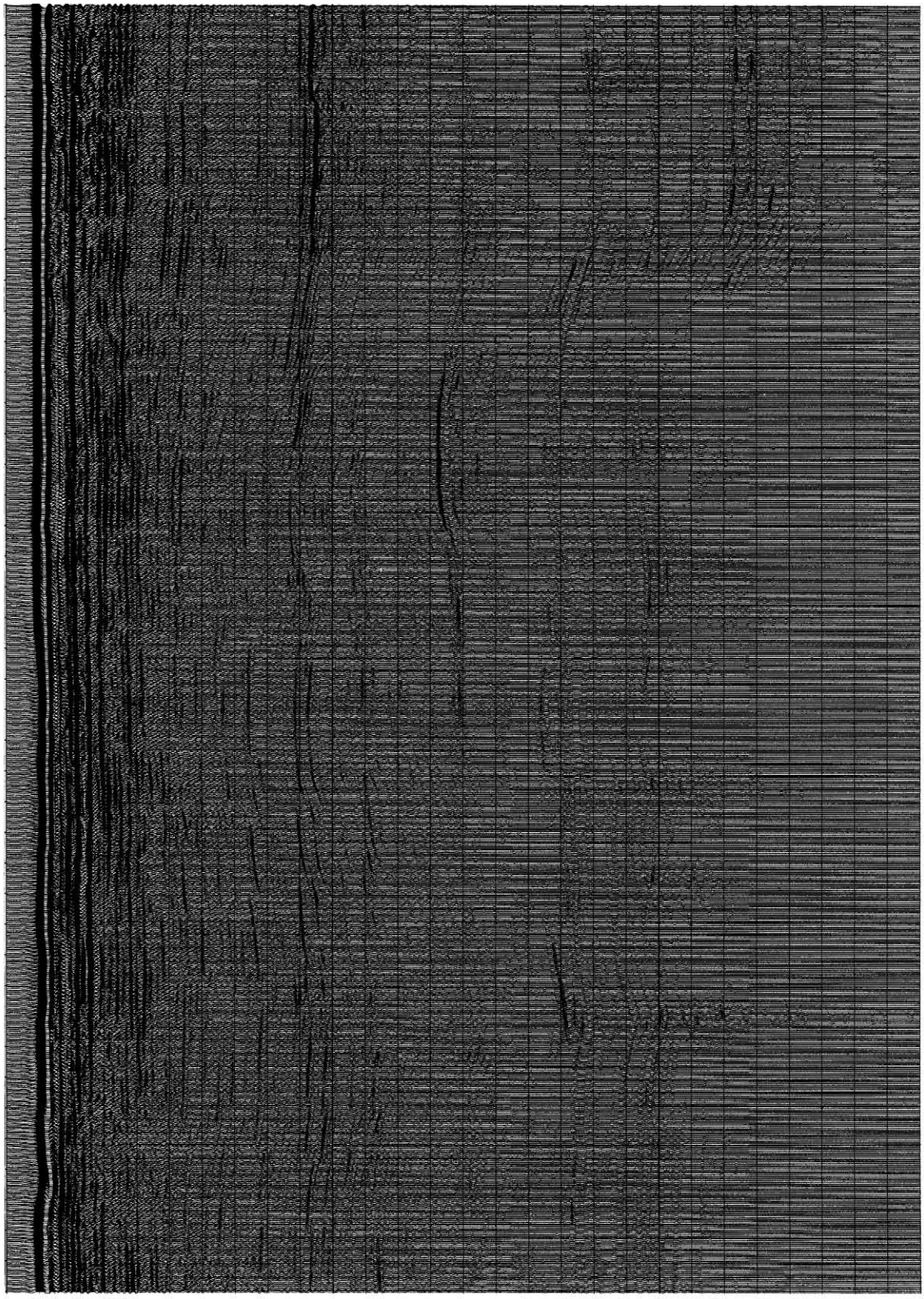

FIG. 1.5-43. CMP stack as in figure 1.5-42 based on a processing sequence that includes low-cut filtering of the prestack data, only.

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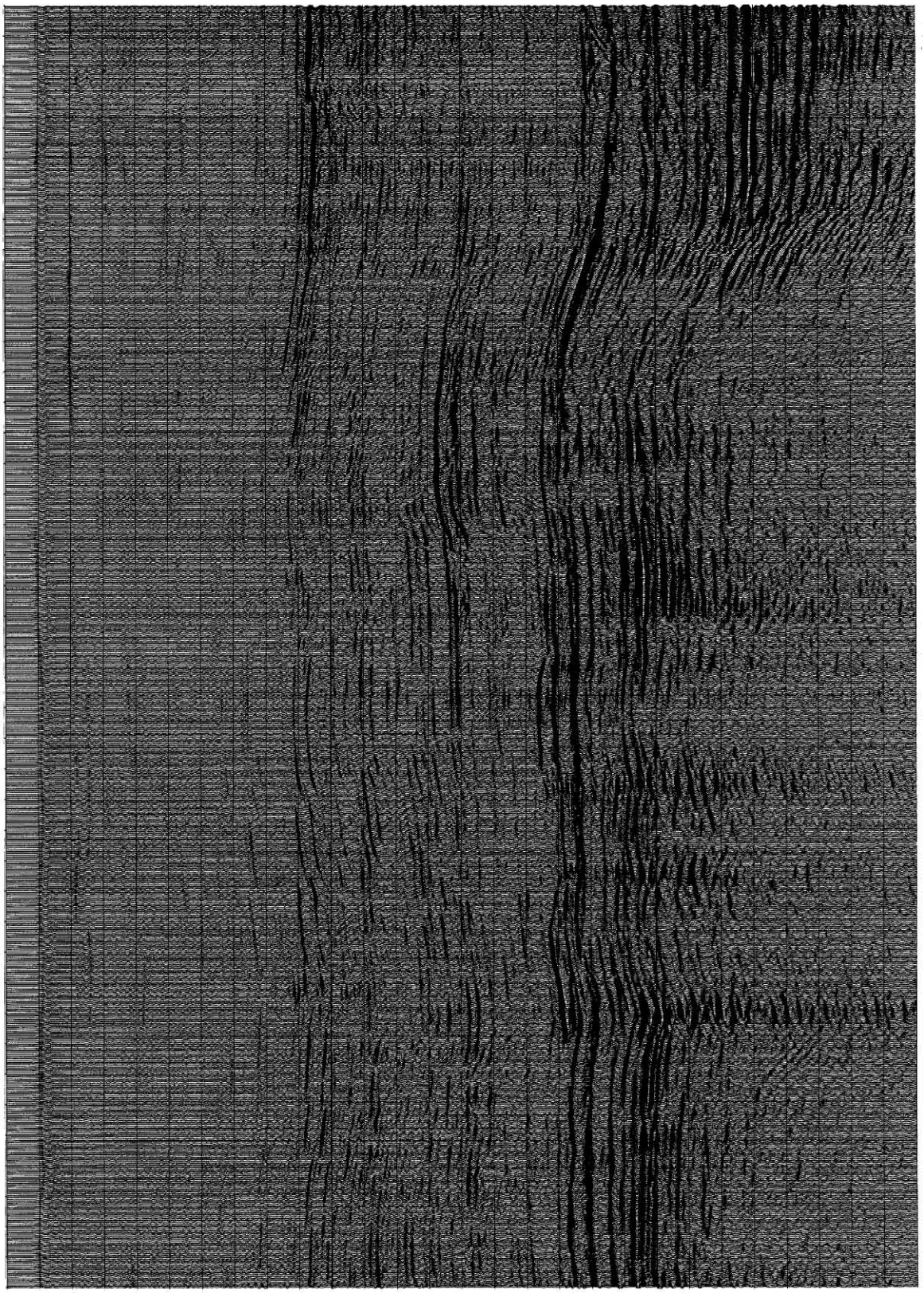

FIG. 1.5-44. CMP stack as in Figure 1.5-43 with the additional step in the processing sequence for t^2 scaling of the prestack data.

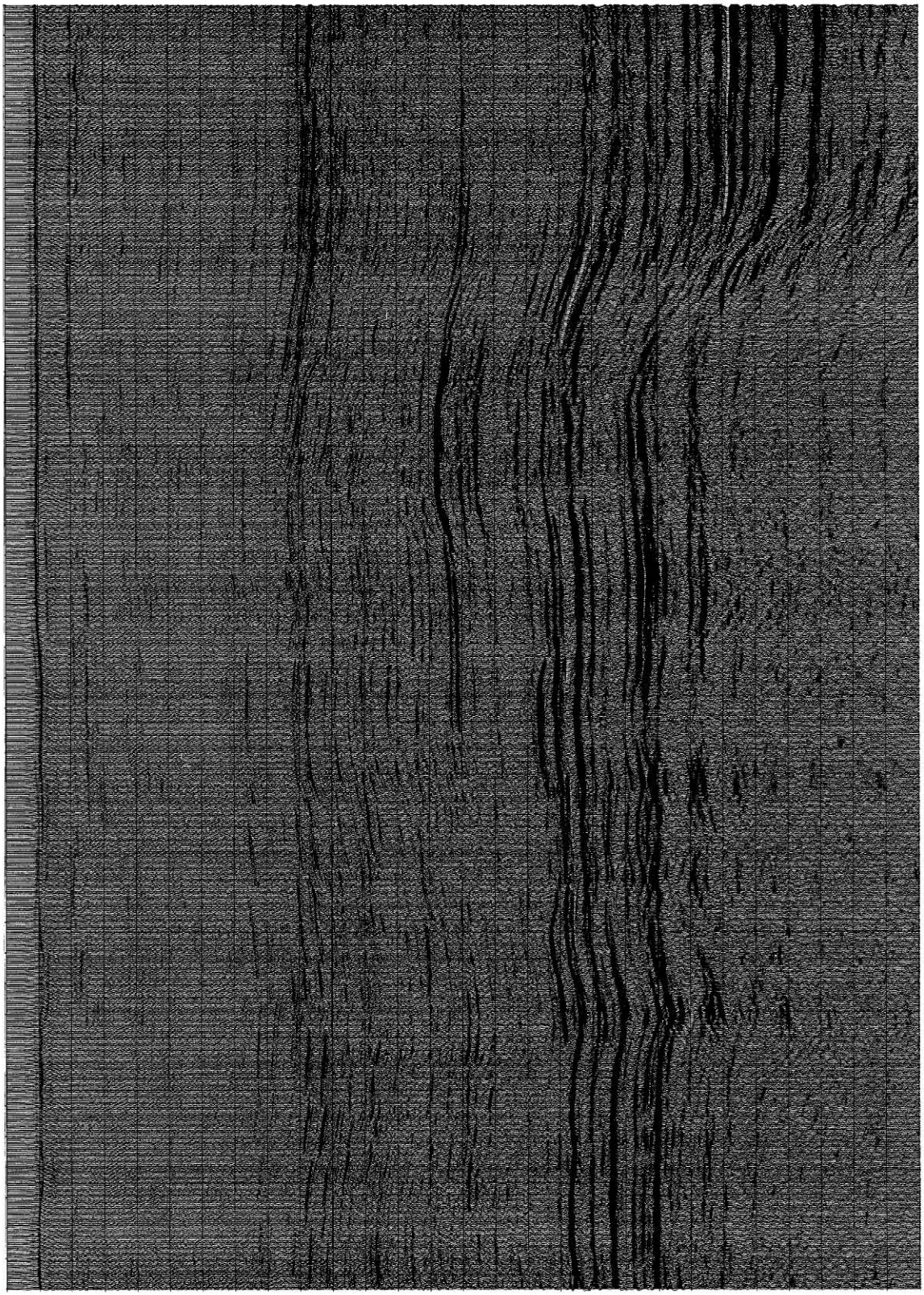

FIG. 1.5-45. CMP stack as in Figure 1.5-44 with the additional step in the processing sequence for spiking deconvolution of the prestack data.

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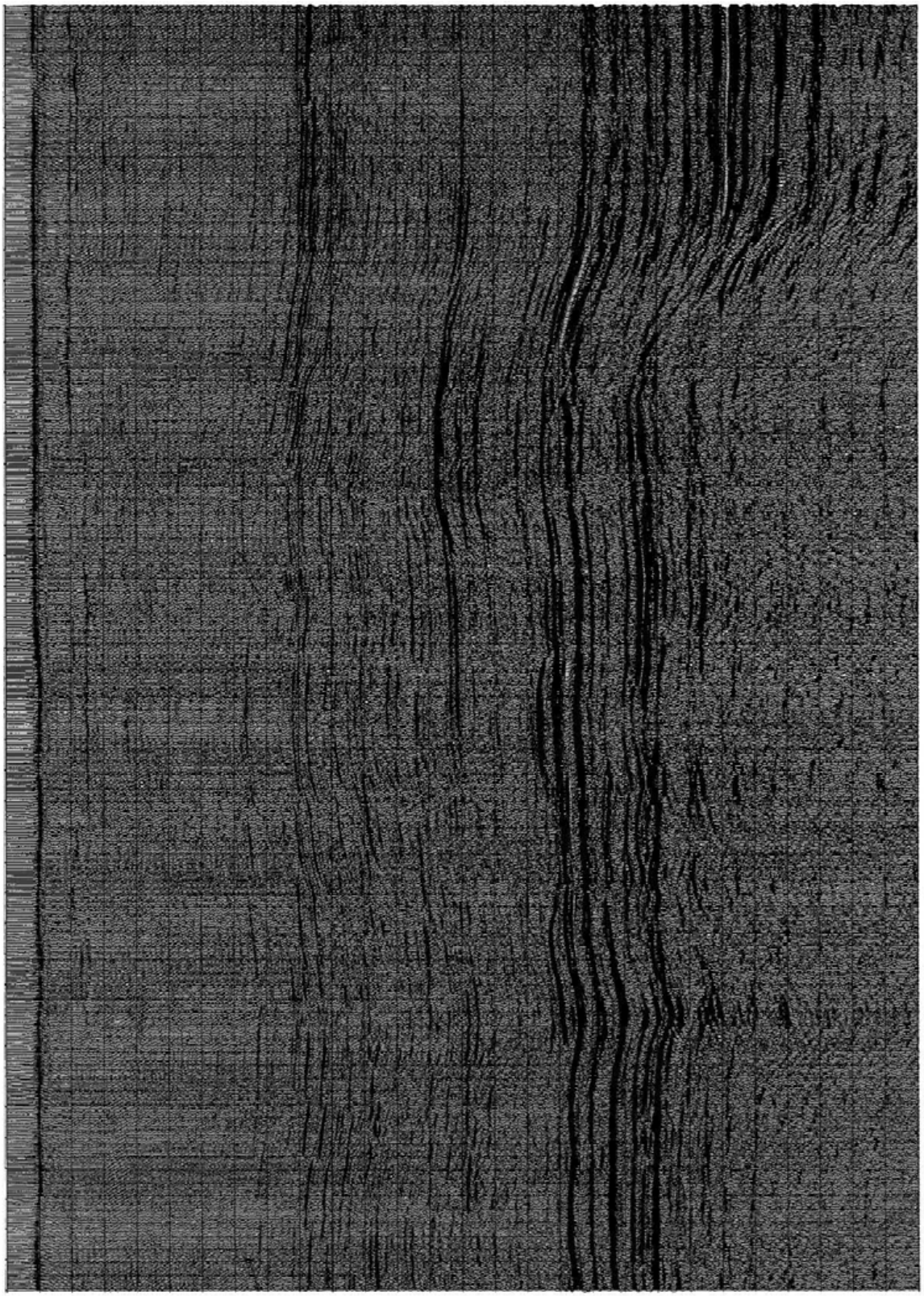

FIG. 1.5-46. CMP stack as in Figure 1.5-45 with the additional step in the processing sequence for wide bandpass filtering of the prestack data.

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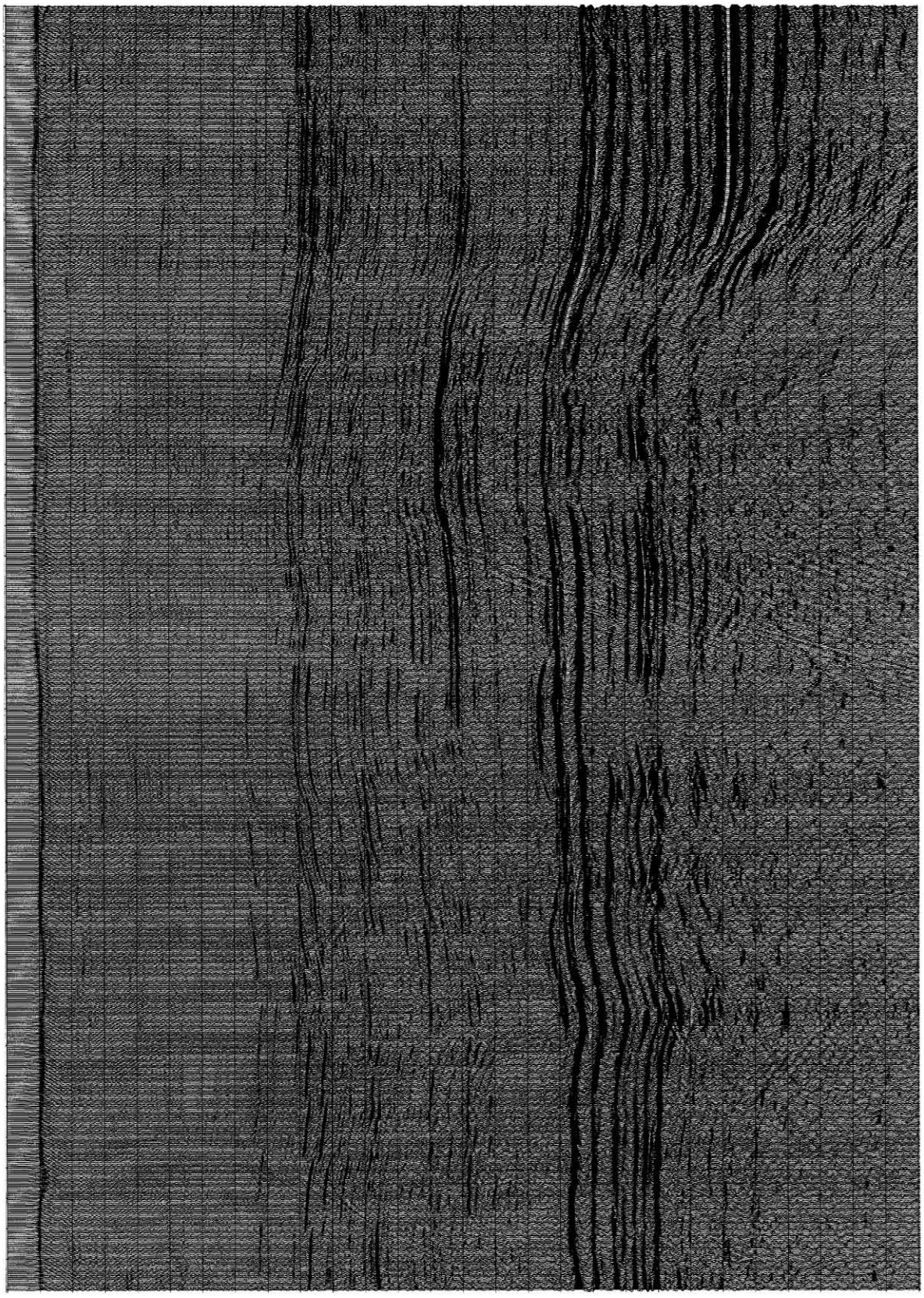

FIG. 1.5-47. CMP stack as in Figure 1.5-46 with the additional step in the processing sequence for dip-moveout correction.

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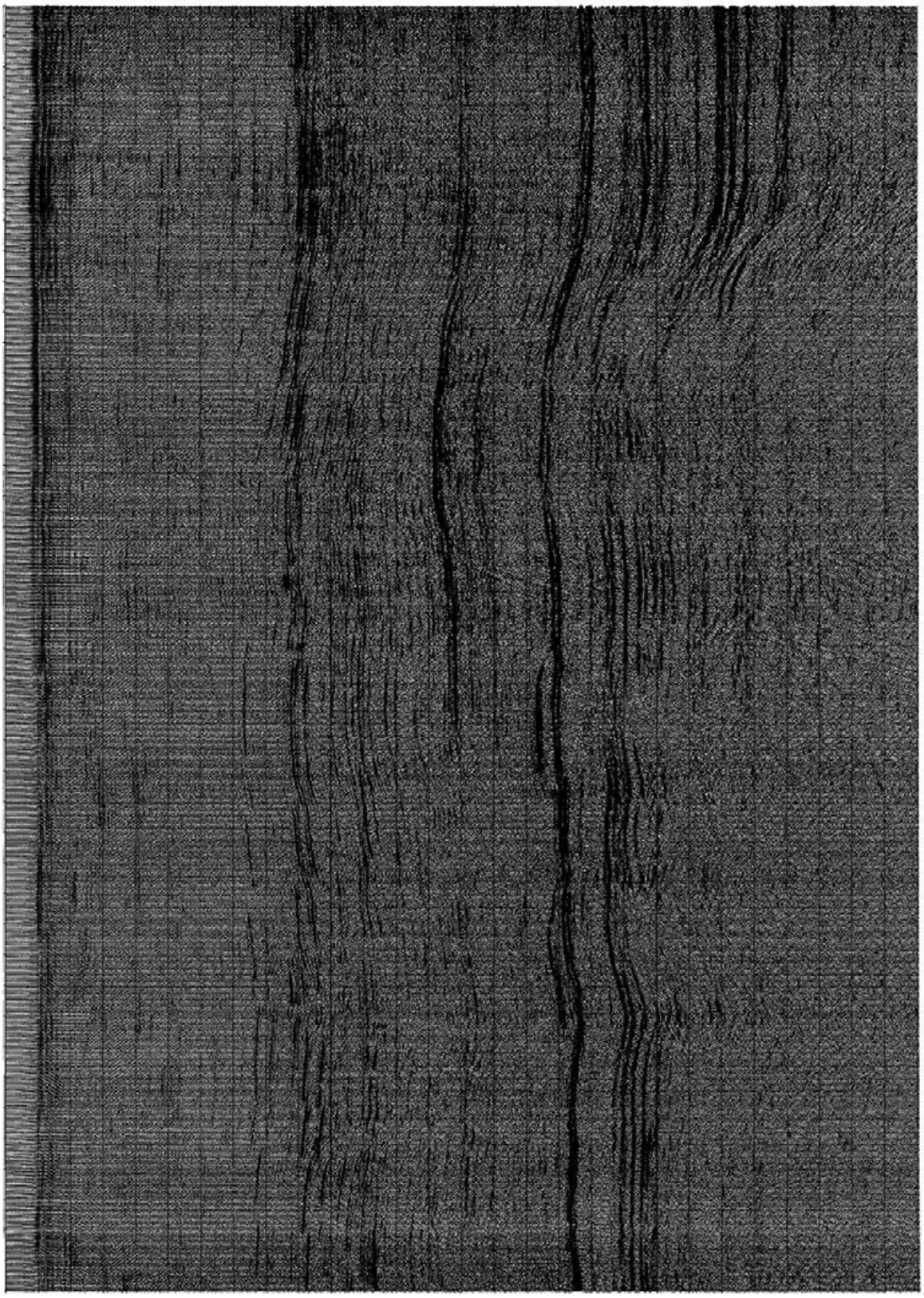

FIG. 1.5-48. CMP stack as in Figure 1.5-47 with the additional step in the processing sequence for spiking deconvolution of the poststack data.

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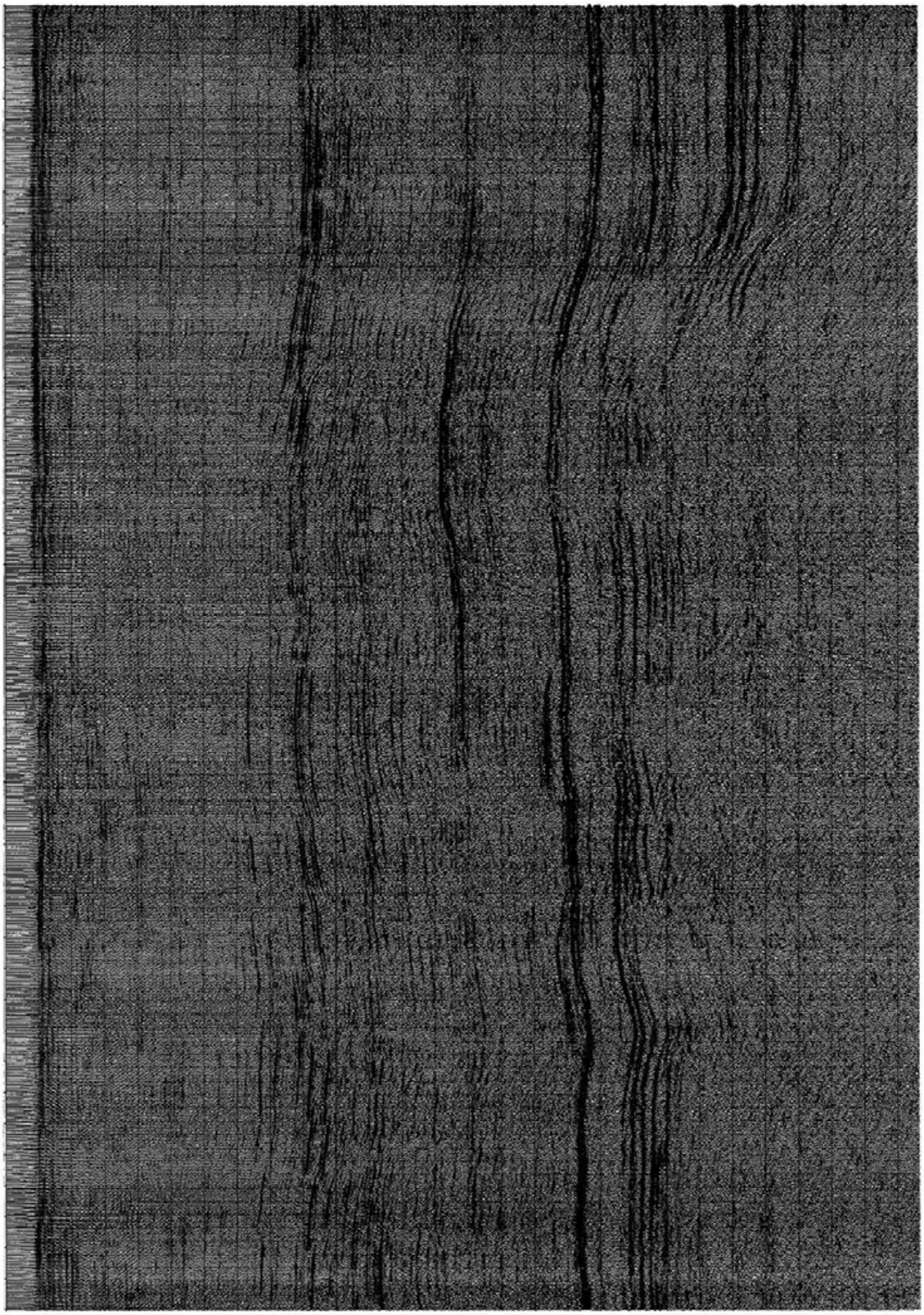

FIG. 1.5-49. CMP stack as in Figure 1.5-48 with the additional step in the processing sequence for time-variant spectral whitening of the poststack data.

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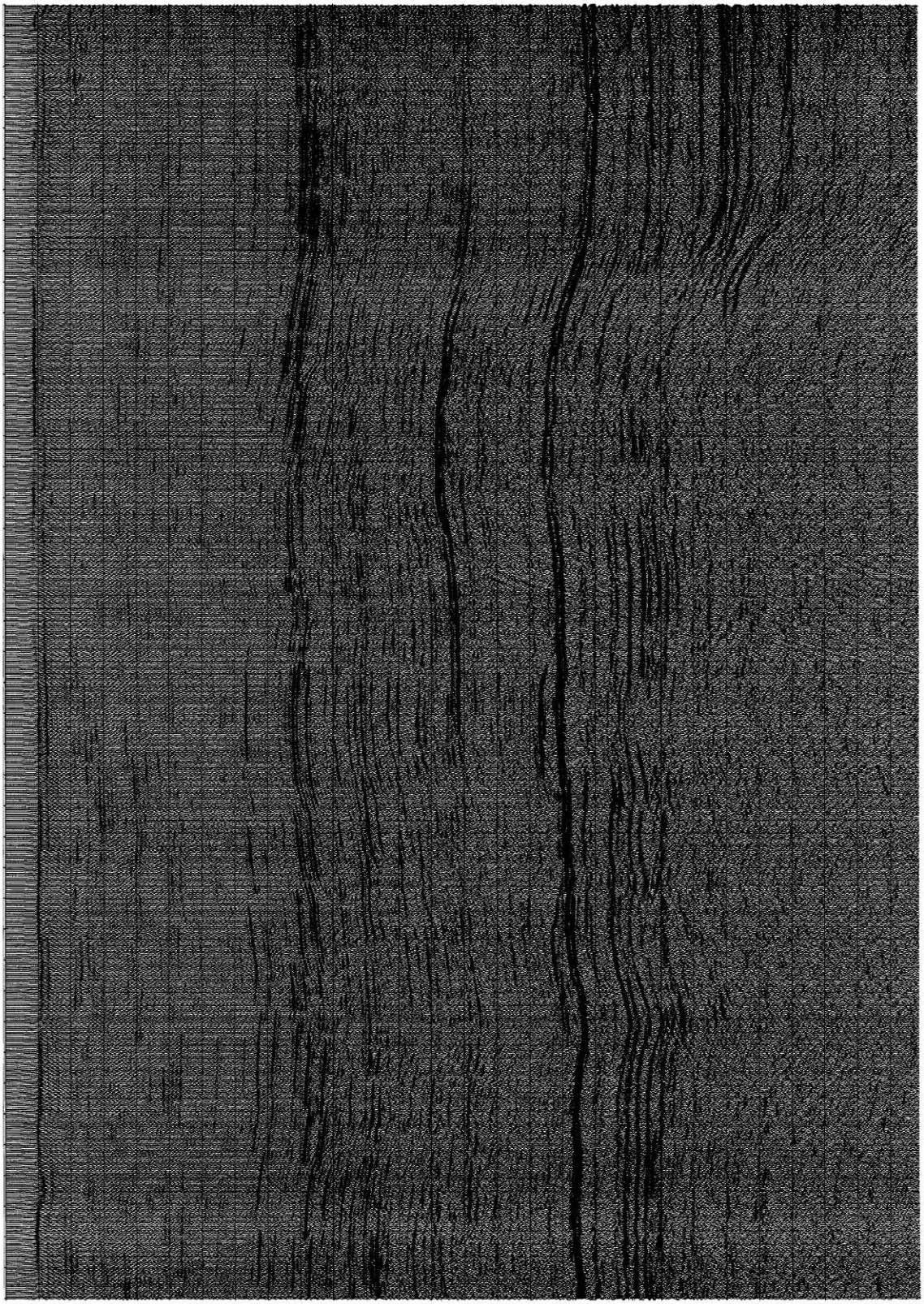

FIG. 1.5-50. CMP stack as in Figure 1.5-49 with the additional step in the processing sequence for wide bandpass filtering of the poststack data.

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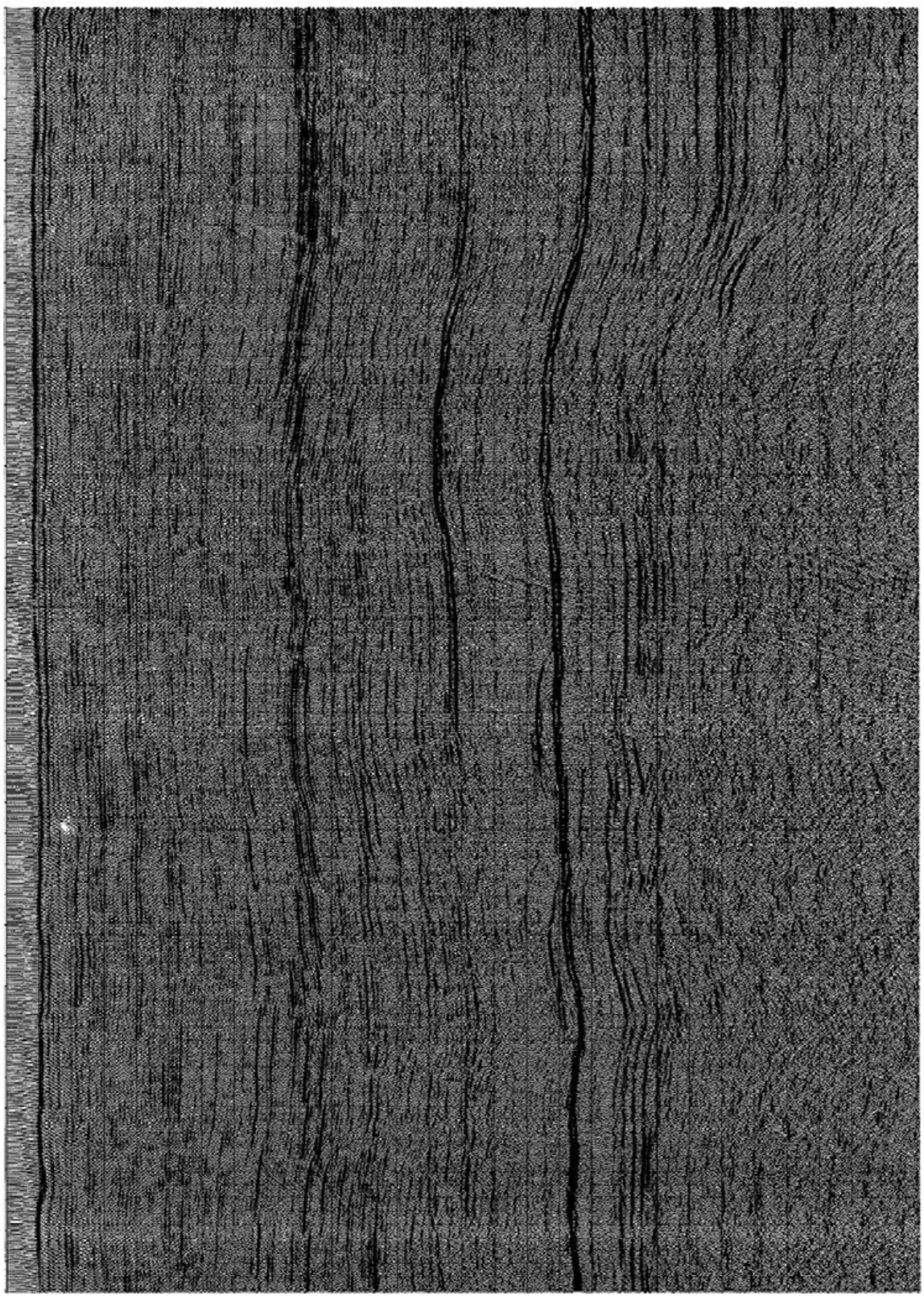

FIG. 1.5-51. CMP stack as in Figure 1.5-50 with the additional step in the processing sequence for AGC scaling of the poststack data.

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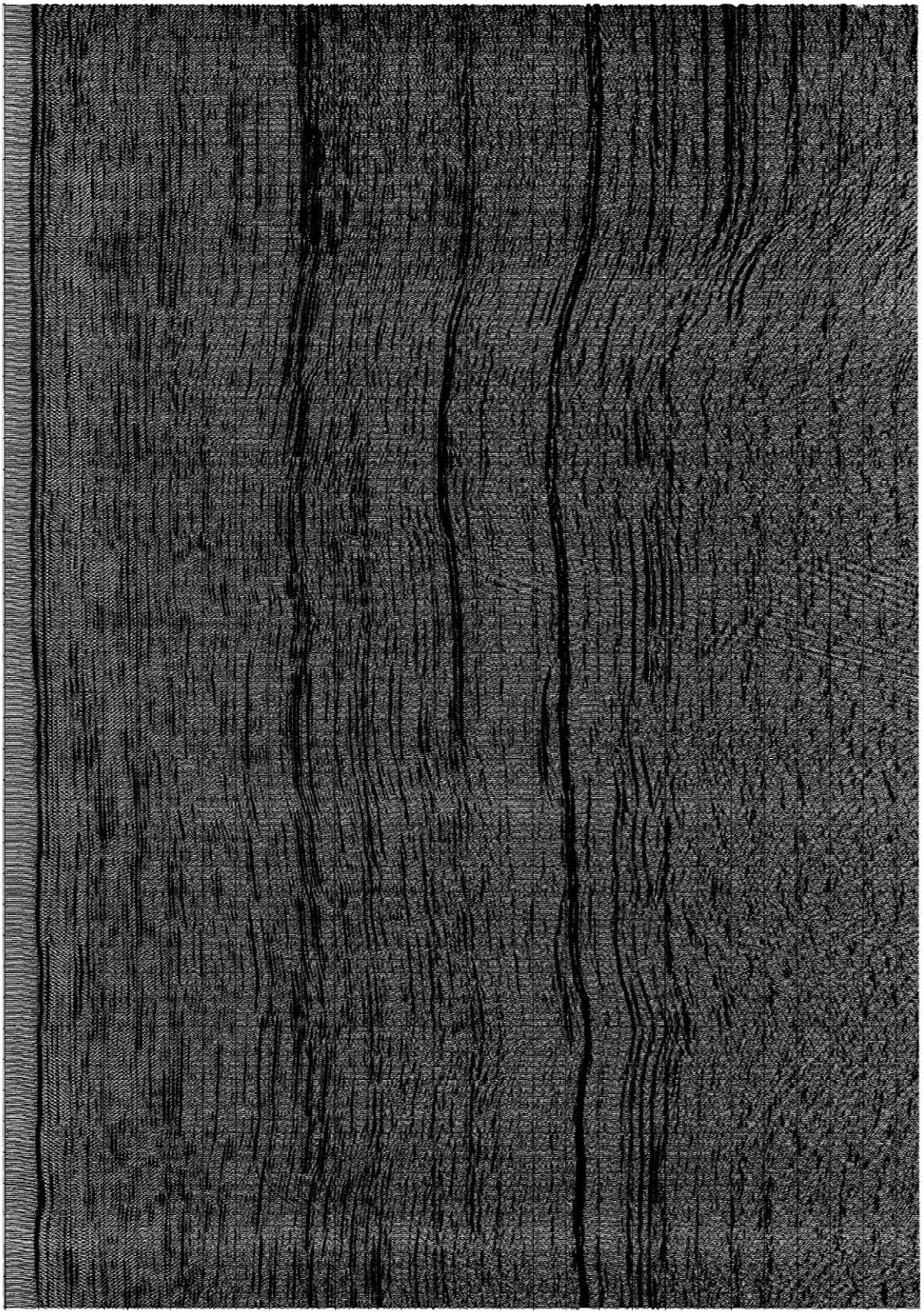

FIG. 1.5-52. CMP stack as in Figure 1.5-51 with the additional step in the processing sequence for $f - x$ deconvolution of poststack data.

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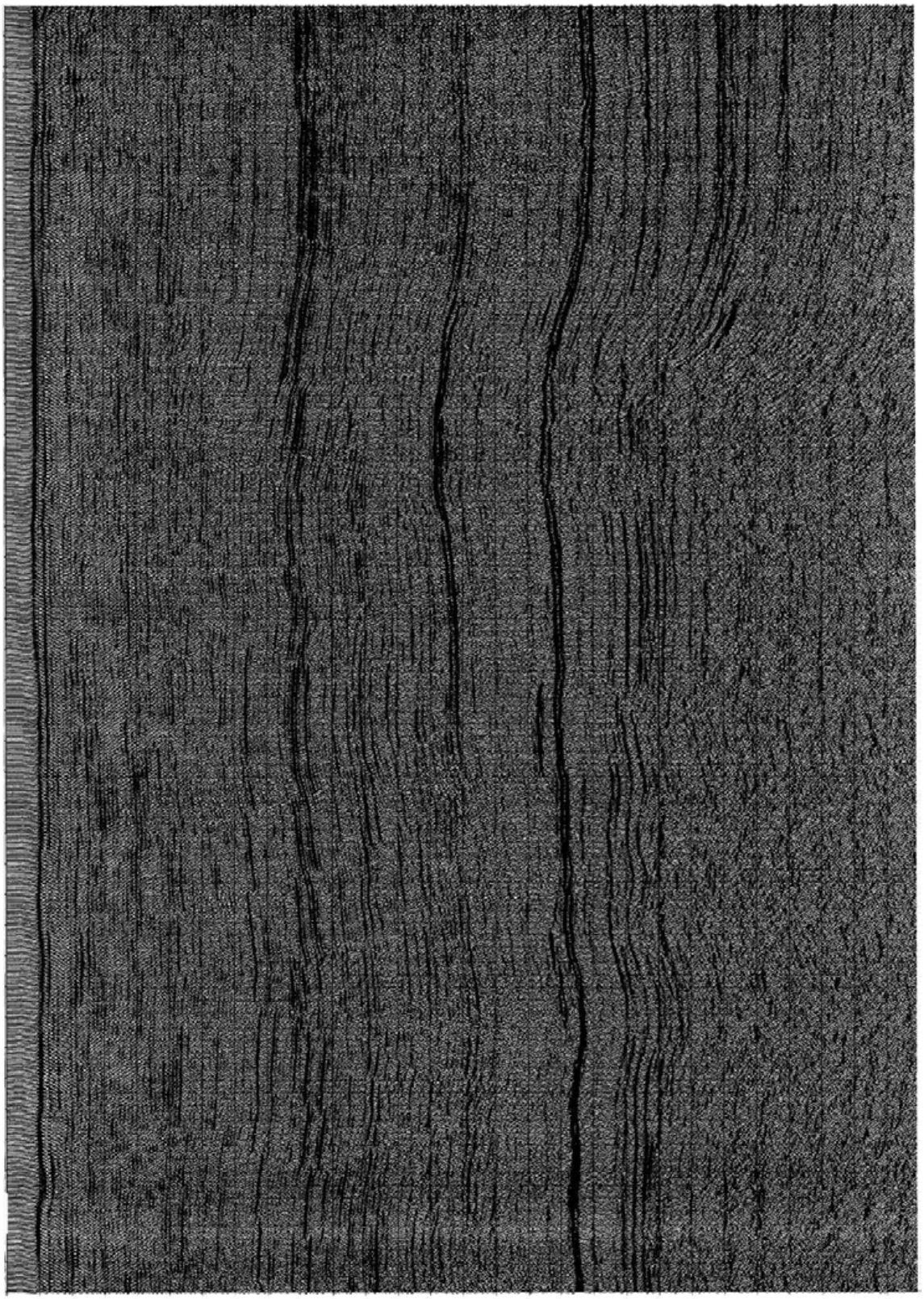

FIG. 1.5-53. Migrated CMP stack as in Figure 1.5-50 with the additional step in the processing sequence for AGC scaling of the migrated data.

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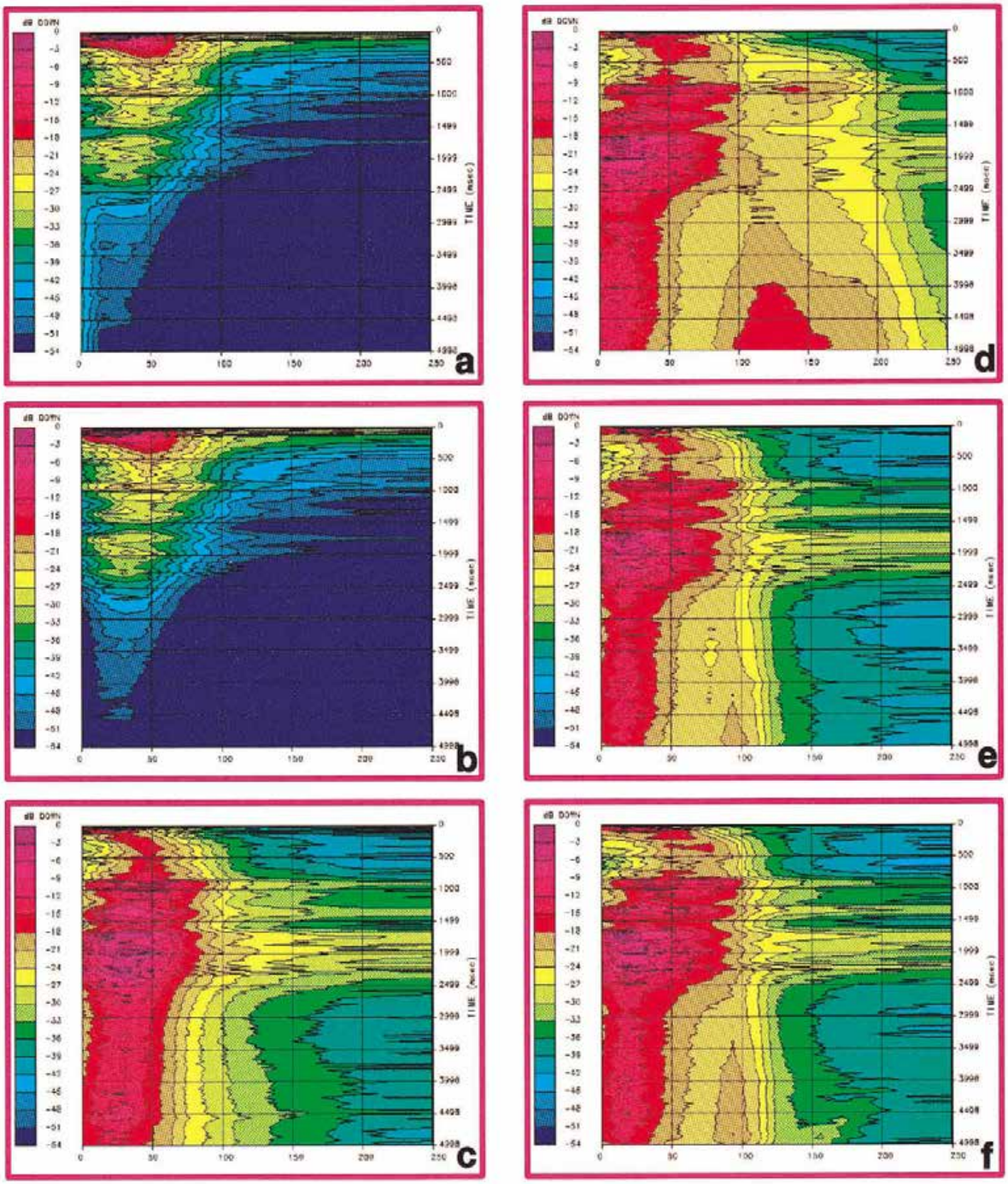

FIG. 1.5-54. Spectral analysis of the sections based on the prestack processing shown in Figures 1.5-42 through 1.5-47. See text for details.

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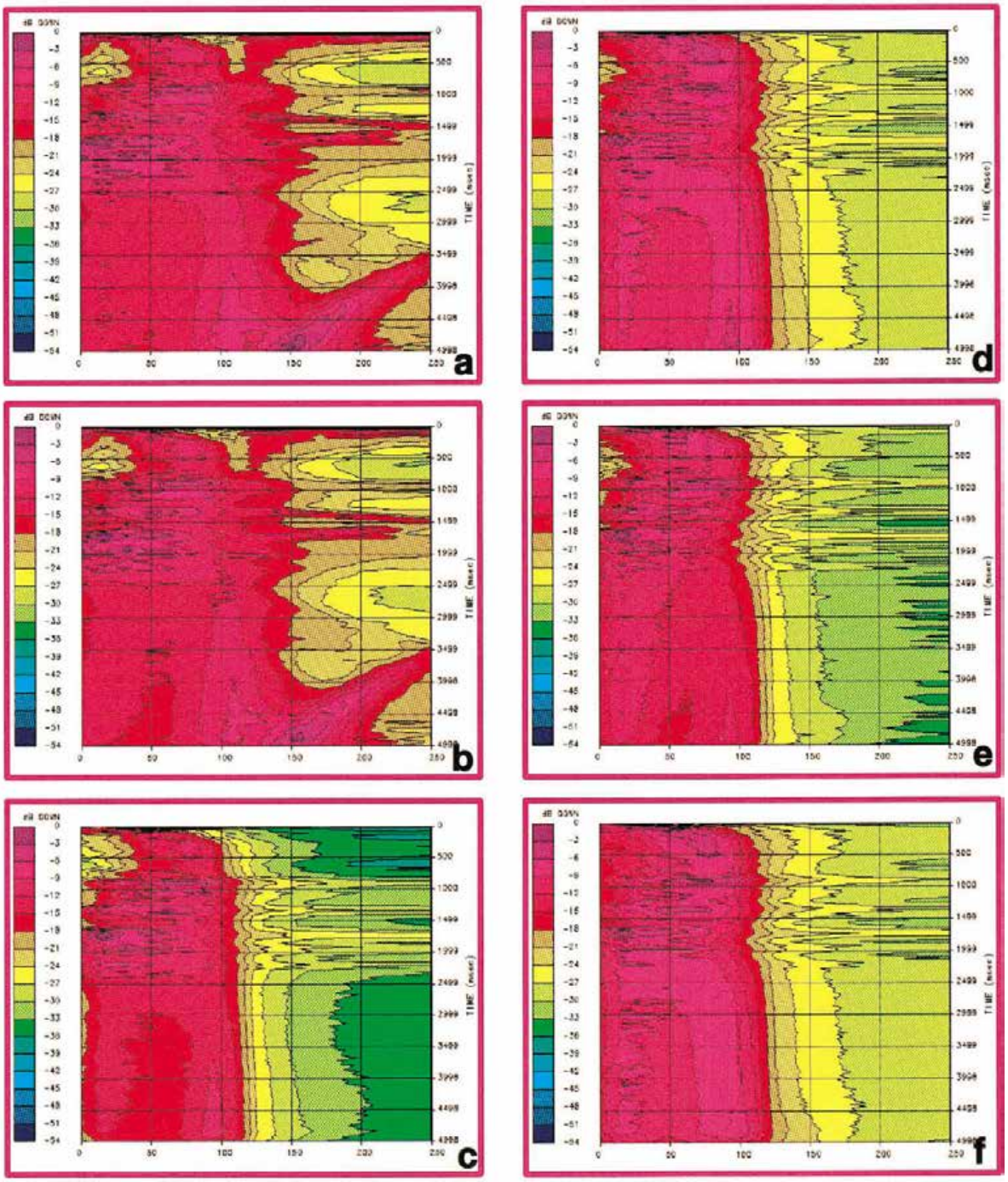

FIG. 1.5-55. Spectral analysis of the sections based on the poststack processing shown in Figures 1.5-48 through 1.5-53. See text for details.

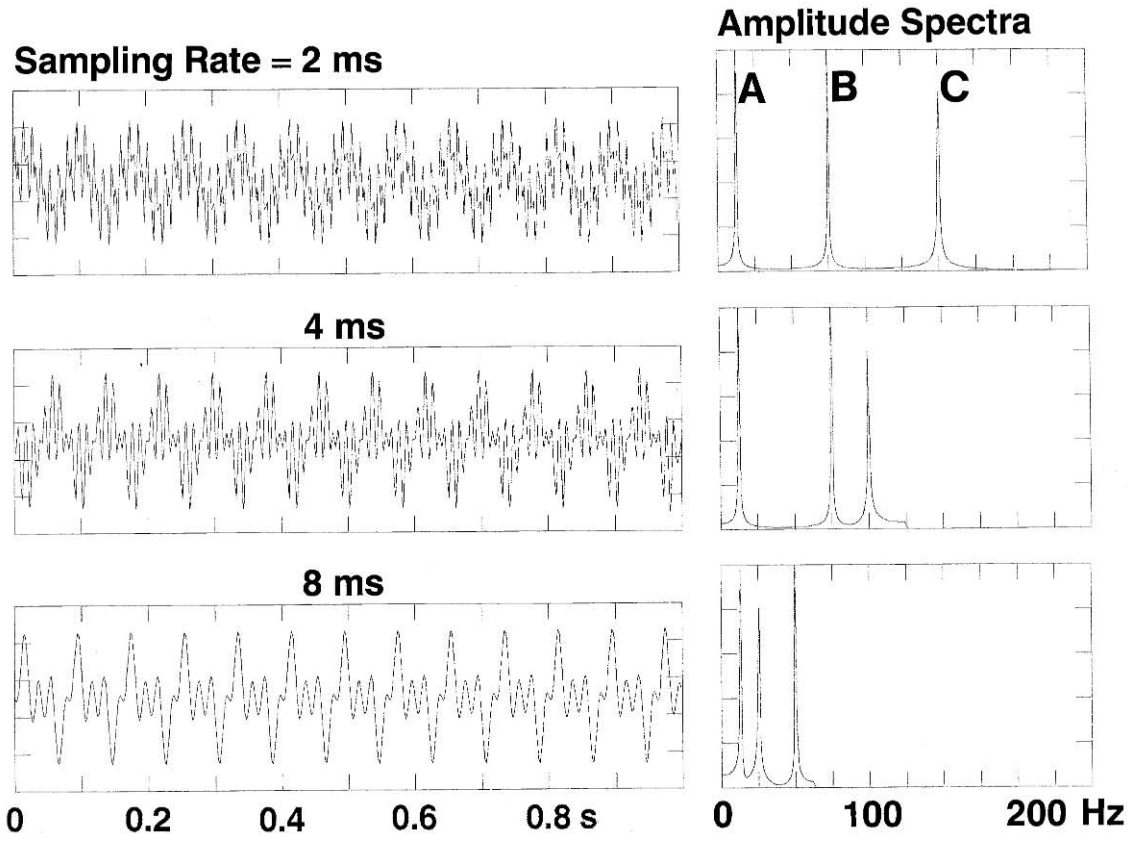

FIG. 1.E-1. A signal with three frequency components, A, B, and C, sampled at three different rates, 2, 4, and 8 ms. Frequency aliasing occurs at coarser sampling intervals (see Exercise 1-1).

The stacked sections in Figures 1.5-48 through 1.5-53 were created based on the following poststack processing sequence:

(a) Figure 1.5-48: stack as in (f) of the prestack processing sequence described above with the additional step for poststack spiking deconvolution; note the increase in vertical resolution as a result of wavelet compression.
(b) Figure 1.5-49: stack as in (a) with the additional step for time-variant spectral whitening to account for nonstationarity.
(c) Figure 1.5-50: stack as in (b) with the additional step for wide bandpass filtering.
(d) Figure 1.5-51: stack as in (c) with the additional step for AGC scaling.
(e) Figure 1.5-52: stack as in (d) with the additional step for attenuation of random noise uncorrelated from trace to trace using $f-x$ deconvolution (Section 6.5).
(f) Figure 1.5-53: migrated stack as in (c) with the additional step for AGC scaling.

Scan through the stacked sections starting with Figure 1.5-42 and observe the effect of each processing step on the result. Additonally, examine the spectra labeled as (a) through (f) in Figure 1.5-54 which correspond to the stacked sections in Figures 1.5-42 through Figure 1.5-47 involving the prestack sequence, respectively, and the spectra labeled as (a) through (f) in Figure 1.5-55 which correspond to the stacked sections in Figures 1.5-48 through Figure 1.5-53 involving the poststack sequence, respectively. Observe the change in the spectral content induced by each process, and note that the ultimate objectives in processing are aimed at preserving the bandwidth of the recorded signal and flattening the spectrum within the signal passband to attain the maximum possible vertical and lateral resolutions.

EXERCISES

Exercise 1-1. Refer to Figure 1.E-1. Using equation (1-2), compute the alias frequencies at 4- and 8-ms

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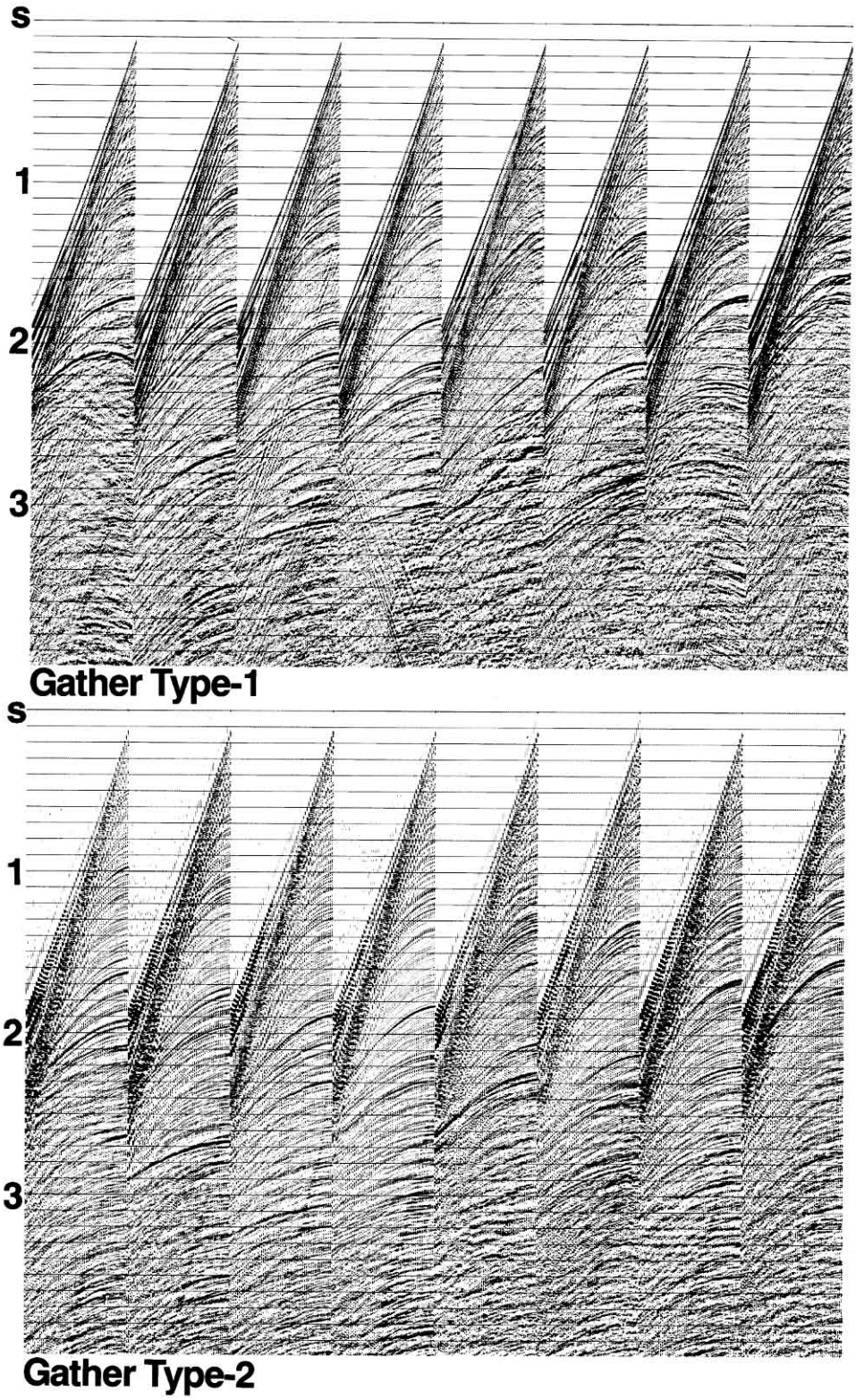

FIG. 1.E-2. A field data set displayed in two different domains — common-shot and common-midpoint (see Exercise 1-12).

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sampling intervals for the three frequency components — A, B and C from the amplitude spectrum of the 2-ms time series. Confirm the results of your computation by the amplitude spectra.

Exercise 1-2. Using the hyperbolic traveltime equation, compute the average velocity down to reflector A in Figure 1.3-8. Assume a constant velocity between A and the surface. The required information about Record 8 is in Table 1-13.

Exercise 1-3. What is the dispersive event C in Figure 1.3-8?

Exercise 1-4. Refer to the walk-away noise test record in Figure 1.3-19. Measure the phase velocity ($\Delta x/\Delta t$) of the ground-roll energy at location A_1. Also measure the dominant frequency at the same location. Then, estimate the dominant wavelength (velocity/dominant frequency) of the ground roll. The receiver array length needed to suppress this energy in the field should be equal to or greater than the longest noise wavelength. The required information about Record 19 that you need for your computations is in Table 1-13.

Exercise 1-5. Measure the group velocity (x/t) of ground-roll energy A in Figure 1.3-25. The required information for Record 25 is in Table 1-13.

Exercise 1-6. Refer to the shot record in Figure 1.3-27. The near-offset 28 channels were recorded using a 50-m group interval. What is the group interval for the remaining 64 channels? The required information for Record 27 is in Table 1-13.

Exercise 1-7. What is event A in Figure 1.3-29? Are events C, D, and E multiples of B?

Exercise 1-8. Refer to Figure 1.3-30. Compute the water velocity using both the direct arrivals A and the water-bottom reflection B. Use one-way time for A and two-way time for B. The required information for Record 30 is in Table 1-13.

Exercise 1-9. Identify events A, B, C, D, E, and F in Figure 1.3-33.

Exercise 1-10. Identify events A, B, C, D, and E in Figure 1.3-34.

Exercise 1-11. Gain application involves multiplying the gain function with the seismic trace (Figure 1.4-8). Use the appropriate entry in Table A-1 to describe the effect of the gain application in the frequency domain.

Exercise 1-12. From Figure 1.E-2, identify the common-shot and common-midpoint gathers.

Exercise 1-13. Use equation (1-10) to compute fold n_f for each of the following recording geometries:

Number of Channels	Shot Spacing, m	Receiver Spacing, m
240	25	25
240	50	25
240	100	25
240	25	50

Exercise 1-14. Consider the recording geometry in Figure 1.5-8. Sketch the traveltime curves on a common-shot gather associated with point scatterers (a) beneath the cable, (b) behind, and (c) in front of the cable. Assume all scatterers are on the plane of recording.

Exercise 1-15. Suppose that the shot associated with gather 1 in Figure 1.5-10 is missing. Identify the midpoints that are affected by this missing shot; that is, the midpoints with a lower fold of coverage. Suppose the receiver associated with gather 2 in Figure 1.5-10 is missing. Identify the midpoints that are affected by this.

Exercise 1-16. Prove the shifting, scaling, and differentiation rules applied to the Fourier transform of a function given by the entires (1), (2), and (3) in Table A-1.

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Appendix A
A MATHEMATICAL REVIEW OF THE FOURIER TRANSFORM

A.1 The 1-D Fourier Transform

Given a continuous function $x(t)$ of a single variable t, its Fourier transform is defined by the integral

$$X(\omega) = \int_{-\infty}^{+\infty} x(t)\ \exp(-i\omega t)\, dt, \tag{A-1}$$

where ω is the Fourier dual of the variable t. If t signifies time, then ω is angular frequency. The temporal frequency f is related to the angular frequency ω by $\omega = 2\pi f$.

The Fourier transform is reversible; that is, given $X(\omega)$, the corresponding time function is

$$x(t) = \int_{-\infty}^{+\infty} X(\omega)\ \exp(i\omega t)\, d\omega. \tag{A-2}$$

Throughout this book, the following sign convention is used for the Fourier transform. For the forward transform, the sign of the argument in the exponent is negative if the variable is time and positive if the variable is space. Of course, the inverse transform has the opposite sign used in the respective forward transform. For convenience, the scale factor 2π in equations (A-1) and (A-2) are omitted.

Generally, $X(\omega)$ is a complex function. By using the properties of the complex functions, $X(\omega)$ is expressed as two other functions of frequency

$$X(\omega) = A(\omega)\ \exp[i\phi(\omega)], \tag{A-3}$$

where $A(\omega)$ and $\phi(\omega)$ are the amplitude and phase spectra, respectively. They are computed by the following equations:

$$A(\omega) = \sqrt{X_r^2(\omega) + X_i^2(\omega)} \tag{A-4a}$$

and

$$\phi(\omega) = \tan^{-1} \frac{X_i(\omega)}{X_r(\omega)}, \tag{A-4b}$$

where $X_r(\omega)$ and $X_i(\omega)$ are the real and imaginary parts of the Fourier transform $X(\omega)$. When $X(\omega)$ is expressed in terms of its real and imaginary components

$$X(\omega) = X_r(\omega) + iX_i(\omega), \tag{A-5}$$

and is compared with equation (A-3), note that

$$X_r(\omega) = A(\omega)\cos\phi(\omega), \tag{A-6a}$$

and

$$X_i(\omega) = A(\omega)\sin\phi(\omega). \tag{A-6b}$$

We now consider two functions — $x(t)$ and $f(t)$. Listed in Table A-1 are basic theorems that are useful in various applications of the Fourier transform.

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Table A-1. Fourier transform theorems (Bracewell, 1965).

Operation	*Time Domain*	*Frequency Domain*
(1) Shifting	$x(t-\tau)$	$\exp(-i\omega\tau)X(\omega)$
(2) Scaling	$x(at)$	$\lvert a\rvert^{-1}X(\omega/a)$
(3) Differentiation	$dx(t)/dt$	$i\omega X(\omega)$
(4) Addition	$f(t)+x(t)$	$F(\omega)+X(\omega)$
(5) Multiplication	$f(t)\,x(t)$	$F(\omega)*X(\omega)$
(6) Convolution	$f(t)*x(t)$	$F(\omega)\,X(\omega)$
(7) Autocorrelation	$x(t)*x(-t)$	$\lvert X(\omega)\rvert^2$
(8) Parseval's theorem	$\int \lvert x(t)\rvert^2 dt$	$\int \lvert X(\omega)\rvert^2 d\omega$

$*$ denotes convolution.

Proofs of these theorems can be found in the classic reference on Fourier transforms by Bracewell (1965). Also, some of the proofs are left to the exercises at the end of this chapter. Here, we shall derive the convolutional relation (6) for continuous functions, and the same relation for discrete functions in Section A.2. Consider convolution of two functions $x(t)$ and $f(t)$ with their Fourier transforms $X(\omega)$ and $F(\omega)$, respectively,

$$y(t) = f(t) * x(t), \tag{A-7a}$$

which is explicitly given by the integral

$$y(t) = \int_{-\infty}^{+\infty} f(t-t')\,x(t')\,dt'. \tag{A-7b}$$

The Fourier transform of the resulting function $y(t)$ is

$$Y(\omega) = \int_{-\infty}^{+\infty} y(t)\ \exp(-i\omega t)\,dt. \tag{A-8a}$$

Substitute the convolution integral of equation (A-7b) into equation (A-8a)

$$Y(\omega) = \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} f(t-t')\,x(t')\,dt'\right] \exp(-i\omega t)\,dt, \tag{A-8b}$$

and interchange the two integrals

$$Y(\omega) = \int_{-\infty}^{+\infty} x(t') \left[\int_{-\infty}^{+\infty} f(t-t')\ \exp(-i\omega t)\,dt\right] dt'. \tag{A-8c}$$

From the shift theorem given by entry (1) of Table A-1, we have

$$\int_{-\infty}^{+\infty} f(t-t')\ \exp(-i\omega t)\,dt = F(\omega)\ \exp(-i\omega t'). \tag{A-9}$$

Use this relation in equation (A-8c) to get

$$Y(\omega) = \int_{-\infty}^{+\infty} x(t') \left[F(\omega) \exp(-i\omega t') \right] dt', \qquad (A-10a)$$

then rearrange the terms to obtain

$$Y(\omega) = F(\omega) \int_{-\infty}^{+\infty} x(t') \exp(-i\omega t')\, dt'. \qquad (A-10b)$$

Note that the integral in equation (A-10b) is the Fourier transform of $x(t)$, and therefore,

$$Y(\omega) = F(\omega)\, X(\omega), \qquad (A-11)$$

which is the desired result given by entry (6) of Table A-1.

A.2 The z-Transform

A discrete time function is called a *time series*. When digitized, the continuous function $x(t)$ takes the form

$$x(t) = \sum_k x_k \delta(t - k\Delta t), \quad k = 0, 1, 2, \ldots, \qquad (A-12)$$

where Δt is the sampling interval and $\delta(t - k\Delta t)$ is the Dirac delta function. The discrete equivalent of the Fourier integral given by equation (A-1) is written as a summation

$$X(\omega) = \sum_k x_k \exp(-i\omega k\Delta t), \quad k = 0, 1, 2, \ldots. \qquad (A-13)$$

A new variable $z = \exp(-i\omega\Delta t)$ now is defined. By substituting into equation (A-13) and explicitly writing the summation, we get

$$X(z) = x_0 + x_1 z + x_2 z^2 + \ldots. \qquad (A-14)$$

Function $X(z)$ in equation (A-14) is called the z-transform of $x(t)$. It is a polynomial of the z variable. The power of z represents the time delay of the discrete samples in the time series $x(t)$.

We now show that convolution of two time series is equivalent to the multiplication of their $z-$transforms. Consider two discrete time series — $x(t) : (x_0, x_1, x_2)$ and $f(t) : (f_0, f_1)$. Convolution of the two series is obtained by using Table 1-5. The results of the convolutional process are displayed in Table A-2.

The discrete output series $y(t) : (y_0, y_1, y_2, y_3)$ is given by

$$\begin{aligned} y_0 &= f_0 x_0 \\ y_1 &= f_1 x_0 + f_0 x_1 \\ y_2 &= f_1 x_1 + f_0 x_2 \\ y_3 &= f_1 x_2. \end{aligned} \qquad (A-15)$$

The $z-$transforms of the two input series are expressed by

$$X(z) = x_0 + x_1 z + x_2 z^2, \qquad (A-16a)$$

and

$$F(z) = f_0 + f_1 z. \qquad (A-16b)$$

By multiplying the two polynomials of equations (A-16a) and (A-16b), we obtain

$$X(z)F(z) = f_0 x_0 + (f_1 x_0 + f_0 x_1)z + (f_1 x_1 + f_0 x_2)z^2 + (f_1 x_2)z^3. \qquad (A-17)$$

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Table A-2. Convolution of (f_0, f_1) with (x_0, x_1, x_2).

	x_0	x_1	x_2		Output
f_1	f_0				y_0
	f_1	f_0			y_1
		f_1	f_0		y_2
			f_1	f_0	y_3

By comparing the coefficients of the polynomial given by equation (A-17) with the output of convolution in equations (A-15), we find that if

$$y(t) = f(t) * x(t), \tag{A-18a}$$

then,

$$Y(z) = F(z)\, X(z), \tag{A-18b}$$

and, since $z = \exp(-i\omega\Delta t)$,

$$Y(\omega) = F(\omega)\, X(\omega). \tag{A-18c}$$

A.3 The 2-D Fourier Transform

The 2-D Fourier transform of a 2-D function, such as a wavefield $P(x, t)$, is given by

$$P(k_x, \omega) = \int\int P(x, t)\ \exp(ik_x x - i\omega t)\, dx\, dt. \tag{A-19}$$

Function $P(x, t)$ can be reconstructed from $P(k_x, \omega)$ by the 2-D inverse Fourier transform:

$$P(x, t) = \int\int P(k_x, \omega)\ \exp(-ik_x x + i\omega t)\, dk_x\, d\omega. \tag{A-20}$$

The integral given by equation (A-19) is evaluated in two steps. First, by Fourier transforming in t,

$$P(x, \omega) = \int P(x, t)\ \exp(-i\omega t)\, dt, \tag{A-21a}$$

then by Fourier transforming in x, we get the 2-D transform:

$$P(k_x, \omega) = \int P(x, \omega)\ \exp(ik_x)\, dx. \tag{A-21b}$$

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